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NOVEL MODULATORS OF THE 5-HYDROXYTRYPTAMINE RECEPTOR 7 AND THEIR METHOD OF USE

Abstract

Pharmaceutical compositions of the invention comprise functionalized lactone derivatives having a disease-modifying action in the treatment of diseases associated with dysregulation of 5-hydroxytryptamine receptor 7 activity.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/709,847, filed Mar. 31, 2022, which is a continuation application of U.S. patent application Ser. No. 17/015,696, filed on Sep. 9, 2020, which issued as U.S. Pat. No. 11,319,327 on May 3, 2022, which is a continuation application of U.S. patent application Ser. No. 16/349,811, filed on May 14, 2019, which issued as U.S. Pat. No. 10,858,368 on Dec. 8, 2020, which is a 35 U.S.C. § 371 National Stage Application of International Application No. PCT/US2017/061677, filed Nov. 15, 2017, which claims the benefit of U.S. provisional application No. 62/422,344, filed Nov. 15, 2016, which are each herein incorporated by reference in their entireties.

FIELD OF INVENTION

[0003] Embodiments of the invention are directed to novel compounds useful as modulators of 5-hydroxytryptamine receptor 7 (5-HT.sub.7) activity and their method of use. Embodiments are further directed to a novel chemotype useful for the treatment diseases that are associated with dysregulation of 5-hydroxytryptamine receptor 7 activity.

BACKGROUND OF THE INVENTION

[0004] Serotonin was discovered in the late 1940s and is present in both the peripheral and central nervous systems [Physiol. Res, 60 (2011) 15-25; Psychopharmacology 213 (2011) 167-169]. Serotonin or 5-hydroxytryptamine (5-HT) is a monoamine neurotransmitter of the indolalkylamine group that acts at synapses of nerve cells. Seven distinct families of serotonin receptors have been identified and at least 20 subpopulations have been cloned on the basis of sequence similarity, signal transduction coupling and pharmacological characteristics. The seven families of 5-HT receptor are named 5-HT.sub.1, 5-HT.sub.2, 5-HT.sub.3, 5-HT.sub.4, 5-HT.sub.5, 5-HT.sub.6, and 5-HT.sub.7 and each of these receptors in turn has subfamilies or subpopulations. The signal transduction mechanism for all seven families have been studied and it is known that activation of 5-HT.sub.1 and 5-HT.sub.5 receptors causes a decrease in intracellular cAMP whereas activation of 5-HT.sub.2, 5-HT.sub.3, 5-HT.sub.4, 5-HT.sub.6, and 5-HT.sub.7 results in an increase in intracellular IP3 and DAG. The 5-HT pathways in the brain are important targets for drug development in the area of CNS disorders. The neurotransmitter binds to its a G-protein coupled receptor and is involved in a wide variety of actions including cognition, mood, anxiety, attention, appetite, cardiovascular function, vasoconstriction, sleep (ACS Medicinal Chemistry Letters, 2011, 2, 929-932; Physiological Research, 2011, 60, 15-25), inflammatory bowel disease (IBD), and intestinal inflammation (WO 2012058769, Khan, W. I., et. al. Journal of Immunology, 2013, 190, 4795-4804), epilepsy, seizure disorders (Epilepsy Research (2007) 75, 39), drug addiction, and alcohol addiction (Hauser, S. R. et. al. Frontiers in Neuroscience, 2015, 8, 1-9) among others.

BRIEF SUMMARY OF THE INVENTION

[0005] The present invention is directed toward novel 5-hydroxytryptamine receptor 7 (5-HT7) activity modulators, compounds of formula (I),

##STR00001##

[0006] Including hydrates, solvates, pharmaceutically acceptable salts, prodrugs and complexes

thereof, wherein: [0007] A is selected from a group consisting of
##STR00002## [0008] X is selected from the group consisting of O, S, SO, SO.sub.2, NR; [0009]
n.sup.1 is 0, 1, 2; [0010] n.sup.2 is 0, 1, 2; [0011] R is selected from the group consisting of H,
C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl,
##STR00003##
COR.sup.2, CO.sub.2R.sup.2a, CONR.sup.2bR.sup.2c, SO.sub.2NR.sup.2bR.sup.2c, and
SO.sub.2R.sup.2a; R.sup.1a, R.sup.1b, R.sup.1c, R.sup.1d, and R.sup.1e are at each occurrence
independently selected from the group consisting of H, OH, NO.sub.2, halogen, CN, C.sub.1-6
linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, C.sub.1-6 linear alkoxy, C.sub.3-7
branched alkoxy, C.sub.3-7 cycloalkoxy, C.sub.1-6 linear haloalkyl, C.sub.3-7 branched haloalkyl,
C.sub.1-6 linear haloalkoxy, —S(C.sub.1-6 linear alkyl), S(C.sub.3-7 branched alkyl), —
S(C.sub.3-7 cycloalkyl), COR.sup.6, CO.sub.2R.sup.7, CONR.sup.8aR.sup.8b,
SO.sub.2NR.sup.8aR.sup.8b, NR.sup.9aR.sup.9b, NR.sup.9aCOR.sup.10,
NR.sup.9aSO.sub.2R.sup.11, and NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b; [0012] R.sup.2 is
selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and
C.sub.3-7 cycloalkyl; [0013] R.sup.2a is selected from the group consisting of C.sub.1-6 linear
alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0014] R.sup.2b is selected from the
group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl;
[0015] R.sup.2c is selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7
branched alkyl, and C.sub.3-7 cycloalkyl; [0016] R.sup.21 is selected from the group consisting of
C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, C.sub.1-6 linear haloalkyl,
C.sub.3-7 branched haloalkyl, —(CH.sub.2).sub.qCN, —(CH.sub.2).sub.qSO.sub.2R.sup.13, —
(CH.sub.2).sub.qOR.sup.14,
##STR00004## [0017] R.sup.3 is selected from a group consisting of C.sub.1-6 linear alkyl,
C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, optionally substituted aryl,
##STR00005## [0018] R.sup.4 is an optionally substituted aryl; [0019] R.sup.5a and R.sup.5b are
each independently optionally substituted aryl; [0020] R.sup.6 is at each occurrence independently
selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and
C.sub.3-7 cycloalkyl; [0021] R.sup.7 is at each occurrence independently selected from the group
consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0022]
R.sup.8 is at each occurrence independently selected from the group consisting of H, C.sub.1-6
linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0023] R.sup.8b is at each
occurrence independently selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7
branched alkyl, and C.sub.3-7 cycloalkyl; [0024] R.sup.9a is at each occurrence independently
selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and
C.sub.3-7 cycloalkyl; [0025] R.sup.9b is at each occurrence independently selected from the group
consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0026]
R.sup.10 is at each occurrence independently selected from the group consisting of H, C.sub.1-6
linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0027] R.sup.11 is at each
occurrence independently selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7
branched alkyl, and C.sub.3-7 cycloalkyl; [0028] R.sup.12a is at each occurrence independently
selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7
cycloalkyl; [0029] R.sup.12b is at each occurrence independently selected from the group
consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0030]
R.sup.13 is selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl,
and C.sub.3-7 cycloalkyl; [0031] R.sup.14 is selected from the group consisting of C.sub.1-6 linear
alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0032] n is 1, 2, or 3; [0033] m is 1 or 2;
[0034] and q is 1, 2, or 3;
[0035] The present invention further relates to compositions comprising: an effective amount of
one or more compounds according to the present invention and an excipient.

[0036] The present invention also relates to a method for treating or preventing diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity, including, for example, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, and alcohol addiction said method comprising administering to a subject an effective amount of a compound or composition according to the present invention.

[0037] The present invention yet further relates to a method for treating or preventing diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity, including, for example, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar, disorder inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, and alcohol addiction wherein said method comprises administering to a subject a composition comprising an effective amount of one or more compounds according to the present invention and an excipient.

[0038] The present invention also relates to a method for treating or preventing diseases or conditions associated with circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, alcohol addiction and diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity. Said methods comprise administering to a subject an effective amount of a compound or composition according to the present invention.

[0039] The present invention yet further relates to a method for treating or preventing diseases or conditions associated with circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, and bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, alcohol addiction and diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity, wherein said method comprises administering to a subject a composition comprising an effective amount of one or more compounds according to the present invention and an excipient.

[0040] The present invention also relates to a method for treating or preventing diseases or conditions associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. Said methods comprise administering to a subject an effective amount of a compound or composition according to the present invention.

[0041] The present invention yet further relates to a method for treating or preventing diseases or conditions associated with dysregulation of 5-hydroxytryptamine receptor 7 activity, wherein said

method comprises administering to a subject a composition comprising an effective amount of one or more compounds according to the present invention and an excipient.

[0042] The present invention further relates to a process for preparing the 5-hydroxytryptamine receptor 7 activity modulators of the present invention.

[0043] These and other objects, features, and advantages will become apparent to those of ordinary skill in the art from a reading of the following detailed description and the appended claims. All percentages, ratios and proportions herein are by weight, unless otherwise specified. All temperatures are in degrees Celsius (° C.) unless otherwise specified. All documents cited are in relevant part, incorporated herein by reference; the citation of any document is not to be construed as an admission that it is prior art with respect to the present invention.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0044] There is evidence that suggests a role for the 5-HT₇ receptor in a number of medical disorders. 5-HT₇ receptor activity modulators are likely to have a beneficial effect on patients suffering from these disorders. The disorders in which 5-HT₇ dysregulation plays a role and modulation of 5-HT₇ receptor activity by a therapeutic agent may be a viable approach to therapeutic relief include, but are not limited to, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine (Vanhoenacker, P. et al. Trends in Pharmacological Sciences, 2000, 21, 2, 70-77), neuropathic pain, peripheral pain, allodynia (EP1875899), thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder (WO20100197700) attention deficit/hyperactivity disorder (ADHD) (WO20100069390), anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder (WO20040229874), inflammatory bowel disease (IBD), intestinal inflammation (WO 2012058769, Khan, W. I., et. al. Journal of Immunology, 2013, 190, 4795-4804), epilepsy, seizure disorders (Epilepsy Research (2007) 75, 39), drug addiction, and alcohol addiction (Hauser, S. R. et. al. Frontiers in Neuroscience, 2015, 8, 1-9).

[0045] There is a long felt need for new 5-HT₇ modulators that will provide therapeutic relief from patients suffering from diseases associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. The invention addresses the need to identify novel 5-HT₇ modulators capable of treating disease associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. The present invention addresses the need to develop new therapeutic agents for the treatment and prevention of circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation epilepsy, seizure disorders, drug addiction, and alcohol addiction.

[0046] The 5-hydroxytryptamine receptor 7 activity modulators of the present invention are capable of treating and preventing diseases associated with dysregulation of 5-hydroxytryptamine receptor 7 activity, for example circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease

(IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, and alcohol addiction. It has been discovered that the 5-hydroxytryptamine receptor 7 play a role in a number of medical disorders, and therefore, 5-HT.sub.7 receptor activity modulators are likely to have a beneficial effect on patients suffering from these disorders. The disorders in which 5-HT.sub.7 dysregulation plays a role and modulation of 5-HT.sub.7 receptor activity by a therapeutic agent may be a viable approach to therapeutic relief include, but are not limited to, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine (Vanhoenacker, P. et. al. Trends in Pharmacological Sciences, 2000, 21, 2, 70-77), neuropathic pain, peripheral pain, allodynia (EP1875899), thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder (WO20100197700) attention deficit/hyperactivity disorder (ADHD) (WO2010069390), anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder (WO20040229874), inflammatory bowel disease (IBD), intestinal inflammation (WO 2012058769) epilepsy, seizure disorders (Epilepsy Research (2007) 75, 39), drug addiction, and alcohol addiction (Hauser, S. R. et. al. Frontiers in Neuroscience, 2015, 8, 1-9).

[0047] Without wishing to be limited by theory, it is believed that 5-hydroxytryptamine receptor 7 receptor activity modulators of the present invention can ameliorate, abate, otherwise cause to be controlled, diseases associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. The diseases include, but are not limited to circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, and alcohol addiction.

[0048] Throughout the description, where compositions are described as having, including, or comprising specific components, or where processes are described as having, including, or comprising specific process steps, it is contemplated that compositions of the present teachings also consist essentially of, or consist of, the recited components, and that the processes of the present teachings also consist essentially of, or consist of, the recited processing steps.

[0049] In the application, where an element or component is said to be included in and/or selected from a list of recited elements or components, it should be understood that the element or component can be any one of the recited elements or components and can be selected from a group consisting of two or more of the recited elements or components.

[0050] The use of the singular herein includes the plural (and vice versa) unless specifically stated otherwise. In addition, where the use of the term “about” is before a quantitative value, the present teachings also include the specific quantitative value itself, unless specifically stated otherwise.

[0051] It should be understood that the order of steps or order for performing certain actions is immaterial so long as the present teachings remain operable. Moreover, two or more steps or actions can be conducted simultaneously.

[0052] As used herein, the term “halogen” shall mean chlorine, bromine, fluorine and iodine.

[0053] As used herein, unless otherwise noted, “alkyl” and/or “aliphatic” whether used alone or as part of a substituent group refers to straight and branched carbon chains having 1 to 20 carbon atoms or any number within this range, for example 1 to 6 carbon atoms or 1 to 4 carbon atoms. Designated numbers of carbon atoms (e.g. C.sub.1-6) shall refer independently to the number of carbon atoms in an alkyl moiety or to the alkyl portion of a larger alkyl-containing substituent. Non-limiting examples of alkyl groups include methyl, ethyl, n-propyl, iso-propyl, n-butyl, sec-butyl, iso-butyl, tert-butyl, and the like. Alkyl groups can be optionally substituted. Non-limiting

examples of substituted alkyl groups include hydroxymethyl, chloromethyl, trifluoromethyl, aminomethyl, 1-chloroethyl, 2-hydroxyethyl, 1,2-difluoroethyl, 3-carboxypropyl, and the like. In substituent groups with multiple alkyl groups such as (C.sub.1-6alkyl).sub.2amino, the alkyl groups may be the same or different.

[0054] As used herein, the terms “alkenyl” and “alkynyl” groups, whether used alone or as part of a substituent group, refer to straight and branched carbon chains having 2 or more carbon atoms, preferably 2 to 20, wherein an alkenyl chain has at least one double bond in the chain and an alkynyl chain has at least one triple bond in the chain. Alkenyl and alkynyl groups can be optionally substituted. Nonlimiting examples of alkenyl groups include ethenyl, 3-propenyl, 1-propenyl (also 2-methylethenyl), isopropenyl (also 2-methylethen-2-yl), buten-4-yl, and the like. Nonlimiting examples of substituted alkenyl groups include 2-chloroethenyl (also 2-chlorovinyl), 4-hydroxybuten-1-yl, 7-hydroxy-7-methyloct-4-en-2-yl, 7-hydroxy-7-methyloct-3,5-dien-2-yl, and the like. Nonlimiting examples of alkynyl groups include ethynyl, prop-2-ynyl (also propargyl), propyn-1-yl, and 2-methyl-hex-4-yn-1-yl. Nonlimiting examples of substituted alkynyl groups include, 5-hydroxy-5-methylhex-3-ynyl, 6-hydroxy-6-methylhept-3-yn-2-yl, 5-hydroxy-5-ethylhept-3-ynyl, and the like.

[0055] As used herein, “cycloalkyl,” whether used alone or as part of another group, refers to a non-aromatic carbon-containing ring including cyclized alkyl, alkenyl, and alkynyl groups, e.g., having from 3 to 14 ring carbon atoms, preferably from 3 to 7 or 3 to 6 ring carbon atoms, or even 3 to 4 ring carbon atoms, and optionally containing one or more (e.g., 1, 2, or 3) double or triple bond. Cycloalkyl groups can be monocyclic (e.g., cyclohexyl) or polycyclic (e.g., containing fused, bridged, and/or spiro ring systems), wherein the carbon atoms are located inside or outside of the ring system. Any suitable ring position of the cycloalkyl group can be covalently linked to the defined chemical structure. Cycloalkyl rings can be optionally substituted. Nonlimiting examples of cycloalkyl groups include: cyclopropyl, 2-methyl-cyclopropyl, cyclopropenyl, cyclobutyl, 2,3-dihydroxycyclobutyl, cyclobutenyl, cyclopentyl, cyclopentenyl, cyclopentadienyl, cyclohexyl, cyclohexenyl, cycloheptyl, cyclooctanyl, decalanyl, 2,5-dimethylcyclopentyl, 3,5-dichlorocyclohexyl, 4-hydroxycyclohexyl, 3,3,5-trimethylcyclohex-1-yl, octahydropentalenyl, octahydro-1H-indenyl, 3a,4,5,6,7,7a-hexahydro-3H-inden-4-yl, decahydroazulenyl; bicyclo[6.2.0]decanyl, decahydronaphthalenyl, and dodecahydro-1H-fluorenyl. The term “cycloalkyl” also includes carbocyclic rings which are bicyclic hydrocarbon rings, non-limiting examples of which include, bicyclo-[2.1.1]hexanyl, bicyclo[2.2.1]heptanyl, bicyclo[3.1.1]heptanyl, 1,3-dimethyl[2.2.1]heptan-2-yl, bicyclo[2.2.2]octanyl, and bicyclo[3.3.3]undecanyl.

[0056] “Haloalkyl” is intended to include both branched and straight-chain saturated aliphatic hydrocarbon groups having the specified number of carbon atoms, substituted with 1 or more halogen. Haloalkyl groups include perhaloalkyl groups, wherein all hydrogens of an alkyl group have been replaced with halogens (e.g., —CF.sub.3, —CF.sub.2CF.sub.3). Haloalkyl groups can optionally be substituted with one or more substituents in addition to halogen. Examples of haloalkyl groups include, but are not limited to, fluoromethyl, dichloroethyl, trifluoromethyl, trichloromethyl, pentafluoroethyl, and pentachloroethyl groups.

[0057] The term “alkoxy” refers to the group —O-alkyl, wherein the alkyl group is as defined above. Alkoxy groups optionally may be substituted. The term C.sub.3-C.sub.6 cyclic alkoxy refers to a ring containing 3 to 6 carbon atoms and at least one oxygen atom (e.g., tetrahydrofuran, tetrahydro-2H-pyran). C.sub.3-C.sub.6 cyclic alkoxy groups optionally may be substituted.

[0058] The term “haloalkoxy” refers to the group —O-haloalkyl, wherein the haloalkyl group is as defined above. Examples of haloalkoxy groups include, but are not limited to, fluoromethoxy, difluoromethoxy, trifluoromethoxy, and pentafluoroethoxyl.

[0059] The term “aryl,” wherein used alone or as part of another group, is defined herein as an unsaturated, aromatic monocyclic ring of 6 carbon members or to an unsaturated, aromatic polycyclic ring of from 10 to 14 carbon members. Aryl rings can be, for example, phenyl or

naphthyl ring each optionally substituted with one or more moieties capable of replacing one or more hydrogen atoms. Non-limiting examples of aryl groups include: phenyl, naphthyl-1-yl, naphthyl-2-yl, 4-fluorophenyl, 2-hydroxyphenyl, 3-methylphenyl, 2-amino-4-fluorophenyl, 2-(N,N-diethylamino)phenyl, 2-cyanophenyl, 2,6-di-tert-butylphenyl, 3-methoxyphenyl, 8-hydroxynaphthyl-2-yl, 4,5-dimethoxynaphthyl-1-yl, and 6-cyano-naphthyl-1-yl. Aryl groups also include, for example, phenyl or naphthyl rings fused with one or more saturated or partially saturated carbon rings (e.g., bicyclo[4.2.0]octa-1,3,5-trienyl, indanyl), which can be substituted at one or more carbon atoms of the aromatic and/or saturated or partially saturated rings.

[0060] The term “arylalkyl” or “aralkyl” refers to the group-alkyl-aryl, where the alkyl and aryl groups are as defined herein. Aralkyl groups of the present invention are optionally substituted. Examples of arylalkyl groups include, for example, benzyl, 1-phenylethyl, 2-phenylethyl, 3-phenylpropyl, 2-phenylpropyl, fluorenylmethyl and the like.

[0061] The terms “heterocyclic” and/or “heterocycle” and/or “heterocyclyl,” whether used alone or as part of another group, are defined herein as one or more ring having from 3 to 20 atoms wherein at least one atom in at least one ring is a heteroatom selected from nitrogen (N), oxygen (O), or sulfur (S), and wherein further the ring that includes the heteroatom is non-aromatic. In heterocycle groups that include 2 or more fused rings, the non-heteroatom bearing ring may be aryl (e.g., indolyl, tetrahydroquinolyl, chromanyl). Exemplary heterocycle groups have from 3 to 14 ring atoms of which from 1 to 5 are heteroatoms independently selected from nitrogen (N), oxygen (O), or sulfur (S). One or more N or S atoms in a heterocycle group can be oxidized. Heterocycle groups can be optionally substituted.

[0062] Non-limiting examples of heterocyclic units having a single ring include: diazirinyl, aziridinyl, urazolyl, azetidyl, pyrazolidinyl, imidazolidinyl, oxazolidinyl, isoxazolyl, isoxazolyl, thiazolidinyl, isothiazolyl, isothiazolyl oxathiazolidinonyl, oxazolidinonyl, hydantoinyl, tetrahydrofuranyl, pyrrolidinyl, morpholinyl, piperazinyl, piperidinyl, dihydropyranyl, tetrahydropyranyl, piperidin-2-onyl (valerolactam), 2,3,4,5-tetrahydro-1H-azepinyl, 2,3-dihydro-1H-indole, and 1,2,3,4-tetrahydro-quinoline. Non-limiting examples of heterocyclic units having 2 or more rings include: hexahydro-1H-pyrrolizyl, 3a,4,5,6,7,7a-hexahydro-1H-benzo[d]imidazolyl, 3a,4,5,6,7,7a-hexahydro-1H-indolyl, 1,2,3,4-tetrahydroquinolyl, chromanyl, isochromanyl, indolyl, isoindolyl, and decahydro-1H-cycloocta[b]pyrrolyl.

[0063] The term “heteroaryl,” whether used alone or as part of another group, is defined herein as one or more rings having from 5 to 20 atoms wherein at least one atom in at least one ring is a heteroatom chosen from nitrogen (N), oxygen (O), or sulfur (S), and wherein further at least one of the rings that includes a heteroatom is aromatic. In heteroaryl groups that include 2 or more fused rings, the non-heteroatom bearing ring may be a carbocycle (e.g., 6,7-Dihydro-5H-cyclopentapyrimidine) or aryl (e.g., benzofuranyl, benzothiophenyl, indolyl). Exemplary heteroaryl groups have from 5 to 14 ring atoms and contain from 1 to 5 ring heteroatoms independently selected from nitrogen (N), oxygen (O), or sulfur (S). One or more N or S atoms in a heteroaryl group can be oxidized. Heteroaryl groups can be substituted. Non-limiting examples of heteroaryl rings containing a single ring include: 1,2,3,4-tetrazolyl, [1,2,3]triazolyl, [1,2,4]triazolyl, triazinyl, thiazolyl, 1H-imidazolyl, oxazolyl, furanyl, thiophenyl, pyrimidinyl, 2-phenylpyrimidinyl, pyridinyl, 3-methylpyridinyl, and 4-dimethylaminopyridinyl. Non-limiting examples of heteroaryl rings containing 2 or more fused rings include: benzofuranyl, benzothiophenyl, benzoxazolyl, benzthiazolyl, benztriazolyl, cinnolyl, naphthyridinyl, phenanthridinyl, 7H-purinyl, 9H-purinyl, 6-amino-9H-purinyl, 5H-pyrrolo[3,2-d]pyrimidinyl, 7H-pyrrolo[2,3-d]pyrimidinyl, pyrido[2,3-d]pyrimidinyl, 2-phenylbenzo[d]thiazolyl, 1H-indolyl, 4,5,6,7-tetrahydro-1-H-indolyl, quinoxalinyl, 5-methylquinoxalinyl, quinazolinyl, quinolyl, 8-hydroxy-quinolyl, 1H-benzo[d]imidazol-2(3H)-onyl, 1H-benzo[d]imidazolyl, and isoquinolyl.

[0064] One non-limiting example of a heteroaryl group as described above is C.sub.1-C.sub.5 heteroaryl, which has 1 to 5 carbon ring atoms and at least one additional ring atom that is a

heteroatom (preferably 1 to 4 additional ring atoms that are heteroatoms) independently selected from nitrogen (N), oxygen (O), or sulfur (S). Examples of C.sub.1-C.sub.5 heteroaryl include, but are not limited to, triazinyl, thiazol-2-yl, thiazol-4-yl, imidazol-1-yl, 1H-imidazol-2-yl, 1H-imidazol-4-yl, isoxazolin-5-yl, furan-2-yl, furan-3-yl, thiophen-2-yl, thiophen-4-yl, pyrimidin-2-yl, pyrimidin-4-yl, pyrimidin-5-yl, pyridin-2-yl, pyridin-3-yl, and pyridin-4-yl.

[0065] Unless otherwise noted, when two substituents are taken together to form a ring having a specified number of ring atoms (e.g., R.sup.2 and R.sup.3 taken together with the nitrogen (N) to which they are attached to form a ring having from 3 to 7 ring members), the ring can have carbon atoms and optionally one or more (e.g., 1 to 3) additional heteroatoms independently selected from nitrogen (N), oxygen (O), or sulfur (S). The ring can be saturated or partially saturated and can be optionally substituted.

[0066] For the purposes of the present invention fused ring units, as well as spirocyclic rings, bicyclic rings and the like, which comprise a single heteroatom will be considered to belong to the cyclic family corresponding to the heteroatom containing ring. For example, 1,2,3,4-tetrahydroquinoline having the formula:

##STR00006##

is, for the purposes of the present invention, considered a heterocyclic unit. 6,7-Dihydro-5H-cyclopentapyrimidine having the formula:

##STR00007##

is, for the purposes of the present invention, considered a heteroaryl unit. When a fused ring unit contains heteroatoms in both a saturated and an aryl ring, the aryl ring will predominate and determine the type of category to which the ring is assigned. For example, 1,2,3,4-tetrahydro-[1,8]naphthyridine having the formula:

##STR00008##

is, for the purposes of the present invention, considered a heteroaryl unit.

[0067] Whenever a term or either of their prefix roots appear in a name of a substituent the name is to be interpreted as including those limitations provided herein. For example, whenever the term “alkyl” or “aryl” or either of their prefix roots appear in a name of a substituent (e.g., arylalkyl, alkylamino) the name is to be interpreted as including those limitations given above for “alkyl” and “aryl.”

[0068] The term “substituted” is used throughout the specification. The term “substituted” is defined herein as a moiety, whether acyclic or cyclic, which has one or more hydrogen atoms replaced by a substituent or several (e.g., 1 to 10) substituents as defined herein below. The substituents are capable of replacing one or two hydrogen atoms of a single moiety at a time. In addition, these substituents can replace two hydrogen atoms on two adjacent carbons to form said substituent, new moiety or unit. For example, a substituted unit that requires a single hydrogen atom replacement includes halogen, hydroxyl, and the like. A two hydrogen atom replacement includes carbonyl, oximino, and the like. A two hydrogen atom replacement from adjacent carbon atoms includes epoxy, and the like. The term “substituted” is used throughout the present specification to indicate that a moiety can have one or more of the hydrogen atoms replaced by a substituent. When a moiety is described as “substituted” any number of the hydrogen atoms may be replaced. For example, difluoromethyl is a substituted C.sub.1 alkyl; trifluoromethyl is a substituted C.sub.1 alkyl; 4-hydroxyphenyl is a substituted aromatic ring; (N,N-dimethyl-5-amino)octanyl is a substituted C.sub.8 alkyl; 3-guanidinopropyl is a substituted C.sub.3 alkyl; and 2-carboxypyridinyl is a substituted heteroaryl.

[0069] The variable groups defined herein, e.g., alkyl, alkenyl, alkynyl, cycloalkyl, alkoxy, aryloxy, aryl, heterocycle and heteroaryl groups defined herein, whether used alone or as part of another group, can be optionally substituted. Optionally substituted groups will be so indicated.

[0070] The following are non-limiting examples of substituents which can substitute for hydrogen atoms on a moiety: halogen (chlorine (C.sub.1), bromine (Br), fluorine (F) and iodine(I)), —CN,

—NO.sub.2, oxo (=O), —OR.sup.15, —SR.sup.15, —N(R.sup.15).sub.2, —NR.sup.15C(O)R.sup.15, —SO.sub.2R.sup.15, —SO.sub.2OR.sup.15, —SO.sub.2N(R.sup.15).sub.2, —C(O)R.sup.15, —C(O)OR.sup.15, —C(O)N(R.sup.15).sub.2, C.sub.1-6 alkyl, C.sub.1-6 haloalkyl, C.sub.1-6 alkoxy, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-14 cycloalkyl, aryl, heterocycle, or heteroaryl, wherein each of the alkyl, haloalkyl, alkenyl, alkynyl, alkoxy, cycloalkyl, aryl, heterocycle, and heteroaryl groups is optionally substituted with 1-10 (e.g., 1-6 or 1-4) groups selected independently from halogen, —CN, —NO.sub.2, oxo, and R.sup.15; wherein R.sup.15, at each occurrence, independently is hydrogen, —OR.sup.16, —SR.sup.16, —C(O)R.sup.16, —C(O)OR.sup.16, —C(O)N(R.sup.16).sub.2, —SO.sub.2R.sup.16, —S(O).sub.2OR.sup.16, —N(R.sup.16).sub.2, —NR.sup.16C(O)R.sup.16, C.sub.1-6 alkyl, C.sub.1-6 haloalkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, cycloalkyl (e.g., C.sub.3-6 cycloalkyl), aryl, heterocycle, or heteroaryl, or two R.sup.15 units taken together with the atom(s) to which they are bound form an optionally substituted carbocycle or heterocycle wherein said carbocycle or heterocycle has 3 to 7 ring atoms; wherein R.sup.16, at each occurrence, independently is hydrogen, C.sub.1-6 alkyl, C.sub.1-6 haloalkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, cycloalkyl (e.g., C.sub.3-6 cycloalkyl), aryl, heterocycle, or heteroaryl, or two R.sup.16 units taken together with the atom(s) to which they are bound form an optionally substituted carbocycle or heterocycle wherein said carbocycle or heterocycle preferably has 3 to 7 ring atoms.

[0071] In some embodiments, the substituents are selected from [0072] i) —OR.sup.17; for example, —OH, —OCH.sub.3, —OCH.sub.2CH.sub.3, —OCH.sub.2CH.sub.2CH.sub.3; [0073] ii) —C(O)R.sup.17; for example, —COCH.sub.3, —COCH.sub.2CH.sub.3, —COCH.sub.2CH.sub.2CH.sub.3; [0074] iii) —C(O)OR.sup.17; for example, —CO.sub.2CH.sub.3, —CO.sub.2CH.sub.2CH.sub.3, —CO.sub.2CH.sub.2CH.sub.2CH.sub.3; [0075] iv) —C(O)N(R.sup.17).sub.2; for example, —CONH.sub.2, —CONHCH.sub.3, —CON(CH.sub.3).sub.2; [0076] v) —N(R.sup.17).sub.2; for example, —NH.sub.2, —NHCH.sub.3, —N(CH.sub.3).sub.2, —NH(CH.sub.2CH.sub.3); [0077] vi) halogen: —F, —Cl, —Br, and —I; [0078] vii) —CH.sub.eX.sub.g; wherein X is halogen, m is from 0 to 2, e+g=3; for example, —CH.sub.2F, —CHF.sub.2, —CF.sub.3, —CCl.sub.3, or —CBr.sub.3; [0079] viii) —SO.sub.2R.sup.17; for example, —SO.sub.2H; —SO.sub.2CH.sub.3; —SO.sub.2C.sub.6H.sub.5; [0080] ix) C.sub.1-C.sub.6 linear, branched, or cyclic alkyl; [0081] x) Cyano [0082] xi) Nitro; [0083] xii) N(R.sup.17)C(O)R.sup.17; [0084] xiii) Oxo (=O); [0085] xiv) Heterocycle; and [0086] xv) Heteroaryl.

wherein each R.sup.17 is independently hydrogen, optionally substituted C.sub.1-C.sub.6 linear or branched alkyl (e.g., optionally substituted C.sub.1-C.sub.4 linear or branched alkyl), or optionally substituted C.sub.3-C.sub.6 cycloalkyl (e.g. optionally substituted C.sub.3-C.sub.4 cycloalkyl); or two R.sup.17 units can be taken together to form a ring comprising 3-7 ring atoms. In certain aspects, each R.sup.17 is independently hydrogen, C.sub.1-C.sub.6 linear or branched alkyl optionally substituted with halogen or C.sub.3-C.sub.6 cycloalkyl or C.sub.3-C.sub.6 cycloalkyl.

[0087] At various places in the present specification, substituents of compounds are disclosed in groups or in ranges. It is specifically intended that the description include each and every individual subcombination of the members of such groups and ranges. For example, the term “C.sub.1-6 alkyl” is specifically intended to individually disclose C.sub.1, C.sub.2, C.sub.3, C.sub.4, C.sub.5, C.sub.6, C.sub.1-C.sub.6, C.sub.1-C.sub.5, C.sub.1-C.sub.4, C.sub.1-C.sub.3, C.sub.1-C.sub.2, C.sub.2-C.sub.6, C.sub.2-C.sub.5, C.sub.2-C.sub.4, C.sub.2-C.sub.3, C.sub.3-C.sub.6, C.sub.3-C.sub.5, C.sub.3-C.sub.4, C.sub.4-C.sub.6, C.sub.4-C.sub.5, and C.sub.5-C.sub.6, alkyl.

[0088] For the purposes of the present invention the terms “compound,” “analog,” and “composition of matter” stand equally well for the 5-hydroxytryptamine receptor 7 activity modulators described herein, including all enantiomeric forms, diastereomeric forms, salts, and the like, and the terms “compound,” “analog,” and “composition of matter” are used interchangeably

throughout the present specification.

[0089] Compounds described herein can contain an asymmetric atom (also referred as a chiral center), and some of the compounds can contain one or more asymmetric atoms or centers, which can thus give rise to optical isomers (enantiomers) and diastereomers. The present teachings and compounds disclosed herein include such enantiomers and diastereomers, as well as the racemic and resolved, enantiomerically pure R and S stereoisomers, as well as other mixtures of the R and S stereoisomers and pharmaceutically acceptable salts thereof. Optical isomers can be obtained in pure form by standard procedures known to those skilled in the art, which include, but are not limited to, diastereomeric salt formation, kinetic resolution, and asymmetric synthesis. The present teachings also encompass cis and trans isomers of compounds containing alkenyl moieties (e.g., alkenes and imines). It is also understood that the present teachings encompass all possible regioisomers, and mixtures thereof, which can be obtained in pure form by standard separation procedures known to those skilled in the art, and include, but are not limited to, column chromatography, thin-layer chromatography, and high-performance liquid chromatography.

[0090] Pharmaceutically acceptable salts of compounds of the present teachings, which can have an acidic moiety, can be formed using organic and inorganic bases. Both mono and polyanionic salts are contemplated, depending on the number of acidic hydrogens available for deprotonation. Suitable salts formed with bases include metal salts, such as alkali metal or alkaline earth metal salts, for example sodium, potassium, or magnesium salts; ammonia salts and organic amine salts, such as those formed with morpholine, thiomorpholine, piperidine, pyrrolidine, a mono-, di- or tri-lower alkylamine (e.g., ethyl-tert-butyl-, diethyl-, diisopropyl-, triethyl-, tributyl- or dimethylpropylamine), or a mono-, di-, or trihydroxy lower alkylamine (e.g., mono-, di- or triethanolamine). Specific non-limiting examples of inorganic bases include NaHCO_3 , Na_2CO_3 , KHCO_3 , K_2CO_3 , Cs_2CO_3 , LiOH , NaOH , KOH , NaH_2PO_4 , Na_2HPO_4 , and Na_3PO_4 . Internal salts also can be formed. Similarly, when a compound disclosed herein contains a basic moiety, salts can be formed using organic and inorganic acids. For example, salts can be formed from the following acids: acetic, propionic, lactic, benzenesulfonic, benzoic, camphorsulfonic, citric, tartaric, succinic, dichloroacetic, ethenesulfonic, formic, fumaric, gluconic, glutamic, hippuric, hydrobromic, hydrochloric, isethionic, lactic, maleic, malic, malonic, mandelic, methanesulfonic, mucic, naphthalenesulfonic, nitric, oxalic, pantoic, pantothenic, phosphoric, phthalic, propionic, succinic, sulfuric, tartaric, toluenesulfonic, and camphorsulfonic as well as other known pharmaceutically acceptable acids.

[0091] When any variable occurs more than one time in any constituent or in any formula, its definition in each occurrence is independent of its definition at every other occurrence (e.g., in $\text{N}(\text{R}^9)_2$, each R^9 may be the same or different than the other). Combinations of substituents and/or variables are permissible only if such combinations result in stable compounds.

[0092] The terms “treat” and “treating” and “treatment” as used herein, refer to partially or completely alleviating, inhibiting, ameliorating and/or relieving a condition from which a patient is suspected to suffer.

[0093] As used herein, “therapeutically effective” and “effective dose” refer to a substance or an amount that elicits a desirable biological activity or effect.

[0094] Except when noted, the terms “subject” or “patient” are used interchangeably and refer to mammals such as human patients and non-human primates, as well as experimental animals such as rabbits, rats, and mice, and other animals. Accordingly, the term “subject” or “patient” as used herein means any mammalian patient or subject to which the compounds of the invention can be administered. In an exemplary embodiment of the present invention, to identify subject patients for treatment according to the methods of the invention, accepted screening methods are employed to determine risk factors associated with a targeted or suspected disease or condition or to determine the status of an existing disease or condition in a subject. These screening methods include, for

example, conventional work-ups to determine risk factors that may be associated with the targeted or suspected disease or condition. These and other routine methods allow the clinician to select patients in need of therapy using the methods and compounds of the present invention.

The 5-Hydroxytryptamine Receptor 7 Activity Modulators

[0095] The 5-hydroxytryptamine receptor 7 activity modulators of the present invention include all enantiomeric and diastereomeric formsalts thereof having the formula

##STR00009##

Including hydrates, solvates, pharmaceutically acceptable salts, prodrugs and complexes thereof, wherein: [0096] A is selected from a group consisting of

##STR00010## [0097] X is selected from the group consisting of O, S, SO, SO.sub.2, NR; [0098] n.sup.1 is 0, 1, 2; [0099] n.sup.2 is 0, 1, 2; [0100] R is selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl,

##STR00011##

COR.sup.2, CO.sub.2R.sup.2a, CONR.sup.2bR.sup.2c, SO.sub.2NR.sup.2bR.sup.2c, and SO.sub.2R.sup.2d; [0101] R.sup.1a, R.sup.1b, R.sup.1c, R.sup.1d, and R.sup.1e are at each occurrence independently selected from the group consisting of H, OH, NO.sub.2, halogen, CN, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, C.sub.1-6 linear alkoxy, C.sub.3-7 branched alkoxy, C.sub.3-7 cycloalkoxy, C.sub.1-6 linear haloalkyl, C.sub.3-7 branched haloalkyl, C.sub.1-6 linear haloalkoxy, —S(C.sub.1-6 linear alkyl), S(C.sub.3-7 branched alkyl), —S(C.sub.3-7 cycloalkyl), COR.sup.6, CO.sub.2R.sup.7, CONR.sup.8aR.sup.8b, SO.sub.2NR.sup.8aR.sup.8b, NR.sup.9aR.sup.9b, NR.sup.9aCOR.sup.10, NR.sup.9aSO.sub.2R.sup.11, and NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b; [0102] R.sup.2 is selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0103] R.sup.2a is selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0104] R.sup.2b is selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0105] R.sup.2c is selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0106] R.sup.2d is selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, C.sub.1-6 linear haloalkyl, C.sub.3-7 branched haloalkyl, —(CH.sub.2).sub.qCN, —(CH.sub.2).sub.qSO.sub.2R.sup.13, —(CH.sub.2).sub.qOR.sup.14,

##STR00012## [0107] R.sup.3 is selected from a group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, optionally substituted aryl,

##STR00013## [0108] R.sup.4 is an optionally substituted aryl; [0109] R.sup.5a and R.sup.5b are each independently optionally substituted aryl; [0110] R.sup.6 is at each occurrence independently selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0111] R.sup.7 is at each occurrence independently selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0112] R.sup.8a is at each occurrence independently selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0113] R.sup.8b is at each occurrence independently selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0114] R.sup.9a is at each occurrence independently selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0115] R.sup.9b is at each occurrence independently selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0116] R.sup.10 is at each occurrence independently selected from the group consisting of H, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0117] R.sup.11 is at each occurrence independently selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0118] R.sup.12a is at each occurrence independently selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-

7 cycloalkyl; [0119] R.sup.12b is at each occurrence independently selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0120] R.sup.13 is selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0121] R.sup.14 is selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; [0122] n is 1, 2, or 3; [0123] m is 1 or 2; [0124] and q is 1, 2, or 3;

[0125] The embodiments of the present invention include compounds having formula (II):

##STR00014##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0126] The embodiments of the present invention include compounds having formula (III):

##STR00015##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0127] The embodiments of the present invention include compounds having formula (IV):

##STR00016##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0128] The embodiments of the present invention include compounds having formula (V):

##STR00017##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0129] The embodiments of the present invention include compounds having formula (VI):

##STR00018##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0130] The embodiments of the present invention include compounds having formula (VII):

##STR00019##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0131] The embodiments of the present invention include compounds having formula (VIII):

##STR00020##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0132] The embodiments of the present invention include compounds having formula (IX):

##STR00021##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0133] The embodiments of the present invention include compounds having formula (X):

##STR00022##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0134] The embodiments of the present invention include compounds having formula (XI):

##STR00023##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0135] The embodiments of the present invention include compounds having formula (XII):

##STR00024##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0136] The embodiments of the present invention include compounds having formula (XIII):

##STR00025##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0137] The embodiments of the present invention include compounds having formula (XIV):

##STR00026##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0138] The embodiments of the present invention include compounds having formula (XV):

##STR00027##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0139] The embodiments of the present invention include compounds having formula (XVI):

##STR00028##

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0140] In some embodiments A is

##STR00029##

[0141] In some embodiments A is

##STR00030##

[0142] In some embodiments A is

##STR00031##

[0143] In some embodiments A is

##STR00032##

[0144] In some embodiments A is

##STR00033##

[0145] In some embodiments X is O.

[0146] In some embodiments X is S.

[0147] In some embodiments X is SO.

[0148] In some embodiments X is SO.sub.2.

[0149] In some embodiments X is NR.

[0150] In some embodiments n.sup.1 is 0.

[0151] In some embodiments n.sup.1 is 1.

[0152] In some embodiments n.sup.1 is 2.

[0153] In some embodiments n.sup.1 is 0.

[0154] In some embodiments n.sup.2 is 1.

[0155] In some embodiments n.sup.2 is 2.

[0156] In some embodiments n.sup.2 is H.

[0157] In some embodiments R is H.

[0158] In some embodiments R is C.sub.1-6 linear alkyl.

[0159] In some embodiments R is C.sub.3-7 branched alkyl.

[0160] In some embodiments R is C.sub.3-7 cycloalkyl.

[0161] In some embodiments R is,

##STR00034##

[0162] In some embodiments R is COR.sup.2.

[0163] In some embodiments R is CO.sub.2R.sup.2a.

[0164] In some embodiments R is CONR.sup.2bR.sup.2c.

[0165] In some embodiments R is SO.sub.2NR.sup.2bR.sup.2c.

[0166] In some embodiments R is SO.sub.2R.sup.2d.

[0167] In some embodiments R.sup.1a is H.

[0168] In some embodiments R.sup.1a is OH.
[0169] In some embodiments R.sup.1a is NO.sub.2.
[0170] In some embodiments R.sup.1a is halogen.
[0171] In some embodiments R.sup.1a is CN.
[0172] In some embodiments R.sup.1a is C.sub.1-6 linear alkyl.
[0173] In some embodiments R.sup.1a is C.sub.3-7 branched alkyl.
[0174] In some embodiments R.sup.1a is C.sub.3-7 cycloalkyl.
[0175] In some embodiments R.sup.1a is C.sub.1-6 linear alkoxy.
[0176] In some embodiments R.sup.1a is C.sub.3-7 branched alkoxy.
[0177] In some embodiments R.sup.1a is C.sub.3-7 cycloalkoxy.
[0178] In some embodiments R.sup.1a is C.sub.1-6 linear haloalkyl.
[0179] In some embodiments R.sup.1a is C.sub.3-7 branched haloalkyl.
[0180] In some embodiments R.sup.1a is C.sub.1-6 linear haloalkoxy.
[0181] In some embodiments R.sup.1a is —S(C.sub.1-6 linear alkyl).
[0182] In some embodiments R.sup.1a is S(C.sub.3-7 branched alkyl).
[0183] In some embodiments R.sup.1a is —S(C.sub.3-7 cycloalkyl).
[0184] In some embodiments R.sup.1a is COR.sup.6.
[0185] In some embodiments R.sup.1a is CO.sub.2R.sup.7.
[0186] In some embodiments R.sup.1a is CONR.sup.8aR.sup.8b.
[0187] In some embodiments R.sup.1a is SO.sub.2NR.sup.8aR.sup.8b.
[0188] In some embodiments R.sup.1a is NR.sup.9aR.sup.9b.
[0189] In some embodiments R.sup.1a is NR.sup.9aCOR.sup.10.
[0190] In some embodiments R.sup.1a is NR.sup.9aSO.sub.2R.sup.11.
[0191] In some embodiments R.sup.1a is NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b.
[0192] In some embodiments R.sup.1b is H.
[0193] In some embodiments R.sup.1b is OH.
[0194] In some embodiments R.sup.1b is NO.sub.2.
[0195] In some embodiments R.sup.1b is halogen.
[0196] In some embodiments R.sup.1b is CN.
[0197] In some embodiments R.sup.1b is C.sub.1-6 linear alkyl.
[0198] In some embodiments R.sup.1b is C.sub.3-7 branched alkyl.
[0199] In some embodiments R.sup.1b is C.sub.3-7 cycloalkyl.
[0200] In some embodiments R.sup.1b is C.sub.1-6 linear alkoxy.
[0201] In some embodiments R.sup.1b is C.sub.3-7 branched alkoxy.
[0202] In some embodiments R.sup.1b is C.sub.3-7 cycloalkoxy.
[0203] In some embodiments R.sup.1b is C.sub.1-6 linear haloalkyl.
[0204] In some embodiments R.sup.1b is C.sub.3-7 branched haloalkyl.
[0205] In some embodiments R.sup.1b is C.sub.1-6 linear haloalkoxy.
[0206] In some embodiments R.sup.1b is —S(C.sub.1-6 linear alkyl).
[0207] In some embodiments R.sup.1b is S(C.sub.3-7 branched alkyl).
[0208] In some embodiments R.sup.1b is —S(C.sub.3-7 cycloalkyl).
[0209] In some embodiments R.sup.1b is COR.sup.6.
[0210] In some embodiments R.sup.1b is CO.sub.2R.sup.7.
[0211] In some embodiments R.sup.1b is CONR.sup.8aR.sup.8b.
[0212] In some embodiments R.sup.1b is SO.sub.2NR.sup.8aR.sup.8b.
[0213] In some embodiments R.sup.1b is NR.sup.9aR.sup.9b.
[0214] In some embodiments R.sup.1b is NR.sup.9aCOR.sup.11.
[0215] In some embodiments R.sup.1b is NR.sup.9aSO.sub.2R.sup.11.
[0216] In some embodiments R.sup.1b is NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b.
[0217] In some embodiments R.sup.1c is H.

[0218] In some embodiments R.sup.1c is OH.
[0219] In some embodiments R.sup.1c is NO.sub.2.
[0220] In some embodiments R.sup.1c is halogen.
[0221] In some embodiments R.sup.1c is CN.
[0222] In some embodiments R.sup.1c is C.sub.1-6 linear alkyl.
[0223] In some embodiments R.sup.1c is C.sub.3-7 branched alkyl.
[0224] In some embodiments R.sup.1c is C.sub.3-7 cycloalkyl.
[0225] In some embodiments R.sup.1c is C.sub.1-6 linear alkoxy.
[0226] In some embodiments R.sup.1c is C.sub.3-7 branched alkoxy.
[0227] In some embodiments R.sup.1c is C.sub.3-7 cycloalkoxy.
[0228] In some embodiments R.sup.1c is C.sub.1-6 linear haloalkyl.
[0229] In some embodiments R.sup.1c is C.sub.3-7 branched haloalkyl.
[0230] In some embodiments R.sup.1c is C.sub.1-6 linear haloalkoxy.
[0231] In some embodiments R.sup.1c is —S(C.sub.1-6 linear alkyl).
[0232] In some embodiments R.sup.1c is S(C.sub.3-7 branched alkyl).
[0233] In some embodiments R.sup.1c is —S(C.sub.3-7 cycloalkyl).
[0234] In some embodiments R.sup.1c is COR.sup.6.
[0235] In some embodiments R.sup.1c is CO.sub.2R.sup.7.
[0236] In some embodiments R.sup.1c is CONR.sup.8aR.sup.8b.
[0237] In some embodiments R.sup.1c is SO.sub.2NR.sup.8aR.sup.8b.
[0238] In some embodiments R.sup.1c is NR.sup.9aR.sup.9b.
[0239] In some embodiments R.sup.1c is NR.sup.9aCOR.sup.10.
[0240] In some embodiments R.sup.1c is NR.sup.9aSO.sub.2R.sup.11.
[0241] In some embodiments R.sup.1c is NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b.
[0242] In some embodiments R.sup.1d is H.
[0243] In some embodiments R.sup.1d is OH.
[0244] In some embodiments R.sup.1d is NO.sub.2.
[0245] In some embodiments R.sup.1d is halogen.
[0246] In some embodiments R.sup.1d is CN.
[0247] In some embodiments R.sup.1d is C.sub.1-6 linear alkyl.
[0248] In some embodiments R.sup.1d is C.sub.3-7 branched alkyl.
[0249] In some embodiments R.sup.1d is C.sub.3-7 cycloalkyl.
[0250] In some embodiments R.sup.1d is C.sub.1-6 linear alkoxy.
[0251] In some embodiments R.sup.1d is C.sub.3-7 branched alkoxy.
[0252] In some embodiments R.sup.1d is C.sub.3-7 cycloalkoxy.
[0253] In some embodiments R.sup.1d is C.sub.1-6 linear haloalkyl.
[0254] In some embodiments R.sup.1d is C.sub.3-7 branched haloalkyl.
[0255] In some embodiments R.sup.1d is C.sub.1-6 linear haloalkoxy.
[0256] In some embodiments R.sup.1d is —S(C.sub.1-6 linear alkyl).
[0257] In some embodiments R.sup.1d is S(C.sub.3-7 branched alkyl).
[0258] In some embodiments R.sup.1d is —S(C.sub.3-7 cycloalkyl).
[0259] In some embodiments R.sup.1d is COR.sup.6.
[0260] In some embodiments R.sup.1d is CO.sub.2R.sup.7.
[0261] In some embodiments R.sup.1d is CONR.sup.8aR.sup.8b.
[0262] In some embodiments R.sup.1d is SO.sub.2NR.sup.8aR.sup.8b.
[0263] In some embodiments R.sup.1d is NR.sup.9aR.sup.9b.
[0264] In some embodiments R.sup.1d is NR.sup.9aCOR.sup.10.
[0265] In some embodiments R.sup.1d is NR.sup.9aSO.sub.2R.sup.11.
[0266] In some embodiments R.sup.1d is NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b.
[0267] In some embodiments R.sup.1e is H.

[0268] In some embodiments R.sup.1e is OH.
[0269] In some embodiments R.sup.1e is NO.sub.2.
[0270] In some embodiments R.sup.1e is halogen.
[0271] In some embodiments R.sup.1e is CN.
[0272] In some embodiments R.sup.1e is C.sub.1-6 linear alkyl.
[0273] In some embodiments R.sup.1e is C.sub.3-7 branched alkyl.
[0274] In some embodiments R.sup.1e is C.sub.3-7 cycloalkyl.
[0275] In some embodiments R.sup.1e is C.sub.1-6 linear alkoxy.
[0276] In some embodiments R.sup.1e is C.sub.3-7 branched alkoxy.
[0277] In some embodiments R.sup.1e is C.sub.3-7 cycloalkoxy.
[0278] In some embodiments R.sup.1e is C.sub.1-6 linear haloalkyl.
[0279] In some embodiments R.sup.1e is C.sub.3-7 branched haloalkyl.
[0280] In some embodiments R.sup.1e is C.sub.1-6 linear haloalkoxy.
[0281] In some embodiments R.sup.1e is —S(C.sub.1-6 linear alkyl).
[0282] In some embodiments R.sup.1e is S(C.sub.3-7 branched alkyl).
[0283] In some embodiments R.sup.1e is —S(C.sub.3-7 cycloalkyl).
[0284] In some embodiments R.sup.1e is COR.sup.6.
[0285] In some embodiments R.sup.1e is CO.sub.2R.sup.7.
[0286] In some embodiments R.sup.1e is CONR.sup.8aR.sup.8b.
[0287] In some embodiments R.sup.1e is SO.sub.2NR.sup.8aR.sup.8b.
[0288] In some embodiments R.sup.1e is NR.sup.9aR.sup.9b.
[0289] In some embodiments R.sup.1e is NR.sup.9aCOR.sup.10.
[0290] In some embodiments R.sup.1e is NR.sup.9aSO.sub.2R.sup.11.
[0291] In some embodiments R.sup.1e is NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b.
[0292] In some embodiments R.sup.2 is H.
[0293] In some embodiments R.sup.2 is C.sub.1-6 linear alkyl.
[0294] In some embodiments R.sup.2 is, C.sub.3-7 branched alkyl.
[0295] In some embodiments R.sup.2 is C.sub.3-7 cycloalkyl.
[0296] In some embodiments R.sup.2a is C.sub.1-6 linear alkyl.
[0297] In some embodiments R.sup.2a is C.sub.3-7 branched alkyl.
[0298] In some embodiments R.sup.2a is C.sub.3-7 cycloalkyl.
[0299] In some embodiments R.sup.2b is H.
[0300] In some embodiments R.sup.2b is C.sub.1-6 linear alkyl.
[0301] In some embodiments R.sup.2b is, C.sub.3-7 branched alkyl.
[0302] In some embodiments R.sup.2b is C.sub.3-7 cycloalkyl.
[0303] In some embodiments R.sup.2c is H.
[0304] In some embodiments R.sup.2c is C.sub.1-6 linear alkyl.
[0305] In some embodiments R.sup.2c is, C.sub.3-7 branched alkyl.
[0306] In some embodiments R.sup.2c is C.sub.3-7 cycloalkyl.
[0307] In some embodiments R.sup.2d is C.sub.1-6 linear alkyl.
[0308] In some embodiments R.sup.2d is C.sub.3-7 branched alkyl.
[0309] In some embodiments R.sup.2d is C.sub.3-7 cycloalkyl.
[0310] In some embodiments R.sup.2d is C.sub.1-6 linear haloalkyl.
[0311] In some embodiments R.sup.2d is C.sub.3-7 branched haloalkyl.
[0312] In some embodiments R.sup.2d is —(CH.sub.2).sub.qCN.
[0313] In some embodiments R.sup.2d is —(CH.sub.2).sub.qSO.sub.2R.sup.13.
[0314] In some embodiments R.sup.2d is —(CH.sub.2).sub.qOR.sup.14.
[0315] In some embodiments R.sup.2d is

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[0316] In some embodiments R.sup.2d is

##STR00036##

[0317] In some embodiments R.sup.2d is

##STR00037##

[0318] In some embodiments R.sup.2d is

##STR00038##

[0319] In some embodiments R.sup.2d is

##STR00039##

[0320] In some embodiments R.sup.2d is

##STR00040##

[0321] In some embodiments R.sup.2d is

##STR00041##

[0322] In some embodiments R.sup.2d is

##STR00042##

[0323] In some embodiments R.sup.2d is

##STR00043##

[0324] In some embodiments R.sup.2d is

##STR00044##

[0325] In some embodiments R.sup.2d is

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[0326] In some embodiments R.sup.2d is

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[0327] In some embodiments R.sup.3 is C.sub.1-6 linear alkyl.

[0328] In some embodiments R.sup.3 is C.sub.3-7 branched alkyl.

[0329] In some embodiments R.sup.3 is C.sub.3-7 cycloalkyl.

[0330] In some embodiments R.sup.3 is optionally substituted aryl.

[0331] In some embodiments R.sup.3 is phenyl.

[0332] In some embodiments R.sup.3 is an optionally aryl substituted with 1 to 4 units independently selected from the group consisting of OH, NO.sub.2, halogen, CN, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, C.sub.1-6 linear alkoxy, C.sub.3-7 branched alkoxy, C.sub.3-7 cycloalkoxy, C.sub.1-6 linear haloalkyl, C.sub.3-7 branched haloalkyl, C.sub.1-6 linear haloalkoxy, —S(C.sub.1-6 linear alkyl), S(C.sub.3-7 branched alkyl), —S(C.sub.3-7 cycloalkyl), COR.sup.6, CO.sub.2R.sup.7, CONR.sup.8aR.sup.8b, SO.sub.2NR.sup.8aR.sup.8b, NR.sup.9aR.sup.9b, NR.sup.9aCOR.sup.10, NR.sup.9aSO.sub.2R.sup.11, NR.sup.9aSO.sub.2NR.sup.12aR.sup.12b,

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[0333] In some embodiments R.sup.3 is

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[0334] In some embodiments R.sup.3 is

##STR00049##

[0335] In some embodiments R.sup.3 is

##STR00050##

[0336] In some embodiments R.sup.3 is

##STR00051##

[0337] In some embodiments R.sup.3 is

##STR00052##

[0338] In some embodiments R.sup.3 is

##STR00053##

[0339] In some embodiments R.sup.3 is

##STR00054##

[0340] In some embodiments R.sup.3 is

##STR00055##

[0341] In some embodiments R^{sup.3} is

##STR00056##

[0342] In some embodiments R^{sup.3} is

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[0343] In some embodiments R^{sup.4} is optionally substituted aryl.

[0344] In some embodiments R^{sup.4} is an optionally substituted aryl substituted with 1 to 4 units independently selected from the group consisting of OH, NO₂, halogen, CN, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, C₃₋₇ cycloalkyl, C₁₋₆ linear alkoxy, C₃₋₇ branched alkoxy, C₃₋₇ cycloalkoxy, C₁₋₆ linear haloalkyl, C₃₋₇ branched haloalkyl, C₁₋₆ linear haloalkoxy, —S(C₁₋₆ linear alkyl), S(C₃₋₇ branched alkyl), —S(C₃₋₇ cycloalkyl), COR^{sup.6}, CO₂R^{sup.7}, CONR^{sup.8a}R^{sup.8b}, SO₂NR^{sup.8a}R^{sup.8b}, NR^{sup.9a}R^{sup.9b}, NR^{sup.9a}COR^{sup.10}, NR^{sup.9a}SO₂R^{sup.11}, NR^{sup.9a}SO₂NR^{sup.12a}R^{sup.12b},

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[0345] In some embodiments R^{sup.5} is optionally substituted aryl.

[0346] In some embodiments R^{sup.5} is optionally substituted aryl.

[0347] In some embodiments R^{sup.6} is H.

[0348] In some embodiments R^{sup.6} is C₁₋₆ linear alkyl.

[0349] In some embodiments R^{sup.6} is C₃₋₇ branched alkyl.

[0350] In some embodiments R^{sup.6} is C₃₋₇ cycloalkyl.

[0351] In some embodiments R^{sup.7} is C₁₋₆ linear alkyl.

[0352] In some embodiments R^{sup.7} is C₃₋₇ branched alkyl.

[0353] In some embodiments R^{sup.7} is C₃₋₇ cycloalkyl.

[0354] In some embodiments R^{sup.8a} is H.

[0355] In some embodiments R^{sup.8} is C₁₋₆ linear alkyl.

[0356] In some embodiments R^{sup.8a} is C₃₋₇ branched alkyl.

[0357] In some embodiments R^{sup.8a} is C₃₋₇ cycloalkyl.

[0358] In some embodiments R^{sup.8b} is H.

[0359] In some embodiments R^{sup.8b} is C₁₋₆ linear alkyl.

[0360] In some embodiments R^{sup.8b} is C₃₋₇ branched alkyl.

[0361] In some embodiments R^{sup.8b} is C₃₋₇ cycloalkyl.

[0362] In some embodiments R^{sup.9a} is H.

[0363] In some embodiments R^{sup.9a} is C₁₋₆ linear alkyl.

[0364] In some embodiments R^{sup.9a} is C₃₋₇ branched alkyl.

[0365] In some embodiments R^{sup.9a} is C₃₋₇ cycloalkyl.

[0366] In some embodiments R^{sup.9b} is H.

[0367] In some embodiments R^{sup.9b} is C₁₋₆ linear alkyl.

[0368] In some embodiments R^{sup.9b} is C₃₋₇ branched alkyl.

[0369] In some embodiments R^{sup.9b} is C₃₋₇ cycloalkyl.

[0370] In some embodiments R^{sup.10} is H.

[0371] In some embodiments R^{sup.10} is C₁₋₆ linear alkyl.

[0372] In some embodiments R^{sup.10} is C₃₋₇ branched alkyl.

[0373] In some embodiments R^{sup.10} is C₃₋₇ cycloalkyl.

[0374] In some embodiments R^{sup.11} is C₁₋₆ linear alkyl.

[0375] In some embodiments R^{sup.11} is C₃₋₇ branched alkyl.

[0376] In some embodiments R^{sup.11} is C₃₋₇ cycloalkyl.

[0377] In some embodiments R^{sup.10} is C₁₋₆ linear alkyl.

[0378] In some embodiments R^{sup.10} is C₃₋₇ branched alkyl.

[0379] In some embodiments R^{sup.10} is C₃₋₇ cycloalkyl.

- [0380] In some embodiments R.sup.12a is C.sub.1-6 linear alkyl.
- [0381] In some embodiments R.sup.12a is C.sub.3-7 branched alkyl.
- [0382] In some embodiments R.sup.12a is C.sub.3-7 cycloalkyl.
- [0383] In some embodiments R.sup.12b is C.sub.1-6 linear alkyl.
- [0384] In some embodiments R.sup.12b is C.sub.3-7 branched alkyl.
- [0385] In some embodiments R.sup.12b is C.sub.3-7 cycloalkyl.
- [0386] In some embodiments R.sup.13 is C.sub.1-6 linear alkyl.
- [0387] In some embodiments R.sup.13 is C.sub.3-7 branched alkyl.
- [0388] In some embodiments R.sup.13 is C.sub.3-7 cycloalkyl.
- [0389] In some embodiments R.sup.14 is C.sub.1-6 linear alkyl.
- [0390] In some embodiments R.sup.14 is C.sub.3-7 branched alkyl.
- [0391] In some embodiments R.sup.14 is C.sub.3-7 cycloalkyl.
- [0392] In some embodiments n is 1.
- [0393] In some embodiments n is 2.
- [0394] In some embodiments n is 3.
- [0395] In some embodiments m is 1.
- [0396] In some embodiments m is 2.
- [0397] In some embodiments q is 1.
- [0398] In some embodiments q is 2.
- [0399] In some embodiments q is 3.
- [0400] Exemplary embodiments include compounds having the formula (XVII)

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or a pharmaceutically acceptable salt form thereof defined herein below in Table 1.

TABLE-US-00001 TABLE 1 Entry n R R.sup.3 1 1 H Phenyl 2 2 H Phenyl 3 3 H Phenyl 4 1 Me Phenyl 5 2 Me Phenyl 6 3 Me Phenyl 7 1 CH.sub.2Ph Phenyl 8 2 CH.sub.2Ph Phenyl 9 3 CH.sub.2Ph Phenyl 10 1 COMe Phenyl 11 2 COMe Phenyl 12 3 COMe Phenyl 13 1 CO.sub.2Me Phenyl 14 2 CO.sub.2Me Phenyl 15 3 CO.sub.2Me Phenyl 16 1 CO.sub.2tBu Phenyl 17 2 CO.sub.2tBu Phenyl 18 3 CO.sub.2tBu Phenyl 19 1 CONHMe Phenyl 20 2 CONHMe Phenyl 21 3 CONHMe Phenyl 22 1 SO.sub.2Me Phenyl 23 2 SO.sub.2Me Phenyl 24 3 SO.sub.2Me Phenyl 25 1 SO.sub.2NH.sub.2 Phenyl 26 2 SO.sub.2NH.sub.2 Phenyl 27 3 SO.sub.2NH.sub.2 Phenyl 28 1 H 3-OH-Phenyl 29 2 H 3-OH-Phenyl 30 3 H 3-OH-Phenyl 31 1 Me 3-OH-Phenyl 32 2 Me 3-OH-Phenyl 33 3 Me 3-OH-Phenyl 34 1 CH.sub.2Ph 3-OH-Phenyl 35 2 CH.sub.2Ph 3-OH-Phenyl 36 3 CH.sub.2Ph 3-OH-Phenyl 37 1 COMe 3-OH-Phenyl 38 2 COMe 3-OH-Phenyl 39 3 COMe 3-OH-Phenyl 40 1 CO.sub.2Me 3-OH-Phenyl 41 2 CO.sub.2Me 3-OH-Phenyl 42 3 CO.sub.2Me 3-OH-Phenyl 43 1 CO.sub.2tBu 3-OH-Phenyl 44 2 CO.sub.2tBu 3-OH-Phenyl 45 3 CO.sub.2tBu 3-OH-Phenyl 46 1 CONHMe 3-OH-Phenyl 47 2 CONHMe 3-OH-Phenyl 48 3 CONHMe 3-OH-Phenyl 49 1 SO.sub.2Me 3-OH-Phenyl 50 2 SO.sub.2Me 3-OH-Phenyl 51 3 SO.sub.2Me 3-OH-Phenyl 52 1 SO.sub.2NH.sub.2 3-OH-Phenyl 53 2 SO.sub.2NH.sub.2 3-OH-Phenyl 54 3 SO.sub.2NH.sub.2 3-OH-Phenyl 55 1 H 4-NO.sub.2-Phenyl 56 2 H 4-NO.sub.2-Phenyl 57 3 H 4-NO.sub.2-Phenyl 58 1 Me 4-NO.sub.2-Phenyl 59 2 Me 4-NO.sub.2-Phenyl 60 3 Me 4-NO.sub.2-Phenyl 61 1 CH.sub.2Ph 4-NO.sub.2-Phenyl 62 2 CH.sub.2Ph 4-NO.sub.2-Phenyl 63 3 CH.sub.2Ph 4-NO.sub.2-Phenyl 64 1 COMe 4-NO.sub.2-Phenyl 65 2 COMe 4-NO.sub.2-Phenyl 66 3 COMe 4-NO.sub.2-Phenyl 67 1 CO.sub.2Me 4-NO.sub.2-Phenyl 68 2 CO.sub.2Me 4-NO.sub.2-Phenyl 69 3 CO.sub.2Me 4-NO.sub.2-Phenyl 70 1 CO.sub.2tBu 4-NO.sub.2-Phenyl 71 2 CO.sub.2tBu 4-NO.sub.2-Phenyl 72 3 CO.sub.2tBu 4-NO.sub.2-Phenyl 73 1 CONHMe 4-NO.sub.2-Phenyl 74 2 CONHMe 4-NO.sub.2-Phenyl 75 3 CONHMe 4-NO.sub.2-Phenyl 76 1 SO.sub.2Me 4-NO.sub.2-Phenyl 77 2 SO.sub.2Me 4-NO.sub.2-Phenyl 78 3 SO.sub.2Me 4-NO.sub.2-Phenyl 79 1 SO.sub.2NH.sub.2 4-NO.sub.2-Phenyl 80 2 SO.sub.2NH.sub.2 4-NO.sub.2-Phenyl 81 3 SO.sub.2NH.sub.2 4-NO.sub.2-Phenyl 82 1 H 3-OMe-Phenyl 83 2 H 3-OMe-Phenyl 84 3 H 3-OMe-Phenyl 85 1 Me 3-OMe-Phenyl 86 2 Me 3-OMe-Phenyl 87 3 Me 3-OMe-Phenyl 88 1

CH.sub.2Ph 3-OMe-Phenyl 89 2 CH.sub.2Ph 3-OMe-Phenyl 90 3 CH.sub.2Ph 3-OMe-Phenyl 91 1
COMe 3-OMe-Phenyl 92 2 COMe 3-OMe-Phenyl 93 3 COMe 3-OMe-Phenyl 94 1 CO.sub.2Me 3-
OMe-Phenyl 95 2 CO.sub.2Me 3-OMe-Phenyl 96 3 CO.sub.2Me 3-OMe-Phenyl 97 1 CO.sub.2tBu
3-OMe-Phenyl 98 2 CO.sub.2tBu 3-OMe-Phenyl 99 3 CO.sub.2tBu 3-OMe-Phenyl 100 1
CONHMe 3-OMe-Phenyl 101 2 CONHMe 3-OMe-Phenyl 102 3 CONHMe 3-OMe-Phenyl 103 1
SO.sub.2Me 3-OMe-Phenyl 104 2 SO.sub.2Me 3-OMe-Phenyl 105 3 SO.sub.2Me 3-OMe-Phenyl
106 1 SO.sub.2NH.sub.2 3-OMe-Phenyl 107 2 SO.sub.2NH.sub.2 3-OMe-Phenyl 108 3
SO.sub.2NH.sub.2 3-OMe-Phenyl 109 1 H 4-CN-Phenyl 110 2 H 4-CN-Phenyl 111 3 H 4-CN-
Phenyl 112 1 Me 4-CN-Phenyl 113 2 Me 4-CN-Phenyl 114 3 Me 4-CN-Phenyl 115 1 CH.sub.2Ph
4-CN-Phenyl 116 2 CH.sub.2Ph 4-CN-Phenyl 117 3 CH.sub.2Ph 4-CN-Phenyl 118 1 COMe 4-CN-
Phenyl 119 2 COMe 4-CN-Phenyl 120 3 COMe 4-CN-Phenyl 121 1 CO.sub.2Me 4-CN-Phenyl
122 2 CO.sub.2Me 4-CN-Phenyl 123 3 CO.sub.2Me 4-CN-Phenyl 124 1 CO.sub.2tBu 4-CN-
Phenyl 125 2 CO.sub.2tBu 4-CN-Phenyl 126 3 CO.sub.2tBu 4-CN-Phenyl 127 1 CONHMe 4-CN-
Phenyl 128 2 CONHMe 4-CN-Phenyl 129 3 CONHMe 4-CN-Phenyl 130 1 SO.sub.2Me 4-CN-
Phenyl 131 2 SO.sub.2Me 4-CN-Phenyl 132 3 SO.sub.2Me 4-CN-Phenyl 133 1 SO.sub.2NH.sub.2
4-CN-Phenyl 134 2 SO.sub.2NH.sub.2 4-CN-Phenyl 135 3 SO.sub.2NH.sub.2 4-CN-Phenyl 136 1
H 2-CN-Phenyl 137 2 H 2-CN-Phenyl 138 3 H 2-CN-Phenyl 139 1 Me 2-CN-Phenyl 140 2 Me 2-
CN-Phenyl 141 3 Me 2-CN-Phenyl 142 1 CH.sub.2Ph 2-CN-Phenyl 143 2 CH.sub.2Ph 2-CN-
Phenyl 144 3 CH.sub.2Ph 2-CN-Phenyl 145 1 COMe 2-CN-Phenyl 146 2 COMe 2-CN-Phenyl 147
3 COMe 2-CN-Phenyl 148 1 CO.sub.2Me 2-CN-Phenyl 149 2 CO.sub.2Me 2-CN-Phenyl 150 3
CO.sub.2Me 2-CN-Phenyl 151 1 CO.sub.2tBu 2-CN-Phenyl 152 2 CO.sub.2tBu 2-CN-Phenyl 153
3 CO.sub.2tBu 2-CN-Phenyl 154 1 CONHMe 2-CN-Phenyl 155 2 CONHMe 2-CN-Phenyl 156 3
CONHMe 2-CN-Phenyl 157 1 SO.sub.2Me 2-CN-Phenyl 158 2 SO.sub.2Me 2-CN-Phenyl 159 3
SO.sub.2Me 2-CN-Phenyl 160 1 SO.sub.2NH.sub.2 2-CN-Phenyl 161 2 SO.sub.2NH.sub.2 2-CN-
Phenyl 162 3 SO.sub.2NH.sub.2 2-CN-Phenyl 163 1 H 3-Me-Phenyl 164 2 H 3-Me-Phenyl 165 3
H 3-Me-Phenyl 166 1 Me 3-Me-Phenyl 167 2 Me 3-Me-Phenyl 168 3 Me 3-Me-Phenyl 169 1
CH.sub.2Ph 3-Me-Phenyl 170 2 CH.sub.2Ph 3-Me-Phenyl 171 3 CH.sub.2Ph 3-Me-Phenyl 172 1
COMe 3-Me-Phenyl 173 2 COMe 3-Me-Phenyl 174 3 COMe 3-Me-Phenyl 175 1 CO.sub.2Me 3-
Me-Phenyl 176 2 CO.sub.2Me 3-Me-Phenyl 177 3 CO.sub.2Me 3-Me-Phenyl 178 1 CO.sub.2tBu
3-Me-Phenyl 179 2 CO.sub.2tBu 3-Me-Phenyl 180 3 CO.sub.2tBu 3-Me-Phenyl 181 1 CONHMe
3-Me-Phenyl 182 2 CONHMe 3-Me-Phenyl 183 3 CONHMe 3-Me-Phenyl 184 1 SO.sub.2Me 3-
Me-Phenyl 185 2 SO.sub.2Me 3-Me-Phenyl 186 3 SO.sub.2Me 3-Me-Phenyl 187 1
SO.sub.2NH.sub.2 3-Me-Phenyl 188 2 SO.sub.2NH.sub.2 3-Me-Phenyl 189 3 SO.sub.2NH.sub.2
3-Me-Phenyl 190 1 H 2-F-Phenyl 191 2 H 2-F-Phenyl 192 3 H 2-F-Phenyl 193 1 Me 2-F-Phenyl
194 2 Me 2-F-Phenyl 195 3 Me 2-F-Phenyl 196 1 CH.sub.2Ph 2-F-Phenyl 197 2 CH.sub.2Ph 2-F-
Phenyl 198 3 CH.sub.2Ph 2-F-Phenyl 199 1 COMe 2-F-Phenyl 200 2 COMe 2-F-Phenyl 201 3
COMe 2-F-Phenyl 202 1 CO.sub.2Me 2-F-Phenyl 203 2 CO.sub.2Me 2-F-Phenyl 204 3
CO.sub.2Me 2-F-Phenyl 205 1 CO.sub.2tBu 2-F-Phenyl 206 2 CO.sub.2tBu 2-F-Phenyl 207 3
CO.sub.2tBu 2-F-Phenyl 208 1 CONHMe 2-F-Phenyl 209 2 CONHMe 2-F-Phenyl 210 3
CONHMe 2-F-Phenyl 211 1 SO.sub.2Me 2-F-Phenyl 212 2 SO.sub.2Me 2-F-Phenyl 213 3
SO.sub.2Me 2-F-Phenyl 214 1 SO.sub.2NH.sub.2 2-F-Phenyl 215 2 SO.sub.2NH.sub.2 2-F-Phenyl
216 3 SO.sub.2NH.sub.2 2-F-Phenyl 217 1 H 4-F-Phenyl 218 2 H 4-F-Phenyl 219 3 H 4-F-Phenyl
220 1 Me 4-F-Phenyl 221 2 Me 4-F-Phenyl 222 3 Me 4-F-Phenyl 223 1 CH.sub.2Ph 4-F-Phenyl
224 2 CH.sub.2Ph 4-F-Phenyl 225 3 CH.sub.2Ph 4-F-Phenyl 226 1 COMe 4-F-Phenyl 227 2
COMe 4-F-Phenyl 228 3 COMe 4-F-Phenyl 229 1 CO.sub.2Me 4-F-Phenyl 230 2 CO.sub.2Me 4-
F-Phenyl 231 3 CO.sub.2Me 4-F-Phenyl 232 1 CO.sub.2tBu 4-F-Phenyl 233 2 CO.sub.2tBu 4-F-
Phenyl 234 3 CO.sub.2tBu 4-F-Phenyl 235 1 CONHMe 4-F-Phenyl 236 2 CONHMe 4-F-Phenyl
237 3 CONHMe 4-F-Phenyl 238 1 SO.sub.2Me 4-F-Phenyl 239 2 SO.sub.2Me 4-F-Phenyl 240 3
SO.sub.2Me 4-F-Phenyl 241 1 SO.sub.2NH.sub.2 4-F-Phenyl 242 2 SO.sub.2NH.sub.2 4-F-Phenyl
243 3 SO.sub.2NH.sub.2 4-F-Phenyl 244 1 H 3-Cl-Phenyl 245 2 H 3-Cl-Phenyl 246 3 H 3-Cl-

Phenyl 247 1 Me 3-Cl-Phenyl 248 2 Me 3-Cl-Phenyl 249 3 Me 3-Cl-Phenyl 250 1 CH.sub.2Ph 3-Cl-Phenyl 251 2 CH.sub.2Ph 3-Cl-Phenyl 252 3 CH.sub.2Ph 3-Cl-Phenyl 253 1 COMe 3-Cl-Phenyl 254 2 COMe 3-Cl-Phenyl 255 3 COMe 3-Cl-Phenyl 256 1 CO.sub.2Me 3-Cl-Phenyl 257 2 CO.sub.2Me 3-Cl-Phenyl 258 3 CO.sub.2Me 3-Cl-Phenyl 259 1 CO.sub.2tBu 3-Cl-Phenyl 260 2 CO.sub.2tBu 3-Cl-Phenyl 261 3 CO.sub.2tBu 3-Cl-Phenyl 262 1 CONHMe 3-Cl-Phenyl 263 2 CONHMe 3-Cl-Phenyl 264 3 CONHMe 3-Cl-Phenyl 265 1 SO.sub.2Me 3-Cl-Phenyl 266 2 SO.sub.2Me 3-Cl-Phenyl 267 3 SO.sub.2Me 3-Cl-Phenyl 268 1 SO.sub.2NH.sub.2 3-Cl-Phenyl 269 2 SO.sub.2NH.sub.2 3-Cl-Phenyl 270 3 SO.sub.2NH.sub.2 3-Cl-Phenyl 271 1 H 2-Br-Phenyl 272 2 H 2-Br-Phenyl 273 3 H 2-Br-Phenyl 274 1 Me 2-Br-Phenyl 275 2 Me 2-Br-Phenyl 276 3 Me 2-Br-Phenyl 277 1 CH.sub.2Ph 2-Br-Phenyl 278 2 CH.sub.2Ph 2-Br-Phenyl 279 3 CH.sub.2Ph 2-Br-Phenyl 280 1 COMe 2-Br-Phenyl 281 2 COMe 2-Br-Phenyl 282 3 COMe 2-Br-Phenyl 283 1 CO.sub.2Me 2-Br-Phenyl 284 2 CO.sub.2Me 2-Br-Phenyl 285 3 CO.sub.2Me 2-Br-Phenyl 286 1 CO.sub.2tBu 2-Br-Phenyl 287 2 CO.sub.2tBu 2-Br-Phenyl 288 3 CO.sub.2tBu 2-Br-Phenyl 289 1 CONHMe 2-Br-Phenyl 290 2 CONHMe 2-Br-Phenyl 291 3 CONHMe 2-Br-Phenyl 292 1 SO.sub.2Me 2-Br-Phenyl 293 2 SO.sub.2Me 2-Br-Phenyl 294 3 SO.sub.2Me 2-Br-Phenyl 295 1 SO.sub.2NH.sub.2 2-Br-Phenyl 296 2 SO.sub.2NH.sub.2 2-Br-Phenyl 297 3 SO.sub.2NH.sub.2 2-Br-Phenyl 298 1 H 4-Br-Phenyl 299 2 H 4-Br-Phenyl 300 3 H 4-Br-Phenyl 301 1 Me 4-Br-Phenyl 302 2 Me 4-Br-Phenyl 303 3 Me 4-Br-Phenyl 304 1 CH.sub.2Ph 4-Br-Phenyl 305 2 CH.sub.2Ph 4-Br-Phenyl 306 3 CH.sub.2Ph 4-Br-Phenyl 307 1 COMe 4-Br-Phenyl 308 2 COMe 4-Br-Phenyl 309 3 COMe 4-Br-Phenyl 310 1 CO.sub.2Me 4-Br-Phenyl 311 2 CO.sub.2Me 4-Br-Phenyl 312 3 CO.sub.2Me 4-Br-Phenyl 313 1 CO.sub.2tBu 4-Br-Phenyl 314 2 CO.sub.2tBu 4-Br-Phenyl 315 3 CO.sub.2tBu 4-Br-Phenyl 316 1 CONHMe 4-Br-Phenyl 317 2 CONHMe 4-Br-Phenyl 318 3 CONHMe 4-Br-Phenyl 319 1 SO.sub.2Me 4-Br-Phenyl 320 2 SO.sub.2Me 4-Br-Phenyl 321 3 SO.sub.2Me 4-Br-Phenyl 322 1 SO.sub.2NH.sub.2 4-Br-Phenyl 323 2 SO.sub.2NH.sub.2 4-Br-Phenyl 324 3 SO.sub.2NH.sub.2 4-Br-Phenyl 325 1 H 3-CF.sub.3-Phenyl 326 2 H 3-CF.sub.3-Phenyl 327 3 H 3-CF.sub.3-Phenyl 328 1 Me 3-CF.sub.3-Phenyl 329 2 Me 3-CF.sub.3-Phenyl 330 3 Me 3-CF.sub.3-Phenyl 331 1 CH.sub.2Ph 3-CF.sub.3-Phenyl 332 2 CH.sub.2Ph 3-CF.sub.3-Phenyl 333 3 CH.sub.2Ph 3-CF.sub.3-Phenyl 334 1 COMe 3-CF.sub.3-Phenyl 335 2 COMe 3-CF.sub.3-Phenyl 336 3 COMe 3-CF.sub.3-Phenyl 337 1 CO.sub.2Me 3-CF.sub.3-Phenyl 338 2 CO.sub.2Me 3-CF.sub.3-Phenyl 339 3 CO.sub.2Me 3-CF.sub.3-Phenyl 340 1 CO.sub.2tBu 3-CF.sub.3-Phenyl 341 2 CO.sub.2tBu 3-CF.sub.3-Phenyl 342 3 CO.sub.2tBu 3-CF.sub.3-Phenyl 343 1 CONHMe 3-CF.sub.3-Phenyl 344 2 CONHMe 3-CF.sub.3-Phenyl 345 3 CONHMe 3-CF.sub.3-Phenyl 346 1 SO.sub.2Me 3-CF.sub.3-Phenyl 347 2 SO.sub.2Me 3-CF.sub.3-Phenyl 348 3 SO.sub.2Me 3-CF.sub.3-Phenyl 349 1 SO.sub.2NH.sub.2 3-CF.sub.3-Phenyl 350 2 SO.sub.2NH.sub.2 3-CF.sub.3-Phenyl 351 3 SO.sub.2NH.sub.2 3-CF.sub.3-Phenyl 352 1 H 2-iPr-Phenyl 353 2 H 2-iPr-Phenyl 354 3 H 2-iPr-Phenyl 355 1 Me 2-iPr-Phenyl 356 2 Me 2-iPr-Phenyl 357 3 Me 2-iPr-Phenyl 358 1 CH.sub.2Ph 2-iPr-Phenyl 359 2 CH.sub.2Ph 2-iPr-Phenyl 360 3 CH.sub.2Ph 2-iPr-Phenyl 361 1 COMe 2-iPr-Phenyl 362 2 COMe 2-iPr-Phenyl 363 3 COMe 2-iPr-Phenyl 364 1 CO.sub.2Me 2-iPr-Phenyl 365 2 CO.sub.2Me 2-iPr-Phenyl 366 3 CO.sub.2Me 2-iPr-Phenyl 367 1 CO.sub.2tBu 2-iPr-Phenyl 368 2 CO.sub.2tBu 2-iPr-Phenyl 369 3 CO.sub.2tBu 2-iPr-Phenyl 370 1 CONHMe 2-iPr-Phenyl 371 2 CONHMe 2-iPr-Phenyl 372 3 CONHMe 2-iPr-Phenyl 373 1 SO.sub.2Me 2-iPr-Phenyl 374 2 SO.sub.2Me 2-iPr-Phenyl 375 3 SO.sub.2Me 2-iPr-Phenyl 376 1 SO.sub.2NH.sub.2 2-iPr-Phenyl 377 2 SO.sub.2NH.sub.2 2-iPr-Phenyl 378 3 SO.sub.2NH.sub.2 2-iPr-Phenyl 379 1 H 4-iPr-Phenyl 380 2 H 4-iPr-Phenyl 381 3 H 4-iPr-Phenyl 382 1 Me 4-iPr-Phenyl 383 2 Me 4-iPr-Phenyl 384 3 Me 4-iPr-Phenyl 385 1 CH.sub.2Ph 4-iPr-Phenyl 386 2 CH.sub.2Ph 4-iPr-Phenyl 387 3 CH.sub.2Ph 4-iPr-Phenyl 388 1 COMe 4-iPr-Phenyl 389 2 COMe 4-iPr-Phenyl 390 3 COMe 4-iPr-Phenyl 391 1 CO.sub.2Me 4-iPr-Phenyl 392 2 CO.sub.2Me 4-iPr-Phenyl 393 3 CO.sub.2Me 4-iPr-Phenyl 394 1 CO.sub.2tBu 4-iPr-Phenyl 395 2 CO.sub.2tBu 4-iPr-Phenyl 396 3 CO.sub.2tBu 4-iPr-Phenyl 397 1 CONHMe 4-iPr-Phenyl 398 2 CONHMe 4-iPr-Phenyl 399 3 CONHMe 4-iPr-Phenyl 400 1 SO.sub.2Me 4-iPr-Phenyl 401 2

SO.sub.2Me 4-iPr-Phenyl 402 3 SO.sub.2Me 4-iPr-Phenyl 403 1 SO.sub.2NH.sub.2 4-iPr-Phenyl 404 2 SO.sub.2NH.sub.2 4-iPr-Phenyl 405 3 SO.sub.2NH.sub.2 4-iPr-Phenyl 406 1 H 3-NH.sub.2-Phenyl 407 2 H 3-NH.sub.2-Phenyl 408 3 H 3-NH.sub.2-Phenyl 409 1 Me 3-NH.sub.2-Phenyl 410 2 Me 3-NH.sub.2-Phenyl 411 3 Me 3-NH.sub.2-Phenyl 412 1 CH.sub.2Ph 3-NH.sub.2-Phenyl 413 2 CH.sub.2Ph 3-NH.sub.2-Phenyl 414 3 CH.sub.2Ph 3-NH.sub.2-Phenyl 415 1 COMe 3-NH.sub.2-Phenyl 416 2 COMe 3-NH.sub.2-Phenyl 417 3 COMe 3-NH.sub.2-Phenyl 418 1 CO.sub.2Me 3-NH.sub.2-Phenyl 419 2 CO.sub.2Me 3-NH.sub.2-Phenyl 420 3 CO.sub.2Me 3-NH.sub.2-Phenyl 421 1 CO.sub.2tBu 3-NH.sub.2-Phenyl 422 2 CO.sub.2tBu 3-NH.sub.2-Phenyl 423 3 CO.sub.2tBu 3-NH.sub.2-Phenyl 424 1 CONHMe 3-NH.sub.2-Phenyl 425 2 CONHMe 3-NH.sub.2-Phenyl 426 3 CONHMe 3-NH.sub.2-Phenyl 427 1 SO.sub.2Me 3-NH.sub.2-Phenyl 428 2 SO.sub.2Me 3-NH.sub.2-Phenyl 429 3 SO.sub.2Me 3-NH.sub.2-Phenyl 430 1 SO.sub.2NH.sub.2 3-NH.sub.2-Phenyl 431 2 SO.sub.2NH.sub.2 3-NH.sub.2-Phenyl 432 3 SO.sub.2NH.sub.2 3-NH.sub.2-Phenyl 433 1 H 2,4-di-Me-Phenyl 434 2 H 2,4-di-Me-Phenyl 435 3 H 2,4-di-Me-Phenyl 436 1 Me 2,4-di-Me-Phenyl 437 2 Me 2,4-di-Me-Phenyl 438 3 Me 2,4-di-Me-Phenyl 439 1 CH.sub.2Ph 2,4-di-Me-Phenyl 440 2 CH.sub.2Ph 2,4-di-Me-Phenyl 441 3 CH.sub.2Ph 2,4-di-Me-Phenyl 442 1 COMe 2,4-di-Me-Phenyl 443 2 COMe 2,4-di-Me-Phenyl 444 3 COMe 2,4-di-Me-Phenyl 445 1 CO.sub.2Me 2,4-di-Me-Phenyl 446 2 CO.sub.2Me 2,4-di-Me-Phenyl 447 3 CO.sub.2Me 2,4-di-Me-Phenyl 448 1 CO.sub.2tBu 2,4-di-Me-Phenyl 449 2 CO.sub.2tBu 2,4-di-Me-Phenyl 450 3 CO.sub.2tBu 2,4-di-Me-Phenyl 451 1 CONHMe 2,4-di-Me-Phenyl 452 2 CONHMe 2,4-di-Me-Phenyl 453 3 CONHMe 2,4-di-Me-Phenyl 454 1 SO.sub.2Me 2,4-di-Me-Phenyl 455 2 SO.sub.2Me 2,4-di-Me-Phenyl 456 3 SO.sub.2Me 2,4-di-Me-Phenyl 457 1 SO.sub.2NH.sub.2 2,4-di-Me-Phenyl 458 2 SO.sub.2NH.sub.2 2,4-di-Me-Phenyl 459 3 SO.sub.2NH.sub.2 2,4-di-Me-Phenyl 460 1 H 2,6-di-iPr-Phenyl 461 2 H 2,6-di-iPr-Phenyl 462 3 H 2,6-di-iPr-Phenyl 463 1 Me 2,6-di-iPr-Phenyl 464 2 Me 2,6-di-iPr-Phenyl 465 3 Me 2,6-di-iPr-Phenyl 466 1 CH.sub.2Ph 2,6-di-iPr-Phenyl 467 2 CH.sub.2Ph 2,6-di-iPr-Phenyl 468 3 CH.sub.2Ph 2,6-di-iPr-Phenyl 469 1 COMe 2,6-di-iPr-Phenyl 470 2 COMe 2,6-di-iPr-Phenyl 471 3 COMe 2,6-di-iPr-Phenyl 472 1 CO.sub.2Me 2,6-di-iPr-Phenyl 473 2 CO.sub.2Me 2,6-di-iPr-Phenyl 474 3 CO.sub.2Me 2,6-di-iPr-Phenyl 475 1 CO.sub.2tBu 2,6-di-iPr-Phenyl 476 2 CO.sub.2tBu 2,6-di-iPr-Phenyl 477 3 CO.sub.2tBu 2,6-di-iPr-Phenyl 478 1 CONHMe 2,6-di-iPr-Phenyl 479 2 CONHMe 2,6-di-iPr-Phenyl 480 3 CONHMe 2,6-di-iPr-Phenyl 481 1 SO.sub.2Me 2,6-di-iPr-Phenyl 482 2 SO.sub.2Me 2,6-di-iPr-Phenyl 483 3 SO.sub.2Me 2,6-di-iPr-Phenyl 484 1 SO.sub.2NH.sub.2 2,6-di-iPr-Phenyl 485 2 SO.sub.2NH.sub.2 2,6-di-iPr-Phenyl 486 3 SO.sub.2NH.sub.2 2,6-di-iPr-Phenyl 487 1 H 3-Ph-Phenyl 488 2 H 3-Ph-Phenyl 489 3 H 3-Ph-Phenyl 490 1 Me 3-Ph-Phenyl 491 2 Me 3-Ph-Phenyl 492 3 Me 3-Ph-Phenyl 493 1 CH.sub.2Ph 3-Ph-Phenyl 494 2 CH.sub.2Ph 3-Ph-Phenyl 495 3 CH.sub.2Ph 3-Ph-Phenyl 496 1 COMe 3-Ph-Phenyl 497 2 COMe 3-Ph-Phenyl 498 3 COMe 3-Ph-Phenyl 499 1 CO.sub.2Me 3-Ph-Phenyl 500 2 CO.sub.2Me 3-Ph-Phenyl 501 3 CO.sub.2Me 3-Ph-Phenyl 502 1 CO.sub.2tBu 3-Ph-Phenyl 503 2 CO.sub.2tBu 3-Ph-Phenyl 504 3 CO.sub.2tBu 3-Ph-Phenyl 505 1 CONHMe 3-Ph-Phenyl 506 2 CONHMe 3-Ph-Phenyl 507 3 CONHMe 3-Ph-Phenyl 508 1 SO.sub.2Me 3-Ph-Phenyl 509 2 SO.sub.2Me 3-Ph-Phenyl 510 3 SO.sub.2Me 3-Ph-Phenyl 511 1 SO.sub.2NH.sub.2 3-Ph-Phenyl 512 2 SO.sub.2NH.sub.2 3-Ph-Phenyl 513 3 SO.sub.2NH.sub.2 3-Ph-Phenyl 514 1 H 2-morpholino-phenyl 515 2 H 2-morpholino-phenyl 516 3 H 2-morpholino-phenyl 517 1 Me 2-morpholino-phenyl 518 2 Me 2-morpholino-phenyl 519 3 Me 2-morpholino-phenyl 520 1 CH.sub.2Ph 2-morpholino-phenyl 521 2 CH.sub.2Ph 2-morpholino-phenyl 522 3 CH.sub.2Ph 2-morpholino-phenyl 523 1 COMe 2-morpholino-phenyl 524 2 COMe 2-morpholino-phenyl 525 3 COMe 2-morpholino-phenyl 526 1 CO.sub.2Me 2-morpholino-phenyl 527 2 CO.sub.2Me 2-morpholino-phenyl 528 3 CO.sub.2Me 2-morpholino-phenyl 529 1 CO.sub.2tBu 2-morpholino-phenyl 530 2 CO.sub.2tBu 2-morpholino-phenyl 531 3 CO.sub.2tBu 2-morpholino-phenyl 532 1 CONHMe 2-morpholino-phenyl 533 2 CONHMe 2-morpholino-phenyl 534 3 CONHMe 2-morpholino-phenyl 535 1 SO.sub.2Me 2-morpholino-phenyl 536 2 SO.sub.2Me 2-morpholino-phenyl 537 3 SO.sub.2Me 2-morpholino-

phenyl 538 1 SO.sub.2NH.sub.2 2-morpholino-phenyl 539 2 SO.sub.2NH.sub.2 2-morpholino-phenyl 540 3 SO.sub.2NH.sub.2 2-morpholino-phenyl 541 1 H 4-morpholino-phenyl 542 2 H 4-morpholino-phenyl 543 3 H 4-morpholino-phenyl 544 1 Me 4-morpholino-phenyl 545 2 Me 4-morpholino-phenyl 546 3 Me 4-morpholino-phenyl 547 1 CH.sub.2Ph 4-morpholino-phenyl 548 2 CH.sub.2Ph 4-morpholino-phenyl 549 3 CH.sub.2Ph 4-morpholino-phenyl 550 1 COMe 4-morpholino-phenyl 551 2 COMe 4-morpholino-phenyl 552 3 COMe 4-morpholino-phenyl 553 1 CO.sub.2Me 4-morpholino-phenyl 554 2 CO.sub.2Me 4-morpholino-phenyl 555 3 CO.sub.2Me 4-morpholino-phenyl 556 1 CO.sub.2tBu 4-morpholino-phenyl 557 2 CO.sub.2tBu 4-morpholino-phenyl 558 3 CO.sub.2tBu 4-morpholino-phenyl 559 1 CONHMe 4-morpholino-phenyl 560 2 CONHMe 4-morpholino-phenyl 561 3 CONHMe 4-morpholino-phenyl 562 1 SO.sub.2Me 4-morpholino-phenyl 563 2 SO.sub.2Me 4-morpholino-phenyl 564 3 SO.sub.2Me 4-morpholino-phenyl 565 1 SO.sub.2NH.sub.2 4-morpholino-phenyl 566 2 SO.sub.2NH.sub.2 4-morpholino-phenyl 567 3 SO.sub.2NH.sub.2 4-morpholino-phenyl 568 1 H 2-pyrimidinyl 569 2 H 2-pyrimidinyl 570 3 H 2-pyrimidinyl 571 1 Me 2-pyrimidinyl 572 2 Me 2-pyrimidinyl 573 3 Me 2-pyrimidinyl 574 1 CH.sub.2Ph 2-pyrimidinyl 575 2 CH.sub.2Ph 2-pyrimidinyl 576 3 CH.sub.2Ph 2-pyrimidinyl 577 1 COMe 2-pyrimidinyl 578 2 COMe 2-pyrimidinyl 579 3 COMe 2-pyrimidinyl 580 1 CO.sub.2Me 2-pyrimidinyl 581 2 CO.sub.2Me 2-pyrimidinyl 582 3 CO.sub.2Me 2-pyrimidinyl 583 1 CO.sub.2tBu 2-pyrimidinyl 584 2 CO.sub.2tBu 2-pyrimidinyl 585 3 CO.sub.2tBu 2-pyrimidinyl 586 1 CONHMe 2-pyrimidinyl 587 2 CONHMe 2-pyrimidinyl 588 3 CONHMe 2-pyrimidinyl 589 1 SO.sub.2Me 2-pyrimidinyl 590 2 SO.sub.2Me 2-pyrimidinyl 591 3 SO.sub.2Me 2-pyrimidinyl 592 1 SO.sub.2NH.sub.2 2-pyrimidinyl 593 2 SO.sub.2NH.sub.2 2-pyrimidinyl 594 3 SO.sub.2NH.sub.2 2-pyrimidinyl 595 1 H 2-methyl-1H- benzo[d]imidazol-4-yl 596 2 H 2-methyl-1H- benzo[d]imidazol-4-yl 597 3 H 2-methyl-1H- benzo[d]imidazol-4-yl 598 1 Me 2-methyl-1H- benzo[d]imidazol-4-yl 599 2 Me 2-methyl-1H- benzo[d]imidazol-4-yl 600 3 Me 2-methyl-1H- benzo[d]imidazol-4-yl 601 1 CH.sub.2Ph 2-methyl-1H- benzo[d]imidazol-4-yl 602 2 CH.sub.2Ph 2-methyl-1H- benzo[d]imidazol-4-yl 603 3 CH.sub.2Ph 2-methyl-1H- benzo[d]imidazol-4-yl 604 1 COMe 2-methyl-1H- benzo[d]imidazol-4-yl 605 2 COMe 2-methyl-1H- benzo[d]imidazol-4-yl 606 3 COMe 2-methyl-1H- benzo[d]imidazol-4-yl 607 1 CO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 608 2 CO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 609 3 CO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 610 1 CO.sub.2tBu 2-methyl-1H- benzo[d]imidazol-4-yl 611 2 CO.sub.2tBu 2-methyl-1H- benzo[d]imidazol-4-yl 612 3 CO.sub.2tBu 2-methyl-1H- benzo[d]imidazol-4-yl 613 1 CONHMe 2-methyl-1H- benzo[d]imidazol-4-yl 614 2 CONHMe 2-methyl-1H- benzo[d]imidazol-4-yl 615 3 CONHMe 2-methyl-1H- benzo[d]imidazol-4-yl 616 1 SO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 617 2 SO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 618 3 SO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 619 1 SO.sub.2NH.sub.2 2-methyl-1H- benzo[d]imidazol-4-yl 620 2 SO.sub.2NH.sub.2 2-methyl-1H- benzo[d]imidazol-4-yl 621 3 SO.sub.2NH.sub.2 2-methyl-1H- benzo[d]imidazol-4-yl 622 1 H 4-OH-Phenyl 623 2 H 4-OH-Phenyl 624 3 H 4-OH-Phenyl 625 1 Me 4-OH-Phenyl 626 2 Me 4-OH-Phenyl 627 3 Me 4-OH-Phenyl 628 1 CH.sub.2Ph 4-OH-Phenyl 629 2 CH.sub.2Ph 4-OH-Phenyl 630 3 CH.sub.2Ph 4-OH-Phenyl 631 1 COMe 4-OH-Phenyl 632 2 COMe 4-OH-Phenyl 633 3 COMe 4-OH-Phenyl 634 1 CO.sub.2Me 4-OH-Phenyl 635 2 CO.sub.2Me 4-OH-Phenyl 636 3 CO.sub.2Me 4-OH-Phenyl 637 1 CO.sub.2tBu 4-OH-Phenyl 638 2 CO.sub.2tBu 4-OH-Phenyl 639 3 CO.sub.2tBu 4-OH-Phenyl 640 1 CONHMe 4-OH-Phenyl 641 2 CONHMe 4-OH-Phenyl 642 3 CONHMe 4-OH-Phenyl 643 1 SO.sub.2Me 4-OH-Phenyl 644 2 SO.sub.2Me 4-OH-Phenyl 645 3 SO.sub.2Me 4-OH-Phenyl 646 1 SO.sub.2NH.sub.2 4-OH-Phenyl 647 2 SO.sub.2NH.sub.2 4-OH-Phenyl 648 3 SO.sub.2NH.sub.2 4-OH-Phenyl 649 1 H 2-OH-Phenyl 650 2 H 2-OH-Phenyl 651 3 H 2-OH-Phenyl 652 1 Me 2-OH-Phenyl 653 2 Me 2-OH-Phenyl 654 3 Me 2-OH-Phenyl 655 1 CH.sub.2Ph 2-OH-Phenyl 656 2 CH.sub.2Ph 2-OH-Phenyl 657 3 CH.sub.2Ph 2-OH-Phenyl 658 1 COMe 2-OH-Phenyl 659 2 COMe 2-OH-Phenyl 660 3 COMe 2-OH-Phenyl 661 1 CO.sub.2Me 2-OH-Phenyl 662 2 CO.sub.2Me 2-OH-Phenyl 663 3 CO.sub.2Me 2-OH-Phenyl 664 1 CO.sub.2tBu

2-OH-Phenyl 665 2 CO.sub.2tBu 2-OH-Phenyl 666 1 CONHMe
2-OH-Phenyl 668 2 CONHMe 2-OH-Phenyl 669 3 CONHMe 2-OH-Phenyl 670 1 SO.sub.2Me 2-
OH-Phenyl 671 2 SO.sub.2Me 2-OH-Phenyl 672 3 SO.sub.2Me 2-OH-Phenyl 673 1
SO.sub.2NH.sub.2 2-OH-Phenyl 674 2 SO.sub.2NH.sub.2 2-OH-Phenyl 675 3 SO.sub.2NH.sub.2
2-OH-Phenyl 676 1 H 4-OMe-Phenyl 677 2 H 4-OMe-Phenyl 678 3 H 4-OMe-Phenyl 679 1 Me 4-
OMe-Phenyl 680 2 Me 4-OMe-Phenyl 681 3 Me 4-OMe-Phenyl 682 1 CH.sub.2Ph 4-OMe-Phenyl
683 2 CH.sub.2Ph 4-OMe-Phenyl 684 3 CH.sub.2Ph 4-OMe-Phenyl 685 1 COMe 4-OMe-Phenyl
686 2 COMe 4-OMe-Phenyl 687 3 COMe 4-OMe-Phenyl 688 1 CO.sub.2Me 4-OMe-Phenyl 689 2
CO.sub.2Me 4-OMe-Phenyl 690 3 CO.sub.2Me 4-OMe-Phenyl 691 1 CO.sub.2tBu 4-OMe-Phenyl
692 2 CO.sub.2tBu 4-OMe-Phenyl 693 3 CO.sub.2tBu 4-OMe-Phenyl 694 1 CONHMe 4-OMe-
Phenyl 695 2 CONHMe 4-OMe-Phenyl 696 3 CONHMe 4-OMe-Phenyl 697 1 SO.sub.2Me 4-
OMe-Phenyl 698 2 SO.sub.2Me 4-OMe-Phenyl 699 3 SO.sub.2Me 4-OMe-Phenyl 700 1
SO.sub.2NH.sub.2 4-OMe-Phenyl 701 2 SO.sub.2NH.sub.2 4-OMe-Phenyl 702 3
SO.sub.2NH.sub.2 4-OMe-Phenyl 703 1 H 2-OMe-Phenyl 704 2 H 2-OMe-Phenyl 705 3 H 2-
OMe-Phenyl 706 1 Me 2-OMe-Phenyl 707 2 Me 2-OMe-Phenyl 708 3 Me 2-OMe-Phenyl 709 1
CH.sub.2Ph 2-OMe-Phenyl 710 2 CH.sub.2Ph 2-OMe-Phenyl 711 3 CH.sub.2Ph 2-OMe-Phenyl
712 1 COMe 2-OMe-Phenyl 713 2 COMe 2-OMe-Phenyl 714 3 COMe 2-OMe-Phenyl 715 1
CO.sub.2Me 2-OMe-Phenyl 716 2 CO.sub.2Me 2-OMe-Phenyl 717 3 CO.sub.2Me 2-OMe-Phenyl
718 1 CO.sub.2tBu 2-OMe-Phenyl 719 2 CO.sub.2tBu 2-OMe-Phenyl 720 3 CO.sub.2tBu 2-OMe-
Phenyl 721 1 CONHMe 2-OMe-Phenyl 722 2 CONHMe 2-OMe-Phenyl 723 3 CONHMe 2-OMe-
Phenyl 724 1 SO.sub.2Me 2-OMe-Phenyl 725 2 SO.sub.2Me 2-OMe-Phenyl 726 3 SO.sub.2Me 2-
OMe-Phenyl 727 1 SO.sub.2NH.sub.2 2-OMe-Phenyl 728 2 SO.sub.2NH.sub.2 2-OMe-Phenyl 729
3 SO.sub.2NH.sub.2 2-OMe-Phenyl 730 1 H 3-CN-Phenyl 731 2 H 3-CN-Phenyl 732 3 H 3-CN-
Phenyl 733 1 Me 3-CN-Phenyl 734 2 Me 3-CN-Phenyl 735 3 Me 3-CN-Phenyl 736 1 CH.sub.2Ph
3-CN-Phenyl 737 2 CH.sub.2Ph 3-CN-Phenyl 738 3 CH.sub.2Ph 3-CN-Phenyl 739 1 COMe 3-
CN-Phenyl 740 2 COMe 3-CN-Phenyl 741 3 COMe 3-CN-Phenyl 742 1 CO.sub.2Me 3-CN-
Phenyl 743 2 CO.sub.2Me 3-CN-Phenyl 744 3 CO.sub.2Me 3-CN-Phenyl 745 1 CO.sub.2tBu 3-
CN-Phenyl 746 2 CO.sub.2tBu 3-CN-Phenyl 747 3 CO.sub.2tBu 3-CN-Phenyl 748 1 CONHMe 3-
CN-Phenyl 749 2 CONHMe 3-CN-Phenyl 750 3 CONHMe 3-CN-Phenyl 751 1 SO.sub.2Me 3-
CN-Phenyl 752 2 SO.sub.2Me 3-CN-Phenyl 753 3 SO.sub.2Me 3-CN-Phenyl 754 1
SO.sub.2NH.sub.2 3-CN-Phenyl 755 2 SO.sub.2NH.sub.2 3-CN-Phenyl 756 3 SO.sub.2NH.sub.2
3-CN-Phenyl 757 1 H 2-Me-Phenyl 758 2 H 2-Me-Phenyl 759 3 H 2-Me-Phenyl 760 1 Me 2-Me-
Phenyl 761 2 Me 2-Me-Phenyl 762 3 Me 2-Me-Phenyl 763 1 CH.sub.2Ph 2-Me-Phenyl 764 2
CH.sub.2Ph 2-Me-Phenyl 765 3 CH.sub.2Ph 2-Me-Phenyl 766 1 COMe 2-Me-Phenyl 767 2
COMe 2-Me-Phenyl 768 3 COMe 2-Me-Phenyl 769 1 CO.sub.2Me 2-Me-Phenyl 770 2
CO.sub.2Me 2-Me-Phenyl 771 3 CO.sub.2Me 2-Me-Phenyl 772 1 CO.sub.2tBu 2-Me-Phenyl 773
2 CO.sub.2tBu 2-Me-Phenyl 774 3 CO.sub.2tBu 2-Me-Phenyl 775 1 CONHMe 2-Me-Phenyl 776
2 CONHMe 2-Me-Phenyl 777 3 CONHMe 2-Me-Phenyl 778 1 SO.sub.2Me 2-Me-Phenyl 779 2
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Phenyl 785 2 H 4-Me-Phenyl 786 3 H 4-Me-Phenyl 787 1 Me 4-Me-Phenyl 788 2 Me 4-Me-
Phenyl 789 3 Me 4-Me-Phenyl 790 1 CH.sub.2Ph 4-Me-Phenyl 791 2 CH.sub.2Ph 4-Me-Phenyl
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COMe 4-Me-Phenyl 796 1 CO.sub.2Me 4-Me-Phenyl 797 2 CO.sub.2Me 4-Me-Phenyl 798 3
CO.sub.2Me 4-Me-Phenyl 799 1 CO.sub.2tBu 4-Me-Phenyl 800 2 CO.sub.2tBu 4-Me-Phenyl 801
3 CO.sub.2tBu 4-Me-Phenyl 802 1 CONHMe 4-Me-Phenyl 803 2 CONHMe 4-Me-Phenyl 804 3
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Phenyl 810 3 SO.sub.2NH.sub.2 4-Me-Phenyl 811 1 H 3-F-Phenyl 812 2 H 3-F-Phenyl 813 3 H 3-
F-Phenyl 814 1 Me 3-F-Phenyl 815 2 Me 3-F-Phenyl 816 3 Me 3-F-Phenyl 817 1 CH.sub.2Ph 3-F-

Phenyl 818 2 CH.sub.2Ph 3-F-Phenyl 819 3 CH.sub.2Ph 3-F-Phenyl 820 1 COMe 3-F-Phenyl 821
2 COMe 3-F-Phenyl 822 3 COMe 3-F-Phenyl 823 1 CO.sub.2Me 3-F-Phenyl 824 2 CO.sub.2Me
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Phenyl 828 3 CO.sub.2tBu 3-F-Phenyl 829 1 CONHMe 3-F-Phenyl 830 2 CONHMe 3-F-Phenyl
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837 3 SO.sub.2NH.sub.2 3-F-Phenyl 838 1 H 2-Cl-Phenyl 839 2 H 2-Cl-Phenyl 840 3 H 2-Cl-
Phenyl 841 1 Me 2-Cl-Phenyl 842 2 Me 2-Cl-Phenyl 843 3 Me 2-Cl-Phenyl 844 1 CH.sub.2Ph 2-
Cl-Phenyl 845 2 CH.sub.2Ph 2-Cl-Phenyl 846 3 CH.sub.2Ph 2-Cl-Phenyl 847 1 COMe 2-Cl-
Phenyl 848 2 COMe 2-Cl-Phenyl 849 3 COMe 2-Cl-Phenyl 850 1 CO.sub.2Me 2-Cl-Phenyl 851 2
CO.sub.2Me 2-Cl-Phenyl 852 3 CO.sub.2Me 2-Cl-Phenyl 853 1 CO.sub.2tBu 2-Cl-Phenyl 854 2
CO.sub.2tBu 2-Cl-Phenyl 855 3 CO.sub.2tBu 2-Cl-Phenyl 856 1 CONHMe 2-Cl-Phenyl 857 2
CONHMe 2-Cl-Phenyl 858 3 CONHMe 2-Cl-Phenyl 859 1 SO.sub.2Me 2-Cl-Phenyl 860 2
SO.sub.2Me 2-Cl-Phenyl 861 3 SO.sub.2Me 2-Cl-Phenyl 862 1 SO.sub.2NH.sub.2 2-Cl-Phenyl
863 2 SO.sub.2NH.sub.2 2-Cl-Phenyl 864 3 SO.sub.2NH.sub.2 2-Cl-Phenyl 865 1 H 4-Cl-Phenyl
866 2 H 4-Cl-Phenyl 867 3 H 4-Cl-Phenyl 868 1 Me 4-Cl-Phenyl 869 2 Me 4-Cl-Phenyl 870 3 Me
4-Cl-Phenyl 871 1 CH.sub.2Ph 4-Cl-Phenyl 872 2 CH.sub.2Ph 4-Cl-Phenyl 873 3 CH.sub.2Ph 4-
Cl-Phenyl 874 1 COMe 4-Cl-Phenyl 875 2 COMe 4-Cl-Phenyl 876 3 COMe 4-Cl-Phenyl 877 1
CO.sub.2Me 4-Cl-Phenyl 878 2 CO.sub.2Me 4-Cl-Phenyl 879 3 CO.sub.2Me 4-Cl-Phenyl 880 1
CO.sub.2tBu 4-Cl-Phenyl 881 2 CO.sub.2tBu 4-Cl-Phenyl 882 3 CO.sub.2tBu 4-Cl-Phenyl 883 1
CONHMe 4-Cl-Phenyl 884 2 CONHMe 4-Cl-Phenyl 885 3 CONHMe 4-Cl-Phenyl 886 1
SO.sub.2Me 4-Cl-Phenyl 887 2 SO.sub.2Me 4-Cl-Phenyl 888 3 SO.sub.2Me 4-Cl-Phenyl 889 1
SO.sub.2NH.sub.2 4-Cl-Phenyl 890 2 SO.sub.2NH.sub.2 4-Cl-Phenyl 891 3 SO.sub.2NH.sub.2 4-
Cl-Phenyl 892 1 H 3-Br-Phenyl 893 2 H 3-Br-Phenyl 894 3 H 3-Br-Phenyl 895 1 Me 3-Br-Phenyl
896 2 Me 3-Br-Phenyl 897 3 Me 3-Br-Phenyl 898 1 CH.sub.2Ph 3-Br-Phenyl 899 2 CH.sub.2Ph 3-
Br-Phenyl 900 3 CH.sub.2Ph 3-Br-Phenyl 901 1 COMe 3-Br-Phenyl 902 2 COMe 3-Br-Phenyl 903
3 COMe 3-Br-Phenyl 904 1 CO.sub.2Me 3-Br-Phenyl 905 2 CO.sub.2Me 3-Br-Phenyl 906 3
CO.sub.2Me 3-Br-Phenyl 907 1 CO.sub.2tBu 3-Br-Phenyl 908 2 CO.sub.2tBu 3-Br-Phenyl 909 3
CO.sub.2tBu 3-Br-Phenyl 910 1 CONHMe 3-Br-Phenyl 911 2 CONHMe 3-Br-Phenyl 912 3
CONHMe 3-Br-Phenyl 913 1 SO.sub.2Me 3-Br-Phenyl 914 2 SO.sub.2Me 3-Br-Phenyl 915 3
SO.sub.2Me 3-Br-Phenyl 916 1 SO.sub.2NH.sub.2 3-Br-Phenyl 917 2 SO.sub.2NH.sub.2 3-Br-
Phenyl 918 3 SO.sub.2NH.sub.2 3-Br-Phenyl 919 1 H 2-CF.sub.3-Phenyl 920 2 H 2-CF.sub.3-
Phenyl 921 3 H 2-CF.sub.3-Phenyl 922 1 Me 2-CF.sub.3-Phenyl 923 2 Me 2-CF.sub.3-Phenyl 924
3 Me 2-CF.sub.3-Phenyl 925 1 CH.sub.2Ph 2-CF.sub.3-Phenyl 926 2 CH.sub.2Ph 2-CF.sub.3-
Phenyl 927 3 CH.sub.2Ph 2-CF.sub.3-Phenyl 928 1 COMe 2-CF.sub.3-Phenyl 929 2 COMe 2-
CF.sub.3-Phenyl 930 3 COMe 2-CF.sub.3-Phenyl 931 1 CO.sub.2Me 2-CF.sub.3-Phenyl 932 2
CO.sub.2Me 2-CF.sub.3-Phenyl 933 3 CO.sub.2Me 2-CF.sub.3-Phenyl 934 1 CO.sub.2tBu 2-
CF.sub.3-Phenyl 935 2 CO.sub.2tBu 2-CF.sub.3-Phenyl 936 3 CO.sub.2tBu 2-CF.sub.3-Phenyl 937
1 CONHMe 2-CF.sub.3-Phenyl 938 2 CONHMe 2-CF.sub.3-Phenyl 939 3 CONHMe 2-CF.sub.3-
Phenyl 940 1 SO.sub.2Me 2-CF.sub.3-Phenyl 941 2 SO.sub.2Me 2-CF.sub.3-Phenyl 942 3
SO.sub.2Me 2-CF.sub.3-Phenyl 943 1 SO.sub.2NH.sub.2 2-CF.sub.3-Phenyl 944 2
SO.sub.2NH.sub.2 2-CF.sub.3-Phenyl 945 3 SO.sub.2NH.sub.2 2-CF.sub.3-Phenyl 946 1 H 4-
CF.sub.3-Phenyl 947 2 H 4-CF.sub.3-Phenyl 948 3 H 4-CF.sub.3-Phenyl 949 1 Me 4-CF.sub.3-
Phenyl 950 2 Me 4-CF.sub.3-Phenyl 951 3 Me 4-CF.sub.3-Phenyl 952 1 CH.sub.2Ph 4-CF.sub.3-
Phenyl 953 2 CH.sub.2Ph 4-CF.sub.3-Phenyl 954 3 CH.sub.2Ph 4-CF.sub.3-Phenyl 955 1 COMe
4-CF.sub.3-Phenyl 956 2 COMe 4-CF.sub.3-Phenyl 957 3 COMe 4-CF.sub.3-Phenyl 958 1
CO.sub.2Me 4-CF.sub.3-Phenyl 959 2 CO.sub.2Me 4-CF.sub.3-Phenyl 960 3 CO.sub.2Me 4-
CF.sub.3-Phenyl 961 1 CO.sub.2tBu 4-CF.sub.3-Phenyl 962 2 CO.sub.2tBu 4-CF.sub.3-Phenyl 963
3 CO.sub.2tBu 4-CF.sub.3-Phenyl 964 1 CONHMe 4-CF.sub.3-Phenyl 965 2 CONHMe 4-
CF.sub.3-Phenyl 966 3 CONHMe 4-CF.sub.3-Phenyl 967 1 SO.sub.2Me 4-CF.sub.3-Phenyl 968 2

SO.sub.2Me 4-CF.sub.3-Phenyl 969 3 SO.sub.2Me 4-CF.sub.3-Phenyl 970 1 SO.sub.2NH.sub.2 4-CF.sub.3-Phenyl 971 2 SO.sub.2NH.sub.2 4-CF.sub.3-Phenyl 972 3 SO.sub.2NH.sub.2 4-CF.sub.3-Phenyl 973 1 H 3-iPr-Phenyl 974 2 H 3-iPr-Phenyl 975 3 H 3-iPr-Phenyl 976 1 Me 3-iPr-Phenyl 977 2 Me 3-iPr-Phenyl 978 3 Me 3-iPr-Phenyl 979 1 CH.sub.2Ph 3-iPr-Phenyl 980 2 CH.sub.2Ph 3-iPr-Phenyl 981 3 CH.sub.2Ph 3-iPr-Phenyl 982 1 COMe 3-iPr-Phenyl 983 2 COMe 3-iPr-Phenyl 984 3 COMe 3-iPr-Phenyl 985 1 CO.sub.2Me 3-iPr-Phenyl 986 2 CO.sub.2Me 3-iPr-Phenyl 987 3 CO.sub.2Me 3-iPr-Phenyl 988 1 CO.sub.2tBu 3-iPr-Phenyl 989 2 CO.sub.2tBu 3-iPr-Phenyl 990 3 CO.sub.2tBu 3-iPr-Phenyl 991 1 CONHMe 3-iPr-Phenyl 992 2 CONHMe 3-iPr-Phenyl 993 3 CONHMe 3-iPr-Phenyl 994 1 SO.sub.2Me 3-iPr-Phenyl 995 2 SO.sub.2Me 3-iPr-Phenyl 996 3 SO.sub.2Me 3-iPr-Phenyl 997 1 SO.sub.2NH.sub.2 3-iPr-Phenyl 998 2 SO.sub.2NH.sub.2 3-iPr-Phenyl 999 3 SO.sub.2NH.sub.2 3-iPr-Phenyl 1000 1 H 4-NH.sub.2-Phenyl 1001 2 H 4-NH.sub.2-Phenyl 1002 3 H 4-NH.sub.2-Phenyl 1003 1 Me 4-NH.sub.2-Phenyl 1004 2 Me 4-NH.sub.2-Phenyl 1005 3 Me 4-NH.sub.2-Phenyl 1006 1 CH.sub.2Ph 4-NH.sub.2-Phenyl 1007 2 CH.sub.2Ph 4-NH.sub.2-Phenyl 1008 3 CH.sub.2Ph 4-NH.sub.2-Phenyl 1009 1 COMe 4-NH.sub.2-Phenyl 1010 2 COMe 4-NH.sub.2-Phenyl 1011 3 COMe 4-NH.sub.2-Phenyl 1012 1 CO.sub.2Me 4-NH.sub.2-Phenyl 1013 2 CO.sub.2Me 4-NH.sub.2-Phenyl 1014 3 CO.sub.2Me 4-NH.sub.2-Phenyl 1015 1 CO.sub.2tBu 4-NH.sub.2-Phenyl 1016 2 CO.sub.2tBu 4-NH.sub.2-Phenyl 1017 3 CO.sub.2tBu 4-NH.sub.2-Phenyl 1018 1 CONHMe 4-NH.sub.2-Phenyl 1019 2 CONHMe 4-NH.sub.2-Phenyl 1020 3 CONHMe 4-NH.sub.2-Phenyl 1021 1 SO.sub.2Me 4-NH.sub.2-Phenyl 1022 2 SO.sub.2Me 4-NH.sub.2-Phenyl 1023 3 SO.sub.2Me 4-NH.sub.2-Phenyl 1024 1 SO.sub.2NH.sub.2 4-NH.sub.2-Phenyl 1025 2 SO.sub.2NH.sub.2 4-NH.sub.2-Phenyl 1026 3 SO.sub.2NH.sub.2 4-NH.sub.2-Phenyl 1027 1 H 2-NH.sub.2-Phenyl 1028 2 H 2-NH.sub.2-Phenyl 1029 3 H 2-NH.sub.2-Phenyl 1030 1 Me 2-NH.sub.2-Phenyl 1031 2 Me 2-NH.sub.2-Phenyl 1032 3 Me 2-NH.sub.2-Phenyl 1033 1 CH.sub.2Ph 2-NH.sub.2-Phenyl 1034 2 CH.sub.2Ph 2-NH.sub.2-Phenyl 1035 3 CH.sub.2Ph 2-NH.sub.2-Phenyl 1036 1 COMe 2-NH.sub.2-Phenyl 1037 2 COMe 2-NH.sub.2-Phenyl 1038 3 COMe 2-NH.sub.2-Phenyl 1039 1 CO.sub.2Me 2-NH.sub.2-Phenyl 1040 2 CO.sub.2Me 2-NH.sub.2-Phenyl 1041 3 CO.sub.2Me 2-NH.sub.2-Phenyl 1042 1 CO.sub.2tBu 2-NH.sub.2-Phenyl 1043 2 CO.sub.2tBu 2-NH.sub.2-Phenyl 1044 3 CO.sub.2tBu 2-NH.sub.2-Phenyl 1045 1 CONHMe 2-NH.sub.2-Phenyl 1046 2 CONHMe 2-NH.sub.2-Phenyl 1047 3 CONHMe 2-NH.sub.2-Phenyl 1048 1 SO.sub.2Me 2-NH.sub.2-Phenyl 1049 2 SO.sub.2Me 2-NH.sub.2-Phenyl 1050 3 SO.sub.2Me 2-NH.sub.2-Phenyl 1051 1 SO.sub.2NH.sub.2 2-NH.sub.2-Phenyl 1052 2 SO.sub.2NH.sub.2 2-NH.sub.2-Phenyl 1053 3 SO.sub.2NH.sub.2 2-NH.sub.2-Phenyl 1054 1 H 2,6-di-Me-Phenyl 1055 2 H 2,6-di-Me-Phenyl 1056 3 H 2,6-di-Me-Phenyl 1057 1 Me 2,6-di-Me-Phenyl 1058 2 Me 2,6-di-Me-Phenyl 1059 3 Me 2,6-di-Me-Phenyl 1060 1 CH.sub.2Ph 2,6-di-Me-Phenyl 1061 2 CH.sub.2Ph 2,6-di-Me-Phenyl 1062 3 CH.sub.2Ph 2,6-di-Me-Phenyl 1063 1 COMe 2,6-di-Me-Phenyl 1064 2 COMe 2,6-di-Me-Phenyl 1065 3 COMe 2,6-di-Me-Phenyl 1066 1 CO.sub.2Me 2,6-di-Me-Phenyl 1067 2 CO.sub.2Me 2,6-di-Me-Phenyl 1068 3 CO.sub.2Me 2,6-di-Me-Phenyl 1069 1 CO.sub.2tBu 2,6-di-Me-Phenyl 1070 2 CO.sub.2tBu 2,6-di-Me-Phenyl 1071 3 CO.sub.2tBu 2,6-di-Me-Phenyl 1072 1 CONHMe 2,6-di-Me-Phenyl 1073 2 CONHMe 2,6-di-Me-Phenyl 1074 3 CONHMe 2,6-di-Me-Phenyl 1075 1 SO.sub.2Me 2,6-di-Me-Phenyl 1076 2 SO.sub.2Me 2,6-di-Me-Phenyl 1077 3 SO.sub.2Me 2,6-di-Me-Phenyl 1078 1 SO.sub.2NH.sub.2 2,6-di-Me-Phenyl 1079 2 SO.sub.2NH.sub.2 2,6-di-Me-Phenyl 1080 3 SO.sub.2NH.sub.2 2,6-di-Me-Phenyl 1081 1 H 2-Ph-Phenyl 1082 2 H 2-Ph-Phenyl 1083 3 H 2-Ph-Phenyl 1084 1 Me 2-Ph-Phenyl 1085 2 Me 2-Ph-Phenyl 1086 3 Me 2-Ph-Phenyl 1087 1 CH.sub.2Ph 2-Ph-Phenyl 1088 2 CH.sub.2Ph 2-Ph-Phenyl 1089 3 CH.sub.2Ph 2-Ph-Phenyl 1090 1 COMe 2-Ph-Phenyl 1091 2 COMe 2-Ph-Phenyl 1092 3 COMe 2-Ph-Phenyl 1093 1 CO.sub.2Me 2-Ph-Phenyl 1094 2 CO.sub.2Me 2-Ph-Phenyl 1095 3 CO.sub.2Me 2-Ph-Phenyl 1096 1 CO.sub.2tBu 2-Ph-Phenyl 1097 2 CO.sub.2tBu 2-Ph-Phenyl 1098 3 CO.sub.2tBu 2-Ph-Phenyl 1099 1 CONHMe 2-Ph-Phenyl 1100 2 CONHMe 2-Ph-Phenyl 1101 3 CONHMe 2-Ph-Phenyl 1102 1 SO.sub.2Me 2-Ph-Phenyl 1103 2 SO.sub.2Me 2-Ph-Phenyl 1104 3 SO.sub.2Me 2-Ph-Phenyl 1105 1

SO.sub.2NH.sub.2 2-Ph-Phenyl 1106 2 SO.sub.2NH.sub.2 2-Ph-Phenyl 1107 3 SO.sub.2NH.sub.2 2-Ph-Phenyl 1108 1 H 4-Ph-Phenyl 1109 2 H 4-Ph-Phenyl 1110 3 H 4-Ph-Phenyl 1111 1 Me 4-Ph-Phenyl 1112 2 Me 4-Ph-Phenyl 1113 3 Me 4-Ph-Phenyl 1114 1 CH.sub.2Ph 4-Ph-Phenyl 1115 2 CH.sub.2Ph 4-Ph-Phenyl 1116 3 CH.sub.2Ph 4-Ph-Phenyl 1117 1 COMe 4-Ph-Phenyl 1118 2 COMe 4-Ph-Phenyl 1119 3 COMe 4-Ph-Phenyl 1120 1 CO.sub.2Me 4-Ph-Phenyl 1121 2 CO.sub.2Me 4-Ph-Phenyl 1122 3 CO.sub.2Me 4-Ph-Phenyl 1123 1 CO.sub.2tBu 4-Ph-Phenyl 1124 2 CO.sub.2tBu 4-Ph-Phenyl 1125 3 CO.sub.2tBu 4-Ph-Phenyl 1126 1 CONHMe 4-Ph-Phenyl 1127 2 CONHMe 4-Ph-Phenyl 1128 3 CONHMe 4-Ph-Phenyl 1129 1 SO.sub.2Me 4-Ph-Phenyl 1130 2 SO.sub.2Me 4-Ph-Phenyl 1131 3 SO.sub.2Me 4-Ph-Phenyl 1132 1 SO.sub.2NH.sub.2 4-Ph-Phenyl 1133 2 SO.sub.2NH.sub.2 4-Ph-Phenyl 1134 3 SO.sub.2NH.sub.2 4-Ph-Phenyl 1135 1 H 3-morpholino-phenyl 1136 2 H 3-morpholino-phenyl 1137 3 H 3-morpholino-phenyl 1138 1 Me 3-morpholino-phenyl 1139 2 Me 3-morpholino-phenyl 1140 3 Me 3-morpholino-phenyl 1141 1 CH.sub.2Ph 3-morpholino-phenyl 1142 2 CH.sub.2Ph 3-morpholino-phenyl 1143 3 CH.sub.2Ph 3-morpholino-phenyl 1144 1 COMe 3-morpholino-phenyl 1145 2 COMe 3-morpholino-phenyl 1146 3 COMe 3-morpholino-phenyl 1147 1 CO.sub.2Me 3-morpholino-phenyl 1148 2 CO.sub.2Me 3-morpholino-phenyl 1149 3 CO.sub.2Me 3-morpholino-phenyl 1150 1 CO.sub.2tBu 3-morpholino-phenyl 1151 2 CO.sub.2tBu 3-morpholino-phenyl 1152 3 CO.sub.2tBu 3-morpholino-phenyl 1153 1 CONHMe 3-morpholino-phenyl 1154 2 CONHMe 3-morpholino-phenyl 1155 3 CONHMe 3-morpholino-phenyl 1156 1 SO.sub.2Me 3-morpholino-phenyl 1157 2 SO.sub.2Me 3-morpholino-phenyl 1158 3 SO.sub.2Me 3-morpholino-phenyl 1159 1 SO.sub.2NH.sub.2 3-morpholino-phenyl 1160 2 SO.sub.2NH.sub.2 3-morpholino-phenyl 1161 3 SO.sub.2NH.sub.2 3-morpholino-phenyl 1162 1 H 2-pyrazinyl 1163 2 H 2-pyrazinyl 1164 3 H 2-pyrazinyl 1165 1 Me 2-pyrazinyl 1166 2 Me 2-pyrazinyl 1167 3 Me 2-pyrazinyl 1168 1 CH.sub.2Ph 2-pyrazinyl 1169 2 CH.sub.2Ph 2-pyrazinyl 1170 3 CH.sub.2Ph 2-pyrazinyl 1171 1 COMe 2-pyrazinyl 1172 2 COMe 2-pyrazinyl 1173 3 COMe 2-pyrazinyl 1174 1 CO.sub.2Me 2-pyrazinyl 1175 2 CO.sub.2Me 2-pyrazinyl 1176 3 CO.sub.2Me 2-pyrazinyl 1177 1 CO.sub.2tBu 2-pyrazinyl 1178 2 CO.sub.2tBu 2-pyrazinyl 1179 3 CO.sub.2tBu 2-pyrazinyl 1180 1 CONHMe 2-pyrazinyl 1181 2 CONHMe 2-pyrazinyl 1182 3 CONHMe 2-pyrazinyl 1183 1 SO.sub.2Me 2-pyrazinyl 1184 2 SO.sub.2Me 2-pyrazinyl 1185 3 SO.sub.2Me 2-pyrazinyl 1186 1 SO.sub.2NH.sub.2 2-pyrazinyl 1187 2 SO.sub.2NH.sub.2 2-pyrazinyl 1188 3 SO.sub.2NH.sub.2 2-pyrazinyl 1189 1 H 5-indolyl 1190 2 H 5-indolyl 1191 3 H 5-indolyl 1192 1 Me 5-indolyl 1193 2 Me 5-indolyl 1194 3 Me 5-indolyl 1195 1 CH.sub.2Ph 5-indolyl 1196 2 CH.sub.2Ph 5-indolyl 1197 3 CH.sub.2Ph 5-indolyl 1198 1 COMe 5-indolyl 1199 2 COMe 5-indolyl 1200 3 COMe 5-indolyl 1201 1 CO.sub.2Me 5-indolyl 1202 2 CO.sub.2Me 5-indolyl 1203 3 CO.sub.2Me 5-indolyl 1204 1 CO.sub.2tBu 5-indolyl 1205 2 CO.sub.2tBu 5-indolyl 1206 3 CO.sub.2tBu 5-indolyl 1207 1 CONHMe 5-indolyl 1208 2 CONHMe 5-indolyl 1209 3 CONHMe 5-indolyl 1210 1 SO.sub.2Me 5-indolyl 1211 2 SO.sub.2Me 5-indolyl 1212 3 SO.sub.2Me 5-indolyl 1213 1 SO.sub.2NH.sub.2 5-indolyl 1214 2 SO.sub.2NH.sub.2 5-indolyl 1215 3 SO.sub.2NH.sub.2 5-indolyl 1216 1 H 1H-benzo[d]imidazol- 4-yl 1217 2 H 1H-benzo[d]imidazol- 4-yl 1218 3 H 1H-benzo[d]imidazol- 4-yl 1219 1 Me 1H-benzo[d]imidazol- 4-yl 1220 2 Me 1H-benzo[d]imidazol- 4-yl 1221 3 Me 1H-benzo[d]imidazol- 4-yl 1222 1 CH.sub.2Ph 1H-benzo[d]imidazol- 4-yl 1223 2 CH.sub.2Ph 1H-benzo[d]imidazol- 4-yl 1224 3 CH.sub.2Ph 1H-benzo[d]imidazol- 4-yl 1225 1 COMe 1H-benzo[d]imidazol- 4-yl 1226 2 COMe 1H-benzo[d]imidazol- 4-yl 1227 3 COMe 1H-benzo[d]imidazol- 4-yl 1228 1 CO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1229 2 CO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1230 3 CO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1231 1 CO.sub.2tBu 1H-benzo[d]imidazol- 4-yl 1232 2 CO.sub.2tBu 1H-benzo[d]imidazol- 4-yl 1233 3 CO.sub.2tBu 1H-benzo[d]imidazol- 4-yl 1234 1 CONHMe 1H-benzo[d]imidazol- 4-yl 1235 2 CONHMe 1H-benzo[d]imidazol- 4-yl 1236 3 CONHMe 1H-benzo[d]imidazol- 4-yl 1237 1 SO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1238 2 SO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1239 3 SO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1240 1 SO.sub.2NH.sub.2 1H-benzo[d]imidazol- 4-yl 1241 2 SO.sub.2NH.sub.2 1H-benzo[d]imidazol- 4-yl 1242 3

SO.sub.2NH.sub.2 1H-benzo[d]imidazol- 4-yl

[0401] Exemplary embodiments include compounds having the formula (XVIII)

##STR00060##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 2.

TABLE-US-00002 TABLE 2 Entry n R R.sup.3 1 1 H Phenyl 2 2 H Phenyl 3 3 H Phenyl 4 1 Me Phenyl 5 2 Me Phenyl 6 3 Me Phenyl 7 1 CH.sub.2Ph Phenyl 8 2 CH.sub.2Ph Phenyl 9 3 CH.sub.2Ph Phenyl 10 1 COMe Phenyl 11 2 COMe Phenyl 12 3 COMe Phenyl 13 1 CO.sub.2Me Phenyl 14 2 CO.sub.2Me Phenyl 15 3 CO.sub.2Me Phenyl 16 1 CO.sub.2tBu Phenyl 17 2 CO.sub.2tBu Phenyl 18 3 CO.sub.2tBu Phenyl 19 1 CONHMe Phenyl 20 2 CONHMe Phenyl 21 3 CONHMe Phenyl 22 1 SO.sub.2Me Phenyl 23 2 SO.sub.2Me Phenyl 24 3 SO.sub.2Me Phenyl 25 1 SO.sub.2NH.sub.2 Phenyl 26 2 SO.sub.2NH.sub.2 Phenyl 27 3 SO.sub.2NH.sub.2 Phenyl 28 1 H 3-OH-Phenyl 29 2 H 3-OH-Phenyl 30 3 H 3-OH-Phenyl 31 1 Me 3-OH-Phenyl 32 2 Me 3-OH-Phenyl 33 3 Me 3-OH-Phenyl 34 1 CH.sub.2Ph 3-OH-Phenyl 35 2 CH.sub.2Ph 3-OH-Phenyl 36 3 CH.sub.2Ph 3-OH-Phenyl 37 1 COMe 3-OH-Phenyl 38 2 COMe 3-OH-Phenyl 39 3 COMe 3-OH-Phenyl 40 1 CO.sub.2Me 3-OH-Phenyl 41 2 CO.sub.2Me 3-OH-Phenyl 42 3 CO.sub.2Me 3-OH-Phenyl 43 1 CO.sub.2tBu 3-OH-Phenyl 44 2 CO.sub.2tBu 3-OH-Phenyl 45 3 CO.sub.2tBu 3-OH-Phenyl 46 1 CONHMe 3-OH-Phenyl 47 2 CONHMe 3-OH-Phenyl 48 3 CONHMe 3-OH-Phenyl 49 1 SO.sub.2Me 3-OH-Phenyl 50 2 SO.sub.2Me 3-OH-Phenyl 51 3 SO.sub.2Me 3-OH-Phenyl 52 1 SO.sub.2NH.sub.2 3-OH-Phenyl 53 2 SO.sub.2NH.sub.2 3-OH-Phenyl 54 3 SO.sub.2NH.sub.2 3-OH-Phenyl 55 1 H 4-NO.sub.2-Phenyl 56 2 H 4-NO.sub.2-Phenyl 57 3 H 4-NO.sub.2-Phenyl 58 1 Me 4-NO.sub.2-Phenyl 59 2 Me 4-NO.sub.2-Phenyl 60 3 Me 4-NO.sub.2-Phenyl 61 1 CH.sub.2Ph 4-NO.sub.2-Phenyl 62 2 CH.sub.2Ph 4-NO.sub.2-Phenyl 63 3 CH.sub.2Ph 4-NO.sub.2-Phenyl 64 1 COMe 4-NO.sub.2-Phenyl 65 2 COMe 4-NO.sub.2-Phenyl 66 3 COMe 4-NO.sub.2-Phenyl 67 1 CO.sub.2Me 4-NO.sub.2-Phenyl 68 2 CO.sub.2Me 4-NO.sub.2-Phenyl 69 3 CO.sub.2Me 4-NO.sub.2-Phenyl 70 1 CO.sub.2tBu 4-NO.sub.2-Phenyl 71 2 CO.sub.2tBu 4-NO.sub.2-Phenyl 72 3 CO.sub.2tBu 4-NO.sub.2-Phenyl 73 1 CONHMe 4-NO.sub.2-Phenyl 74 2 CONHMe 4-NO.sub.2-Phenyl 75 3 CONHMe 4-NO.sub.2-Phenyl 76 1 SO.sub.2Me 4-NO.sub.2-Phenyl 77 2 SO.sub.2Me 4-NO.sub.2-Phenyl 78 3 SO.sub.2Me 4-NO.sub.2-Phenyl 79 1 SO.sub.2NH.sub.2 4-NO.sub.2-Phenyl 80 2 SO.sub.2NH.sub.2 4-NO.sub.2-Phenyl 81 3 SO.sub.2NH.sub.2 4-NO.sub.2-Phenyl 82 1 H 3-OMe-Phenyl 83 2 H 3-OMe-Phenyl 84 3 H 3-OMe-Phenyl 85 1 Me 3-OMe-Phenyl 86 2 Me 3-OMe-Phenyl 87 3 Me 3-OMe-Phenyl 88 1 CH.sub.2Ph 3-OMe-Phenyl 89 2 CH.sub.2Ph 3-OMe-Phenyl 90 3 CH.sub.2Ph 3-OMe-Phenyl 91 1 COMe 3-OMe-Phenyl 92 2 COMe 3-OMe-Phenyl 93 3 COMe 3-OMe-Phenyl 94 1 CO.sub.2Me 3-OMe-Phenyl 95 2 CO.sub.2Me 3-OMe-Phenyl 96 3 CO.sub.2Me 3-OMe-Phenyl 97 1 CO.sub.2tBu 3-OMe-Phenyl 98 2 CO.sub.2tBu 3-OMe-Phenyl 99 3 CO.sub.2tBu 3-OMe-Phenyl 100 1 CONHMe 3-OMe-Phenyl 101 2 CONHMe 3-OMe-Phenyl 102 3 CONHMe 3-OMe-Phenyl 103 1 SO.sub.2Me 3-OMe-Phenyl 104 2 SO.sub.2Me 3-OMe-Phenyl 105 3 SO.sub.2Me 3-OMe-Phenyl 106 1 SO.sub.2NH.sub.2 3-OMe-Phenyl 107 2 SO.sub.2NH.sub.2 3-OMe-Phenyl 108 3 SO.sub.2NH.sub.2 3-OMe-Phenyl 109 1 H 4-CN-Phenyl 110 2 H 4-CN-Phenyl 111 3 H 4-CN-Phenyl 112 1 Me 4-CN-Phenyl 113 2 Me 4-CN-Phenyl 114 3 Me 4-CN-Phenyl 115 1 CH.sub.2Ph 4-CN-Phenyl 116 2 CH.sub.2Ph 4-CN-Phenyl 117 3 CH.sub.2Ph 4-CN-Phenyl 118 1 COMe 4-CN-Phenyl 119 2 COMe 4-CN-Phenyl 120 3 COMe 4-CN-Phenyl 121 1 CO.sub.2Me 4-CN-Phenyl 122 2 CO.sub.2Me 4-CN-Phenyl 123 3 CO.sub.2Me 4-CN-Phenyl 124 1 CO.sub.2tBu 4-CN-Phenyl 125 2 CO.sub.2tBu 4-CN-Phenyl 126 3 CO.sub.2tBu 4-CN-Phenyl 127 1 CONHMe 4-CN-Phenyl 128 2 CONHMe 4-CN-Phenyl 129 3 CONHMe 4-CN-Phenyl 130 1 SO.sub.2Me 4-CN-Phenyl 131 2 SO.sub.2Me 4-CN-Phenyl 132 3 SO.sub.2Me 4-CN-Phenyl 133 1 SO.sub.2NH.sub.2 4-CN-Phenyl 134 2 SO.sub.2NH.sub.2 4-CN-Phenyl 135 3 SO.sub.2NH.sub.2 4-CN-Phenyl 136 1 H 2-CN-Phenyl 137 2 H 2-CN-Phenyl 138 3 H 2-CN-Phenyl 139 1 Me 2-CN-Phenyl 140 2 Me 2-CN-Phenyl 141 3 Me 2-CN-Phenyl 142 1 CH.sub.2Ph 2-CN-Phenyl 143 2 CH.sub.2Ph 2-CN-Phenyl 144 3 CH.sub.2Ph 2-CN-Phenyl 145 1 COMe 2-CN-Phenyl 146 2 COMe 2-CN-Phenyl 147

3 COMe 2-CN-Phenyl 148 1 CO.sub.2Me 2-CN-Phenyl 149 2 CO.sub.2Me 2-CN-Phenyl 150 3
CO.sub.2Me 2-CN-Phenyl 151 1 CO.sub.2tBu 2-CN-Phenyl 152 2 CO.sub.2tBu 2-CN-Phenyl 153
3 CO.sub.2tBu 2-CN-Phenyl 154 1 CONHMe 2-CN-Phenyl 155 2 CONHMe 2-CN-Phenyl 156 3
CONHMe 2-CN-Phenyl 157 1 SO.sub.2Me 2-CN-Phenyl 158 2 SO.sub.2Me 2-CN-Phenyl 159 3
SO.sub.2Me 2-CN-Phenyl 160 1 SO.sub.2NH.sub.2 2-CN-Phenyl 161 2 SO.sub.2NH.sub.2 2-CN-
Phenyl 162 3 SO.sub.2NH.sub.2 2-CN-Phenyl 163 1 H 3-Me-Phenyl 164 2 H 3-Me-Phenyl 165 3
H 3-Me-Phenyl 166 1 Me 3-Me-Phenyl 167 2 Me 3-Me-Phenyl 168 3 Me 3-Me-Phenyl 169 1
CH.sub.2Ph 3-Me-Phenyl 170 2 CH.sub.2Ph 3-Me-Phenyl 171 3 CH.sub.2Ph 3-Me-Phenyl 172 1
COMe 3-Me-Phenyl 173 2 COMe 3-Me-Phenyl 174 3 COMe 3-Me-Phenyl 175 1 CO.sub.2Me 3-
Me-Phenyl 176 2 CO.sub.2Me 3-Me-Phenyl 177 3 CO.sub.2Me 3-Me-Phenyl 178 1 CO.sub.2tBu
3-Me-Phenyl 179 2 CO.sub.2tBu 3-Me-Phenyl 180 3 CO.sub.2tBu 3-Me-Phenyl 181 1 CONHMe
3-Me-Phenyl 182 2 CONHMe 3-Me-Phenyl 183 3 CONHMe 3-Me-Phenyl 184 1 SO.sub.2Me 3-
Me-Phenyl 185 2 SO.sub.2Me 3-Me-Phenyl 186 3 SO.sub.2Me 3-Me-Phenyl 187 1
SO.sub.2NH.sub.2 3-Me-Phenyl 188 2 SO.sub.2NH.sub.2 3-Me-Phenyl 189 3 SO.sub.2NH.sub.2
3-Me-Phenyl 190 1 H 2-F-Phenyl 191 2 H 2-F-Phenyl 192 3 H 2-F-Phenyl 193 1 Me 2-F-Phenyl
194 2 Me 2-F-Phenyl 195 3 Me 2-F-Phenyl 196 1 CH.sub.2Ph 2-F-Phenyl 197 2 CH.sub.2Ph 2-F-
Phenyl 198 3 CH.sub.2Ph 2-F-Phenyl 199 1 COMe 2-F-Phenyl 200 2 COMe 2-F-Phenyl 201 3
COMe 2-F-Phenyl 202 1 CO.sub.2Me 2-F-Phenyl 203 2 CO.sub.2Me 2-F-Phenyl 204 3
CO.sub.2Me 2-F-Phenyl 205 1 CO.sub.2tBu 2-F-Phenyl 206 2 CO.sub.2tBu 2-F-Phenyl 207 3
CO.sub.2tBu 2-F-Phenyl 208 1 CONHMe 2-F-Phenyl 209 2 CONHMe 2-F-Phenyl 210 3
CONHMe 2-F-Phenyl 211 1 SO.sub.2Me 2-F-Phenyl 212 2 SO.sub.2Me 2-F-Phenyl 213 3
SO.sub.2Me 2-F-Phenyl 214 1 SO.sub.2NH.sub.2 2-F-Phenyl 215 2 SO.sub.2NH.sub.2 2-F-Phenyl
216 3 SO.sub.2NH.sub.2 2-F-Phenyl 217 1 H 4-F-Phenyl 218 2 H 4-F-Phenyl 219 3 H 4-F-Phenyl
220 1 Me 4-F-Phenyl 221 2 Me 4-F-Phenyl 222 3 Me 4-F-Phenyl 223 1 CH.sub.2Ph 4-F-Phenyl
224 2 CH.sub.2Ph 4-F-Phenyl 225 3 CH.sub.2Ph 4-F-Phenyl 226 1 COMe 4-F-Phenyl 227 2
COMe 4-F-Phenyl 228 3 COMe 4-F-Phenyl 229 1 CO.sub.2Me 4-F-Phenyl 230 2 CO.sub.2Me 4-
F-Phenyl 231 3 CO.sub.2Me 4-F-Phenyl 232 1 CO.sub.2tBu 4-F-Phenyl 233 2 CO.sub.2tBu 4-F-
Phenyl 234 3 CO.sub.2tBu 4-F-Phenyl 235 1 CONHMe 4-F-Phenyl 236 2 CONHMe 4-F-Phenyl
237 3 CONHMe 4-F-Phenyl 238 1 SO.sub.2Me 4-F-Phenyl 239 2 SO.sub.2Me 4-F-Phenyl 240 3
SO.sub.2Me 4-F-Phenyl 241 1 SO.sub.2NH.sub.2 4-F-Phenyl 242 2 SO.sub.2NH.sub.2 4-F-Phenyl
243 3 SO.sub.2NH.sub.2 4-F-Phenyl 244 1 H 3-Cl-Phenyl 245 2 H 3-Cl-Phenyl 246 3 H 3-Cl-
Phenyl 247 1 Me 3-Cl-Phenyl 248 2 Me 3-Cl-Phenyl 249 3 Me 3-Cl-Phenyl 250 1 CH.sub.2Ph 3-
Cl-Phenyl 251 2 CH.sub.2Ph 3-Cl-Phenyl 252 3 CH.sub.2Ph 3-Cl-Phenyl 253 1 COMe 3-Cl-
Phenyl 254 2 COMe 3-Cl-Phenyl 255 3 COMe 3-Cl-Phenyl 256 1 CO.sub.2Me 3-Cl-Phenyl 257 2
CO.sub.2Me 3-Cl-Phenyl 258 3 CO.sub.2Me 3-Cl-Phenyl 259 1 CO.sub.2tBu 3-Cl-Phenyl 260 2
CO.sub.2tBu 3-Cl-Phenyl 261 3 CO.sub.2tBu 3-Cl-Phenyl 262 1 CONHMe 3-Cl-Phenyl 263 2
CONHMe 3-Cl-Phenyl 264 3 CONHMe 3-Cl-Phenyl 265 1 SO.sub.2Me 3-Cl-Phenyl 266 2
SO.sub.2Me 3-Cl-Phenyl 267 3 SO.sub.2Me 3-Cl-Phenyl 268 1 SO.sub.2NH.sub.2 3-Cl-Phenyl
269 2 SO.sub.2NH.sub.2 3-Cl-Phenyl 270 3 SO.sub.2NH.sub.2 3-Cl-Phenyl 271 1 H 2-Br-Phenyl
272 2 H 2-Br-Phenyl 273 3 H 2-Br-Phenyl 274 1 Me 2-Br-Phenyl 275 2 Me 2-Br-Phenyl 276 3 Me
2-Br-Phenyl 277 1 CH.sub.2Ph 2-Br-Phenyl 278 2 CH.sub.2Ph 2-Br-Phenyl 279 3 CH.sub.2Ph 2-
Br-Phenyl 280 1 COMe 2-Br-Phenyl 281 2 COMe 2-Br-Phenyl 282 3 COMe 2-Br-Phenyl 283 1
CO.sub.2Me 2-Br-Phenyl 284 2 CO.sub.2Me 2-Br-Phenyl 285 3 CO.sub.2Me 2-Br-Phenyl 286 1
CO.sub.2tBu 2-Br-Phenyl 287 2 CO.sub.2tBu 2-Br-Phenyl 288 3 CO.sub.2tBu 2-Br-Phenyl 289 1
CONHMe 2-Br-Phenyl 290 2 CONHMe 2-Br-Phenyl 291 3 CONHMe 2-Br-Phenyl 292 1
SO.sub.2Me 2-Br-Phenyl 293 2 SO.sub.2Me 2-Br-Phenyl 294 3 SO.sub.2Me 2-Br-Phenyl 295 1
SO.sub.2NH.sub.2 2-Br-Phenyl 296 2 SO.sub.2NH.sub.2 2-Br-Phenyl 297 3 SO.sub.2NH.sub.2 2-
Br-Phenyl 298 1 H 4-Br-Phenyl 299 2 H 4-Br-Phenyl 300 3 H 4-Br-Phenyl 301 1 Me 4-Br-Phenyl
302 2 Me 4-Br-Phenyl 303 3 Me 4-Br-Phenyl 304 1 CH.sub.2Ph 4-Br-Phenyl 305 2 CH.sub.2Ph 4-
Br-Phenyl 306 3 CH.sub.2Ph 4-Br-Phenyl 307 1 COMe 4-Br-Phenyl 308 2 COMe 4-Br-Phenyl 309

3 COMe 4-Br-Phenyl 310 1 CO.sub.2Me 4-Br-Phenyl 311 2 CO.sub.2Me 4-Br-Phenyl 312 3
CO.sub.2Me 4-Br-Phenyl 313 1 CO.sub.2tBu 4-Br-Phenyl 314 2 CO.sub.2tBu 4-Br-Phenyl 315 3
CO.sub.2tBu 4-Br-Phenyl 316 1 CONHMe 4-Br-Phenyl 317 2 CONHMe 4-Br-Phenyl 318 3
CONHMe 4-Br-Phenyl 319 1 SO.sub.2Me 4-Br-Phenyl 320 2 SO.sub.2Me 4-Br-Phenyl 321 3
SO.sub.2Me 4-Br-Phenyl 322 1 SO.sub.2NH.sub.2 4-Br-Phenyl 323 2 SO.sub.2NH.sub.2 4-Br-
Phenyl 324 3 SO.sub.2NH.sub.2 4-Br-Phenyl 325 1 H 3-CF.sub.3-Phenyl 326 2 H 3-CF.sub.3-
Phenyl 327 3 H 3-CF.sub.3-Phenyl 328 1 Me 3-CF.sub.3-Phenyl 329 2 Me 3-CF.sub.3-Phenyl 330
3 Me 3-CF.sub.3-Phenyl 331 1 CH.sub.2Ph 3-CF.sub.3-Phenyl 332 2 CH.sub.2Ph 3-CF.sub.3-
Phenyl 333 3 CH.sub.2Ph 3-CF.sub.3-Phenyl 334 1 COMe 3-CF.sub.3-Phenyl 335 2 COMe 3-
CF.sub.3-Phenyl 336 3 COMe 3-CF.sub.3-Phenyl 337 1 CO.sub.2Me 3-CF.sub.3-Phenyl 338 2
CO.sub.2Me 3-CF.sub.3-Phenyl 339 3 CO.sub.2Me 3-CF.sub.3-Phenyl 340 1 CO.sub.2tBu 3-
CF.sub.3-Phenyl 341 2 CO.sub.2tBu 3-CF.sub.3-Phenyl 342 3 CO.sub.2tBu 3-CF.sub.3-Phenyl 343
1 CONHMe 3-CF.sub.3-Phenyl 344 2 CONHMe 3-CF.sub.3-Phenyl 345 3 CONHMe 3-CF.sub.3-
Phenyl 346 1 SO.sub.2Me 3-CF.sub.3-Phenyl 347 2 SO.sub.2Me 3-CF.sub.3-Phenyl 348 3
SO.sub.2Me 3-CF.sub.3-Phenyl 349 1 SO.sub.2NH.sub.2 3-CF.sub.3-Phenyl 350 2
SO.sub.2NH.sub.2 3-CF.sub.3-Phenyl 351 3 SO.sub.2NH.sub.2 3-CF.sub.3-Phenyl 352 1 H 2-iPr-
Phenyl 353 2 H 2-iPr-Phenyl 354 3 H 2-iPr-Phenyl 355 1 Me 2-iPr-Phenyl 356 2 Me 2-iPr-Phenyl
357 3 Me 2-iPr-Phenyl 358 1 CH.sub.2Ph 2-iPr-Phenyl 359 2 CH.sub.2Ph 2-iPr-Phenyl 360 3
CH.sub.2Ph 2-iPr-Phenyl 361 1 COMe 2-iPr-Phenyl 362 2 COMe 2-iPr-Phenyl 363 3 COMe 2-iPr-
Phenyl 364 1 CO.sub.2Me 2-iPr-Phenyl 365 2 CO.sub.2Me 2-iPr-Phenyl 366 3 CO.sub.2Me 2-iPr-
Phenyl 367 1 CO.sub.2tBu 2-iPr-Phenyl 368 2 CO.sub.2tBu 2-iPr-Phenyl 369 3 CO.sub.2tBu 2-
iPr-Phenyl 370 1 CONHMe 2-iPr-Phenyl 371 2 CONHMe 2-iPr-Phenyl 372 3 CONHMe 2-iPr-
Phenyl 373 1 SO.sub.2Me 2-iPr-Phenyl 374 2 SO.sub.2Me 2-iPr-Phenyl 375 3 SO.sub.2Me 2-iPr-
Phenyl 376 1 SO.sub.2NH.sub.2 2-iPr-Phenyl 377 2 SO.sub.2NH.sub.2 2-iPr-Phenyl 378 3
SO.sub.2NH.sub.2 2-iPr-Phenyl 379 1 H 4-iPr-Phenyl 380 2 H 4-iPr-Phenyl 381 3 H 4-iPr-Phenyl
382 1 Me 4-iPr-Phenyl 383 2 Me 4-iPr-Phenyl 384 3 Me 4-iPr-Phenyl 385 1 CH.sub.2Ph 4-iPr-
Phenyl 386 2 CH.sub.2Ph 4-iPr-Phenyl 387 3 CH.sub.2Ph 4-iPr-Phenyl 388 1 COMe 4-iPr-Phenyl
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CO.sub.2Me 4-iPr-Phenyl 393 3 CO.sub.2Me 4-iPr-Phenyl 394 1 CO.sub.2tBu 4-iPr-Phenyl 395 2
CO.sub.2tBu 4-iPr-Phenyl 396 3 CO.sub.2tBu 4-iPr-Phenyl 397 1 CONHMe 4-iPr-Phenyl 398 2
CONHMe 4-iPr-Phenyl 399 3 CONHMe 4-iPr-Phenyl 400 1 SO.sub.2Me 4-iPr-Phenyl 401 2
SO.sub.2Me 4-iPr-Phenyl 402 3 SO.sub.2Me 4-iPr-Phenyl 403 1 SO.sub.2NH.sub.2 4-iPr-Phenyl
404 2 SO.sub.2NH.sub.2 4-iPr-Phenyl 405 3 SO.sub.2NH.sub.2 4-iPr-Phenyl 406 1 H 3-NH.sub.2-
Phenyl 407 2 H 3-NH.sub.2-Phenyl 408 3 H 3-NH.sub.2-Phenyl 409 1 Me 3-NH.sub.2-Phenyl 410
2 Me 3-NH.sub.2-Phenyl 411 3 Me 3-NH.sub.2-Phenyl 412 1 CH.sub.2Ph 3-NH.sub.2-Phenyl 413
2 CH.sub.2Ph 3-NH.sub.2-Phenyl 414 3 CH.sub.2Ph 3-NH.sub.2-Phenyl 415 1 COMe 3-
NH.sub.2-Phenyl 416 2 COMe 3-NH.sub.2-Phenyl 417 3 COMe 3-NH.sub.2-Phenyl 418 1
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2 SO.sub.2Me 3-NH.sub.2-Phenyl 429 3 SO.sub.2Me 3-NH.sub.2-Phenyl 430 1 SO.sub.2NH.sub.2
3-NH.sub.2-Phenyl 431 2 SO.sub.2NH.sub.2 3-NH.sub.2-Phenyl 432 3 SO.sub.2NH.sub.2 3-
NH.sub.2-Phenyl 433 1 H 2,4-di-Me-Phenyl 434 2 H 2,4-di-Me-Phenyl 435 3 H 2,4-di-Me-Phenyl
436 1 Me 2,4-di-Me-Phenyl 437 2 Me 2,4-di-Me-Phenyl 438 3 Me 2,4-di-Me-Phenyl 439 1
CH.sub.2Ph 2,4-di-Me-Phenyl 440 2 CH.sub.2Ph 2,4-di-Me-Phenyl 441 3 CH.sub.2Ph 2,4-di-Me-
Phenyl 442 1 COMe 2,4-di-Me-Phenyl 443 2 COMe 2,4-di-Me-Phenyl 444 3 COMe 2,4-di-Me-
Phenyl 445 1 CO.sub.2Me 2,4-di-Me-Phenyl 446 2 CO.sub.2Me 2,4-di-Me-Phenyl 447 3
CO.sub.2Me 2,4-di-Me-Phenyl 448 1 CO.sub.2tBu 2,4-di-Me-Phenyl 449 2 CO.sub.2tBu 2,4-di-
Me-Phenyl 450 3 CO.sub.2tBu 2,4-di-Me-Phenyl 451 1 CONHMe 2,4-di-Me-Phenyl 452 2

CONHMe 2,4-di-Me-Phenyl 453 3 CONHMe 2,4-di-Me-Phenyl 454 1 SO.sub.2Me 2,4-di-Me-Phenyl 455 2 SO.sub.2Me 2,4-di-Me-Phenyl 456 3 SO.sub.2Me 2,4-di-Me-Phenyl 457 1 SO.sub.2NH.sub.2 2,4-di-Me-Phenyl 458 2 SO.sub.2NH.sub.2 2,4-di-Me-Phenyl 459 3 SO.sub.2NH.sub.2 2,4-di-Me-Phenyl 460 1 H 2,6-di-iPr-Phenyl 461 2 H 2,6-di-iPr-Phenyl 462 3 H 2,6-di-iPr-Phenyl 463 1 Me 2,6-di-iPr-Phenyl 464 2 Me 2,6-di-iPr-Phenyl 465 3 Me 2,6-di-iPr-Phenyl 466 1 CH.sub.2Ph 2,6-di-iPr-Phenyl 467 2 CH.sub.2Ph 2,6-di-iPr-Phenyl 468 3 CH.sub.2Ph 2,6-di-iPr-Phenyl 469 1 COMe 2,6-di-iPr-Phenyl 470 2 COMe 2,6-di-iPr-Phenyl 471 3 COMe 2,6-di-iPr-Phenyl 472 1 CO.sub.2Me 2,6-di-iPr-Phenyl 473 2 CO.sub.2Me 2,6-di-iPr-Phenyl 474 3 CO.sub.2Me 2,6-di-iPr-Phenyl 475 1 CO.sub.2tBu 2,6-di-iPr-Phenyl 476 2 CO.sub.2tBu 2,6-di-iPr-Phenyl 477 3 CO.sub.2tBu 2,6-di-iPr-Phenyl 478 1 CONHMe 2,6-di-iPr-Phenyl 479 2 CONHMe 2,6-di-iPr-Phenyl 480 3 CONHMe 2,6-di-iPr-Phenyl 481 1 SO.sub.2Me 2,6-di-iPr-Phenyl 482 2 SO.sub.2Me 2,6-di-iPr-Phenyl 483 3 SO.sub.2Me 2,6-di-iPr-Phenyl 484 1 SO.sub.2NH.sub.2 2,6-di-iPr-Phenyl 485 2 SO.sub.2NH.sub.2 2,6-di-iPr-Phenyl 486 3 SO.sub.2NH.sub.2 2,6-di-iPr-Phenyl 487 1 H 3-Ph-Phenyl 488 2 H 3-Ph-Phenyl 489 3 H 3-Ph-Phenyl 490 1 Me 3-Ph-Phenyl 491 2 Me 3-Ph-Phenyl 492 3 Me 3-Ph-Phenyl 493 1 CH.sub.2Ph 3-Ph-Phenyl 494 2 CH.sub.2Ph 3-Ph-Phenyl 495 3 CH.sub.2Ph 3-Ph-Phenyl 496 1 COMe 3-Ph-Phenyl 497 2 COMe 3-Ph-Phenyl 498 3 COMe 3-Ph-Phenyl 499 1 CO.sub.2Me 3-Ph-Phenyl 500 2 CO.sub.2Me 3-Ph-Phenyl 501 3 CO.sub.2Me 3-Ph-Phenyl 502 1 CO.sub.2tBu 3-Ph-Phenyl 503 2 CO.sub.2tBu 3-Ph-Phenyl 504 3 CO.sub.2tBu 3-Ph-Phenyl 505 1 CONHMe 3-Ph-Phenyl 506 2 CONHMe 3-Ph-Phenyl 507 3 CONHMe 3-Ph-Phenyl 508 1 SO.sub.2Me 3-Ph-Phenyl 509 2 SO.sub.2Me 3-Ph-Phenyl 510 3 SO.sub.2Me 3-Ph-Phenyl 511 1 SO.sub.2NH.sub.2 3-Ph-Phenyl 512 2 SO.sub.2NH.sub.2 3-Ph-Phenyl 513 3 SO.sub.2NH.sub.2 3-Ph-Phenyl 514 1 H 2-morpholino-phenyl 515 2 H 2-morpholino-phenyl 516 3 H 2-morpholino-phenyl 517 1 Me 2-morpholino-phenyl 518 2 Me 2-morpholino-phenyl 519 3 Me 2-morpholino-phenyl 520 1 CH.sub.2Ph 2-morpholino-phenyl 521 2 CH.sub.2Ph 2-morpholino-phenyl 522 3 CH.sub.2Ph 2-morpholino-phenyl 523 1 COMe 2-morpholino-phenyl 524 2 COMe 2-morpholino-phenyl 525 3 COMe 2-morpholino-phenyl 526 1 CO.sub.2Me 2-morpholino-phenyl 527 2 CO.sub.2Me 2-morpholino-phenyl 528 3 CO.sub.2Me 2-morpholino-phenyl 529 1 CO.sub.2tBu 2-morpholino-phenyl 530 2 CO.sub.2tBu 2-morpholino-phenyl 531 3 CO.sub.2tBu 2-morpholino-phenyl 532 1 CONHMe 2-morpholino-phenyl 533 2 CONHMe 2-morpholino-phenyl 534 3 CONHMe 2-morpholino-phenyl 535 1 SO.sub.2Me 2-morpholino-phenyl 536 2 SO.sub.2Me 2-morpholino-phenyl 537 3 SO.sub.2Me 2-morpholino-phenyl 538 1 SO.sub.2NH.sub.2 2-morpholino-phenyl 539 2 SO.sub.2NH.sub.2 2-morpholino-phenyl 540 3 SO.sub.2NH.sub.2 2-morpholino-phenyl 541 1 H 4-morpholino-phenyl 542 2 H 4-morpholino-phenyl 543 3 H 4-morpholino-phenyl 544 1 Me 4-morpholino-phenyl 545 2 Me 4-morpholino-phenyl 546 3 Me 4-morpholino-phenyl 547 1 CH.sub.2Ph 4-morpholino-phenyl 548 2 CH.sub.2Ph 4-morpholino-phenyl 549 3 CH.sub.2Ph 4-morpholino-phenyl 550 1 COMe 4-morpholino-phenyl 551 2 COMe 4-morpholino-phenyl 552 3 COMe 4-morpholino-phenyl 553 1 CO.sub.2Me 4-morpholino-phenyl 554 2 CO.sub.2Me 4-morpholino-phenyl 555 3 CO.sub.2Me 4-morpholino-phenyl 556 1 CO.sub.2tBu 4-morpholino-phenyl 557 2 CO.sub.2tBu 4-morpholino-phenyl 558 3 CO.sub.2tBu 4-morpholino-phenyl 559 1 CONHMe 4-morpholino-phenyl 560 2 CONHMe 4-morpholino-phenyl 561 3 CONHMe 4-morpholino-phenyl 562 1 SO.sub.2Me 4-morpholino-phenyl 563 2 SO.sub.2Me 4-morpholino-phenyl 564 3 SO.sub.2Me 4-morpholino-phenyl 565 1 SO.sub.2NH.sub.2 4-morpholino-phenyl 566 2 SO.sub.2NH.sub.2 4-morpholino-phenyl 567 3 SO.sub.2NH.sub.2 4-morpholino-phenyl 568 1 H 2-pyrimidinyl 569 2 H 2-pyrimidinyl 570 3 H 2-pyrimidinyl 571 1 Me 2-pyrimidinyl 572 2 Me 2-pyrimidinyl 573 3 Me 2-pyrimidinyl 574 1 CH.sub.2Ph 2-pyrimidinyl 575 2 CH.sub.2Ph 2-pyrimidinyl 576 3 CH.sub.2Ph 2-pyrimidinyl 577 1 COMe 2-pyrimidinyl 578 2 COMe 2-pyrimidinyl 579 3 COMe 2-pyrimidinyl 580 1 CO.sub.2Me 2-pyrimidinyl 581 2 CO.sub.2Me 2-pyrimidinyl 582 3 CO.sub.2Me 2-pyrimidinyl 583 1 CO.sub.2tBu 2-pyrimidinyl 584 2 CO.sub.2tBu 2-pyrimidinyl 585 3 CO.sub.2tBu 2-pyrimidinyl 586 1 CONHMe 2-pyrimidinyl 587 2 CONHMe 2-pyrimidinyl 588 3

CONHMe 2-pyrimidinyl 589 1 SO.sub.2Me 2-pyrimidinyl 590 2 SO.sub.2Me 2-pyrimidinyl 591 3
SO.sub.2Me 2-pyrimidinyl 592 1 SO.sub.2NH.sub.2 2-pyrimidinyl 593 2 SO.sub.2NH.sub.2 2-
pyrimidinyl 594 3 SO.sub.2NH.sub.2 2-pyrimidinyl 595 1 H 2-methyl-1H- benzo[d]imidazol-4-yl
596 2 H 2-methyl-1H- benzo[d]imidazol-4-yl 597 3 H 2-methyl-1H- benzo[d]imidazol-4-yl 598 1
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2-methyl-1H- benzo[d]imidazol-4-yl 601 1 CH.sub.2Ph 2-methyl-1H- benzo[d]imidazol-4-yl 602 2
CH.sub.2Ph 2-methyl-1H- benzo[d]imidazol-4-yl 603 3 CH.sub.2Ph 2-methyl-1H-
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1H- benzo[d]imidazol-4-yl 606 3 COMe 2-methyl-1H- benzo[d]imidazol-4-yl 607 1 CO.sub.2Me
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3 CO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 610 1 CO.sub.2tBu 2-methyl-1H-
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CONHMe 2-methyl-1H- benzo[d]imidazol-4-yl 615 3 CONHMe 2-methyl-1H- benzo[d]imidazol-
4-yl 616 1 SO.sub.2Me 2-methyl-1H- benzo[d]imidazol-4-yl 617 2 SO.sub.2Me 2-methyl-1H-
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SO.sub.2NH.sub.2 2-methyl-1H- benzo[d]imidazol-4-yl 620 2 SO.sub.2NH.sub.2 2-methyl-1H-
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CO.sub.2Me 4-OH-Phenyl 637 1 CO.sub.2tBu 4-OH-Phenyl 638 2 CO.sub.2tBu 4-OH-Phenyl 639
3 CO.sub.2tBu 4-OH-Phenyl 640 1 CONHMe 4-OH-Phenyl 641 2 CONHMe 4-OH-Phenyl 642 3
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SO.sub.2Me 4-OH-Phenyl 646 1 SO.sub.2NH.sub.2 4-OH-Phenyl 647 2 SO.sub.2NH.sub.2 4-OH-
Phenyl 648 3 SO.sub.2NH.sub.2 4-OH-Phenyl 649 1 H 2-OH-Phenyl 650 2 H 2-OH-Phenyl 651 3
H 2-OH-Phenyl 652 1 Me 2-OH-Phenyl 653 2 Me 2-OH-Phenyl 654 3 Me 2-OH-Phenyl 655 1
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2-OH-Phenyl 665 2 CO.sub.2tBu 2-OH-Phenyl 666 3 CO.sub.2tBu 2-OH-Phenyl 667 1 CONHMe
2-OH-Phenyl 668 2 CONHMe 2-OH-Phenyl 669 3 CONHMe 2-OH-Phenyl 670 1 SO.sub.2Me 2-
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2-OH-Phenyl 676 1 H 4-OMe-Phenyl 677 2 H 4-OMe-Phenyl 678 3 H 4-OMe-Phenyl 679 1 Me 4-
OMe-Phenyl 680 2 Me 4-OMe-Phenyl 681 3 Me 4-OMe-Phenyl 682 1 CH.sub.2Ph 4-OMe-Phenyl
683 2 CH.sub.2Ph 4-OMe-Phenyl 684 3 CH.sub.2Ph 4-OMe-Phenyl 685 1 COMe 4-OMe-Phenyl
686 2 COMe 4-OMe-Phenyl 687 3 COMe 4-OMe-Phenyl 688 1 CO.sub.2Me 4-OMe-Phenyl 689 2
CO.sub.2Me 4-OMe-Phenyl 690 3 CO.sub.2Me 4-OMe-Phenyl 691 1 CO.sub.2tBu 4-OMe-Phenyl
692 2 CO.sub.2tBu 4-OMe-Phenyl 693 3 CO.sub.2tBu 4-OMe-Phenyl 694 1 CONHMe 4-OMe-
Phenyl 695 2 CONHMe 4-OMe-Phenyl 696 3 CONHMe 4-OMe-Phenyl 697 1 SO.sub.2Me 4-
OMe-Phenyl 698 2 SO.sub.2Me 4-OMe-Phenyl 699 3 SO.sub.2Me 4-OMe-Phenyl 700 1
SO.sub.2NH.sub.2 4-OMe-Phenyl 701 2 SO.sub.2NH.sub.2 4-OMe-Phenyl 702 3
SO.sub.2NH.sub.2 4-OMe-Phenyl 703 1 H 2-OMe-Phenyl 704 2 H 2-OMe-Phenyl 705 3 H 2-
OMe-Phenyl 706 1 Me 2-OMe-Phenyl 707 2 Me 2-OMe-Phenyl 708 3 Me 2-OMe-Phenyl 709 1
CH.sub.2Ph 2-OMe-Phenyl 710 2 CH.sub.2Ph 2-OMe-Phenyl 711 3 CH.sub.2Ph 2-OMe-Phenyl
712 1 COMe 2-OMe-Phenyl 713 2 COMe 2-OMe-Phenyl 714 3 COMe 2-OMe-Phenyl 715 1
CO.sub.2Me 2-OMe-Phenyl 716 2 CO.sub.2Me 2-OMe-Phenyl 717 3 CO.sub.2Me 2-OMe-Phenyl
718 1 CO.sub.2tBu 2-OMe-Phenyl 719 2 CO.sub.2tBu 2-OMe-Phenyl 720 3 CO.sub.2tBu 2-OMe-

Phenyl 721 1 CONHMe 2-OMe-Phenyl 722 2 CONHMe 2-OMe-Phenyl 723 3 CONHMe 2-OMe-Phenyl 724 1 SO.sub.2Me 2-OMe-Phenyl 725 2 SO.sub.2Me 2-OMe-Phenyl 726 3 SO.sub.2Me 2-OMe-Phenyl 727 1 SO.sub.2NH.sub.2 2-OMe-Phenyl 728 2 SO.sub.2NH.sub.2 2-OMe-Phenyl 729 3 SO.sub.2NH.sub.2 2-OMe-Phenyl 730 1 H 3-CN-Phenyl 731 2 H 3-CN-Phenyl 732 3 H 3-CN-Phenyl 733 1 Me 3-CN-Phenyl 734 2 Me 3-CN-Phenyl 735 3 Me 3-CN-Phenyl 736 1 CH.sub.2Ph 3-CN-Phenyl 737 2 CH.sub.2Ph 3-CN-Phenyl 738 3 CH.sub.2Ph 3-CN-Phenyl 739 1 COMe 3-CN-Phenyl 740 2 COMe 3-CN-Phenyl 741 3 COMe 3-CN-Phenyl 742 1 CO.sub.2Me 3-CN-Phenyl 743 2 CO.sub.2Me 3-CN-Phenyl 744 3 CO.sub.2Me 3-CN-Phenyl 745 1 CO.sub.2tBu 3-CN-Phenyl 746 2 CO.sub.2tBu 3-CN-Phenyl 747 3 CO.sub.2tBu 3-CN-Phenyl 748 1 CONHMe 3-CN-Phenyl 749 2 CONHMe 3-CN-Phenyl 750 3 CONHMe 3-CN-Phenyl 751 1 SO.sub.2Me 3-CN-Phenyl 752 2 SO.sub.2Me 3-CN-Phenyl 753 3 SO.sub.2Me 3-CN-Phenyl 754 1 SO.sub.2NH.sub.2 3-CN-Phenyl 755 2 SO.sub.2NH.sub.2 3-CN-Phenyl 756 3 SO.sub.2NH.sub.2 3-CN-Phenyl 757 1 H 2-Me-Phenyl 758 2 H 2-Me-Phenyl 759 3 H 2-Me-Phenyl 760 1 Me 2-Me-Phenyl 761 2 Me 2-Me-Phenyl 762 3 Me 2-Me-Phenyl 763 1 CH.sub.2Ph 2-Me-Phenyl 764 2 CH.sub.2Ph 2-Me-Phenyl 765 3 CH.sub.2Ph 2-Me-Phenyl 766 1 COMe 2-Me-Phenyl 767 2 COMe 2-Me-Phenyl 768 3 COMe 2-Me-Phenyl 769 1 CO.sub.2Me 2-Me-Phenyl 770 2 CO.sub.2Me 2-Me-Phenyl 771 3 CO.sub.2Me 2-Me-Phenyl 772 1 CO.sub.2tBu 2-Me-Phenyl 773 2 CO.sub.2tBu 2-Me-Phenyl 774 3 CO.sub.2tBu 2-Me-Phenyl 775 1 CONHMe 2-Me-Phenyl 776 2 CONHMe 2-Me-Phenyl 777 3 CONHMe 2-Me-Phenyl 778 1 SO.sub.2Me 2-Me-Phenyl 779 2 SO.sub.2Me 2-Me-Phenyl 780 3 SO.sub.2Me 2-Me-Phenyl 781 1 SO.sub.2NH.sub.2 2-Me-Phenyl 782 2 SO.sub.2NH.sub.2 2-Me-Phenyl 783 3 SO.sub.2NH.sub.2 2-Me-Phenyl 784 1 H 4-Me-Phenyl 785 2 H 4-Me-Phenyl 786 3 H 4-Me-Phenyl 787 1 Me 4-Me-Phenyl 788 2 Me 4-Me-Phenyl 789 3 Me 4-Me-Phenyl 790 1 CH.sub.2Ph 4-Me-Phenyl 791 2 CH.sub.2Ph 4-Me-Phenyl 792 3 CH.sub.2Ph 4-Me-Phenyl 793 1 COMe 4-Me-Phenyl 794 2 COMe 4-Me-Phenyl 795 3 COMe 4-Me-Phenyl 796 1 CO.sub.2Me 4-Me-Phenyl 797 2 CO.sub.2Me 4-Me-Phenyl 798 3 CO.sub.2Me 4-Me-Phenyl 799 1 CO.sub.2tBu 4-Me-Phenyl 800 2 CO.sub.2tBu 4-Me-Phenyl 801 3 CO.sub.2tBu 4-Me-Phenyl 802 1 CONHMe 4-Me-Phenyl 803 2 CONHMe 4-Me-Phenyl 804 3 CONHMe 4-Me-Phenyl 805 1 SO.sub.2Me 4-Me-Phenyl 806 2 SO.sub.2Me 4-Me-Phenyl 807 3 SO.sub.2Me 4-Me-Phenyl 808 1 SO.sub.2NH.sub.2 4-Me-Phenyl 809 2 SO.sub.2NH.sub.2 4-Me-Phenyl 810 3 SO.sub.2NH.sub.2 4-Me-Phenyl 811 1 H 3-F-Phenyl 812 2 H 3-F-Phenyl 813 3 H 3-F-Phenyl 814 1 Me 3-F-Phenyl 815 2 Me 3-F-Phenyl 816 3 Me 3-F-Phenyl 817 1 CH.sub.2Ph 3-F-Phenyl 818 2 CH.sub.2Ph 3-F-Phenyl 819 3 CH.sub.2Ph 3-F-Phenyl 820 1 COMe 3-F-Phenyl 821 2 COMe 3-F-Phenyl 822 3 COMe 3-F-Phenyl 823 1 CO.sub.2Me 3-F-Phenyl 824 2 CO.sub.2Me 3-F-Phenyl 825 3 CO.sub.2Me 3-F-Phenyl 826 1 CO.sub.2tBu 3-F-Phenyl 827 2 CO.sub.2tBu 3-F-Phenyl 828 3 CO.sub.2tBu 3-F-Phenyl 829 1 CONHMe 3-F-Phenyl 830 2 CONHMe 3-F-Phenyl 831 3 CONHMe 3-F-Phenyl 832 1 SO.sub.2Me 3-F-Phenyl 833 2 SO.sub.2Me 3-F-Phenyl 834 3 SO.sub.2Me 3-F-Phenyl 835 1 SO.sub.2NH.sub.2 3-F-Phenyl 836 2 SO.sub.2NH.sub.2 3-F-Phenyl 837 3 SO.sub.2NH.sub.2 3-F-Phenyl 838 1 H 2-Cl-Phenyl 839 2 H 2-Cl-Phenyl 840 3 H 2-Cl-Phenyl 841 1 Me 2-Cl-Phenyl 842 2 Me 2-Cl-Phenyl 843 3 Me 2-Cl-Phenyl 844 1 CH.sub.2Ph 2-Cl-Phenyl 845 2 CH.sub.2Ph 2-Cl-Phenyl 846 3 CH.sub.2Ph 2-Cl-Phenyl 847 1 COMe 2-Cl-Phenyl 848 2 COMe 2-Cl-Phenyl 849 3 COMe 2-Cl-Phenyl 850 1 CO.sub.2Me 2-Cl-Phenyl 851 2 CO.sub.2Me 2-Cl-Phenyl 852 3 CO.sub.2Me 2-Cl-Phenyl 853 1 CO.sub.2tBu 2-Cl-Phenyl 854 2 CO.sub.2tBu 2-Cl-Phenyl 855 3 CO.sub.2tBu 2-Cl-Phenyl 856 1 CONHMe 2-Cl-Phenyl 857 2 CONHMe 2-Cl-Phenyl 858 3 CONHMe 2-Cl-Phenyl 859 1 SO.sub.2Me 2-Cl-Phenyl 860 2 SO.sub.2Me 2-Cl-Phenyl 861 3 SO.sub.2Me 2-Cl-Phenyl 862 1 SO.sub.2NH.sub.2 2-Cl-Phenyl 863 2 SO.sub.2NH.sub.2 2-Cl-Phenyl 864 3 SO.sub.2NH.sub.2 2-Cl-Phenyl 865 1 H 4-Cl-Phenyl 866 2 H 4-Cl-Phenyl 867 3 H 4-Cl-Phenyl 868 1 Me 4-Cl-Phenyl 869 2 Me 4-Cl-Phenyl 870 3 Me 4-Cl-Phenyl 871 1 CH.sub.2Ph 4-Cl-Phenyl 872 2 CH.sub.2Ph 4-Cl-Phenyl 873 3 CH.sub.2Ph 4-Cl-Phenyl 874 1 COMe 4-Cl-Phenyl 875 2 COMe 4-Cl-Phenyl 876 3 COMe 4-Cl-Phenyl 877 1 CO.sub.2Me 4-Cl-Phenyl 878 2 CO.sub.2Me 4-Cl-Phenyl 879 3 CO.sub.2Me 4-Cl-Phenyl 880 1

CO.sub.2tBu 4-Cl-Phenyl 881 2 CO.sub.2tBu 4-Cl-Phenyl 882 3 CO.sub.2tBu 4-Cl-Phenyl 883 1
CONHMe 4-Cl-Phenyl 884 2 CONHMe 4-Cl-Phenyl 885 3 CONHMe 4-Cl-Phenyl 886 1
SO.sub.2Me 4-Cl-Phenyl 887 2 SO.sub.2Me 4-Cl-Phenyl 888 3 SO.sub.2Me 4-Cl-Phenyl 889 1
SO.sub.2NH.sub.2 4-Cl-Phenyl 890 2 SO.sub.2NH.sub.2 4-Cl-Phenyl 891 3 SO.sub.2NH.sub.2 4-
Cl-Phenyl 892 1 H 3-Br-Phenyl 893 2 H 3-Br-Phenyl 894 3 H 3-Br-Phenyl 895 1 Me 3-Br-Phenyl
896 2 Me 3-Br-Phenyl 897 3 Me 3-Br-Phenyl 898 1 CH.sub.2Ph 3-Br-Phenyl 899 2 CH.sub.2Ph 3-
Br-Phenyl 900 3 CH.sub.2Ph 3-Br-Phenyl 901 1 COMe 3-Br-Phenyl 902 2 COMe 3-Br-Phenyl 903
3 COMe 3-Br-Phenyl 904 1 CO.sub.2Me 3-Br-Phenyl 905 2 CO.sub.2Me 3-Br-Phenyl 906 3
CO.sub.2Me 3-Br-Phenyl 907 1 CO.sub.2tBu 3-Br-Phenyl 908 2 CO.sub.2tBu 3-Br-Phenyl 909 3
CO.sub.2tBu 3-Br-Phenyl 910 1 CONHMe 3-Br-Phenyl 911 2 CONHMe 3-Br-Phenyl 912 3
CONHMe 3-Br-Phenyl 913 1 SO.sub.2Me 3-Br-Phenyl 914 2 SO.sub.2Me 3-Br-Phenyl 915 3
SO.sub.2Me 3-Br-Phenyl 916 1 SO.sub.2NH.sub.2 3-Br-Phenyl 917 2 SO.sub.2NH.sub.2 3-Br-
Phenyl 918 3 SO.sub.2NH.sub.2 3-Br-Phenyl 919 1 H 2-CF.sub.3-Phenyl 920 2 H 2-CF.sub.3-
Phenyl 921 3 H 2-CF.sub.3-Phenyl 922 1 Me 2-CF.sub.3-Phenyl 923 2 Me 2-CF.sub.3-Phenyl 924
3 Me 2-CF.sub.3-Phenyl 925 1 CH.sub.2Ph 2-CF.sub.3-Phenyl 926 2 CH.sub.2Ph 2-CF.sub.3-
Phenyl 927 3 CH.sub.2Ph 2-CF.sub.3-Phenyl 928 1 COMe 2-CF.sub.3-Phenyl 929 2 COMe 2-
CF.sub.3-Phenyl 930 3 COMe 2-CF.sub.3-Phenyl 931 1 CO.sub.2Me 2-CF.sub.3-Phenyl 932 2
CO.sub.2Me 2-CF.sub.3-Phenyl 933 3 CO.sub.2Me 2-CF.sub.3-Phenyl 934 1 CO.sub.2tBu 2-
CF.sub.3-Phenyl 935 2 CO.sub.2tBu 2-CF.sub.3-Phenyl 936 3 CO.sub.2tBu 2-CF.sub.3-Phenyl 937
1 CONHMe 2-CF.sub.3-Phenyl 938 2 CONHMe 2-CF.sub.3-Phenyl 939 3 CONHMe 2-CF.sub.3-
Phenyl 940 1 SO.sub.2Me 2-CF.sub.3-Phenyl 941 2 SO.sub.2Me 2-CF.sub.3-Phenyl 942 3
SO.sub.2Me 2-CF.sub.3-Phenyl 943 1 SO.sub.2NH.sub.2 2-CF.sub.3-Phenyl 944 2
SO.sub.2NH.sub.2 2-CF.sub.3-Phenyl 945 3 SO.sub.2NH.sub.2 2-CF.sub.3-Phenyl 946 1 H 4-
CF.sub.3-Phenyl 947 2 H 4-CF.sub.3-Phenyl 948 3 H 4-CF.sub.3-Phenyl 949 1 Me 4-CF.sub.3-
Phenyl 950 2 Me 4-CF.sub.3-Phenyl 951 3 Me 4-CF.sub.3-Phenyl 952 1 CH.sub.2Ph 4-CF.sub.3-
Phenyl 953 2 CH.sub.2Ph 4-CF.sub.3-Phenyl 954 3 CH.sub.2Ph 4-CF.sub.3-Phenyl 955 1 COMe
4-CF.sub.3-Phenyl 956 2 COMe 4-CF.sub.3-Phenyl 957 3 COMe 4-CF.sub.3-Phenyl 958 1
CO.sub.2Me 4-CF.sub.3-Phenyl 959 2 CO.sub.2Me 4-CF.sub.3-Phenyl 960 3 CO.sub.2Me 4-
CF.sub.3-Phenyl 961 1 CO.sub.2tBu 4-CF.sub.3-Phenyl 962 2 CO.sub.2tBu 4-CF.sub.3-Phenyl 963
3 CO.sub.2tBu 4-CF.sub.3-Phenyl 964 1 CONHMe 4-CF.sub.3-Phenyl 965 2 CONHMe 4-
CF.sub.3-Phenyl 966 3 CONHMe 4-CF.sub.3-Phenyl 967 1 SO.sub.2Me 4-CF.sub.3-Phenyl 968 2
SO.sub.2Me 4-CF.sub.3-Phenyl 969 3 SO.sub.2Me 4-CF.sub.3-Phenyl 970 1 SO.sub.2NH.sub.2 4-
CF.sub.3-Phenyl 971 2 SO.sub.2NH.sub.2 4-CF.sub.3-Phenyl 972 3 SO.sub.2NH.sub.2 4-CF.sub.3-
Phenyl 973 1 H 3-iPr-Phenyl 974 2 H 3-iPr-Phenyl 975 3 H 3-iPr-Phenyl 976 1 Me 3-iPr-Phenyl
977 2 Me 3-iPr-Phenyl 978 3 Me 3-iPr-Phenyl 979 1 CH.sub.2Ph 3-iPr-Phenyl 980 2 CH.sub.2Ph
3-iPr-Phenyl 981 3 CH.sub.2Ph 3-iPr-Phenyl 982 1 COMe 3-iPr-Phenyl 983 2 COMe 3-iPr-Phenyl
984 3 COMe 3-iPr-Phenyl 985 1 CO.sub.2Me 3-iPr-Phenyl 986 2 CO.sub.2Me 3-iPr-Phenyl 987 3
CO.sub.2Me 3-iPr-Phenyl 988 1 CO.sub.2tBu 3-iPr-Phenyl 989 2 CO.sub.2tBu 3-iPr-Phenyl 990 3
CO.sub.2tBu 3-iPr-Phenyl 991 1 CONHMe 3-iPr-Phenyl 992 2 CONHMe 3-iPr-Phenyl 993 3
CONHMe 3-iPr-Phenyl 994 1 SO.sub.2Me 3-iPr-Phenyl 995 2 SO.sub.2Me 3-iPr-Phenyl 996 3
SO.sub.2Me 3-iPr-Phenyl 997 1 SO.sub.2NH.sub.2 3-iPr-Phenyl 998 2 SO.sub.2NH.sub.2 3-iPr-
Phenyl 999 3 SO.sub.2NH.sub.2 3-iPr-Phenyl 1000 1 H 4-NH.sub.2-Phenyl 1001 2 H 4-NH.sub.2-
Phenyl 1002 3 H 4-NH.sub.2-Phenyl 1003 1 Me 4-NH.sub.2-Phenyl 1004 2 Me 4-NH.sub.2-
Phenyl 1005 3 Me 4-NH.sub.2-Phenyl 1006 1 CH.sub.2Ph 4-NH.sub.2-Phenyl 1007 2 CH.sub.2Ph
4-NH.sub.2-Phenyl 1008 3 CH.sub.2Ph 4-NH.sub.2-Phenyl 1009 1 COMe 4-NH.sub.2-Phenyl
1010 2 COMe 4-NH.sub.2-Phenyl 1011 3 COMe 4-NH.sub.2-Phenyl 1012 1 CO.sub.2Me 4-
NH.sub.2-Phenyl 1013 2 CO.sub.2Me 4-NH.sub.2-Phenyl 1014 3 CO.sub.2Me 4-NH.sub.2-Phenyl
1015 1 CO.sub.2tBu 4-NH.sub.2-Phenyl 1016 2 CO.sub.2tBu 4-NH.sub.2-Phenyl 1017 3
CO.sub.2tBu 4-NH.sub.2-Phenyl 1018 1 CONHMe 4-NH.sub.2-Phenyl 1019 2 CONHMe 4-
NH.sub.2-Phenyl 1020 3 CONHMe 4-NH.sub.2-Phenyl 1021 1 SO.sub.2Me 4-NH.sub.2-Phenyl

1022 2 SO.sub.2Me 4-NH.sub.2-Phenyl 1023 3 SO.sub.2Me 4-NH.sub.2-Phenyl 1024 1
SO.sub.2NH.sub.2 4-NH.sub.2-Phenyl 1025 2 SO.sub.2NH.sub.2 4-NH.sub.2-Phenyl 1026 3
SO.sub.2NH.sub.2 4-NH.sub.2-Phenyl 1027 1 H 2-NH.sub.2-Phenyl 1028 2 H 2-NH.sub.2-Phenyl
1029 3 H 2-NH.sub.2-Phenyl 1030 1 Me 2-NH.sub.2-Phenyl 1031 2 Me 2-NH.sub.2-Phenyl 1032
3 Me 2-NH.sub.2-Phenyl 1033 1 CH.sub.2Ph 2-NH.sub.2-Phenyl 1034 2 CH.sub.2Ph 2-NH.sub.2-
Phenyl 1035 3 CH.sub.2Ph 2-NH.sub.2-Phenyl 1036 1 COMe 2-NH.sub.2-Phenyl 1037 2 COMe
2-NH.sub.2-Phenyl 1038 3 COMe 2-NH.sub.2-Phenyl 1039 1 CO.sub.2Me 2-NH.sub.2-Phenyl
1040 2 CO.sub.2Me 2-NH.sub.2-Phenyl 1041 3 CO.sub.2Me 2-NH.sub.2-Phenyl 1042 1
CO.sub.2tBu 2-NH.sub.2-Phenyl 1043 2 CO.sub.2tBu 2-NH.sub.2-Phenyl 1044 3 CO.sub.2tBu 2-
NH.sub.2-Phenyl 1045 1 CONHMe 2-NH.sub.2-Phenyl 1046 2 CONHMe 2-NH.sub.2-Phenyl
1047 3 CONHMe 2-NH.sub.2-Phenyl 1048 1 SO.sub.2Me 2-NH.sub.2-Phenyl 1049 2 SO.sub.2Me
2-NH.sub.2-Phenyl 1050 3 SO.sub.2Me 2-NH.sub.2-Phenyl 1051 1 SO.sub.2NH.sub.2 2-
NH.sub.2-Phenyl 1052 2 SO.sub.2NH.sub.2 2-NH.sub.2-Phenyl 1053 3 SO.sub.2NH.sub.2 2-
NH.sub.2-Phenyl 1054 1 H 2,6-di-Me-Phenyl 1055 2 H 2,6-di-Me-Phenyl 1056 3 H 2,6-di-Me-
Phenyl 1057 1 Me 2,6-di-Me-Phenyl 1058 2 Me 2,6-di-Me-Phenyl 1059 3 Me 2,6-di-Me-Phenyl
1060 1 CH.sub.2Ph 2,6-di-Me-Phenyl 1061 2 CH.sub.2Ph 2,6-di-Me-Phenyl 1062 3 CH.sub.2Ph
2,6-di-Me-Phenyl 1063 1 COMe 2,6-di-Me-Phenyl 1064 2 COMe 2,6-di-Me-Phenyl 1065 3 COMe
2,6-di-Me-Phenyl 1066 1 CO.sub.2Me 2,6-di-Me-Phenyl 1067 2 CO.sub.2Me 2,6-di-Me-Phenyl
1068 3 CO.sub.2Me 2,6-di-Me-Phenyl 1069 1 CO.sub.2tBu 2,6-di-Me-Phenyl 1070 2 CO.sub.2tBu
2,6-di-Me-Phenyl 1071 3 CO.sub.2tBu 2,6-di-Me-Phenyl 1072 1 CONHMe 2,6-di-Me-Phenyl
1073 2 CONHMe 2,6-di-Me-Phenyl 1074 3 CONHMe 2,6-di-Me-Phenyl 1075 1 SO.sub.2Me 2,6-
di-Me-Phenyl 1076 2 SO.sub.2Me 2,6-di-Me-Phenyl 1077 3 SO.sub.2Me 2,6-di-Me-Phenyl 1078 1
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SO.sub.2NH.sub.2 2,6-di-Me-Phenyl 1081 1 H 2-Ph-Phenyl 1082 2 H 2-Ph-Phenyl 1083 3 H 2-Ph-
Phenyl 1084 1 Me 2-Ph-Phenyl 1085 2 Me 2-Ph-Phenyl 1086 3 Me 2-Ph-Phenyl 1087 1
CH.sub.2Ph 2-Ph-Phenyl 1088 2 CH.sub.2Ph 2-Ph-Phenyl 1089 3 CH.sub.2Ph 2-Ph-Phenyl 1090 1
COMe 2-Ph-Phenyl 1091 2 COMe 2-Ph-Phenyl 1092 3 COMe 2-Ph-Phenyl 1093 1 CO.sub.2Me 2-
Ph-Phenyl 1094 2 CO.sub.2Me 2-Ph-Phenyl 1095 3 CO.sub.2Me 2-Ph-Phenyl 1096 1 CO.sub.2tBu
2-Ph-Phenyl 1097 2 CO.sub.2tBu 2-Ph-Phenyl 1098 3 CO.sub.2tBu 2-Ph-Phenyl 1099 1 CONHMe
2-Ph-Phenyl 1100 2 CONHMe 2-Ph-Phenyl 1101 3 CONHMe 2-Ph-Phenyl 1102 1 SO.sub.2Me 2-
Ph-Phenyl 1103 2 SO.sub.2Me 2-Ph-Phenyl 1104 3 SO.sub.2Me 2-Ph-Phenyl 1105 1
SO.sub.2NH.sub.2 2-Ph-Phenyl 1106 2 SO.sub.2NH.sub.2 2-Ph-Phenyl 1107 3 SO.sub.2NH.sub.2
2-Ph-Phenyl 1108 1 H 4-Ph-Phenyl 1109 2 H 4-Ph-Phenyl 1110 3 H 4-Ph-Phenyl 1111 1 Me 4-Ph-
Phenyl 1112 2 Me 4-Ph-Phenyl 1113 3 Me 4-Ph-Phenyl 1114 1 CH.sub.2Ph 4-Ph-Phenyl 1115 2
CH.sub.2Ph 4-Ph-Phenyl 1116 3 CH.sub.2Ph 4-Ph-Phenyl 1117 1 COMe 4-Ph-Phenyl 1118 2
COMe 4-Ph-Phenyl 1119 3 COMe 4-Ph-Phenyl 1120 1 CO.sub.2Me 4-Ph-Phenyl 1121 2
CO.sub.2Me 4-Ph-Phenyl 1122 3 CO.sub.2Me 4-Ph-Phenyl 1123 1 CO.sub.2tBu 4-Ph-Phenyl 1124
2 CO.sub.2tBu 4-Ph-Phenyl 1125 3 CO.sub.2tBu 4-Ph-Phenyl 1126 1 CONHMe 4-Ph-Phenyl 1127
2 CONHMe 4-Ph-Phenyl 1128 3 CONHMe 4-Ph-Phenyl 1129 1 SO.sub.2Me 4-Ph-Phenyl 1130 2
SO.sub.2Me 4-Ph-Phenyl 1131 3 SO.sub.2Me 4-Ph-Phenyl 1132 1 SO.sub.2NH.sub.2 4-Ph-Phenyl
1133 2 SO.sub.2NH.sub.2 4-Ph-Phenyl 1134 3 SO.sub.2NH.sub.2 4-Ph-Phenyl 1135 1 H 3-
morpholino-phenyl 1136 2 H 3-morpholino-phenyl 1137 3 H 3-morpholino-phenyl 1138 1 Me 3-
morpholino-phenyl 1139 2 Me 3-morpholino-phenyl 1140 3 Me 3-morpholino-phenyl 1141 1
CH.sub.2Ph 3-morpholino-phenyl 1142 2 CH.sub.2Ph 3-morpholino-phenyl 1143 3 CH.sub.2Ph 3-
morpholino-phenyl 1144 1 COMe 3-morpholino-phenyl 1145 2 COMe 3-morpholino-phenyl 1146
3 COMe 3-morpholino-phenyl 1147 1 CO.sub.2Me 3-morpholino-phenyl 1148 2 CO.sub.2Me 3-
morpholino-phenyl 1149 3 CO.sub.2Me 3-morpholino-phenyl 1150 1 CO.sub.2tBu 3-morpholino-
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1 CONHMe 3-morpholino-phenyl 1154 2 CONHMe 3-morpholino-phenyl 1155 3 CONHMe 3-
morpholino-phenyl 1156 1 SO.sub.2Me 3-morpholino-phenyl 1157 2 SO.sub.2Me 3-morpholino-

phenyl 1158 3 SO.sub.2Me 3-morpholino-phenyl 1159 1 SO.sub.2NH.sub.2 3-morpholino-phenyl 1160 2 SO.sub.2NH.sub.2 3-morpholino-phenyl 1161 3 SO.sub.2NH.sub.2 3-morpholino-phenyl 1162 1 H 2-pyrazinyl 1163 2 H 2-pyrazinyl 1164 3 H 2-pyrazinyl 1165 1 Me 2-pyrazinyl 1166 2 Me 2-pyrazinyl 1167 3 Me 2-pyrazinyl 1168 1 CH.sub.2Ph 2-pyrazinyl 1169 2 CH.sub.2Ph 2-pyrazinyl 1170 3 CH.sub.2Ph 2-pyrazinyl 1171 1 COMe 2-pyrazinyl 1172 2 COMe 2-pyrazinyl 1173 3 COMe 2-pyrazinyl 1174 1 CO.sub.2Me 2-pyrazinyl 1175 2 CO.sub.2Me 2-pyrazinyl 1176 3 CO.sub.2Me 2-pyrazinyl 1177 1 CO.sub.2tBu 2-pyrazinyl 1178 2 CO.sub.2tBu 2-pyrazinyl 1179 3 CO.sub.2tBu 2-pyrazinyl 1180 1 CONHMe 2-pyrazinyl 1181 2 CONHMe 2-pyrazinyl 1182 3 CONHMe 2-pyrazinyl 1183 1 SO.sub.2Me 2-pyrazinyl 1184 2 SO.sub.2Me 2-pyrazinyl 1185 3 SO.sub.2Me 2-pyrazinyl 1186 1 SO.sub.2NH.sub.2 2-pyrazinyl 1187 2 SO.sub.2NH.sub.2 2-pyrazinyl 1188 3 SO.sub.2NH.sub.2 2-pyrazinyl 1189 1 H 5-indolyl 1190 2 H 5-indolyl 1191 3 H 5-indolyl 1192 1 Me 5-indolyl 1193 2 Me 5-indolyl 1194 3 Me 5-indolyl 1195 1 CH.sub.2Ph 5-indolyl 1196 2 CH.sub.2Ph 5-indolyl 1197 3 CH.sub.2Ph 5-indolyl 1198 1 COMe 5-indolyl 1199 2 COMe 5-indolyl 1200 3 COMe 5-indolyl 1201 1 CO.sub.2Me 5-indolyl 1202 2 CO.sub.2Me 5-indolyl 1203 3 CO.sub.2Me 5-indolyl 1204 1 CO.sub.2tBu 5-indolyl 1205 2 CO.sub.2tBu 5-indolyl 1206 3 CO.sub.2tBu 5-indolyl 1207 1 CONHMe 5-indolyl 1208 2 CONHMe 5-indolyl 1209 3 CONHMe 5-indolyl 1210 1 SO.sub.2Me 5-indolyl 1211 2 SO.sub.2Me 5-indolyl 1212 3 SO.sub.2Me 5-indolyl 1213 1 SO.sub.2NH.sub.2 5-indolyl 1214 2 SO.sub.2NH.sub.2 5-indolyl 1215 3 SO.sub.2NH.sub.2 5-indolyl 1216 1 H 1H-benzo[d]imidazol- 4-yl 1217 2 H 1H-benzo[d]imidazol- 4-yl 1218 3 H 1H-benzo[d]imidazol- 4-yl 1219 1 Me 1H-benzo[d]imidazol- 4-yl 1220 2 Me 1H-benzo[d]imidazol- 4-yl 1221 3 Me 1H-benzo[d]imidazol- 4-yl 1222 1 CH.sub.2Ph 1H-benzo[d]imidazol- 4-yl 1223 2 CH.sub.2Ph 1H-benzo[d]imidazol- 4-yl 1224 3 CH.sub.2Ph 1H-benzo[d]imidazol- 4-yl 1225 1 COMe 1H-benzo[d]imidazol- 4-yl 1226 2 COMe 1H-benzo[d]imidazol- 4-yl 1227 3 COMe 1H-benzo[d]imidazol- 4-yl 1228 1 CO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1229 2 CO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1230 3 CO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1231 1 CO.sub.2tBu 1H-benzo[d]imidazol- 4-yl 1232 2 CO.sub.2tBu 1H-benzo[d]imidazol- 4-yl 1233 3 CO.sub.2tBu 1H-benzo[d]imidazol- 4-yl 1234 1 CONHMe 1H-benzo[d]imidazol- 4-yl 1235 2 CONHMe 1H-benzo[d]imidazol- 4-yl 1236 3 CONHMe 1H-benzo[d]imidazol- 4-yl 1237 1 SO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1238 2 SO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1239 3 SO.sub.2Me 1H-benzo[d]imidazol- 4-yl 1240 1 SO.sub.2NH.sub.2 1H-benzo[d]imidazol- 4-yl 1241 2 SO.sub.2NH.sub.2 1H-benzo[d]imidazol- 4-yl 1242 3 SO.sub.2NH.sub.2 1H-benzo[d]imidazol- 4-yl

[0402] Exemplary embodiments include compounds having the formula (XIX)

##STR00061##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 3.

TABLE-US-00003 TABLE 3 Entry n R R.sup.1a R.sup.1b R.sup.1c R.sup.1d 1 1 H H H H H 2 2 H H H H H 3 3 H H H H H 4 1 Me H H H H 5 2 Me H H H H 6 3 Me H H H H 7 1 CH.sub.2Ph H H H H 8 2 CH.sub.2Ph H H H H 9 3 CH.sub.2Ph H H H H 10 1 COMe H H H H 11 2 COMe H H H H 12 3 COMe H H H H 13 1 CO.sub.2Me H H H H 14 2 CO.sub.2Me H H H H 15 3 CO.sub.2Me H H H H 16 1 CO.sub.2tBu H H H H 17 2 CO.sub.2tBu H H H H 18 3 CO.sub.2tBu H H H H 19 1 CONHMe H H H H 20 2 CONHMe H H H H 21 3 CONHMe H H H H 22 1 SO.sub.2Me H H H H 23 2 SO.sub.2Me H H H H 24 3 SO.sub.2Me H H H H 25 1 SO.sub.2NH.sub.2 H H H H 26 2 SO.sub.2NH.sub.2 H H H H 27 3 SO.sub.2NH.sub.2 H H H H 28 1 H H H OH H 29 2 H H H OH H 30 3 H H H OH H 31 1 Me H H OH H 32 2 Me H H OH H 33 3 Me H H OH H 34 1 CH.sub.2Ph H H OH H 35 2 CH.sub.2Ph H H OH H 36 3 CH.sub.2Ph H H OH H 37 1 COMe H H OH H 38 2 COMe H H OH H 39 3 COMe H H OH H 40 1 CO.sub.2Me H H OH H 41 2 CO.sub.2Me H H OH H 42 3 CO.sub.2Me H H OH H 43 1 CO.sub.2tBu H H OH H 44 2 CO.sub.2tBu H H OH H 45 3 CO.sub.2tBu H H OH H 46 1 CONHMe H H OH H 47 2 CONHMe H H OH H 48 3 CONHMe H H OH H 49 1 SO.sub.2Me H H OH H 50 2 SO.sub.2Me H H OH H 51 3 SO.sub.2Me H H OH H 52 1 SO.sub.2NH.sub.2 H H OH H 53 2 SO.sub.2NH.sub.2 H H OH

H 54 3 SO.sub.2NH.sub.2 H H OH H 55 1 H H OH H 56 2 H H OH H 57 3 H H OH Me H
58 1 Me H H OMe H 59 2 Me H H OMe H 60 3 Me H H OMe H 61 1 CH.sub.2Ph H H OMe H 62
2 CH.sub.2Ph H H OMe H 63 3 CH.sub.2Ph H H OMe H 64 1 COMe H H OMe H 65 2 COMe H
H OMe H 66 3 COMe H H OMe H 67 1 CO.sub.2Me H H OMe H 68 2 CO.sub.2Me H H OMe H
69 3 CO.sub.2Me H H OMe H 70 1 CO.sub.2tBu H H OMe H 71 2 CO.sub.2tBu H H OMe H 72 3
CO.sub.2tBu H H OMe H 73 1 CONHMe H H OMe H 74 2 CONHMe H H OMe H 75 3
CONHMe H H OMe H 76 1 SO.sub.2Me H H OMe H 77 2 SO.sub.2Me H H OMe H 78 3
SO.sub.2Me H H OMe H 79 1 SO.sub.2NH.sub.2 H H OMe H 80 2 SO.sub.2NH.sub.2 H H OMe
H 81 3 SO.sub.2NH.sub.2 H H OMe H 82 1 H H H Me H 83 2 H H H Me H 84 3 H H H Me H 85
1 Me H H Me H 86 2 Me H H Me H 87 3 Me H H Me H 88 1 CH.sub.2Ph H H Me H 89 2
CH.sub.2Ph H H Me H 90 3 CH.sub.2Ph H H Me H 91 1 COMe H H Me H 92 2 COMe H H Me H
93 3 COMe H H Me H 94 1 CO.sub.2Me H H Me H 95 2 CO.sub.2Me H H Me H 96 3
CO.sub.2Me H H Me H 97 1 CO.sub.2tBu H H Me H 98 2 CO.sub.2tBu H H Me H 99 3
CO.sub.2tBu H H Me H 100 1 CONHMe H H Me H 101 2 CONHMe H H Me H 102 3 CONHMe
H H Me H 103 1 SO.sub.2Me H H Me H 104 2 SO.sub.2Me H H Me H 105 3 SO.sub.2Me H H
Me H 106 1 SO.sub.2NH.sub.2 H H Me H 107 2 SO.sub.2NH.sub.2 H H Me H 108 3
SO.sub.2NH.sub.2 H H Me H 109 1 H H H CF.sub.3 H 110 2 H H H CF.sub.3 H 111 3 H H H
CF.sub.3 H 112 1 Me H H CF.sub.3 H 113 2 Me H H CF.sub.3 H 114 3 Me H H CF.sub.3 H 115 1
CH.sub.2Ph H H CF.sub.3 H 116 2 CH.sub.2Ph H H CF.sub.3 H 117 3 CH.sub.2Ph H H CF.sub.3
H 118 1 COMe H H CF.sub.3 H 119 2 COMe H H CF.sub.3 H 120 3 COMe H H CF.sub.3 H 121 1
CO.sub.2Me H H CF.sub.3 H 122 2 CO.sub.2Me H H CF.sub.3 H 123 3 CO.sub.2Me H H
CF.sub.3 H 124 1 CO.sub.2tBu H H CF.sub.3 H 125 2 CO.sub.2tBu H H CF.sub.3 H 126 3
CO.sub.2tBu H H CF.sub.3 H 127 1 CONHMe H H CF.sub.3 H 128 2 CONHMe H H CF.sub.3 H
129 3 CONHMe H H CF.sub.3 H 130 1 SO.sub.2Me H H CF.sub.3 H 131 2 SO.sub.2Me H H
CF.sub.3 H 132 3 SO.sub.2Me H H CF.sub.3 H 133 1 SO.sub.2NH.sub.2 H H CF.sub.3 H 134 2
SO.sub.2NH.sub.2 H H CF.sub.3 H 135 3 SO.sub.2NH.sub.2 H H CF.sub.3 H 136 1 H H H F H
137 2 H H H F H 138 3 H H H F H 139 1 Me H H F H 140 2 Me H H F H 141 3 Me H H F H 142
1 CH.sub.2Ph H H F H 143 2 CH.sub.2Ph H H F H 144 3 CH.sub.2Ph H H F H 145 1 COMe H H
F H 146 2 COMe H H F H 147 3 COMe H H F H 148 1 CO.sub.2Me H H F H 149 2 CO.sub.2Me
H H F H 150 3 CO.sub.2Me H H F H 151 1 CO.sub.2tBu H H F H 152 2 CO.sub.2tBu H H F H
153 3 CO.sub.2tBu H H F H 154 1 CONHMe H H F H 155 2 CONHMe H H F H 156 3 CONHMe
H H F H 157 1 SO.sub.2Me H H F H 158 2 SO.sub.2Me H H F H 159 3 SO.sub.2Me H H F H 160
1 SO.sub.2NH.sub.2 H H F H 161 2 SO.sub.2NH.sub.2 H H F H 162 3 SO.sub.2NH.sub.2 H H F H
163 1 H H H Cl H 164 2 H H H Cl H 165 3 H H H Cl H 166 1 Me H H Cl H 167 2 Me H H Cl H
168 3 Me H H Cl H 169 1 CH.sub.2Ph H H Cl H 170 2 CH.sub.2Ph H H Cl H 171 3 CH.sub.2Ph H
H Cl H 172 1 COMe H H Cl H 173 2 COMe H H Cl H 174 3 COMe H H Cl H 175 1 CO.sub.2Me
H H Cl H 176 2 CO.sub.2Me H H Cl H 177 3 CO.sub.2Me H H Cl H 178 1 CO.sub.2tBu H H Cl H
179 2 CO.sub.2tBu H H Cl H 180 3 CO.sub.2tBu H H Cl H 181 1 CONHMe H H Cl H 182 2
CONHMe H H Cl H 183 3 CONHMe H H Cl H 184 1 SO.sub.2Me H H Cl H 185 2 SO.sub.2Me H
H Cl H 186 3 SO.sub.2Me H H Cl H 187 1 SO.sub.2NH.sub.2 H H Cl H 188 2 SO.sub.2NH.sub.2
H H Cl H 189 3 SO.sub.2NH.sub.2 H H Cl H 190 1 H H H CN H 191 2 H H H CN H 192 3 H H H
CN H 193 1 Me H H CN H 194 2 Me H H CN H 195 3 Me H H CN H 196 1 CH.sub.2Ph H H CN
H 197 2 CH.sub.2Ph H H CN H 198 3 CH.sub.2Ph H H CN H 199 1 COMe H H CN H 200 2
COMe H H CN H 201 3 COMe H H CN H 202 1 CO.sub.2Me H H CN H 203 2 CO.sub.2Me H H
CN H 204 3 CO.sub.2Me H H CN H 205 1 CO.sub.2tBu H H CN H 206 2 CO.sub.2tBu H H CN H
207 3 CO.sub.2tBu H H CN H 208 1 CONHMe H H CN H 209 2 CONHMe H H CN H 210 3
CONHMe H H CN H 211 1 SO.sub.2Me H H CN H 212 2 SO.sub.2Me H H CN H 213 3
SO.sub.2Me H H CN H 214 1 SO.sub.2NH.sub.2 H H CN H 215 2 SO.sub.2NH.sub.2 H H CN H
216 3 SO.sub.2NH.sub.2 H H CN H 217 1 H H OH H H 218 2 H H OH H H 219 3 H H OH H H
220 1 Me H OH H H 221 2 Me H OH H H 222 3 Me H OH H H 223 1 CH.sub.2Ph H OH H H 224

2 CH.sub.2Ph H OH H H 225 3 CH.sub.2Ph H OH H H 226 1 COMe H OH H H 227 2 COMe H OH H H 228 3 COMe H OH H H 229 1 CO.sub.2Me H OH H H 230 2 CO.sub.2Me H OH H H 231 3 CO.sub.2Me H OH H H 232 1 CO.sub.2tBu H OH H H 233 2 CO.sub.2tBu H OH H H 234 3 CO.sub.2tBu H OH H H 235 1 CONHMe H OH H H 236 2 CONHMe H OH H H 237 3 CONHMe H OH H H 238 1 SO.sub.2Me H OH H H 239 2 SO.sub.2Me H OH H H 240 3 SO.sub.2Me H OH H H 241 1 SO.sub.2NH.sub.2 H OH H H 242 2 SO.sub.2NH.sub.2 H OH H H 243 3 SO.sub.2NH.sub.2 H OH H H 244 1 H H OMe H H 245 2 H H OMe H H 246 3 H H OMe H H 247 1 Me H OMe H H 248 2 Me H OMe H H 249 3 Me H OMe H H 250 1 CH.sub.2Ph H OMe H H 251 2 CH.sub.2Ph H OMe H H 252 3 CH.sub.2Ph H OMe H H 253 1 COMe H OMe H H 254 2 COMe H OMe H H 255 3 COMe H OMe H H 256 1 CO.sub.2Me H OMe H H 257 2 CO.sub.2Me H OMe H H 258 3 CO.sub.2Me H OMe H H 259 1 CO.sub.2tBu H OMe H H 260 2 CO.sub.2tBu H OMe H H 261 3 CO.sub.2tBu H OMe H H 262 1 CONHMe H OMe H H 263 2 CONHMe H OMe H H 264 3 CONHMe H OMe H H 265 1 SO.sub.2Me H OMe H H 266 2 SO.sub.2Me H OMe H H 267 3 SO.sub.2Me H OMe H H 268 1 SO.sub.2NH.sub.2 H OMe H H 269 2 SO.sub.2NH.sub.2 H OMe H H 270 3 SO.sub.2NH.sub.2 H OMe H H 271 1 H H Me H H 272 2 H H Me H H 273 3 H H Me H H 274 1 Me H Me H H 275 2 Me H Me H H 276 3 Me H Me H H 277 1 CH.sub.2Ph H Me H H 278 2 CH.sub.2Ph H Me H H 279 3 CH.sub.2Ph H Me H H 280 1 COMe H Me H H 281 2 COMe H Me H H 282 3 COMe H Me H H 283 1 CO.sub.2Me H Me H H 284 2 CO.sub.2Me H Me H H 285 3 CO.sub.2Me H Me H H 286 1 CO.sub.2tBu H Me H H 287 2 CO.sub.2tBu H Me H H 288 3 CO.sub.2tBu H Me H H 289 1 CONHMe H Me H H 290 2 CONHMe H Me H H 291 3 CONHMe H Me H H 292 1 SO.sub.2Me H Me H H 293 2 SO.sub.2Me H Me H H 294 3 SO.sub.2Me H Me H H 295 1 SO.sub.2NH.sub.2 H Me H H 296 2 SO.sub.2NH.sub.2 H Me H H 297 3 SO.sub.2NH.sub.2 H Me H H 298 1 H H CF.sub.3 H H 299 2 H H CF.sub.3 H H 300 3 H H CF.sub.3 H H 301 1 Me H CF.sub.3 H H 302 2 Me H CF.sub.3 H H 303 3 Me H CF.sub.3 H H 304 1 CH.sub.2Ph H CF.sub.3 H H 305 2 CH.sub.2Ph H CF.sub.3 H H 306 3 CH.sub.2Ph H CF.sub.3 H H 307 1 COMe H CF.sub.3 H H 308 2 COMe H CF.sub.3 H H 309 3 COMe H CF.sub.3 H H 310 1 CO.sub.2Me H CF.sub.3 H H 311 2 CO.sub.2Me H CF.sub.3 H H 312 3 CO.sub.2Me H CF.sub.3 H H 313 1 CO.sub.2tBu H CF.sub.3 H H 314 2 CO.sub.2tBu H CF.sub.3 H H 315 3 CO.sub.2tBu H CF.sub.3 H H 316 1 CONHMe H CF.sub.3 H H 317 2 CONHMe H CF.sub.3 H H 318 3 CONHMe H CF.sub.3 H H 319 1 SO.sub.2Me H CF.sub.3 H H 320 2 SO.sub.2Me H CF.sub.3 H H 321 3 SO.sub.2Me H CF.sub.3 H H 322 1 SO.sub.2NH.sub.2 H CF.sub.3 H H 323 2 SO.sub.2NH.sub.2 H CF.sub.3 H H 324 3 SO.sub.2NH.sub.2 H CF.sub.3 H H 325 1 H H F H H 326 2 H H F H H 327 3 H H F H H 328 1 Me H F H H 329 2 Me H F H H 330 3 Me H F H H 331 1 CH.sub.2Ph H F H H 332 2 CH.sub.2Ph H F H H 333 3 CH.sub.2Ph H F H H 334 1 COMe H F H H 335 2 COMe H F H H 336 3 COMe H F H H 337 1 CO.sub.2Me H F H H 338 2 CO.sub.2Me H F H H 339 3 CO.sub.2Me H F H H 340 1 CO.sub.2tBu H F H H 341 2 CO.sub.2tBu H F H H 342 3 CO.sub.2tBu H F H H 343 1 CONHMe H F H H 344 2 CONHMe H F H H 345 3 CONHMe H F H H 346 1 SO.sub.2Me H F H H 347 2 SO.sub.2Me H F H H 348 3 SO.sub.2Me H F H H 349 1 SO.sub.2NH.sub.2 H F H H 350 2 SO.sub.2NH.sub.2 H F H H 351 3 SO.sub.2NH.sub.2 H F H H 352 1 H H Cl H H 353 2 H H Cl H H 354 3 H H Cl H H 355 1 Me H Cl H H 356 2 Me H Cl H H 357 3 Me H Cl H H 358 1 CH.sub.2Ph H Cl H H 359 2 CH.sub.2Ph H Cl H H 360 3 CH.sub.2Ph H Cl H H 361 1 COMe H Cl H H 362 2 COMe H Cl H H 363 3 COMe H Cl H H 364 1 CO.sub.2Me H Cl H H 365 2 CO.sub.2Me H Cl H H 366 3 CO.sub.2Me H Cl H H 367 1 CO.sub.2tBu H Cl H H 368 2 CO.sub.2tBu H Cl H H 369 3 CO.sub.2tBu H Cl H H 370 1 CONHMe H Cl H H 371 2 CONHMe H Cl H H 372 3 CONHMe H Cl H H 373 1 SO.sub.2Me H Cl H H 374 2 SO.sub.2Me H Cl H H 375 3 SO.sub.2Me H Cl H H 376 1 SO.sub.2NH.sub.2 H Cl H H 377 2 SO.sub.2NH.sub.2 H Cl H H 378 3 SO.sub.2NH.sub.2 H Cl H H 379 1 H H CN H H 380 2 H H CN H H 381 3 H H CN H H 382 1 Me H CN H H 383 2 Me H CN H H 384 3 Me H CN H H 385 1 CH.sub.2Ph H CN H H 386 2 CH.sub.2Ph H CN H H 387 3 CH.sub.2Ph H CN H H 388 1 COMe H CN H H 389 2 COMe H CN H H 390 3 COMe H CN H H 391 1 CO.sub.2Me H CN H

H 392 2 CO.sub.2Me H CN H H 393 3 CO.sub.2Me H CN H H 394 1 CO.sub.2tBu H CN H H 395
2 CO.sub.2tBu H CN H H 396 3 CO.sub.2tBu H CN H H 397 1 CONHMe H CN H H 398 2
CONHMe H CN H H 399 3 CONHMe H CN H H 400 1 SO.sub.2Me H CN H H 401 2
SO.sub.2Me H CN H H 402 3 SO.sub.2Me H CN H H 403 1 SO.sub.2NH.sub.2 H CN H H 404 2
SO.sub.2NH.sub.2 H CN H H 405 3 SO.sub.2NH.sub.2 H CN H H
[0403] Exemplary embodiments include compounds having the formula (XX)
##STR00062##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 4.

TABLE-US-00004 TABLE 4 Entry n R R.sup.1a R.sup.1b R.sup.1c R.sup.1d 1 1 H H H H H 2 2 H
H H H H 3 3 H H H H H 4 1 Me H H H H 5 2 Me H H H H 6 3 Me H H H H 7 1 CH.sub.2Ph H H
H H 8 2 CH.sub.2Ph H H H H 9 3 CH.sub.2Ph H H H H 10 1 COMe H H H H 11 2 COMe H H H
H 12 3 COMe H H H H 13 1 CO.sub.2Me H H H H 14 2 CO.sub.2Me H H H H 15 3 CO.sub.2Me
H H H H 16 1 CO.sub.2tBu H H H H 17 2 CO.sub.2tBu H H H H 18 3 CO.sub.2tBu H H H H 19 1
CONHMe H H H H 20 2 CONHMe H H H H 21 3 CONHMe H H H H 22 1 SO.sub.2Me H H H H
23 2 SO.sub.2Me H H H H 24 3 SO.sub.2Me H H H H 25 1 SO.sub.2NH.sub.2 H H H H 26 2
SO.sub.2NH.sub.2 H H H H 27 3 SO.sub.2NH.sub.2 H H H H 28 1 H H H OH H 29 2 H H H OH
H 30 3 H H H OH H 31 1 Me H H OH H 32 2 Me H H OH H 33 3 Me H H OH H 34 1
CH.sub.2Ph H H OH H 35 2 CH.sub.2Ph H H OH H 36 3 CH.sub.2Ph H H OH H 37 1 COMe H H
OH H 38 2 COMe H H OH H 39 3 COMe H H OH H 40 1 CO.sub.2Me H H OH H 41 2
CO.sub.2Me H H OH H 42 3 CO.sub.2Me H H OH H 43 1 CO.sub.2tBu H H OH H 44 2
CO.sub.2tBu H H OH H 45 3 CO.sub.2tBu H H OH H 46 1 CONHMe H H OH H 47 2 CONHMe
H H OH H 48 3 CONHMe H H OH H 49 1 SO.sub.2Me H H OH H 50 2 SO.sub.2Me H H OH H
51 3 SO.sub.2Me H H OH H 52 1 SO.sub.2NH.sub.2 H H OH H 53 2 SO.sub.2NH.sub.2 H H OH
H 54 3 SO.sub.2NH.sub.2 H H OH H 55 1 H H H OMe H 56 2 H H H OMe H 57 3 H H H OMe H
58 1 Me H H OMe H 59 2 Me H H OMe H 60 3 Me H H OMe H 61 1 CH.sub.2Ph H H OMe H 62
2 CH.sub.2Ph H H OMe H 63 3 CH.sub.2Ph H H OMe H 64 1 COMe H H OMe H 65 2 COMe H
H OMe H 66 3 COMe H H OMe H 67 1 CO.sub.2Me H H OMe H 68 2 CO.sub.2Me H H OMe H
69 3 CO.sub.2Me H H OMe H 70 1 CO.sub.2tBu H H OMe H 71 2 CO.sub.2tBu H H OMe H 72 3
CO.sub.2tBu H H OMe H 73 1 CONHMe H H OMe H 74 2 CONHMe H H OMe H 75 3
CONHMe H H OMe H 76 1 SO.sub.2Me H H OMe H 77 2 SO.sub.2Me H H OMe H 78 3
SO.sub.2Me H H OMe H 79 1 SO.sub.2NH.sub.2 H H OMe H 80 2 SO.sub.2NH.sub.2 H H OMe
H 81 3 SO.sub.2NH.sub.2 H H OMe H 82 1 H H H Me H 83 2 H H H Me H 84 3 H H H Me H 85
1 Me H H Me H 86 2 Me H H Me H 87 3 Me H H Me H 88 1 CH.sub.2Ph H H Me H 89 2
CH.sub.2Ph H H Me H 90 3 CH.sub.2Ph H H Me H 91 1 COMe H H Me H 92 2 COMe H H Me H
93 3 COMe H H Me H 94 1 CO.sub.2Me H H Me H 95 2 CO.sub.2Me H H Me H 96 3
CO.sub.2Me H H Me H 97 1 CO.sub.2tBu H H Me H 98 2 CO.sub.2tBu H H Me H 99 3
CO.sub.2tBu H H Me H 100 1 CONHMe H H Me H 101 2 CONHMe H H Me H 102 3 CONHMe
H H Me H 103 1 SO.sub.2Me H H Me H 104 2 SO.sub.2Me H H Me H 105 3 SO.sub.2Me H H
Me H 106 1 SO.sub.2NH.sub.2 H H Me H 107 2 SO.sub.2NH.sub.2 H H Me H 108 3
SO.sub.2NH.sub.2 H H Me H 109 1 H H H CF.sub.3 H 110 2 H H H CF.sub.3 H 111 3 H H H
CF.sub.3 H 112 1 Me H H CF.sub.3 H 113 2 Me H H CF.sub.3 H 114 3 Me H H CF.sub.3 H 115 1
CH.sub.2Ph H H CF.sub.3 H 116 2 CH.sub.2Ph H H CF.sub.3 H 117 3 CH.sub.2Ph H H CF.sub.3
H 118 1 COMe H H CF.sub.3 H 119 2 COMe H H CF.sub.3 H 120 3 COMe H H CF.sub.3 H 121 1
CO.sub.2Me H H CF.sub.3 H 122 2 CO.sub.2Me H H CF.sub.3 H 123 3 CO.sub.2Me H H
CF.sub.3 H 124 1 CO.sub.2tBu H H CF.sub.3 H 125 2 CO.sub.2tBu H H CF.sub.3 H 126 3
CO.sub.2tBu H H CF.sub.3 H 127 1 CONHMe H H CF.sub.3 H 128 2 CONHMe H H CF.sub.3 H
129 3 CONHMe H H CF.sub.3 H 130 1 SO.sub.2Me H H CF.sub.3 H 131 2 SO.sub.2Me H H
CF.sub.3 H 132 3 SO.sub.2Me H H CF.sub.3 H 133 1 SO.sub.2NH.sub.2 H H CF.sub.3 H 134 2
SO.sub.2NH.sub.2 H H CF.sub.3 H 135 3 SO.sub.2NH.sub.2 H H CF.sub.3 H 136 1 H H H F H
137 2 H H H F H 138 3 H H H F H 139 1 Me H H F H 140 2 Me H H F H 141 3 Me H H F H 142

1 CH.sub.2Ph H H F H 143 2 CH.sub.2Ph H H F H 144 3 CH.sub.2Ph H H F H 145 1 COMe H H F H 146 2 COMe H H F H 147 3 COMe H H F H 148 1 CO.sub.2Me H H F H 149 2 CO.sub.2Me H H F H 150 3 CO.sub.2Me H H F H 151 1 CO.sub.2tBu H H F H 152 2 CO.sub.2tBu H H F H 153 3 CO.sub.2tBu H H F H 154 1 CONHMe H H F H 155 2 CONHMe H H F H 156 3 CONHMe H H F H 157 1 SO.sub.2Me H H F H 158 2 SO.sub.2Me H H F H 159 3 SO.sub.2Me H H F H 160 1 SO.sub.2NH.sub.2 H H F H 161 2 SO.sub.2NH.sub.2 H H F H 162 3 SO.sub.2NH.sub.2 H H F H 163 1 H H H Cl H 164 2 H H H Cl H 165 3 H H H Cl H 166 1 Me H H Cl H 167 2 Me H H Cl H 168 3 Me H H Cl H 169 1 CH.sub.2Ph H H Cl H 170 2 CH.sub.2Ph H H Cl H 171 3 CH.sub.2Ph H H Cl H 172 1 COMe H H Cl H 173 2 COMe H H Cl H 174 3 COMe H H Cl H 175 1 CO.sub.2Me H H Cl H 176 2 CO.sub.2Me H H Cl H 177 3 CO.sub.2Me H H Cl H 178 1 CO.sub.2tBu H H Cl H 179 2 CO.sub.2tBu H H Cl H 180 3 CO.sub.2tBu H H Cl H 181 1 CONHMe H H Cl H 182 2 CONHMe H H Cl H 183 3 CONHMe H H Cl H 184 1 SO.sub.2Me H H Cl H 185 2 SO.sub.2Me H H Cl H 186 3 SO.sub.2Me H H Cl H 187 1 SO.sub.2NH.sub.2 H H Cl H 188 2 SO.sub.2NH.sub.2 H H Cl H 189 3 SO.sub.2NH.sub.2 H H Cl H 190 1 H H H CN H 191 2 H H H CN H 192 3 H H H CN H 193 1 Me H H CN H 194 2 Me H H CN H 195 3 Me H H CN H 196 1 CH.sub.2Ph H H CN H 197 2 CH.sub.2Ph H H CN H 198 3 CH.sub.2Ph H H CN H 199 1 COMe H H CN H 200 2 COMe H H CN H 201 3 COMe H H CN H 202 1 CO.sub.2Me H H CN H 203 2 CO.sub.2Me H H CN H 204 3 CO.sub.2Me H H CN H 205 1 CO.sub.2tBu H H CN H 206 2 CO.sub.2tBu H H CN H 207 3 CO.sub.2tBu H H CN H 208 1 CONHMe H H CN H 209 2 CONHMe H H CN H 210 3 CONHMe H H CN H 211 1 SO.sub.2Me H H CN H 212 2 SO.sub.2Me H H CN H 213 3 SO.sub.2Me H H CN H 214 1 SO.sub.2NH.sub.2 H H CN H 215 2 SO.sub.2NH.sub.2 H H CN H 216 3 SO.sub.2NH.sub.2 H H CN H 217 1 H H OH H H 218 2 H H OH H H 219 3 H H OH H H 220 1 Me H OH H H 221 2 Me H OH H H 222 3 Me H OH H H 223 1 CH.sub.2Ph H OH H H 224 2 CH.sub.2Ph H OH H H 225 3 CH.sub.2Ph H OH H H 226 1 COMe H OH H H 227 2 COMe H OH H H 228 3 COMe H OH H H 229 1 CO.sub.2Me H OH H H 230 2 CO.sub.2Me H OH H H 231 3 CO.sub.2Me H OH H H 232 1 CO.sub.2tBu H OH H H 233 2 CO.sub.2tBu H OH H H 234 3 CO.sub.2tBu H OH H H 235 1 CONHMe H OH H H 236 2 CONHMe H OH H H 237 3 CONHMe H OH H H 238 1 SO.sub.2Me H OH H H 239 2 SO.sub.2Me H OH H H 240 3 SO.sub.2Me H OH H H 241 1 SO.sub.2NH.sub.2 H OH H H 242 2 SO.sub.2NH.sub.2 H OH H H 243 3 SO.sub.2NH.sub.2 H OH H H 244 1 H H OMe H H 245 2 H H OMe H H 246 3 H H OMe H H 247 1 Me H OMe H H 248 2 Me H OMe H H 249 3 Me H OMe H H 250 1 CH.sub.2Ph H OMe H H 251 2 CH.sub.2Ph H OMe H H 252 3 CH.sub.2Ph H OMe H H 253 1 COMe H OMe H H 254 2 COMe H OMe H H 255 3 COMe H OMe H H 256 1 CO.sub.2Me H OMe H H 257 2 CO.sub.2Me H OMe H H 258 3 CO.sub.2Me H OMe H H 259 1 CO.sub.2tBu H OMe H H 260 2 CO.sub.2tBu H OMe H H 261 3 CO.sub.2tBu H OMe H H 262 1 CONHMe H OMe H H 263 2 CONHMe H OMe H H 264 3 CONHMe H OMe H H 265 1 SO.sub.2Me H OMe H H 266 2 SO.sub.2Me H OMe H H 267 3 SO.sub.2Me H OMe H H 268 1 SO.sub.2NH.sub.2 H OMe H H 269 2 SO.sub.2NH.sub.2 H OMe H H 270 3 SO.sub.2NH.sub.2 H OMe H H 271 1 H H Me H H 272 2 H H Me H H 273 3 H H Me H H 274 1 Me H Me H H 275 2 Me H Me H H 276 3 Me H Me H H 277 1 CH.sub.2Ph H Me H H 278 2 CH.sub.2Ph H Me H H 279 3 CH.sub.2Ph H Me H H 280 1 COMe H Me H H 281 2 COMe H Me H H 282 3 COMe H Me H H 283 1 CO.sub.2Me H Me H H 284 2 CO.sub.2Me H Me H H 285 3 CO.sub.2Me H Me H H 286 1 CO.sub.2tBu H Me H H 287 2 CO.sub.2tBu H Me H H 288 3 CO.sub.2tBu H Me H H 289 1 CONHMe H Me H H 290 2 CONHMe H Me H H 291 3 CONHMe H Me H H 292 1 SO.sub.2Me H Me H H 293 2 SO.sub.2Me H Me H H 294 3 SO.sub.2Me H Me H H 295 1 SO.sub.2NH.sub.2 H Me H H 296 2 SO.sub.2NH.sub.2 H Me H H 297 3 SO.sub.2NH.sub.2 H Me H H 298 1 H H CF.sub.3 H H 299 2 H H CF.sub.3 H H 300 3 H H CF.sub.3 H H 301 1 Me H CF.sub.3 H H 302 2 Me H CF.sub.3 H H 303 3 Me H CF.sub.3 H H 304 1 CH.sub.2Ph H CF.sub.3 H H 305 2 CH.sub.2Ph H CF.sub.3 H H 306 3 CH.sub.2Ph H CF.sub.3 H H 307 1 COMe H CF.sub.3 H H 308 2 COMe H CF.sub.3 H H 309 3 COMe H CF.sub.3 H H 310 1 CO.sub.2Me H CF.sub.3 H H 311 2 CO.sub.2Me H CF.sub.3

H H 312 3 CO.sub.2Me H CF.sub.3 H H 313 1 CO.sub.2tBu H CF.sub.3 H H 314 2 CO.sub.2tBu H CF.sub.3 H H 315 3 CO.sub.2tBu H CF.sub.3 H H 316 1 CONHMe H CF.sub.3 H H 317 2 CONHMe H CF.sub.3 H H 318 3 CONHMe H CF.sub.3 H H 319 1 SO.sub.2Me H CF.sub.3 H H 320 2 SO.sub.2Me H CF.sub.3 H H 321 3 SO.sub.2Me H CF.sub.3 H H 322 1 SO.sub.2NH.sub.2 H CF.sub.3 H H 323 2 SO.sub.2NH.sub.2 H CF.sub.3 H H 324 3 SO.sub.2NH.sub.2 H CF.sub.3 H H 325 1 H H F H H 326 2 H H F H H 327 3 H H F H H 328 1 Me H F H H 329 2 Me H F H H 330 3 Me H F H H 331 1 CH.sub.2Ph H F H H 332 2 CH.sub.2Ph H F H H 333 3 CH.sub.2Ph H F H H 334 1 COMe H F H H 335 2 COMe H F H H 336 3 COMe H F H H 337 1 CO.sub.2Me H F H H 338 2 CO.sub.2Me H F H H 339 3 CO.sub.2Me H F H H 340 1 CO.sub.2tBu H F H H 341 2 CO.sub.2tBu H F H H 342 3 CO.sub.2tBu H F H H 343 1 CONHMe H F H H 344 2 CONHMe H F H H 345 3 CONHMe H F H H 346 1 SO.sub.2Me H F H H 347 2 SO.sub.2Me H F H H 348 3 SO.sub.2Me H F H H 349 1 SO.sub.2NH.sub.2 H F H H 350 2 SO.sub.2NH.sub.2 H F H H 351 3 SO.sub.2NH.sub.2 H F H H 352 1 H H Cl H H 353 2 H H Cl H H 354 3 H H Cl H H 355 1 Me H Cl H H 356 2 Me H Cl H H 357 3 Me H Cl H H 358 1 CH.sub.2Ph H Cl H H 359 2 CH.sub.2Ph H Cl H H 360 3 CH.sub.2Ph H Cl H H 361 1 COMe H Cl H H 362 2 COMe H Cl H H 363 3 COMe H Cl H H 364 1 CO.sub.2Me H Cl H H 365 2 CO.sub.2Me H Cl H H 366 3 CO.sub.2Me H Cl H H 367 1 CO.sub.2tBu H Cl H H 368 2 CO.sub.2tBu H Cl H H 369 3 CO.sub.2tBu H Cl H H 370 1 CONHMe H Cl H H 371 2 CONHMe H Cl H H 372 3 CONHMe H Cl H H 373 1 SO.sub.2Me H Cl H H 374 2 SO.sub.2Me H Cl H H 375 3 SO.sub.2Me H Cl H H 376 1 SO.sub.2NH.sub.2 H Cl H H 377 2 SO.sub.2NH.sub.2 H Cl H H 378 3 SO.sub.2NH.sub.2 H Cl H H 379 1 H H CN H H 380 2 H H CN H H 381 3 H H CN H H 382 1 Me H CN H H 383 2 Me H CN H H 384 3 Me H CN H H 385 1 CH.sub.2Ph H CN H H 386 2 CH.sub.2Ph H CN H H 387 3 CH.sub.2Ph H CN H H 388 1 COMe H CN H H 389 2 COMe H CN H H 390 3 COMe H CN H H 391 1 CO.sub.2Me H CN H H 392 2 CO.sub.2Me H CN H H 393 3 CO.sub.2Me H CN H H 394 1 CO.sub.2tBu H CN H H 395 2 CO.sub.2tBu H CN H H 396 3 CO.sub.2tBu H CN H H 397 1 CONHMe H CN H H 398 2 CONHMe H CN H H 399 3 CONHMe H CN H H 400 1 SO.sub.2Me H CN H H 401 2 SO.sub.2Me H CN H H 402 3 SO.sub.2Me H CN H H 403 1 SO.sub.2NH.sub.2 H CN H H 404 2 SO.sub.2NH.sub.2 H CN H H 405 3 SO.sub.2NH.sub.2 H CN H H

[0404] Exemplary embodiments include compounds having the formula (XXI)

##STR00063##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 5.

TABLE-US-00005 TABLE 5 Entry n R R.sup.1a R.sup.1b R.sup.1c R.sup.1d 1 1 H H H H H 2 2 H H H H H 3 3 H H H H H 4 1 Me H H H H 5 2 Me H H H H 6 3 Me H H H H 7 1 CH.sub.2Ph H H H 8 2 CH.sub.2Ph H H H H 9 3 CH.sub.2Ph H H H H 10 1 COMe H H H H 11 2 COMe H H H H 12 3 COMe H H H H 13 1 CO.sub.2Me H H H H 14 2 CO.sub.2Me H H H H 15 3 CO.sub.2Me H H H H 16 1 CO.sub.2tBu H H H H 17 2 CO.sub.2tBu H H H H 18 3 CO.sub.2tBu H H H H 19 1 CONHMe H H H H 20 2 CONHMe H H H H 21 3 CONHMe H H H H 22 1 SO.sub.2Me H H H H 23 2 SO.sub.2Me H H H H 24 3 SO.sub.2Me H H H H 25 1 SO.sub.2NH.sub.2 H H H H 26 2 SO.sub.2NH.sub.2 H H H H 27 3 SO.sub.2NH.sub.2 H H H H

[0405] Exemplary embodiments include compounds having the formula (XXII)

##STR00064##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 6.

TABLE-US-00006 TABLE 6 Entry n R R.sup.1a R.sup.1b R.sup.1c R.sup.1d 1 1 H H H H H 2 2 H H H H H 3 3 H H H H H 4 1 Me H H H H 5 2 Me H H H H 6 3 Me H H H H 7 1 CH.sub.2Ph H H H 8 2 CH.sub.2Ph H H H H 9 3 CH.sub.2Ph H H H H 10 1 COMe H H H H 11 2 COMe H H H H 12 3 COMe H H H H 13 1 CO.sub.2Me H H H H 14 2 CO.sub.2Me H H H H 15 3 CO.sub.2Me H H H H 16 1 CO.sub.2tBu H H H H 17 2 CO.sub.2tBu H H H H 18 3 CO.sub.2tBu H H H H 19 1 CONHMe H H H H 20 2 CONHMe H H H H 21 3 CONHMe H H H H 22 1 SO.sub.2Me H H H H 23 2 SO.sub.2Me H H H H 24 3 SO.sub.2Me H H H H 25 1 SO.sub.2NH.sub.2 H H H H 26 2 SO.sub.2NH.sub.2 H H H H 27 3 SO.sub.2NH.sub.2 H H H H

[0406] Exemplary embodiments include compounds having the formula (XXIII)

##STR00065##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 7.

TABLE-US-00007 TABLE 7 Entry n X R.sup.3 1 1 O Phenyl 2 2 O Phenyl 3 3 O Phenyl 4 1 S Phenyl 5 2 S Phenyl 6 3 S Phenyl 7 1 SO Phenyl 8 2 SO Phenyl 9 3 SO Phenyl 10 1 SO.sub.2 Phenyl 11 2 SO.sub.2 Phenyl 12 3 SO.sub.2 Phenyl 13 1 O 3-OH-Phenyl 14 2 O 3-OH-Phenyl 15 3 O 3-OH-Phenyl 16 1 S 3-OH-Phenyl 17 2 S 3-OH-Phenyl 18 3 S 3-OH-Phenyl 19 1 SO 3-OH-Phenyl 20 2 SO 3-OH-Phenyl 21 3 SO 3-OH-Phenyl 22 1 SO.sub.2 3-OH-Phenyl 23 2 SO.sub.2 3-OH-Phenyl 24 3 SO.sub.2 3-OH-Phenyl 25 1 O 4-NO.sub.2-Phenyl 26 2 O 4-NO.sub.2-Phenyl 27 3 O 4-NO.sub.2-Phenyl 28 1 S 4-NO.sub.2-Phenyl 29 2 S 4-NO.sub.2-Phenyl 30 3 S 4-NO.sub.2-Phenyl 31 1 SO 4-NO.sub.2-Phenyl 32 2 SO 4-NO.sub.2-Phenyl 33 3 SO 4-NO.sub.2-Phenyl 34 1 SO.sub.2 4-NO.sub.2-Phenyl 35 2 SO.sub.2 4-NO.sub.2-Phenyl 36 3 SO.sub.2 4-NO.sub.2-Phenyl 37 1 O 3-OMe-Phenyl 38 2 O 3-OMe-Phenyl 39 3 O 3-OMe-Phenyl 40 1 S 3-OMe-Phenyl 41 2 S 3-OMe-Phenyl 42 3 S 3-OMe-Phenyl 43 1 SO 3-OMe-Phenyl 44 2 SO 3-OMe-Phenyl 45 3 SO 3-OMe-Phenyl 46 1 SO.sub.2 3-OMe-Phenyl 47 2 SO.sub.2 3-OMe-Phenyl 48 3 SO.sub.2 3-OMe-Phenyl 49 1 O 4-CN-Phenyl 50 2 O 4-CN-Phenyl 51 3 O 4-CN-Phenyl 52 1 S 4-CN-Phenyl 53 2 S 4-CN-Phenyl 54 3 S 4-CN-Phenyl 55 1 SO 4-CN-Phenyl 56 2 SO 4-CN-Phenyl 57 3 SO 4-CN-Phenyl 58 1 SO.sub.2 4-CN-Phenyl 59 2 SO.sub.2 4-CN-Phenyl 60 3 SO.sub.2 4-CN-Phenyl 61 1 O 2-CN-Phenyl 62 2 O 2-CN-Phenyl 63 3 O 2-CN-Phenyl 64 1 S 2-CN-Phenyl 65 2 S 2-CN-Phenyl 66 3 S 2-CN-Phenyl 67 1 SO 2-CN-Phenyl 68 2 SO 2-CN-Phenyl 69 3 SO 2-CN-Phenyl 70 1 SO.sub.2 2-CN-Phenyl 71 2 SO.sub.2 2-CN-Phenyl 72 3 SO.sub.2 2-CN-Phenyl 73 1 O 3-Me-Phenyl 74 2 O 3-Me-Phenyl 75 3 O 3-Me-Phenyl 76 1 S 3-Me-Phenyl 77 2 S 3-Me-Phenyl 78 3 S 3-Me-Phenyl 79 1 SO 3-Me-Phenyl 80 2 SO 3-Me-Phenyl 81 3 SO 3-Me-Phenyl 82 1 SO.sub.2 3-Me-Phenyl 83 2 SO.sub.2 3-Me-Phenyl 84 3 SO.sub.2 3-Me-Phenyl 85 1 O 2-F-Phenyl 86 2 O 2-F-Phenyl 87 3 O 2-F-Phenyl 88 1 S 2-F-Phenyl 89 2 S 2-F-Phenyl 90 3 S 2-F-Phenyl 91 1 SO 2-F-Phenyl 92 2 SO 2-F-Phenyl 93 3 SO 2-F-Phenyl 94 1 SO.sub.2 2-F-Phenyl 95 2 SO.sub.2 2-F-Phenyl 96 3 SO.sub.2 2-F-Phenyl 97 1 O 4-F-Phenyl 98 2 O 4-F-Phenyl 99 3 O 4-F-Phenyl 100 1 S 4-F-Phenyl 101 2 S 4-F-Phenyl 102 3 S 4-F-Phenyl 103 1 SO 4-F-Phenyl 104 2 SO 4-F-Phenyl 105 3 SO 4-F-Phenyl 106 1 SO.sub.2 4-F-Phenyl 107 2 SO.sub.2 4-F-Phenyl 108 3 SO.sub.2 4-F-Phenyl 109 1 O 3-Cl-Phenyl 110 2 O 3-Cl-Phenyl 111 3 O 3-Cl-Phenyl 112 1 S 3-Cl-Phenyl 113 2 S 3-Cl-Phenyl 114 3 S 3-Cl-Phenyl 115 1 SO 3-Cl-Phenyl 116 2 SO 3-Cl-Phenyl 117 3 SO 3-Cl-Phenyl 118 1 SO.sub.2 3-Cl-Phenyl 119 2 SO.sub.2 3-Cl-Phenyl 120 3 SO.sub.2 3-Cl-Phenyl 121 1 O 2-Br-Phenyl 122 2 O 2-Br-Phenyl 123 3 O 2-Br-Phenyl 124 1 S 2-Br-Phenyl 125 2 S 2-Br-Phenyl 126 3 S 2-Br-Phenyl 127 1 SO 2-Br-Phenyl 128 2 SO 2-Br-Phenyl 129 3 SO 2-Br-Phenyl 130 1 SO.sub.2 2-Br-Phenyl 131 2 SO.sub.2 2-Br-Phenyl 132 3 SO.sub.2 2-Br-Phenyl 133 1 O 4-Br-Phenyl 134 2 O 4-Br-Phenyl 135 3 O 4-Br-Phenyl 136 1 S 4-Br-Phenyl 137 2 S 4-Br-Phenyl 138 3 S 4-Br-Phenyl 139 1 SO 4-Br-Phenyl 140 2 SO 4-Br-Phenyl 141 3 SO 4-Br-Phenyl 142 1 SO.sub.2 4-Br-Phenyl 143 2 SO.sub.2 4-Br-Phenyl 144 3 SO.sub.2 4-Br-Phenyl 145 1 O 3-CF.sub.3-Phenyl 146 2 O 3-CF.sub.3-Phenyl 147 3 O 3-CF.sub.3-Phenyl 148 1 S 3-CF.sub.3-Phenyl 149 2 S 3-CF.sub.3-Phenyl 150 3 S 3-CF.sub.3-Phenyl 151 1 SO 3-CF.sub.3-Phenyl 152 2 SO 3-CF.sub.3-Phenyl 153 3 SO 3-CF.sub.3-Phenyl 154 1 SO.sub.2 3-CF.sub.3-Phenyl 155 2 SO.sub.2 3-CF.sub.3-Phenyl 156 3 SO.sub.2 3-CF.sub.3-Phenyl 157 1 O 2-iPr-Phenyl 158 2 O 2-iPr-Phenyl 159 3 O 2-iPr-Phenyl 160 1 S 2-iPr-Phenyl 161 2 S 2-iPr-Phenyl 162 3 S 2-iPr-Phenyl 163 1 SO 2-iPr-Phenyl 164 2 SO 2-iPr-Phenyl 165 3 SO 2-iPr-Phenyl 166 1 SO.sub.2 2-iPr-Phenyl 167 2 SO.sub.2 2-iPr-Phenyl 168 3 SO.sub.2 2-iPr-Phenyl 169 1 O 4-iPr-Phenyl 170 2 O 4-iPr-Phenyl 171 3 O 4-iPr-Phenyl 172 1 S 4-iPr-Phenyl 173 2 S 4-iPr-Phenyl 174 3 S 4-iPr-Phenyl 175 1 SO 4-iPr-Phenyl 176 2 SO 4-iPr-Phenyl 177 3 SO 4-iPr-Phenyl 178 1 SO.sub.2 4-iPr-Phenyl 179 2 SO.sub.2 4-iPr-Phenyl 180 3 SO.sub.2 4-iPr-Phenyl 181 1 O 3-NH.sub.2-Phenyl 182 2 O 3-NH.sub.2-Phenyl 183 3 O 3-NH.sub.2-Phenyl 184 1 S 3-NH.sub.2-Phenyl 185 2 S 3-NH.sub.2-Phenyl 186 3 S 3-NH.sub.2-Phenyl 187 1 SO 3-NH.sub.2-Phenyl 188 2 SO 3-NH.sub.2-Phenyl

189 3 SO 3-NH.sub.2-Phenyl 190 1 SO.sub.2 3-NH.sub.2-Phenyl 191 2 SO.sub.2 3-NH.sub.2-Phenyl 192 3 SO.sub.2 3-NH.sub.2-Phenyl 193 1 O 2,4-di-Me-Phenyl 194 2 O 2,4-di-Me-Phenyl 195 3 O 2,4-di-Me-Phenyl 196 1 S 2,4-di-Me-Phenyl 197 2 S 2,4-di-Me-Phenyl 198 3 S 2,4-di-Me-Phenyl 199 1 SO 2,4-di-Me-Phenyl 200 2 SO 2,4-di-Me-Phenyl 201 3 SO 2,4-di-Me-Phenyl 202 1 SO.sub.2 2,4-di-Me-Phenyl 203 2 SO.sub.2 2,4-di-Me-Phenyl 204 3 SO.sub.2 2,4-di-Me-Phenyl 205 1 O 2,6-di-iPr-Phenyl 206 2 O 2,6-di-iPr-Phenyl 207 3 O 2,6-di-iPr-Phenyl 208 1 S 2,6-di-iPr-Phenyl 209 2 S 2,6-di-iPr-Phenyl 210 3 S 2,6-di-iPr-Phenyl 211 1 SO 2,6-di-iPr-Phenyl 212 2 SO 2,6-di-iPr-Phenyl 213 3 SO 2,6-di-iPr-Phenyl 214 1 SO.sub.2 2,6-di-iPr-Phenyl 215 2 SO.sub.2 2,6-di-iPr-Phenyl 216 3 SO.sub.2 2,6-di-iPr-Phenyl 217 1 O 3-Ph-Phenyl 218 2 O 3-Ph-Phenyl 219 3 O 3-Ph-Phenyl 220 1 S 3-Ph-Phenyl 221 2 S 3-Ph-Phenyl 222 3 S 3-Ph-Phenyl 223 1 SO 3-Ph-Phenyl 224 2 SO 3-Ph-Phenyl 225 3 SO 3-Ph-Phenyl 226 1 SO.sub.2 3-Ph-Phenyl 227 2 SO.sub.2 3-Ph-Phenyl 228 3 SO.sub.2 3-Ph-Phenyl 229 1 O 2-morpholino-phenyl 230 2 O 2-morpholino-phenyl 231 3 O 2-morpholino-phenyl 232 1 S 2-morpholino-phenyl 233 2 S 2-morpholino-phenyl 234 3 S 2-morpholino-phenyl 235 1 SO 2-morpholino-phenyl 236 2 SO 2-morpholino-phenyl 237 3 SO 2-morpholino-phenyl 238 1 SO.sub.2 2-morpholino-phenyl 239 2 SO.sub.2 2-morpholino-phenyl 240 3 SO.sub.2 2-morpholino-phenyl 241 1 O 4-morpholino-phenyl 242 2 O 4-morpholino-phenyl 243 3 O 4-morpholino-phenyl 244 1 S 4-morpholino-phenyl 245 2 S 4-morpholino-phenyl 246 3 S 4-morpholino-phenyl 247 1 SO 4-morpholino-phenyl 248 2 SO 4-morpholino-phenyl 249 3 SO 4-morpholino-phenyl 250 1 SO.sub.2 4-morpholino-phenyl 251 2 SO.sub.2 4-morpholino-phenyl 252 3 SO.sub.2 4-morpholino-phenyl 253 1 O 2-pyrimidinyl 254 2 O 2-pyrimidinyl 255 3 O 2-pyrimidinyl 256 1 S 2-pyrimidinyl 257 2 S 2-pyrimidinyl 258 3 S 2-pyrimidinyl 259 1 SO 2-pyrimidinyl 260 2 SO 2-pyrimidinyl 261 3 SO 2-pyrimidinyl 262 1 SO.sub.2 2-pyrimidinyl 263 2 SO.sub.2 2-pyrimidinyl 264 3 SO.sub.2 2-pyrimidinyl 265 1 O 2-methyl-1H-benzo[d]imidazol-4-yl 266 2 O 2-methyl-1H-benzo[d]imidazol-4-yl 267 3 O 2-methyl-1H-benzo[d]imidazol-4-yl 268 1 S 2-methyl-1H-benzo[d]imidazol-4-yl 269 2 S 2-methyl-1H-benzo[d]imidazol-4-yl 270 3 S 2-methyl-1H-benzo[d]imidazol-4-yl 271 1 SO 2-methyl-1H-benzo[d]imidazol-4-yl 272 2 SO 2-methyl-1H-benzo[d]imidazol-4-yl 273 3 SO 2-methyl-1H-benzo[d]imidazol-4-yl 274 1 SO.sub.2 2-methyl-1H-benzo[d]imidazol-4-yl 275 2 SO.sub.2 2-methyl-1H-benzo[d]imidazol-4-yl 276 3 SO.sub.2 2-methyl-1H-benzo[d]imidazol-4-yl 277 1 O 4-OH-Phenyl 278 2 O 4-OH-Phenyl 279 3 O 4-OH-Phenyl 280 1 S 4-OH-Phenyl 281 2 S 4-OH-Phenyl 282 3 S 4-OH-Phenyl 283 1 SO 4-OH-Phenyl 284 2 SO 4-OH-Phenyl 285 3 SO 4-OH-Phenyl 286 1 SO.sub.2 4-OH-Phenyl 287 2 SO.sub.2 4-OH-Phenyl 288 3 SO.sub.2 4-OH-Phenyl 289 1 O 2-OH-Phenyl 290 2 O 2-OH-Phenyl 291 3 O 2-OH-Phenyl 292 1 S 2-OH-Phenyl 293 2 S 2-OH-Phenyl 294 3 S 2-OH-Phenyl 295 1 SO 2-OH-Phenyl 296 2 SO 2-OH-Phenyl 297 3 SO 2-OH-Phenyl 298 1 SO.sub.2 2-OH-Phenyl 299 2 SO.sub.2 2-OH-Phenyl 300 3 SO.sub.2 2-OH-Phenyl 301 1 O 4-OMe-Phenyl 302 2 O 4-OMe-Phenyl 303 3 O 4-OMe-Phenyl 304 1 S 4-OMe-Phenyl 305 2 S 4-OMe-Phenyl 306 3 S 4-OMe-Phenyl 307 1 SO 4-OMe-Phenyl 308 2 SO 4-OMe-Phenyl 309 3 SO 4-OMe-Phenyl 310 1 SO.sub.2 4-OMe-Phenyl 311 2 SO.sub.2 4-OMe-Phenyl 312 3 SO.sub.2 4-OMe-Phenyl 313 1 O 2-OMe-Phenyl 314 2 O 2-OMe-Phenyl 315 3 O 2-OMe-Phenyl 316 1 S 2-OMe-Phenyl 317 2 S 2-OMe-Phenyl 318 3 S 2-OMe-Phenyl 319 1 SO 2-OMe-Phenyl 320 2 SO 2-OMe-Phenyl 321 3 SO 2-OMe-Phenyl 322 1 SO.sub.2 2-OMe-Phenyl 323 2 SO.sub.2 2-OMe-Phenyl 324 3 SO.sub.2 2-OMe-Phenyl 325 1 O 3-CN-Phenyl 326 2 O 3-CN-Phenyl 327 3 O 3-CN-Phenyl 328 1 S 3-CN-Phenyl 329 2 S 3-CN-Phenyl 330 3 S 3-CN-Phenyl 331 1 SO 3-CN-Phenyl 332 2 SO 3-CN-Phenyl 333 3 SO 3-CN-Phenyl 334 1 SO.sub.2 3-CN-Phenyl 335 2 SO.sub.2 3-CN-Phenyl 336 3 SO.sub.2 3-CN-Phenyl 337 1 O 2-Me-Phenyl 338 2 O 2-Me-Phenyl 339 3 O 2-Me-Phenyl 340 1 S 2-Me-Phenyl 341 2 S 2-Me-Phenyl 342 3 S 2-Me-Phenyl 343 1 SO 2-Me-Phenyl 344 2 SO 2-Me-Phenyl 345 3 SO 2-Me-Phenyl 346 1 SO.sub.2 2-Me-Phenyl 347 2 SO.sub.2 2-Me-Phenyl 348 3 SO.sub.2 2-Me-Phenyl 349 1 O 4-Me-Phenyl 350 2 O 4-Me-Phenyl 351 3 O 4-Me-Phenyl 352 1 S 4-Me-Phenyl 353 2 S 4-Me-Phenyl 354 3 S 4-Me-Phenyl 355 1 SO 4-Me-Phenyl 356 2 SO 4-Me-Phenyl 357 3 SO 4-Me-Phenyl 358 1 SO.sub.2 4-Me-Phenyl 359 2 SO.sub.2 4-Me-Phenyl 360 3 SO.sub.2

4-Me-Phenyl 361 1 O 3-F-Phenyl 362 2 O 3-F-Phenyl 363 3 O 3-F-Phenyl 364 1 S 3-F-Phenyl 365
2 S 3-F-Phenyl 366 3 S 3-F-Phenyl 367 1 SO 3-F-Phenyl 368 2 SO 3-F-Phenyl 369 3 SO 3-F-
Phenyl 370 1 SO.sub.2 3-F-Phenyl 371 2 SO.sub.2 3-F-Phenyl 372 3 SO.sub.2 3-F-Phenyl 373 1 O
2-Cl-Phenyl 374 2 O 2-Cl-Phenyl 375 3 O 2-Cl-Phenyl 376 1 S 2-Cl-Phenyl 377 2 S 2-Cl-Phenyl
378 3 S 2-Cl-Phenyl 379 1 SO 2-Cl-Phenyl 380 2 SO 2-Cl-Phenyl 381 3 SO 2-Cl-Phenyl 382 1
SO.sub.2 2-Cl-Phenyl 383 2 SO.sub.2 2-Cl-Phenyl 384 3 SO.sub.2 2-Cl-Phenyl 385 1 O 4-Cl-
Phenyl 386 2 O 4-Cl-Phenyl 387 3 O 4-Cl-Phenyl 388 1 S 4-Cl-Phenyl 389 2 S 4-Cl-Phenyl 390 3
S 4-Cl-Phenyl 391 1 SO 4-Cl-Phenyl 392 2 SO 4-Cl-Phenyl 393 3 SO 4-Cl-Phenyl 394 1 SO.sub.2
4-Cl-Phenyl 395 2 SO.sub.2 4-Cl-Phenyl 396 3 SO.sub.2 4-Cl-Phenyl 397 1 O 3-Br-Phenyl 398 2
O 3-Br-Phenyl 399 3 O 3-Br-Phenyl 400 1 S 3-Br-Phenyl 401 2 S 3-Br-Phenyl 402 3 S 3-Br-
Phenyl 403 1 SO 3-Br-Phenyl 404 2 SO 3-Br-Phenyl 405 3 SO 3-Br-Phenyl 406 1 SO.sub.2 3-Br-
Phenyl 407 2 SO.sub.2 3-Br-Phenyl 408 3 SO.sub.2 3-Br-Phenyl 409 1 O 2-CF.sub.3-Phenyl 410 2
O 2-CF.sub.3-Phenyl 411 3 O 2-CF.sub.3-Phenyl 412 1 S 2-CF.sub.3-Phenyl 413 2 S 2-CF.sub.3-
Phenyl 414 3 S 2-CF.sub.3-Phenyl 415 1 SO 2-CF.sub.3-Phenyl 416 2 SO 2-CF.sub.3-Phenyl 417 3
SO 2-CF.sub.3-Phenyl 418 1 SO.sub.2 2-CF.sub.3-Phenyl 419 2 SO.sub.2 2-CF.sub.3-Phenyl 420 3
SO.sub.2 2-CF.sub.3-Phenyl 421 1 O 4-CF.sub.3-Phenyl 422 2 O 4-CF.sub.3-Phenyl 423 3 O 4-
CF.sub.3-Phenyl 424 1 S 4-CF.sub.3-Phenyl 425 2 S 4-CF.sub.3-Phenyl 426 3 S 4-CF.sub.3-Phenyl
427 1 SO 4-CF.sub.3-Phenyl 428 2 SO 4-CF.sub.3-Phenyl 429 3 SO 4-CF.sub.3-Phenyl 430 1
SO.sub.2 4-CF.sub.3-Phenyl 431 2 SO.sub.2 4-CF.sub.3-Phenyl 432 3 SO.sub.2 4-CF.sub.3-Phenyl
433 1 O 3-iPr-Phenyl 434 2 O 3-iPr-Phenyl 435 3 O 3-iPr-Phenyl 436 1 S 3-iPr-Phenyl 437 2 S 3-
iPr-Phenyl 438 3 S 3-iPr-Phenyl 439 1 SO 3-iPr-Phenyl 440 2 SO 3-iPr-Phenyl 441 3 SO 3-iPr-
Phenyl 442 1 SO.sub.2 3-iPr-Phenyl 443 2 SO.sub.2 3-iPr-Phenyl 444 3 SO.sub.2 3-iPr-Phenyl 445
1 O 4-NH.sub.2-Phenyl 446 2 O 4-NH.sub.2-Phenyl 447 3 O 4-NH.sub.2-Phenyl 448 1 S 4-
NH.sub.2-Phenyl 449 2 S 4-NH.sub.2-Phenyl 450 3 S 4-NH.sub.2-Phenyl 451 1 SO 4-NH.sub.2-
Phenyl 452 2 SO 4-NH.sub.2-Phenyl 453 3 SO 4-NH.sub.2-Phenyl 454 1 SO.sub.2 4-NH.sub.2-
Phenyl 455 2 SO.sub.2 4-NH.sub.2-Phenyl 456 3 SO.sub.2 4-NH.sub.2-Phenyl 457 1 O 2-
NH.sub.2-Phenyl 458 2 O 2-NH.sub.2-Phenyl 459 3 O 2-NH.sub.2-Phenyl 460 1 S 2-NH.sub.2-
Phenyl 461 2 S 2-NH.sub.2-Phenyl 462 3 S 2-NH.sub.2-Phenyl 463 1 SO 2-NH.sub.2-Phenyl 464
2 SO 2-NH.sub.2-Phenyl 465 3 SO 2-NH.sub.2-Phenyl 466 1 SO.sub.2 2-NH.sub.2-Phenyl 467 2
SO.sub.2 2-NH.sub.2-Phenyl 468 3 SO.sub.2 2-NH.sub.2-Phenyl 469 1 O 2,6-di-Me-Phenyl 470 2
O 2,6-di-Me-Phenyl 471 3 O 2,6-di-Me-Phenyl 472 1 S 2,6-di-Me-Phenyl 473 2 S 2,6-di-Me-
Phenyl 474 3 S 2,6-di-Me-Phenyl 475 1 SO 2,6-di-Me-Phenyl 476 2 SO 2,6-di-Me-Phenyl 477 3
SO 2,6-di-Me-Phenyl 478 1 SO.sub.2 2,6-di-Me-Phenyl 479 2 SO.sub.2 2,6-di-Me-Phenyl 480 3
SO.sub.2 2,6-di-Me-Phenyl 481 1 O 2-Ph-Phenyl 482 2 O 2-Ph-Phenyl 483 3 O 2-Ph-Phenyl 484 1
S 2-Ph-Phenyl 485 2 S 2-Ph-Phenyl 486 3 S 2-Ph-Phenyl 487 1 SO 2-Ph-Phenyl 488 2 SO 2-Ph-
Phenyl 489 3 SO 2-Ph-Phenyl 490 1 SO.sub.2 2-Ph-Phenyl 491 2 SO.sub.2 2-Ph-Phenyl 492 3
SO.sub.2 2-Ph-Phenyl 493 1 O 4-Ph-Phenyl 494 2 O 4-Ph-Phenyl 495 3 O 4-Ph-Phenyl 496 1 S 4-
Ph-Phenyl 497 2 S 4-Ph-Phenyl 498 3 S 4-Ph-Phenyl 499 1 SO 4-Ph-Phenyl 500 2 SO 4-Ph-Phenyl
501 3 SO 4-Ph-Phenyl 502 1 SO.sub.2 4-Ph-Phenyl 503 2 SO.sub.2 4-Ph-Phenyl 504 3 SO.sub.2 4-
Ph-Phenyl 505 1 O 3-morpholino-phenyl 506 2 O 3-morpholino-phenyl 507 3 O 3-morpholino-
phenyl 508 1 S 3-morpholino-phenyl 509 2 S 3-morpholino-phenyl 510 3 S 3-morpholino-phenyl
511 1 SO 3-morpholino-phenyl 512 2 SO 3-morpholino-phenyl 513 3 SO 3-morpholino-phenyl
514 1 SO.sub.2 3-morpholino-phenyl 515 2 SO.sub.2 3-morpholino-phenyl 516 3 SO.sub.2 3-
morpholino-phenyl 517 1 O 2-pyrazinyl 518 2 O 2-pyrazinyl 519 3 O 2-pyrazinyl 520 1 S 2-
pyrazinyl 521 2 S 2-pyrazinyl 522 3 S 2-pyrazinyl 523 1 SO 2-pyrazinyl 524 2 SO 2-pyrazinyl 525
3 SO 2-pyrazinyl 526 1 SO.sub.2 2-pyrazinyl 527 2 SO.sub.2 2-pyrazinyl 528 3 SO.sub.2 2-
pyrazinyl 529 1 O 5-indolyl 530 2 O 5-indolyl 531 3 O 5-indolyl 532 1 S 5-indolyl 533 2 S 5-
indolyl 534 3 S 5-indolyl 535 1 SO 5-indolyl 536 2 SO 5-indolyl 537 3 SO 5-indolyl 538 1
SO.sub.2 5-indolyl 539 2 SO.sub.2 5-indolyl 540 3 SO.sub.2 5-indolyl 541 1 O 1H-
benzo[d]imidazol-4-yl 542 2 O 1H-benzo[d]imidazol-4-yl 543 3 O 1H-benzo[d]imidazol-4-yl 544

1 S 1H-benzo[d]imidazol-4-yl 545 2 S 1H-benzo[d]imidazol-4-yl 546 3 S 1H-benzo[d]imidazol-4-yl 547 1 SO 1H-benzo[d]imidazol-4-yl 548 2 SO 1H-benzo[d]imidazol-4-yl 549 3 SO 1H-benzo[d]imidazol-4-yl 550 1 SO.sub.2 1H-benzo[d]imidazol-4-yl 551 2 SO.sub.2 1H-benzo[d]imidazol-4-yl 552 3 SO.sub.2 1H-benzo[d]imidazol-4-yl

[0407] Exemplary embodiments include compounds having the formula (XXIV)

##STR00066##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 8.

TABLE-US-00008 TABLE 8 Entry n X R.sup.3 1 1 O Phenyl 2 2 O Phenyl 3 3 O Phenyl 4 1 S Phenyl 5 2 S Phenyl 6 3 S Phenyl 7 1 SO Phenyl 8 2 SO Phenyl 9 3 SO Phenyl 10 1 SO.sub.2 Phenyl 11 2 SO.sub.2 Phenyl 12 3 SO.sub.2 Phenyl 13 1 O 3-OH-Phenyl 14 2 O 3-OH-Phenyl 15 3 O 3-OH-Phenyl 16 1 S 3-OH-Phenyl 17 2 S 3-OH-Phenyl 18 3 S 3-OH-Phenyl 19 1 SO 3-OH-Phenyl 20 2 SO 3-OH-Phenyl 21 3 SO 3-OH-Phenyl 22 1 SO.sub.2 3-OH-Phenyl 23 2 SO.sub.2 3-OH-Phenyl 24 3 SO.sub.2 3-OH-Phenyl 25 1 O 4-NO.sub.2-Phenyl 26 2 O 4-NO.sub.2-Phenyl 27 3 O 4-NO.sub.2-Phenyl 28 1 S 4-NO.sub.2-Phenyl 29 2 S 4-NO.sub.2-Phenyl 30 3 S 4-NO.sub.2-Phenyl 31 1 SO 4-NO.sub.2-Phenyl 32 2 SO 4-NO.sub.2-Phenyl 33 3 SO 4-NO.sub.2-Phenyl 34 1 SO.sub.2 4-NO.sub.2-Phenyl 35 2 SO.sub.2 4-NO.sub.2-Phenyl 36 3 SO.sub.2 4-NO.sub.2-Phenyl 37 1 O 3-OMe-Phenyl 38 2 O 3-OMe-Phenyl 39 3 O 3-OMe-Phenyl 40 1 S 3-OMe-Phenyl 41 2 S 3-OMe-Phenyl 42 3 S 3-OMe-Phenyl 43 1 SO 3-OMe-Phenyl 44 2 SO 3-OMe-Phenyl 45 3 SO 3-OMe-Phenyl 46 1 SO.sub.2 3-OMe-Phenyl 47 2 SO.sub.2 3-OMe-Phenyl 48 3 SO.sub.2 3-OMe-Phenyl 49 1 O 4-CN-Phenyl 50 2 O 4-CN-Phenyl 51 3 O 4-CN-Phenyl 52 1 S 4-CN-Phenyl 53 2 S 4-CN-Phenyl 54 3 S 4-CN-Phenyl 55 1 SO 4-CN-Phenyl 56 2 SO 4-CN-Phenyl 57 3 SO 4-CN-Phenyl 58 1 SO.sub.2 4-CN-Phenyl 59 2 SO.sub.2 4-CN-Phenyl 60 3 SO.sub.2 4-CN-Phenyl 61 1 O 2-CN-Phenyl 62 2 O 2-CN-Phenyl 63 3 O 2-CN-Phenyl 64 1 S 2-CN-Phenyl 65 2 S 2-CN-Phenyl 66 3 S 2-CN-Phenyl 67 1 SO 2-CN-Phenyl 68 2 SO 2-CN-Phenyl 69 3 SO 2-CN-Phenyl 70 1 SO.sub.2 2-CN-Phenyl 71 2 SO.sub.2 2-CN-Phenyl 72 3 SO.sub.2 2-CN-Phenyl 73 1 O 3-Me-Phenyl 74 2 O 3-Me-Phenyl 75 3 O 3-Me-Phenyl 76 1 S 3-Me-Phenyl 77 2 S 3-Me-Phenyl 78 3 S 3-Me-Phenyl 79 1 SO 3-Me-Phenyl 80 2 SO 3-Me-Phenyl 81 3 SO 3-Me-Phenyl 82 1 SO.sub.2 3-Me-Phenyl 83 2 SO.sub.2 3-Me-Phenyl 84 3 SO.sub.2 3-Me-Phenyl 85 1 O 2-F-Phenyl 86 2 O 2-F-Phenyl 87 3 O 2-F-Phenyl 88 1 S 2-F-Phenyl 89 2 S 2-F-Phenyl 90 3 S 2-F-Phenyl 91 1 SO 2-F-Phenyl 92 2 SO 2-F-Phenyl 93 3 SO 2-F-Phenyl 94 1 SO.sub.2 2-F-Phenyl 95 2 SO.sub.2 2-F-Phenyl 96 3 SO.sub.2 2-F-Phenyl 97 1 O 4-F-Phenyl 98 2 O 4-F-Phenyl 99 3 O 4-F-Phenyl 100 1 S 4-F-Phenyl 101 2 S 4-F-Phenyl 102 3 S 4-F-Phenyl 103 1 SO 4-F-Phenyl 104 2 SO 4-F-Phenyl 105 3 SO 4-F-Phenyl 106 1 SO.sub.2 4-F-Phenyl 107 2 SO.sub.2 4-F-Phenyl 108 3 SO.sub.2 4-F-Phenyl 109 1 O 3-Cl-Phenyl 110 2 O 3-Cl-Phenyl 111 3 O 3-Cl-Phenyl 112 1 S 3-Cl-Phenyl 113 2 S 3-Cl-Phenyl 114 3 S 3-Cl-Phenyl 115 1 SO 3-Cl-Phenyl 116 2 SO 3-Cl-Phenyl 117 3 SO 3-Cl-Phenyl 118 1 SO.sub.2 3-Cl-Phenyl 119 2 SO.sub.2 3-Cl-Phenyl 120 3 SO.sub.2 3-Cl-Phenyl 121 1 O 2-Br-Phenyl 122 2 O 2-Br-Phenyl 123 3 O 2-Br-Phenyl 124 1 S 2-Br-Phenyl 125 2 S 2-Br-Phenyl 126 3 S 2-Br-Phenyl 127 1 SO 2-Br-Phenyl 128 2 SO 2-Br-Phenyl 129 3 SO 2-Br-Phenyl 130 1 SO.sub.2 2-Br-Phenyl 131 2 SO.sub.2 2-Br-Phenyl 132 3 SO.sub.2 2-Br-Phenyl 133 1 O 4-Br-Phenyl 134 2 O 4-Br-Phenyl 135 3 O 4-Br-Phenyl 136 1 S 4-Br-Phenyl 137 2 S 4-Br-Phenyl 138 3 S 4-Br-Phenyl 139 1 SO 4-Br-Phenyl 140 2 SO 4-Br-Phenyl 141 3 SO 4-Br-Phenyl 142 1 SO.sub.2 4-Br-Phenyl 143 2 SO.sub.2 4-Br-Phenyl 144 3 SO.sub.2 4-Br-Phenyl 145 1 O 3-CF.sub.3-Phenyl 146 2 O 3-CF.sub.3-Phenyl 147 3 O 3-CF.sub.3-Phenyl 148 1 S 3-CF.sub.3-Phenyl 149 2 S 3-CF.sub.3-Phenyl 150 3 S 3-CF.sub.3-Phenyl 151 1 SO 3-CF.sub.3-Phenyl 152 2 SO 3-CF.sub.3-Phenyl 153 3 SO 3-CF.sub.3-Phenyl 154 1 SO.sub.2 3-CF.sub.3-Phenyl 155 2 SO.sub.2 3-CF.sub.3-Phenyl 156 3 SO.sub.2 3-CF.sub.3-Phenyl 157 1 O 2-iPr-Phenyl 158 2 O 2-iPr-Phenyl 159 3 O 2-iPr-Phenyl 160 1 S 2-iPr-Phenyl 161 2 S 2-iPr-Phenyl 162 3 S 2-iPr-Phenyl 163 1 SO 2-iPr-Phenyl 164 2 SO 2-iPr-Phenyl 165 3 SO 2-iPr-Phenyl 166 1 SO.sub.2 2-iPr-Phenyl 167 2 SO.sub.2 2-iPr-Phenyl 168 3 SO.sub.2 2-iPr-Phenyl 169 1 O 4-iPr-Phenyl 170 2 O 4-iPr-Phenyl 171 3 O 4-iPr-Phenyl 172 1 S 4-iPr-Phenyl 173 2 S 4-iPr-Phenyl 174 3 S 4-iPr-Phenyl 175

1 SO 4-iPr-Phenyl 176 2 SO 4-iPr-Phenyl 177 3 SO 4-iPr-Phenyl 178 1 SO.sub.2 4-iPr-Phenyl 179
2 SO.sub.2 4-iPr-Phenyl 180 3 SO.sub.2 4-iPr-Phenyl 181 1 O 3-NH.sub.2-Phenyl 182 2 O 3-
NH.sub.2-Phenyl 183 3 O 3-NH.sub.2-Phenyl 184 1 S 3-NH.sub.2-Phenyl 185 2 S 3-NH.sub.2-
Phenyl 186 3 S 3-NH.sub.2-Phenyl 187 1 SO 3-NH.sub.2-Phenyl 188 2 SO 3-NH.sub.2-Phenyl
189 3 SO 3-NH.sub.2-Phenyl 190 1 SO.sub.2 3-NH.sub.2-Phenyl 191 2 SO.sub.2 3-NH.sub.2-
Phenyl 192 3 SO.sub.2 3-NH.sub.2-Phenyl 193 1 O 2,4-di-Me-Phenyl 194 2 O 2,4-di-Me-Phenyl
195 3 O 2,4-di-Me-Phenyl 196 1 S 2,4-di-Me-Phenyl 197 2 S 2,4-di-Me-Phenyl 198 3 S 2,4-di-Me-
Phenyl 199 1 SO 2,4-di-Me-Phenyl 200 2 SO 2,4-di-Me-Phenyl 201 3 SO 2,4-di-Me-Phenyl 202 1
SO.sub.2 2,4-di-Me-Phenyl 203 2 SO.sub.2 2,4-di-Me-Phenyl 204 3 SO.sub.2 2,4-di-Me-Phenyl
205 1 O 2,6-di-iPr-Phenyl 206 2 O 2,6-di-iPr-Phenyl 207 3 O 2,6-di-iPr-Phenyl 208 1 S 2,6-di-iPr-
Phenyl 209 2 S 2,6-di-iPr-Phenyl 210 3 S 2,6-di-iPr-Phenyl 211 1 SO 2,6-di-iPr-Phenyl 212 2 SO
2,6-di-iPr-Phenyl 213 3 SO 2,6-di-iPr-Phenyl 214 1 SO.sub.2 2,6-di-iPr-Phenyl 215 2 SO.sub.2
2,6-di-iPr-Phenyl 216 3 SO.sub.2 2,6-di-iPr-Phenyl 217 1 O 3-Ph-Phenyl 218 2 O 3-Ph-Phenyl 219
3 O 3-Ph-Phenyl 220 1 S 3-Ph-Phenyl 221 2 S 3-Ph-Phenyl 222 3 S 3-Ph-Phenyl 223 1 SO 3-Ph-
Phenyl 224 2 SO 3-Ph-Phenyl 225 3 SO 3-Ph-Phenyl 226 1 SO.sub.2 3-Ph-Phenyl 227 2 SO.sub.2
3-Ph-Phenyl 228 3 SO.sub.2 3-Ph-Phenyl 229 1 O 2-morpholino-phenyl 230 2 O 2-morpholino-
phenyl 231 3 O 2-morpholino-phenyl 232 1 S 2-morpholino-phenyl 233 2 S 2-morpholino-phenyl
234 3 S 2-morpholino-phenyl 235 1 SO 2-morpholino-phenyl 236 2 SO 2-morpholino-phenyl 237
3 SO 2-morpholino-phenyl 238 1 SO.sub.2 2-morpholino-phenyl 239 2 SO.sub.2 2-morpholino-
phenyl 240 3 SO.sub.2 2-morpholino-phenyl 241 1 O 4-morpholino-phenyl 242 2 O 4-morpholino-
phenyl 243 3 O 4-morpholino-phenyl 244 1 S 4-morpholino-phenyl 245 2 S 4-morpholino-phenyl
246 3 S 4-morpholino-phenyl 247 1 SO 4-morpholino-phenyl 248 2 SO 4-morpholino-phenyl 249
3 SO 4-morpholino-phenyl 250 1 SO.sub.2 4-morpholino-phenyl 251 2 SO.sub.2 4-morpholino-
phenyl 252 3 SO.sub.2 4-morpholino-phenyl 253 1 O 2-pyrimidinyl 254 2 O 2-pyrimidinyl 255 3
O 2-pyrimidinyl 256 1 S 2-pyrimidinyl 257 2 S 2-pyrimidinyl 258 3 S 2-pyrimidinyl 259 1 SO 2-
pyrimidinyl 260 2 SO 2-pyrimidinyl 261 3 SO 2-pyrimidinyl 262 1 SO.sub.2 2-pyrimidinyl 263 2
SO.sub.2 2-pyrimidinyl 264 3 SO.sub.2 2-pyrimidinyl 265 1 O 2-methyl-1H- benzo[d]imidazol-4-
yl 266 2 O 2-methyl-1H- benzo[d]imidazol-4-yl 267 3 O 2-methyl-1H- benzo[d]imidazol-4-yl 268
1 S 2-methyl-1H- benzo[d]imidazol-4-yl 269 2 S 2-methyl-1H- benzo[d]imidazol-4-yl 270 3 S 2-
methyl-1H- benzo[d]imidazol-4-yl 271 1 SO 2-methyl-1H- benzo[d]imidazol-4-yl 272 2 SO 2-
methyl-1H- benzo[d]imidazol-4-yl 273 3 SO 2-methyl-1H- benzo[d]imidazol-4-yl 274 1 SO.sub.2
2-methyl-1H- benzo[d]imidazol-4-yl 275 2 SO.sub.2 2-methyl-1H- benzo[d]imidazol-4-yl 276 3
SO.sub.2 2-methyl-1H- benzo[d]imidazol-4-yl 277 1 O 4-OH-Phenyl 278 2 O 4-OH-Phenyl 279 3
O 4-OH-Phenyl 280 1 S 4-OH-Phenyl 281 2 S 4-OH-Phenyl 282 3 S 4-OH-Phenyl 283 1 SO 4-
OH-Phenyl 284 2 SO 4-OH-Phenyl 285 3 SO 4-OH-Phenyl 286 1 SO.sub.2 4-OH-Phenyl 287 2
SO.sub.2 4-OH-Phenyl 288 3 SO.sub.2 4-OH-Phenyl 289 1 O 2-OH-Phenyl 290 2 O 2-OH-Phenyl
291 3 O 2-OH-Phenyl 292 1 S 2-OH-Phenyl 293 2 S 2-OH-Phenyl 294 3 S 2-OH-Phenyl 295 1 SO
2-OH-Phenyl 296 2 SO 2-OH-Phenyl 297 3 SO 2-OH-Phenyl 298 1 SO.sub.2 2-OH-Phenyl 299 2
SO.sub.2 2-OH-Phenyl 300 3 SO.sub.2 2-OH-Phenyl 301 1 O 4-OMe-Phenyl 302 2 O 4-OMe-
Phenyl 303 3 O 4-OMe-Phenyl 304 1 S 4-OMe-Phenyl 305 2 S 4-OMe-Phenyl 306 3 S 4-OMe-
Phenyl 307 1 SO 4-OMe-Phenyl 308 2 SO 4-OMe-Phenyl 309 3 SO 4-OMe-Phenyl 310 1
SO.sub.2 4-OMe-Phenyl 311 2 SO.sub.2 4-OMe-Phenyl 312 3 SO.sub.2 4-OMe-Phenyl 313 1 O 2-
OMe-Phenyl 314 2 O 2-OMe-Phenyl 315 3 O 2-OMe-Phenyl 316 1 S 2-OMe-Phenyl 317 2 S 2-
OMe-Phenyl 318 3 S 2-OMe-Phenyl 319 1 SO 2-OMe-Phenyl 320 2 SO 2-OMe-Phenyl 321 3 SO
2-OMe-Phenyl 322 1 SO.sub.2 2-OMe-Phenyl 323 2 SO.sub.2 2-OMe-Phenyl 324 3 SO.sub.2 2-
OMe-Phenyl 325 1 O 3-CN-Phenyl 326 2 O 3-CN-Phenyl 327 3 O 3-CN-Phenyl 328 1 S 3-CN-
Phenyl 329 2 S 3-CN-Phenyl 330 3 S 3-CN-Phenyl 331 1 SO 3-CN-Phenyl 332 2 SO 3-CN-Phenyl
333 3 SO 3-CN-Phenyl 334 1 SO.sub.2 3-CN-Phenyl 335 2 SO.sub.2 3-CN-Phenyl 336 3 SO.sub.2
3-CN-Phenyl 337 1 O 2-Me-Phenyl 338 2 O 2-Me-Phenyl 339 3 O 2-Me-Phenyl 340 1 S 2-Me-
Phenyl 341 2 S 2-Me-Phenyl 342 3 S 2-Me-Phenyl 343 1 SO 2-Me-Phenyl 344 2 SO 2-Me-Phenyl

345 3 SO 2-Me-Phenyl 346 1 SO.sub.2 2-Me-Phenyl 347 2 SO.sub.2 2-Me-Phenyl 348 3 SO.sub.2
2-Me-Phenyl 349 1 O 4-Me-Phenyl 350 2 O 4-Me-Phenyl 351 3 O 4-Me-Phenyl 352 1 S 4-Me-
Phenyl 353 2 S 4-Me-Phenyl 354 3 S 4-Me-Phenyl 355 1 SO 4-Me-Phenyl 356 2 SO 4-Me-Phenyl
357 3 SO 4-Me-Phenyl 358 1 SO.sub.2 4-Me-Phenyl 359 2 SO.sub.2 4-Me-Phenyl 360 3 SO.sub.2
4-Me-Phenyl 361 1 O 3-F-Phenyl 362 2 O 3-F-Phenyl 363 3 O 3-F-Phenyl 364 1 S 3-F-Phenyl 365
2 S 3-F-Phenyl 366 3 S 3-F-Phenyl 367 1 SO 3-F-Phenyl 368 2 SO 3-F-Phenyl 369 3 SO 3-F-
Phenyl 370 1 SO.sub.2 3-F-Phenyl 371 2 SO.sub.2 3-F-Phenyl 372 3 SO.sub.2 3-F-Phenyl 373 1 O
2-Cl-Phenyl 374 2 O 2-Cl-Phenyl 375 3 O 2-Cl-Phenyl 376 1 S 2-Cl-Phenyl 377 2 S 2-Cl-Phenyl
378 3 S 2-Cl-Phenyl 379 1 SO 2-Cl-Phenyl 380 2 SO 2-Cl-Phenyl 381 3 SO 2-Cl-Phenyl 382 1
SO.sub.2 2-Cl-Phenyl 383 2 SO.sub.2 2-Cl-Phenyl 384 3 SO.sub.2 2-Cl-Phenyl 385 1 O 4-Cl-
Phenyl 386 2 O 4-Cl-Phenyl 387 3 O 4-Cl-Phenyl 388 1 S 4-Cl-Phenyl 389 2 S 4-Cl-Phenyl 390 3
S 4-Cl-Phenyl 391 1 SO 4-Cl-Phenyl 392 2 SO 4-Cl-Phenyl 393 3 SO 4-Cl-Phenyl 394 1 SO.sub.2
4-Cl-Phenyl 395 2 SO.sub.2 4-Cl-Phenyl 396 3 SO.sub.2 4-Cl-Phenyl 397 1 O 3-Br-Phenyl 398 2
O 3-Br-Phenyl 399 3 O 3-Br-Phenyl 400 1 S 3-Br-Phenyl 401 2 S 3-Br-Phenyl 402 3 S 3-Br-
Phenyl 403 1 SO 3-Br-Phenyl 404 2 SO 3-Br-Phenyl 405 3 SO 3-Br-Phenyl 406 1 SO.sub.2 3-Br-
Phenyl 407 2 SO.sub.2 3-Br-Phenyl 408 3 SO.sub.2 3-Br-Phenyl 409 1 O 2-CF.sub.3-Phenyl 410 2
O 2-CF.sub.3-Phenyl 411 3 O 2-CF.sub.3-Phenyl 412 1 S 2-CF.sub.3-Phenyl 413 2 S 2-CF.sub.3-
Phenyl 414 3 S 2-CF.sub.3-Phenyl 415 1 SO 2-CF.sub.3-Phenyl 416 2 SO 2-CF.sub.3-Phenyl 417 3
SO 2-CF.sub.3-Phenyl 418 1 SO.sub.2 2-CF.sub.3-Phenyl 419 2 SO.sub.2 2-CF.sub.3-Phenyl 420 3
SO.sub.2 2-CF.sub.3-Phenyl 421 1 O 4-CF.sub.3-Phenyl 422 2 O 4-CF.sub.3-Phenyl 423 3 O 4-
CF.sub.3-Phenyl 424 1 S 4-CF.sub.3-Phenyl 425 2 S 4-CF.sub.3-Phenyl 426 3 S 4-CF.sub.3-Phenyl
427 1 SO 4-CF.sub.3-Phenyl 428 2 SO 4-CF.sub.3-Phenyl 429 3 SO 4-CF.sub.3-Phenyl 430 1
SO.sub.2 4-CF.sub.3-Phenyl 431 2 SO.sub.2 4-CF.sub.3-Phenyl 432 3 SO.sub.2 4-CF.sub.3-Phenyl
433 1 O 3-iPr-Phenyl 434 2 O 3-iPr-Phenyl 435 3 O 3-iPr-Phenyl 436 1 S 3-iPr-Phenyl 437 2 S 3-
iPr-Phenyl 438 3 S 3-iPr-Phenyl 439 1 SO 3-iPr-Phenyl 440 2 SO 3-iPr-Phenyl 441 3 SO 3-iPr-
Phenyl 442 1 SO.sub.2 3-iPr-Phenyl 443 2 SO.sub.2 3-iPr-Phenyl 444 3 SO.sub.2 3-iPr-Phenyl 445
1 O 4-NH.sub.2-Phenyl 446 2 O 4-NH.sub.2-Phenyl 447 3 O 4-NH.sub.2-Phenyl 448 1 S 4-
NH.sub.2-Phenyl 449 2 S 4-NH.sub.2-Phenyl 450 3 S 4-NH.sub.2-Phenyl 451 1 SO 4-NH.sub.2-
Phenyl 452 2 SO 4-NH.sub.2-Phenyl 453 3 SO 4-NH.sub.2-Phenyl 454 1 SO.sub.2 4-NH.sub.2-
Phenyl 455 2 SO.sub.2 4-NH.sub.2-Phenyl 456 3 SO.sub.2 4-NH.sub.2-Phenyl 457 1 O 2-
NH.sub.2-Phenyl 458 2 O 2-NH.sub.2-Phenyl 459 3 O 2-NH.sub.2-Phenyl 460 1 S 2-NH.sub.2-
Phenyl 461 2 S 2-NH.sub.2-Phenyl 462 3 S 2-NH.sub.2-Phenyl 463 1 SO 2-NH.sub.2-Phenyl 464
2 SO 2-NH.sub.2-Phenyl 465 3 SO 2-NH.sub.2-Phenyl 466 1 SO.sub.2 2-NH.sub.2-Phenyl 467 2
SO.sub.2 2-NH.sub.2-Phenyl 468 3 SO.sub.2 2-NH.sub.2-Phenyl 469 1 O 2,6-di-Me-Phenyl 470 2
O 2,6-di-Me-Phenyl 471 3 O 2,6-di-Me-Phenyl 472 1 S 2,6-di-Me-Phenyl 473 2 S 2,6-di-Me-
Phenyl 474 3 S 2,6-di-Me-Phenyl 475 1 SO 2,6-di-Me-Phenyl 476 2 SO 2,6-di-Me-Phenyl 477 3
SO 2,6-di-Me-Phenyl 478 1 SO.sub.2 2,6-di-Me-Phenyl 479 2 SO.sub.2 2,6-di-Me-Phenyl 480 3
SO.sub.2 2,6-di-Me-Phenyl 481 1 O 2-Ph-Phenyl 482 2 O 2-Ph-Phenyl 483 3 O 2-Ph-Phenyl 484 1
S 2-Ph-Phenyl 485 2 S 2-Ph-Phenyl 486 3 S 2-Ph-Phenyl 487 1 SO 2-Ph-Phenyl 488 2 SO 2-Ph-
Phenyl 489 3 SO 2-Ph-Phenyl 490 1 SO.sub.2 2-Ph-Phenyl 491 2 SO.sub.2 2-Ph-Phenyl 492 3
SO.sub.2 2-Ph-Phenyl 493 1 O 4-Ph-Phenyl 494 2 O 4-Ph-Phenyl 495 3 O 4-Ph-Phenyl 496 1 S 4-
Ph-Phenyl 497 2 S 4-Ph-Phenyl 498 3 S 4-Ph-Phenyl 499 1 SO 4-Ph-Phenyl 500 2 SO 4-Ph-Phenyl
501 3 SO 4-Ph-Phenyl 502 1 SO.sub.2 4-Ph-Phenyl 503 2 SO.sub.2 4-Ph-Phenyl 504 3 SO.sub.2 4-
Ph-Phenyl 505 1 O 3-morpholino-phenyl 506 2 O 3-morpholino-phenyl 507 3 O 3-morpholino-
phenyl 508 1 S 3-morpholino-phenyl 509 2 S 3-morpholino-phenyl 510 3 S 3-morpholino-phenyl
511 1 SO 3-morpholino-phenyl 512 2 SO 3-morpholino-phenyl 513 3 SO 3-morpholino-phenyl
514 1 SO.sub.2 3-morpholino-phenyl 515 2 SO.sub.2 3-morpholino-phenyl 516 3 SO.sub.2 3-
morpholino-phenyl 517 1 O 2-pyrazinyl 518 2 O 2-pyrazinyl 519 3 O 2-pyrazinyl 520 1 S 2-
pyrazinyl 521 2 S 2-pyrazinyl 522 3 S 2-pyrazinyl 523 1 SO 2-pyrazinyl 524 2 SO 2-pyrazinyl 525
3 SO 2-pyrazinyl 526 1 SO.sub.2 2-pyrazinyl 527 2 SO.sub.2 2-pyrazinyl 528 3 SO.sub.2 2-

pyrazinyl 529 1 O 5-indolyl 530 2 O 5-indolyl 531 3 O 5-indolyl 532 1 S 5-indolyl 533 2 S 5-indolyl 534 3 S 5-indolyl 535 1 SO 5-indolyl 536 2 SO 5-indolyl 537 3 SO 5-indolyl 538 1 SO.sub.2 5-indolyl 539 2 SO.sub.2 5-indolyl 540 3 SO.sub.2 5-indolyl 541 1 O 1H-benzo[d]imidazol-4-yl 542 2 O 1H-benzo[d]imidazol-4-yl 543 3 O 1H-benzo[d]imidazol-4-yl 544 1 S 1H-benzo[d]imidazol-4-yl 545 2 S 1H-benzo[d]imidazol-4-yl 546 3 S 1H-benzo[d]imidazol-4-yl 547 1 SO 1H-benzo[d]imidazol-4-yl 548 2 SO 1H-benzo[d]imidazol-4-yl 549 3 SO 1H-benzo[d]imidazol-4-yl 550 1 SO.sub.2 1H-benzo[d]imidazol-4-yl 551 2 SO.sub.2 1H-benzo[d]imidazol-4-yl 552 3 SO.sub.2 1H-benzo[d]imidazol-4-yl

[0408] Exemplary embodiments include compounds having the formula (XXV)

##STR00067##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 9

TABLE-US-00009 TABLE 9 Entry R.sup.2d R.sup.3 1 ethyl 4-CH.sub.3-phenyl 2 n-propyl 4-CH.sub.3-phenyl 3 isopropyl 4-CH.sub.3-phenyl 4 —CH.sub.2CH(CH.sub.3).sub.2 4-CH.sub.3-phenyl 5 CF.sub.3 4-CH.sub.3-phenyl 6 —CH.sub.2CF.sub.3 4-CH.sub.3-phenyl 7 —CH.sub.2CH.sub.2CF.sub.3 4-CH.sub.3-phenyl 8 cyclopropyl 4-CH.sub.3-phenyl 9 Cyclobutyl 4-CH.sub.3-phenyl 10 cyclopentyl 4-CH.sub.3-phenyl 11 cyclohexyl 4-CH.sub.3-phenyl 12 3-pyridyl 4-CH.sub.3-phenyl 13 1-methyl-1H- 4-CH.sub.3-phenyl pyrazol-4-yl 14 1H-imidazol-4-yl 4-CH.sub.3-phenyl 15 2-furanyl 4-CH.sub.3-phenyl 16 ethyl 3-CH.sub.3-phenyl 17 n-propyl 3-CH.sub.3-phenyl 18 isopropyl 3-CH.sub.3-phenyl 19 —CH.sub.2CH(CH.sub.3).sub.2 3-CH.sub.3-phenyl 20 —CF.sub.3 3-CH.sub.3-phenyl 21 —CH.sub.2CF.sub.3 3-CH.sub.3-phenyl 22 —CH.sub.2CH.sub.2CF.sub.3 3-CH.sub.3-phenyl 23 cyclopropyl 3-CH.sub.3-phenyl 24 Cyclobutyl 3-CH.sub.3-phenyl 25 cyclopentyl 3-CH.sub.3-phenyl 26 cyclohexyl 3-CH.sub.3-phenyl 27 3-pyridyl 3-CH.sub.3-phenyl 28 1-methyl-1H- 3-CH.sub.3-phenyl pyrazol-4-yl 29 1H-imidazol-4-yl 3-CH.sub.3-phenyl 30 2-furanyl 3-CH.sub.3-phenyl 31 ethyl 3-OH-Phenyl 32 n-propyl 3-OH-Phenyl 33 isopropyl 3-OH-Phenyl 34 —CH.sub.2CH(CH.sub.3).sub.2 3-OH-Phenyl 35 —CF.sub.3 3-OH-Phenyl 36 —CH.sub.2CF.sub.3 3-OH-Phenyl 37 —CH.sub.2CH.sub.2CF.sub.3 3-OH-Phenyl 38 cyclopropyl 3-OH-Phenyl 39 Cyclobutyl 3-OH-Phenyl 40 cyclopentyl 3-OH-Phenyl 41 cyclohexyl 3-OH-Phenyl 42 3-pyridyl 3-OH-Phenyl 43 1-methyl-1H- 3-OH-Phenyl pyrazol-4-yl 44 1H-imidazol-4-yl 3-OH-Phenyl 45 2-furanyl 3-OH-Phenyl 46 ethyl 4-OMe-Phenyl 47 n-propyl 4-OMe-Phenyl 48 isopropyl 4-OMe-Phenyl 49 —CH.sub.2CH(CH.sub.3).sub.2 4-OMe-Phenyl 50 —CF.sub.3 4-OMe-Phenyl 51 —CH.sub.2CF.sub.3 4-OMe-Phenyl 52 —CH.sub.2CH.sub.2CF.sub.3 4-OMe-Phenyl 53 cyclopropyl 4-OMe-Phenyl 54 Cyclobutyl 4-OMe-Phenyl 55 cyclopentyl 4-OMe-Phenyl 56 cyclohexyl 4-OMe-Phenyl 57 3-pyridyl 4-OMe-Phenyl 58 1-methyl-1H- 4-OMe-Phenyl pyrazol-4-yl 59 1H-imidazol-4-yl 4-OMe-Phenyl 60 2-furanyl 4-OMe-Phenyl 61 ethyl 2-OMe-Phenyl 62 n-propyl 2-OMe-Phenyl 63 isopropyl 2-OMe-Phenyl 64 —CH.sub.2CH(CH.sub.3).sub.2 2-OMe-Phenyl 65 —CF.sub.3 2-OMe-Phenyl 66 —CH.sub.2CF.sub.3 2-OMe-Phenyl 67 —CH.sub.2CH.sub.2CF.sub.3 2-OMe-Phenyl 68 cyclopropyl 2-OMe-Phenyl 69 Cyclobutyl 2-OMe-Phenyl 70 cyclopentyl 2-OMe-Phenyl 71 cyclohexyl 2-OMe-Phenyl 72 3-pyridyl 2-OMe-Phenyl 73 1-methyl-1H- 2-OMe-Phenyl pyrazol-4-yl 74 1H-imidazol-4-yl 2-OMe-Phenyl 75 2-furanyl 2-OMe-Phenyl 76 ethyl 3-CN-Phenyl 77 n-propyl 3-CN-Phenyl 78 isopropyl 3-CN-Phenyl 79 —CH.sub.2CH(CH.sub.3).sub.2 3-CN-Phenyl 80 —CF.sub.3 3-CN-Phenyl 81 —CH.sub.2CF.sub.3 3-CN-Phenyl 82 —CH.sub.2CH.sub.2CF.sub.3 3-CN-Phenyl 83 cyclopropyl 3-CN-Phenyl 84 Cyclobutyl 3-CN-Phenyl 85 cyclopentyl 3-CN-Phenyl 86 cyclohexyl 3-CN-Phenyl 87 3-pyridyl 3-CN-Phenyl 88 1-methyl-1H- 3-CN-Phenyl pyrazol-4-yl 89 1H-imidazol-4-yl 3-CN-Phenyl 90 2-furanyl 3-CN-Phenyl 91 ethyl 2-F-Phenyl 92 n-propyl 2-F-Phenyl 93 isopropyl 2-F-Phenyl 94 —CH.sub.2CH(CH.sub.3).sub.2 2-F-Phenyl 95 —CF.sub.3 2-F-Phenyl 96 —CH.sub.2CF.sub.3 2-F-Phenyl 97 —CH.sub.2CH.sub.2CF.sub.3 2-F-Phenyl 98 cyclopropyl 2-F-Phenyl 99 Cyclobutyl 2-F-Phenyl 100 cyclopentyl 2-F-Phenyl 101 cyclohexyl 2-

F-Phenyl 102 3-pyridyl 2-F-Phenyl 103 1-methyl-1H- 2-F-Phenyl pyrazol-4-yl 104 1H-imidazol-4-yl 2-F-Phenyl 105 2-furanyl 2-F-Phenyl 106 ethyl 4-F-Phenyl 107 n-propyl 4-F-Phenyl 108 isopropyl 4-F-Phenyl 109 —CH.sub.2CH(CH.sub.3).sub.2 4-F-Phenyl 110 —CF.sub.3 4-F-Phenyl 111 —CH.sub.2CF.sub.3 4-F-Phenyl 112 —CH.sub.2CH.sub.2CF.sub.3 4-F-Phenyl 113 cyclopropyl 4-F-Phenyl 114 Cyclobutyl 4-F-Phenyl 115 cyclopentyl 4-F-Phenyl 116 cyclohexyl 4-F-Phenyl 117 3-pyridyl 4-F-Phenyl 118 1-methyl-1H- 4-F-Phenyl pyrazol-4-yl 119 1H-imidazol-4-yl 4-F-Phenyl 120 2-furanyl 4-F-Phenyl 121 ethyl 3-Cl-Phenyl 122 n-propyl 3-Cl-Phenyl 123 isopropyl 3-Cl-Phenyl 124 —CH.sub.2CH(CH.sub.3).sub.2 3-Cl-Phenyl 125 —CF.sub.3 3-Cl-Phenyl 126 —CH.sub.2CF.sub.3 3-Cl-Phenyl 127 —CH.sub.2CH.sub.2CF.sub.3 3-Cl-Phenyl 128 cyclopropyl 3-Cl-Phenyl 129 Cyclobutyl 3-Cl-Phenyl 130 cyclopentyl 3-Cl-Phenyl 131 cyclohexyl 3-Cl-Phenyl 132 3-pyridyl 3-Cl-Phenyl 133 1-methyl-1H- 3-Cl-Phenyl pyrazol-4-yl 134 1H-imidazol-4-yl 3-Cl-Phenyl 135 2-furanyl 3-Cl-Phenyl 136 ethyl 2-Br-Phenyl 137 n-propyl 2-Br-Phenyl 138 isopropyl 2-Br-Phenyl 139 —CH.sub.2CH(CH.sub.3).sub.2 2-Br-Phenyl 140 —CF.sub.3 2-Br-Phenyl 141 —CH.sub.2CF.sub.3 2-Br-Phenyl 142 —CH.sub.2CH.sub.2CF.sub.3 2-Br-Phenyl 143 cyclopropyl 2-Br-Phenyl 144 Cyclobutyl 2-Br-Phenyl 145 cyclopentyl 2-Br-Phenyl 146 cyclohexyl 2-Br-Phenyl 147 3-pyridyl 2-Br-Phenyl 148 1-methyl-1H- 2-Br-Phenyl pyrazol-4-yl 149 1H-imidazol-4-yl 2-Br-Phenyl 150 2-furanyl 2-Br-Phenyl 151 ethyl 4-Br-Phenyl 152 n-propyl 4-Br-Phenyl 153 isopropyl 4-Br-Phenyl 154 —CH.sub.2CH(CH.sub.3).sub.2 4-Br-Phenyl 155 —CF.sub.3 4-Br-Phenyl 156 —CH.sub.2CF.sub.3 4-Br-Phenyl 157 —CH.sub.2CH.sub.2CF.sub.3 4-Br-Phenyl 158 cyclopropyl 4-Br-Phenyl 159 Cyclobutyl 4-Br-Phenyl 160 cyclopentyl 4-Br-Phenyl 161 cyclohexyl 4-Br-Phenyl 162 3-pyridyl 4-Br-Phenyl 163 1-methyl-1H- 4-Br-Phenyl pyrazol-4-yl 164 1H-imidazol-4-yl 4-Br-Phenyl 165 2-furanyl 4-Br-Phenyl 166 ethyl 3-CF.sub.3-Phenyl 167 n-propyl 3-CF.sub.3-Phenyl 168 isopropyl 3-CF.sub.3-Phenyl 169 —CH.sub.2CH(CH.sub.3).sub.2 3-CF.sub.3-Phenyl 170 —CF.sub.3 3-CF.sub.3-Phenyl 171 —CH.sub.2CF.sub.3 3-CF.sub.3-Phenyl 172 —CH.sub.2CH.sub.2CF.sub.3 3-CF.sub.3-Phenyl 173 cyclopropyl 3-CF.sub.3-Phenyl 174 Cyclobutyl 3-CF.sub.3-Phenyl 175 cyclopentyl 3-CF.sub.3-Phenyl 176 cyclohexyl 3-CF.sub.3-Phenyl 177 3-pyridyl 3-CF.sub.3-Phenyl 178 1-methyl-1H- 3-CF.sub.3-Phenyl pyrazol-4-yl 179 1H-imidazol-4-yl 3-CF.sub.3-Phenyl 180 2-furanyl 3-CF.sub.3-Phenyl 181 ethyl 2-iPr-Phenyl 182 n-propyl 2-iPr-Phenyl 183 isopropyl 2-iPr-Phenyl 184 —CH.sub.2CH(CH.sub.3).sub.2 2-iPr-Phenyl 185 —CF.sub.3 2-iPr-Phenyl 186 —CH.sub.2CF.sub.3 2-iPr-Phenyl 187 —CH.sub.2CH.sub.2CF.sub.3 2-iPr-Phenyl 188 cyclopropyl 2-iPr-Phenyl 189 Cyclobutyl 2-iPr-Phenyl 190 cyclopentyl 2-iPr-Phenyl 191 cyclohexyl 2-iPr-Phenyl 192 3-pyridyl 2-iPr-Phenyl 193 1-methyl-1H- 2-iPr-Phenyl pyrazol-4-yl 194 1H-imidazol-4-yl 2-iPr-Phenyl 195 2-furanyl 2-iPr-Phenyl 196 ethyl 4-iPr-Phenyl 197 n-propyl 4-iPr-Phenyl 198 isopropyl 4-iPr-Phenyl 199 —CH.sub.2CH(CH.sub.3).sub.2 4-iPr-Phenyl 200 —CF.sub.3 4-iPr-Phenyl 201 —CH.sub.2CF.sub.3 4-iPr-Phenyl 202 —CH.sub.2CH.sub.2CF.sub.3 4-iPr-Phenyl 203 cyclopropyl 4-iPr-Phenyl 204 Cyclobutyl 4-iPr-Phenyl 205 cyclopentyl 4-iPr-Phenyl 206 cyclohexyl 4-iPr-Phenyl 207 3-pyridyl 4-iPr-Phenyl 208 1-methyl-1H- 4-iPr-Phenyl pyrazol-4-yl 209 1H-imidazol-4-yl 4-iPr-Phenyl 210 2-furanyl 4-iPr-Phenyl 211 ethyl 3-morpholino-phenyl 212 n-propyl 3-morpholino-phenyl 213 isopropyl 3-morpholino-phenyl 214 —CH.sub.2CH(CH.sub.3).sub.2 3-morpholino-phenyl 215 —CF.sub.3 3-morpholino-phenyl 216 —CH.sub.2CF.sub.3 3-morpholino-phenyl 217 —CH.sub.2CH.sub.2CF.sub.3 3-morpholino-phenyl 218 cyclopropyl 3-morpholino-phenyl 219 Cyclobutyl 3-morpholino-phenyl 220 cyclopentyl 3-morpholino-phenyl 221 cyclohexyl 3-morpholino-phenyl 222 3-pyridyl 3-morpholino-phenyl 223 1-methyl-1H- 3-morpholino-phenyl pyrazol-4-yl 224 1H-imidazol-4-yl 3-morpholino-phenyl 225 2-furanyl 3-morpholino-phenyl 226 ethyl 4-cyano-2-morpholino-phenyl 227 n-propyl 4-cyano-2-morpholino-phenyl 228 isopropyl 4-cyano-2-morpholino-phenyl 229 —CH.sub.2CH(CH.sub.3).sub.2 4-cyano-2-morpholino-phenyl 230 —CF.sub.3 4-cyano-2-morpholino-phenyl 231 —CH.sub.2CF.sub.3 4-cyano-2-morpholino-phenyl 232 —CH.sub.2CH.sub.2CF.sub.3 4-cyano-2-morpholino-phenyl 233 cyclopropyl 4-cyano-2-

morpholino- phenyl 234 Cyclobutyl 4-cyano-2-morpholino- phenyl 235 cyclopentyl 4-cyano-2-morpholino- phenyl 236 cyclohexyl 4-cyano-2-morpholino- phenyl 237 3-pyridyl 4-cyano-2-morpholino- phenyl 238 1-methyl-1H- 4-cyano-2-morpholino- pyrazol-4-yl phenyl 239 1H-imidazol-4-yl 4-cyano-2-morpholino- phenyl 240 2-furanyl 4-cyano-2-morpholino- phenyl 241 ethyl 4-hydroxy-2- morpholino-phenyl 242 n-propyl 4-hydroxy-2- morpholino-phenyl 243 isopropyl 4-hydroxy-2- morpholino-phenyl 244 —CH.sub.2CH(CH.sub.3).sub.2 4-hydroxy-2-morpholino-phenyl 245 —CF.sub.3 4-hydroxy-2- morpholino-phenyl 246 —CH.sub.2CF.sub.3 4-hydroxy-2- morpholino-phenyl 247 —CH.sub.2CH.sub.2CF.sub.3 4-hydroxy-2- 248 cyclopropyl 4-hydroxy-2- morpholino-phenyl 249 Cyclobutyl 4-hydroxy-2- morpholino-phenyl 250 cyclopentyl 4-hydroxy-2- morpholino-phenyl 251 cyclohexyl 4-hydroxy-2- morpholino-phenyl 252 3-pyridyl 4-hydroxy-2- morpholino-phenyl 253 1-methyl-1H- 4-hydroxy-2- pyrazol-4-yl morpholino-phenyl 254 1H-imidazol-4-yl 4-hydroxy-2- morpholino-phenyl 255 2-furanyl 4-hydroxy-2- morpholino-phenyl 256 ethyl 2-CH.sub.3-phenyl 257 n-propyl 2-CH.sub.3-phenyl 258 isopropyl 2-CH.sub.3-phenyl 259 —CH.sub.2CH(CH.sub.3).sub.2 2-CH.sub.3-phenyl 260 CF.sub.3 2-CH.sub.3-phenyl 261 —CH.sub.2CF.sub.3 2-CH.sub.3-phenyl 262 —CH.sub.2CH.sub.2CF.sub.3 2-CH.sub.3-phenyl 263 cyclopropyl 2-CH.sub.3-phenyl 264 Cyclobutyl 2-CH.sub.3-phenyl 265 cyclopentyl 2-CH.sub.3-phenyl 266 cyclohexyl 2-CH.sub.3-phenyl 267 3-pyridyl 2-CH.sub.3-phenyl 268 1-methyl-1H- 2-CH.sub.3-phenyl pyrazol-4-yl 269 1H-imidazol-4-yl 2-CH.sub.3-phenyl 270 2-furanyl 2-CH.sub.3-phenyl 271 ethyl 4-OH-Phenyl 272 n-propyl 4-OH-Phenyl 273 isopropyl 4-OH-Phenyl 274 —CH.sub.2CH(CH.sub.3).sub.2 4-OH-Phenyl 275 —CF.sub.3 4-OH-Phenyl 276 —CH.sub.2CF.sub.3 4-OH-Phenyl 277 —CH.sub.2CH.sub.2CF.sub.3 4-OH-Phenyl 278 cyclopropyl 4-OH-Phenyl 279 Cyclobutyl 4-OH-Phenyl 280 cyclopentyl 4-OH-Phenyl 281 cyclohexyl 4-OH-Phenyl 282 3-pyridyl 4-OH-Phenyl 283 1-methyl-1H- 4-OH-Phenyl pyrazol-4-yl 284 1H-imidazol-4-yl 4-OH-Phenyl 285 2-furanyl 4-OH-Phenyl 286 ethyl 2-OH-Phenyl 287 n-propyl 2-OH-Phenyl 288 isopropyl 2-OH-Phenyl 289 —CH.sub.2CH(CH.sub.3).sub.2 2-OH-Phenyl 290 —CF.sub.3 2-OH-Phenyl 291 —CH.sub.2CF.sub.3 2-OH-Phenyl 292 —CH.sub.2CH.sub.2CF.sub.3 2-OH-Phenyl 293 cyclopropyl 2-OH-Phenyl 294 Cyclobutyl 2-OH-Phenyl 295 cyclopentyl 2-OH-Phenyl 296 cyclohexyl 2-OH-Phenyl 297 3-pyridyl 2-OH-Phenyl 298 1-methyl-1H- 2-OH-Phenyl pyrazol-4-yl 299 1H-imidazol-4-yl 2-OH-Phenyl 300 2-furanyl 2-OH-Phenyl 301 ethyl 3-OMe-Phenyl 302 n-propyl 3-OMe-Phenyl 303 isopropyl 3-OMe-Phenyl 304 —CH.sub.2CH(CH.sub.3).sub.2 3-OMe-Phenyl 305 —CF.sub.3 3-OMe-Phenyl 306 —CH.sub.2CF.sub.3 3-OMe-Phenyl 307 —CH.sub.2CH.sub.2CF.sub.3 3-OMe-Phenyl 308 cyclopropyl 3-OMe-Phenyl 309 Cyclobutyl 3-OMe-Phenyl 310 cyclopentyl 3-OMe-Phenyl 311 cyclohexyl 3-OMe-Phenyl 312 3-pyridyl 3-OMe-Phenyl 313 1-methyl-1H- 3-OMe-Phenyl pyrazol-4-yl 314 1H-imidazol-4-yl 3-OMe-Phenyl 315 2-furanyl 3-OMe-Phenyl 316 ethyl 4-CN-Phenyl 317 n-propyl 4-CN-Phenyl 318 isopropyl 4-CN-Phenyl 319 —CH.sub.2CH(CH.sub.3).sub.2 4-CN-Phenyl 320 —CF.sub.3 4-CN-Phenyl 321 —CH.sub.2CF.sub.3 4-CN-Phenyl 322 —CH.sub.2CH.sub.2CF.sub.3 4-CN-Phenyl 323 cyclopropyl 4-CN-Phenyl 324 Cyclobutyl 4-CN-Phenyl 325 cyclopentyl 4-CN-Phenyl 326 cyclohexyl 4-CN-Phenyl 327 3-pyridyl 4-CN-Phenyl 328 1-methyl-1H- 4-CN-Phenyl pyrazol-4-yl 329 1H-imidazol-4-yl 4-CN-Phenyl 330 2-furanyl 4-CN-Phenyl 331 ethyl 2-CN-Phenyl 332 n-propyl 2-CN-Phenyl 333 isopropyl 2-CN-Phenyl 334 —CH.sub.2CH(CH.sub.3).sub.2 2-CN-Phenyl 335 —CF.sub.3 2-CN-Phenyl 336 —CH.sub.2CF.sub.3 2-CN-Phenyl 337 —CH.sub.2CH.sub.2CF.sub.3 2-CN-Phenyl 338 cyclopropyl 2-CN-Phenyl 339 Cyclobutyl 2-CN-Phenyl 340 cyclopentyl 2-CN-Phenyl 341 cyclohexyl 2-CN-Phenyl 342 3-pyridyl 2-CN-Phenyl 343 1-methyl-1H- 2-CN-Phenyl pyrazol-4-yl 344 1H-imidazol-4-yl 2-CN-Phenyl 345 2-furanyl 2-CN-Phenyl 346 ethyl 3-F-Phenyl 347 n-propyl 3-F-Phenyl 348 isopropyl 3-F-Phenyl 349 —CH.sub.2CH(CH.sub.3).sub.2 3-F-Phenyl 350 —CF.sub.3 3-F-Phenyl 351 —CH.sub.2CF.sub.3 3-F-Phenyl 352 —CH.sub.2CH.sub.2CF.sub.3 3-F-Phenyl 353 cyclopropyl 3-F-Phenyl 354 Cyclobutyl 3-F-Phenyl 355 cyclopentyl 3-F-Phenyl 356 cyclohexyl 3-F-Phenyl 357 3-pyridyl 3-F-Phenyl 358 1-methyl-1H- 3-F-Phenyl pyrazol-4-yl 359

1H-imidazol-4-yl 3-F-Phenyl 360 2-furanyl 3-F-Phenyl 361 ethyl 2-Cl-Phenyl 362 n-propyl 2-Cl-Phenyl 363 isopropyl 2-Cl-Phenyl 364 —CH.sub.2CH(CH.sub.3).sub.2 2-Cl-Phenyl 365 —CF.sub.3 2-Cl-Phenyl 366 —CH.sub.2CF.sub.3 2-Cl-Phenyl 367 —CH.sub.2CH.sub.2CF.sub.3 2-Cl-Phenyl 368 cyclopropyl 2-Cl-Phenyl 369 Cyclobutyl 2-Cl-Phenyl 370 cyclopentyl 2-Cl-Phenyl 371 cyclohexyl 2-Cl-Phenyl 372 3-pyridyl 2-Cl-Phenyl 373 1-methyl-1H- 2-Cl-Phenyl pyrazol-4-yl 374 1H-imidazol-4-yl 2-Cl-Phenyl 375 2-furanyl 2-Cl-Phenyl 376 ethyl 4-Cl-Phenyl 377 n-propyl 4-Cl-Phenyl 378 isopropyl 4-Cl-Phenyl 379 —CH.sub.2CH(CH.sub.3).sub.2 4-Cl-Phenyl 380 —CF.sub.3 4-Cl-Phenyl 381 —CH.sub.2CF.sub.3 4-Cl-Phenyl 382 —CH.sub.2CH.sub.2CF.sub.3 4-Cl-Phenyl 383 cyclopropyl 4-Cl-Phenyl 384 Cyclobutyl 4-Cl-Phenyl 385 cyclopentyl 4-Cl-Phenyl 386 cyclohexyl 4-Cl-Phenyl 387 3-pyridyl 4-Cl-Phenyl 388 1-methyl-1H- 4-Cl-Phenyl pyrazol-4-yl 389 1H-imidazol-4-yl 4-Cl-Phenyl 390 2-furanyl 4-Cl-Phenyl 391 ethyl 3-Br-Phenyl 392 n-propyl 3-Br-Phenyl 393 isopropyl 3-Br-Phenyl 394 —CH.sub.2CH(CH.sub.3).sub.2 3-Br-Phenyl 395 —CF.sub.3 3-Br-Phenyl 396 —CH.sub.2CF.sub.3 3-Br-Phenyl 397 —CH.sub.2CH.sub.2CF.sub.3 3-Br-Phenyl 398 cyclopropyl 3-Br-Phenyl 399 Cyclobutyl 3-Br-Phenyl 400 cyclopentyl 3-Br-Phenyl 401 cyclohexyl 3-Br-Phenyl 402 3-pyridyl 3-Br-Phenyl 403 1-methyl-1H- 3-Br-Phenyl pyrazol-4-yl 404 1H-imidazol-4-yl 3-Br-Phenyl 405 2-furanyl 3-Br-Phenyl 406 ethyl 2-CF.sub.3-Phenyl 407 n-propyl 2-CF.sub.3-Phenyl 408 isopropyl 2-CF.sub.3-Phenyl 409 —CH.sub.2CH(CH.sub.3).sub.2 2-CF.sub.3-Phenyl 410 —CF.sub.3 2-CF.sub.3-Phenyl 411 —CH.sub.2CF.sub.3 2-CF.sub.3-Phenyl 412 —CH.sub.2CH.sub.2CF.sub.3 2-CF.sub.3-Phenyl 413 cyclopropyl 2-CF.sub.3-Phenyl 414 Cyclobutyl 2-CF.sub.3-Phenyl 415 cyclopentyl 2-CF.sub.3-Phenyl 416 cyclohexyl 2-CF.sub.3-Phenyl 417 3-pyridyl 2-CF.sub.3-Phenyl 418 1-methyl-1H- 2-CF.sub.3-Phenyl pyrazol-4-yl 419 1H-imidazol-4-yl 2-CF.sub.3-Phenyl 420 2-furanyl 2-CF.sub.3-Phenyl 421 ethyl 4-CF.sub.3-Phenyl 422 n-propyl 4-CF.sub.3-Phenyl 423 isopropyl 4-CF.sub.3-Phenyl 424 —CH.sub.2CH(CH.sub.3).sub.2 4-CF.sub.3-Phenyl 425 —CF.sub.3 4-CF.sub.3-Phenyl 426 —CH.sub.2CF.sub.3 4-CF.sub.3-Phenyl 427 —CH.sub.2CH.sub.2CF.sub.3 4-CF.sub.3-Phenyl 428 cyclopropyl 4-CF.sub.3-Phenyl 429 Cyclobutyl 4-CF.sub.3-Phenyl 430 cyclopentyl 4-CF.sub.3-Phenyl 431 cyclohexyl 4-CF.sub.3-Phenyl 432 3-pyridyl 4-CF.sub.3-Phenyl 433 1-methyl-1H- 4-CF.sub.3-Phenyl pyrazol-4-yl 434 1H-imidazol-4-yl 4-CF.sub.3-Phenyl 435 2-furanyl 4-CF.sub.3-Phenyl 436 ethyl 3-iPr-Phenyl 437 n-propyl 3-iPr-Phenyl 438 isopropyl 3-iPr-Phenyl 439 —CH.sub.2CH(CH.sub.3).sub.2 3-iPr-Phenyl 440 —CF.sub.3 3-iPr-Phenyl 441 —CH.sub.2CF.sub.3 3-iPr-Phenyl 442 —CH.sub.2CH.sub.2CF.sub.3 3-iPr-Phenyl 443 cyclopropyl 3-iPr-Phenyl 444 Cyclobutyl 3-iPr-Phenyl 445 cyclopentyl 3-iPr-Phenyl 446 cyclohexyl 3-iPr-Phenyl 447 3-pyridyl 3-iPr-Phenyl 448 1-methyl-1H- 3-iPr-Phenyl pyrazol-4-yl 449 1H-imidazol-4-yl 3-iPr-Phenyl 450 2-furanyl 3-iPr-Phenyl 451 ethyl 2-morpholino- 452 n-propyl 2-morpholino- phenyl 453 isopropyl 2-morpholino- phenyl 454 —CH.sub.2CH(CH.sub.3).sub.2 2-morpholino- phenyl 455 —CF.sub.3 2-morpholino- phenyl 456 —CH.sub.2CF.sub.3 2-morpholino- phenyl 457 —CH.sub.2CH.sub.2CF.sub.3 2-morpholino- phenyl 458 cyclopropyl 2-morpholino- phenyl 459 Cyclobutyl 2-morpholino- phenyl 460 cyclopentyl 2-morpholino- phenyl 461 cyclohexyl 2-morpholino- phenyl 462 3-pyridyl 2-morpholino- phenyl 463 1-methyl-1H- 2-morpholino- pyrazol-4-yl phenyl 464 1H-imidazol-4-yl 2-morpholino- phenyl 465 2-furanyl 2-morpholino- phenyl 466 ethyl 4-morpholino- phenyl 467 n-propyl 4-morpholino- phenyl 468 isopropyl 4-morpholino- phenyl 469 —CH.sub.2CH(CH.sub.3).sub.2 4-morpholino- phenyl 470 —CF.sub.3 4-morpholino- phenyl 471 —CH.sub.2CF.sub.3 4-morpholino- phenyl 472 —CH.sub.2CH.sub.2CF.sub.3 4-morpholino- phenyl 473 cyclopropyl 4-morpholino- phenyl 474 Cyclobutyl 4-morpholino- phenyl 475 cyclopentyl 4-morpholino- phenyl 476 cyclohexyl 4-morpholino- phenyl 477 3-pyridyl 4-morpholino- phenyl 478 1-methyl-1H- 4-morpholino- pyrazol-4-yl phenyl 479 1H-imidazol-4-yl 4-morpholino- phenyl 480 2-furanyl 4-morpholino- phenyl 481 ethyl 4-methyl-2- morpholino-phenyl 482 n-propyl 4-methyl-2- morpholino-phenyl 483 isopropyl 4-methyl-2- morpholino-phenyl 484 —CH.sub.2CH(CH.sub.3).sub.2 4-methyl-2- morpholino-phenyl 485 —CF.sub.3 4-methyl-2- morpholino-phenyl 486 —CH.sub.2CF.sub.3 4-methyl-2-

morpholino-phenyl 487 —CH.sub.2CH.sub.2CF.sub.3 4-methyl-2- morpholino-phenyl 488
cyclopropyl 4-methyl-2- morpholino-phenyl 489 Cyclobutyl 4-methyl-2- morpholino-phenyl 490
cyclopentyl 4-methyl-2- morpholino-phenyl 491 cyclohexyl 4-methyl-2- morpholino-phenyl 492 3-
pyridyl 4-methyl-2- morpholino-phenyl 493 1-methyl-1H- 4-methyl-2- pyrazol-4-yl morpholino-
phenyl 494 1H-imidazol-4-yl 4-methyl-2- morpholino-phenyl 495 2-furanyl 4-methyl-2-
morpholino-phenyl

[0409] Exemplary embodiments include compounds having the formula (XXVI)

##STR00068##

or a pharmaceutically acceptable salt form thereof defined herein below in Table 10.

TABLE-US-00010 TABLE 10 Entry R.sup.2d R.sup.3 1 ethyl 4-CH.sub.3-phenyl 2 n-propyl
4-CH.sub.3-phenyl 3 isopropyl 4-CH.sub.3-phenyl 4 —CH.sub.2CH(CH.sub.3).sub.2 4-
CH.sub.3-phenyl 5 CF.sub.3 4-CH.sub.3-phenyl 6 —CH.sub.2CF.sub.3 4-CH.sub.3-phenyl
7 —CH.sub.2CH.sub.2CF.sub.3 4-CH.sub.3-phenyl 8 cyclopropyl 4-CH.sub.3-phenyl 9
Cyclobutyl 4-CH.sub.3-phenyl 10 cyclopentyl 4-CH.sub.3-phenyl 11 cyclohexyl 4-CH.sub.3-
phenyl 12 3-pyridyl 4-CH.sub.3-phenyl 13 1-methyl-1H- 4-CH.sub.3-phenyl pyrazol-4-yl 14
1H-imidazol-4-yl 4-CH.sub.3-phenyl 15 2-furanyl 4-CH.sub.3-phenyl 16 ethyl 3-CH.sub.3-
phenyl 17 n-propyl 3-CH.sub.3-phenyl 18 isopropyl 3-CH.sub.3-phenyl 19 —
CH.sub.2CH(CH.sub.3).sub.2 3-CH.sub.3-phenyl 20 —CF.sub.3 3-CH.sub.3-phenyl 21 —
CH.sub.2CF.sub.3 3-CH.sub.3-phenyl 22 —CH.sub.2CH.sub.2CF.sub.3 3-CH.sub.3-phenyl 23
cyclopropyl 3-CH.sub.3-phenyl 24 Cyclobutyl 3-CH.sub.3-phenyl 25 cyclopentyl 3-CH.sub.3-
phenyl 26 cyclohexyl 3-CH.sub.3-phenyl 27 3-pyridyl 3-CH.sub.3-phenyl 28 1-methyl-1H- 3-
CH.sub.3-phenyl pyrazol-4-yl 29 1H-imidazol-4-yl 3-CH.sub.3-phenyl 30 2-furanyl 3-CH.sub.3-
phenyl 31 ethyl 3-OH-Phenyl 32 n-propyl 3-OH-Phenyl 33 isopropyl 3-OH-Phenyl 34 —
CH.sub.2CH(CH.sub.3).sub.2 3-OH-Phenyl 35 —CF.sub.3 3-OH-Phenyl 36 —
CH.sub.2CF.sub.3 3-OH-Phenyl 37 —CH.sub.2CH.sub.2CF.sub.3 3-OH-Phenyl 38 cyclopropyl
3-OH-Phenyl 39 Cyclobutyl 3-OH-Phenyl 40 cyclopentyl 3-OH-Phenyl 41 cyclohexyl 3-OH-
Phenyl 42 3-pyridyl 3-OH-Phenyl 43 1-methyl-1H- 3-OH-Phenyl pyrazol-4-yl 44 1H-imidazol-
4-yl 3-OH-Phenyl 45 2-furanyl 3-OH-Phenyl 46 ethyl 4-OMe-Phenyl 47 n-propyl 4-OMe-
Phenyl 48 isopropyl 4-OMe-Phenyl 49 —CH.sub.2CH(CH.sub.3).sub.2 4-OMe-Phenyl 50 —
CF.sub.3 4-OMe-Phenyl 51 —CH.sub.2CF.sub.3 4-OMe-Phenyl 52 —
CH.sub.2CH.sub.2CF.sub.3 4-OMe-Phenyl 53 cyclopropyl 4-OMe-Phenyl 54 Cyclobutyl 4-
OMe-Phenyl 55 cyclopentyl 4-OMe-Phenyl 56 cyclohexyl 4-OMe-Phenyl 57 3-pyridyl 4-OMe-
Phenyl 58 1-methyl-1H- 4-OMe-Phenyl pyrazol-4-yl 59 1H-imidazol-4-yl 4-OMe-Phenyl 60 2-
furanyl 4-OMe-Phenyl 61 ethyl 2-OMe-Phenyl 62 n-propyl 2-OMe-Phenyl 63 isopropyl 2-
OMe-Phenyl 64 —CH.sub.2CH(CH.sub.3).sub.2 2-OMe-Phenyl 65 —CF.sub.3 2-OMe-Phenyl
66 —CH.sub.2CF.sub.3 2-OMe-Phenyl 67 —CH.sub.2CH.sub.2CF.sub.3 2-OMe-Phenyl 68
cyclopropyl 2-OMe-Phenyl 69 Cyclobutyl 2-OMe-Phenyl 70 cyclopentyl 2-OMe-Phenyl 71
cyclohexyl 2-OMe-Phenyl 72 3-pyridyl 2-OMe-Phenyl 73 1-methyl-1H- 2-OMe-Phenyl
pyrazol-4-yl 74 1H-imidazol-4-yl 2-OMe-Phenyl 75 2-furanyl 2-OMe-Phenyl 76 ethyl 3-CN-
Phenyl 77 n-propyl 3-CN-Phenyl 78 isopropyl 3-CN-Phenyl 79 —
CH.sub.2CH(CH.sub.3).sub.2 3-CN-Phenyl 80 —CF.sub.3 3-CN-Phenyl 81 —
CH.sub.2CF.sub.3 3-CN-Phenyl 82 —CH.sub.2CH.sub.2CF.sub.3 3-CN-Phenyl 83 cyclopropyl
3-CN-Phenyl 84 Cyclobutyl 3-CN-Phenyl 85 cyclopentyl 3-CN-Phenyl 86 cyclohexyl 3-CN-
Phenyl 87 3-pyridyl 3-CN-Phenyl 88 1-methyl-1H- 3-CN-Phenyl pyrazol-4-yl 89 1H-imidazol-
4-yl 3-CN-Phenyl 90 2-furanyl 3-CN-Phenyl 91 ethyl 2-F-Phenyl 92 n-propyl 2-F-Phenyl 93
isopropyl 2-F-Phenyl 94 —CH.sub.2CH(CH.sub.3).sub.2 2-F-Phenyl 95 —CF.sub.3 2-F-Phenyl
96 —CH.sub.2CF.sub.3 2-F-Phenyl 97 —CH.sub.2CH.sub.2CF.sub.3 2-F-Phenyl 98
cyclopropyl 2-F-Phenyl 99 Cyclobutyl 2-F-Phenyl 100 cyclopentyl 2-F-Phenyl 101 cyclohexyl 2-
F-Phenyl 102 3-pyridyl 2-F-Phenyl 103 1-methyl-1H- 2-F-Phenyl pyrazol-4-yl 104 1H-imidazol-4-
yl 2-F-Phenyl 105 2-furanyl 2-F-Phenyl 106 ethyl 4-F-Phenyl 107 n-propyl 4-F-Phenyl 108

isopropyl 4-F-Phenyl 109 —CH.sub.2CH(CH.sub.3).sub.2 4-F-Phenyl 110 —CF.sub.3 4-F-Phenyl 111 —CH.sub.2CF.sub.3 4-F-Phenyl 112 —CH.sub.2CH.sub.2CF.sub.3 4-F-Phenyl 113 cyclopropyl 4-F-Phenyl 114 Cyclobutyl 4-F-Phenyl 115 cyclopentyl 4-F-Phenyl 116 cyclohexyl 4-F-Phenyl 117 3-pyridyl 4-F-Phenyl 118 1-methyl-1H- 4-F-Phenyl pyrazol-4-yl 119 1H-imidazol-4-yl 4-F-Phenyl 120 2-furanyl 4-F-Phenyl 121 ethyl 3-Cl-Phenyl 122 n-propyl 3-Cl-Phenyl 123 isopropyl 3-Cl-Phenyl 124 —CH.sub.2CH(CH.sub.3).sub.2 3-Cl-Phenyl 125 —CF.sub.3 3-Cl-Phenyl 126 —CH.sub.2CF.sub.3 3-Cl-Phenyl 127 —CH.sub.2CH.sub.2CF.sub.3 3-Cl-Phenyl 128 cyclopropyl 3-Cl-Phenyl 129 Cyclobutyl 3-Cl-Phenyl 130 cyclopentyl 3-Cl-Phenyl 131 cyclohexyl 3-Cl-Phenyl 132 3-pyridyl 3-Cl-Phenyl 133 1-methyl-1H- 3-Cl-Phenyl pyrazol-4-yl 134 1H-imidazol-4-yl 3-Cl-Phenyl 135 2-furanyl 3-Cl-Phenyl 136 ethyl 2-Br-Phenyl 137 n-propyl 2-Br-Phenyl 138 isopropyl 2-Br-Phenyl 139 —CH.sub.2CH(CH.sub.3).sub.2 2-Br-Phenyl 140 —CF.sub.3 2-Br-Phenyl 141 —CH.sub.2CF.sub.3 2-Br-Phenyl 142 —CH.sub.2CH.sub.2CF.sub.3 2-Br-Phenyl 143 cyclopropyl 2-Br-Phenyl 144 Cyclobutyl 2-Br-Phenyl 145 cyclopentyl 2-Br-Phenyl 146 cyclohexyl 2-Br-Phenyl 147 3-pyridyl 2-Br-Phenyl 148 1-methyl-1H- 2-Br-Phenyl pyrazol-4-yl 149 1H-imidazol-4-yl 2-Br-Phenyl 150 2-furanyl 2-Br-Phenyl 151 ethyl 4-Br-Phenyl 152 n-propyl 4-Br-Phenyl 153 isopropyl 4-Br-Phenyl 154 —CH.sub.2CH(CH.sub.3).sub.2 4-Br-Phenyl 155 —CF.sub.3 4-Br-Phenyl 156 —CH.sub.2CF.sub.3 4-Br-Phenyl 157 —CH.sub.2CH.sub.2CF.sub.3 4-Br-Phenyl 158 cyclopropyl 4-Br-Phenyl 159 Cyclobutyl 4-Br-Phenyl 160 cyclopentyl 4-Br-Phenyl 161 cyclohexyl 4-Br-Phenyl 162 3-pyridyl 4-Br-Phenyl 163 1-methyl-1H- 4-Br-Phenyl pyrazol-4-yl 164 1H-imidazol-4-yl 4-Br-Phenyl 165 2-furanyl 4-Br-Phenyl 166 ethyl 3-CF.sub.3-Phenyl 167 n-propyl 3-CF.sub.3-Phenyl 168 isopropyl 3-CF.sub.3-Phenyl 169 —CH.sub.2CH(CH.sub.3).sub.2 3-CF.sub.3-Phenyl 170 —CF.sub.3 3-CF.sub.3-Phenyl 171 —CH.sub.2CF.sub.3 3-CF.sub.3-Phenyl 172 —CH.sub.2CH.sub.2CF.sub.3 3-CF.sub.3-Phenyl 173 cyclopropyl 3-CF.sub.3-Phenyl 174 Cyclobutyl 3-CF.sub.3-Phenyl 175 cyclopentyl 3-CF.sub.3-Phenyl 176 cyclohexyl 3-CF.sub.3-Phenyl 177 3-pyridyl 3-CF.sub.3-Phenyl 178 1-methyl-1H- 3-CF.sub.3-Phenyl pyrazol-4-yl 179 1H-imidazol-4-yl 3-CF.sub.3-Phenyl 180 2-furanyl 3-CF.sub.3-Phenyl 181 ethyl 2-iPr-Phenyl 182 n-propyl 2-iPr-Phenyl 183 isopropyl 2-iPr-Phenyl 184 —CH.sub.2CH(CH.sub.3).sub.2 2-iPr-Phenyl 185 —CF.sub.3 2-iPr-Phenyl 186 —CH.sub.2CF.sub.3 2-iPr-Phenyl 187 —CH.sub.2CH.sub.2CF.sub.3 2-iPr-Phenyl 188 cyclopropyl 2-iPr-Phenyl 189 Cyclobutyl 2-iPr-Phenyl 190 cyclopentyl 2-iPr-Phenyl 191 cyclohexyl 2-iPr-Phenyl 192 3-pyridyl 2-iPr-Phenyl 193 1-methyl-1H- 2-iPr-Phenyl pyrazol-4-yl 194 1H-imidazol-4-yl 2-iPr-Phenyl 195 2-furanyl 2-iPr-Phenyl 196 ethyl 4-iPr-Phenyl 197 n-propyl 4-iPr-Phenyl 198 isopropyl 4-iPr-Phenyl 199 —CH.sub.2CH(CH.sub.3).sub.2 4-iPr-Phenyl 200 —CF.sub.3 4-iPr-Phenyl 201 —CH.sub.2CF.sub.3 4-iPr-Phenyl 202 —CH.sub.2CH.sub.2CF.sub.3 4-iPr-Phenyl 203 cyclopropyl 4-iPr-Phenyl 204 Cyclobutyl 4-iPr-Phenyl 205 cyclopentyl 4-iPr-Phenyl 206 cyclohexyl 4-iPr-Phenyl 207 3-pyridyl 4-iPr-Phenyl 208 1-methyl-1H- 4-iPr-Phenyl pyrazol-4-yl 209 1H-imidazol-4-yl 4-iPr-Phenyl 210 2-furanyl 4-iPr-Phenyl 211 ethyl 3-morpholino- phenyl 212 n-propyl 3-morpholino- phenyl 213 isopropyl 3-morpholino- phenyl 214 —CH.sub.2CH(CH.sub.3).sub.2 3-morpholino- phenyl 215 —CF.sub.3 3-morpholino- phenyl 216 —CH.sub.2CF.sub.3 3-morpholino- phenyl 217 —CH.sub.2CH.sub.2CF.sub.3 3-morpholino- phenyl 218 cyclopropyl 3-morpholino- phenyl 219 Cyclobutyl 3-morpholino- phenyl 220 cyclopentyl 3-morpholino- phenyl 221 cyclohexyl 3-morpholino- phenyl 222 3-pyridyl 3-morpholino- phenyl 223 1-methyl-1H- 3-morpholino- pyrazol-4-yl phenyl 224 1H-imidazol-4-yl 3-morpholino- phenyl 225 2-furanyl 3-morpholino- phenyl 226 ethyl 4-cyano-2- morpholino-phenyl 227 n-propyl 4-cyano-2- morpholino-phenyl 228 isopropyl 4-cyano-2- morpholino-phenyl 229 —CH.sub.2CH(CH.sub.3).sub.2 4-cyano-2- morpholino-phenyl 230 —CF.sub.3 4-cyano-2- morpholino-phenyl 231 —CH.sub.2CF.sub.3 4-cyano-2- morpholino-phenyl 232 —CH.sub.2CH.sub.2CF.sub.3 4-cyano-2- morpholino-phenyl 233 cyclopropyl 4-cyano-2- morpholino-phenyl 234 Cyclobutyl 4-cyano-2- morpholino-phenyl 235 cyclopentyl 4-cyano-2- morpholino-phenyl 236 cyclohexyl 4-cyano-2- morpholino-phenyl 237 3-pyridyl 4-cyano-2-

morpholino-phenyl 238 1-methyl-1H- 4-cyano-2- pyrazol-4-yl morpholino-phenyl 239 1H-
 imidazol-4-yl 4-cyano-2- morpholino-phenyl 240 2-furanyl 4-cyano-2- morpholino-phenyl 241
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 2- pyrazol-4-yl morpholino-phenyl 254 1H-imidazol-4-yl 4-hydroxy-2- morpholino-phenyl 255 2-
 furanyl 4-hydroxy-2- morpholino-phenyl 256 ethyl 2-CH.sub.3-phenyl 257 n-propyl 2-CH.sub.3-
 phenyl 258 isopropyl 2-CH.sub.3-phenyl 259 —CH.sub.2CH(CH.sub.3).sub.2 2-CH.sub.3-phenyl
 260 CF.sub.3 2-CH.sub.3-phenyl 261 —CH.sub.2CF.sub.3 2-CH.sub.3-phenyl 262 —
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 4-yl 2-OH-Phenyl 300 2-furanyl 2-OH-Phenyl 301 ethyl 3-OMe-Phenyl 302 n-propyl 3-OMe-
 Phenyl 303 isopropyl 3-OMe-Phenyl 304 —CH.sub.2CH(CH.sub.3).sub.2 3-OMe-Phenyl 305 —
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 furanyl 3-OMe-Phenyl 316 ethyl 4-CN-Phenyl 317 n-propyl 4-CN-Phenyl 318 isopropyl 4-CN-
 Phenyl 319 —CH.sub.2CH(CH.sub.3).sub.2 4-CN-Phenyl 320 —CF.sub.3 4-CN-Phenyl 321 —
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 4-yl 4-CN-Phenyl 330 2-furanyl 4-CN-Phenyl 331 ethyl 2-CN-Phenyl 332 n-propyl 2-CN-Phenyl
 333 isopropyl 2-CN-Phenyl 334 —CH.sub.2CH(CH.sub.3).sub.2 2-CN-Phenyl 335 —CF.sub.3 2-
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 —CF.sub.3 3-F-Phenyl 351 —CH.sub.2CF.sub.3 3-F-Phenyl 352 —CH.sub.2CH.sub.2CF.sub.3 3-
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 cyclohexyl 3-F-Phenyl 357 3-pyridyl 3-F-Phenyl 358 1-methyl-1H- 3-F-Phenyl pyrazol-4-yl 359
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 Phenyl 363 isopropyl 2-Cl-Phenyl 364 —CH.sub.2CH(CH.sub.3).sub.2 2-Cl-Phenyl 365 —

CF.sub.3 2-Cl-Phenyl 366 —CH.sub.2CF.sub.3 2-Cl-Phenyl 367 —CH.sub.2CF.sub.3 2-Cl-Phenyl 368 cyclopropyl 2-Cl-Phenyl 369 Cyclobutyl 2-Cl-Phenyl 370 cyclopentyl 2-Cl-Phenyl 371 cyclohexyl 2-Cl-Phenyl 372 3-pyridyl 2-Cl-Phenyl 373 1-methyl-1H- 2-Cl-Phenyl pyrazol-4-yl 374 1H-imidazol-4-yl 2-Cl-Phenyl 375 2-furanyl 2-Cl-Phenyl 376 ethyl 4-Cl-Phenyl 377 n-propyl 4-Cl-Phenyl 378 isopropyl 4-Cl-Phenyl 379 —CH.sub.2CH(CH.sub.3).sub.2 4-Cl-Phenyl 380 —CF.sub.3 4-Cl-Phenyl 381 —CH.sub.2CF.sub.3 4-Cl-Phenyl 382 —CH.sub.2CH.sub.2CF.sub.3 4-Cl-Phenyl 383 cyclopropyl 4-Cl-Phenyl 384 Cyclobutyl 4-Cl-Phenyl 385 cyclopentyl 4-Cl-Phenyl 386 cyclohexyl 4-Cl-Phenyl 387 3-pyridyl 4-Cl-Phenyl 388 1-methyl-1H- 4-Cl-Phenyl pyrazol-4-yl 389 1H-imidazol-4-yl 4-Cl-Phenyl 390 2-furanyl 4-Cl-Phenyl 391 ethyl 3-Br-Phenyl 392 n-propyl 3-Br-Phenyl 393 isopropyl 3-Br-Phenyl 394 —CH.sub.2CH(CH.sub.3).sub.2 3-Br-Phenyl 395 —CF.sub.3 3-Br-Phenyl 396 —CH.sub.2CF.sub.3 3-Br-Phenyl 397 —CH.sub.2CH.sub.2CF.sub.3 3-Br-Phenyl 398 cyclopropyl 3-Br-Phenyl 399 Cyclobutyl 3-Br-Phenyl 400 cyclopentyl 3-Br-Phenyl 401 cyclohexyl 3-Br-Phenyl 402 3-pyridyl 3-Br-Phenyl 403 1-methyl-1H- 3-Br-Phenyl pyrazol-4-yl 404 1H-imidazol-4-yl 3-Br-Phenyl 405 2-furanyl 3-Br-Phenyl 406 ethyl 2-CF.sub.3-Phenyl 407 n-propyl 2-CF.sub.3-Phenyl 408 isopropyl 2-CF.sub.3-Phenyl 409 —CH.sub.2CH(CH.sub.3).sub.2 2-CF.sub.3-Phenyl 410 —CF.sub.3 2-CF.sub.3-Phenyl 411 —CH.sub.2CF.sub.3 2-CF.sub.3-Phenyl 412 —CH.sub.2CH.sub.2CF.sub.3 2-CF.sub.3-Phenyl 413 cyclopropyl 2-CF.sub.3-Phenyl 414 Cyclobutyl 2-CF.sub.3-Phenyl 415 cyclopentyl 2-CF.sub.3-Phenyl 416 cyclohexyl 2-CF.sub.3-Phenyl 417 3-pyridyl 2-CF.sub.3-Phenyl 418 1-methyl-1H- 2-CF.sub.3-Phenyl pyrazol-4-yl 419 1H-imidazol-4-yl 2-CF.sub.3-Phenyl 420 2-furanyl 2-CF.sub.3-Phenyl 421 ethyl 4-CF.sub.3-Phenyl 422 n-propyl 4-CF.sub.3-Phenyl 423 isopropyl 4-CF.sub.3-Phenyl 424 —CH.sub.2CH(CH.sub.3).sub.2 4-CF.sub.3-Phenyl 425 —CF.sub.3 4-CF.sub.3-Phenyl 426 —CH.sub.2CF.sub.3 4-CF.sub.3-Phenyl 427 —CH.sub.2CH.sub.2CF.sub.3 4-CF.sub.3-Phenyl 428 cyclopropyl 4-CF.sub.3-Phenyl 429 Cyclobutyl 4-CF.sub.3-Phenyl 430 cyclopentyl 4-CF.sub.3-Phenyl 431 cyclohexyl 4-CF.sub.3-Phenyl 432 3-pyridyl 4-CF.sub.3-Phenyl 433 1-methyl-1H- 4-CF.sub.3-Phenyl pyrazol-4-yl 434 1H-imidazol-4-yl 4-CF.sub.3-Phenyl 435 2-furanyl 4-CF.sub.3-Phenyl 436 ethyl 3-iPr-Phenyl 437 n-propyl 3-iPr-Phenyl 438 isopropyl 3-iPr-Phenyl 439 —CH.sub.2CH(CH.sub.3).sub.2 3-iPr-Phenyl 440 —CF.sub.3 3-iPr-Phenyl 441 —CH.sub.2CF.sub.3 3-iPr-Phenyl 442 —CH.sub.2CH.sub.2CF.sub.3 3-iPr-Phenyl 443 cyclopropyl 3-iPr-Phenyl 444 Cyclobutyl 3-iPr-Phenyl 445 cyclopentyl 3-iPr-Phenyl 446 cyclohexyl 3-iPr-Phenyl 447 3-pyridyl 3-iPr-Phenyl 448 1-methyl-1H- 3-iPr-Phenyl pyrazol-4-yl 449 1H-imidazol-4-yl 3-iPr-Phenyl 450 2-furanyl 3-iPr-Phenyl 451 ethyl 2-morpholino- phenyl 452 n-propyl 2-morpholino- phenyl 453 isopropyl 2-morpholino- phenyl 454 —CH.sub.2CH(CH.sub.3).sub.2 2-morpholino- phenyl 455 —CF.sub.3 2-morpholino- phenyl 456 —CH.sub.2CF.sub.3 2-morpholino- phenyl 457 —CH.sub.2CH.sub.2CF.sub.3 2-morpholino- phenyl 458 cyclopropyl 2-morpholino- phenyl 459 Cyclobutyl 2-morpholino- phenyl 460 cyclopentyl 2-morpholino- phenyl 461 cyclohexyl 2-morpholino- phenyl 462 3-pyridyl 2-morpholino- phenyl 463 1-methyl-1H- 2-morpholino- pyrazol-4-yl phenyl 464 1H-imidazol-4-yl 2-morpholino- phenyl 465 2-furanyl 2-morpholino- phenyl 466 ethyl 4-morpholino- phenyl 467 n-propyl 4-morpholino- phenyl 468 isopropyl 4-morpholino- phenyl 469 —CH.sub.2CH(CH.sub.3).sub.2 4-morpholino- phenyl 470 —CF.sub.3 4-morpholino- phenyl 471 —CH.sub.2CF.sub.3 4-morpholino- phenyl 472 —CH.sub.2CH.sub.2CF.sub.3 4-morpholino- phenyl 473 cyclopropyl 4-morpholino- phenyl 474 Cyclobutyl 4-morpholino- phenyl 475 cyclopentyl 4-morpholino- phenyl 476 cyclohexyl 4-morpholino- phenyl 477 3-pyridyl 4-morpholino- phenyl 478 1-methyl-1H- 4-morpholino- pyrazol-4-yl phenyl 479 1H-imidazol-4-yl 4-morpholino- phenyl 480 2-furanyl 4-morpholino- phenyl 481 ethyl 4-methyl-2- morpholino-phenyl 482 n-propyl 4-methyl-2- morpholino-phenyl 483 isopropyl 4-methyl-2- morpholino-phenyl 484 —CH.sub.2CH(CH.sub.3).sub.2 4-methyl-2- morpholino-phenyl 485 —CF.sub.3 4-methyl-2- morpholino-phenyl 486 —CH.sub.2CF.sub.3 4-methyl-2- morpholino-phenyl 487 —CH.sub.2CH.sub.2CF.sub.3 4-methyl-2- morpholino-phenyl 488 cyclopropyl 4-methyl-2-

morpholino-phenyl 489 Cyclobutyl 4-methyl-2- morpholino-phenyl 490 cyclopentyl 4-methyl-2- morpholino-phenyl 491 cyclohexyl 4-methyl-2- morpholino-phenyl 492 3-pyridyl 4-methyl-2- morpholino-phenyl 493 1-methyl-1H- 4-methyl-2- pyrazol-4-yl morpholino-phenyl 494 1H-imidazol-4-yl 4-methyl-2- morpholino-phenyl 495 2-furanyl 4-methyl-2- morpholino-phenyl [0410] For the purposes of demonstrating the manner in which the compounds of the present invention are named and referred to herein, the compound having the formula:

##STR00069##

has the chemical name 8-(methylsulfonyl)-3-(2-(4-phenylpiperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one.

[0411] For the purposes of demonstrating the manner in which the compounds of the present invention are named and referred to herein, the compound having the formula:

##STR00070##

has the chemical name 8-(methylsulfonyl)-3-(2-(5-phenylhexahydropyrrolo[3,4-c]pyrrol-2(1H)-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one.

[0412] For the purposes of the present invention, a compound depicted by the racemic formula, for example:

##STR00071##

will stand equally well for either of the two enantiomers having the formula:

##STR00072##

or the formula:

##STR00073##

or mixtures thereof, or in the case where a second chiral center is present, all diastereomers.

[0413] In all of the embodiments provided herein, examples of suitable optional substituents are not intended to limit the scope of the claimed invention. The compounds of the invention may contain any of the substituents, or combinations of substituents, provided herein. Process for preparing the 5-hydroxytryptamine receptor 7 activity modulators of the invention

[0414] The present invention further relates to a process for preparing the 5-hydroxytryptamine receptor 7 activity modulators of the present invention.

[0415] Compounds of the present teachings can be prepared in accordance with the procedures outlined herein, from commercially available starting materials, compounds known in the literature, or readily prepared intermediates, by employing standard synthetic methods and procedures known to those skilled in the art. Standard synthetic methods and procedures for the preparation of organic molecules and functional group transformations and manipulations can be readily obtained from the relevant scientific literature or from standard textbooks in the field. It will be appreciated that where typical or preferred process conditions (i.e., reaction temperatures, times, mole ratios of reactants, solvents, pressures, etc.) are given, other process conditions can also be used unless otherwise stated. Optimum reaction conditions can vary with the particular reactants or solvent used, but such conditions can be determined by one skilled in the art by routine optimization procedures. Those skilled in the art of organic synthesis will recognize that the nature and order of the synthetic steps presented can be varied for the purpose of optimizing the formation of the compounds described herein.

[0416] The processes described herein can be monitored according to any suitable method known in the art. For example, product formation can be monitored by spectroscopic means, such as nuclear magnetic resonance spectroscopy (e.g., ¹H or ¹³C), infrared spectroscopy, spectrophotometry (e.g., UV-visible), mass spectrometry, or by chromatography such as high pressure liquid chromatography (HPLC), gas chromatography (GC), gel-permeation chromatography (GPC), or thin layer chromatography (TLC).

[0417] Preparation of the compounds can involve protection and deprotection of various chemical groups. The need for protection and deprotection and the selection of appropriate protecting groups can be readily determined by one skilled in the art. The chemistry of protecting groups can be

found, for example, in Greene et al., Protective Groups in Organic Synthesis, 2d. Ed. (Wiley & Sons, 1991), the entire disclosure of which is incorporated by reference herein for all purposes. [0418] The reactions or the processes described herein can be carried out in suitable solvents which can be readily selected by one skilled in the art of organic synthesis. Suitable solvents typically are substantially nonreactive with the reactants, intermediates, and/or products at the temperatures at which the reactions are carried out, i.e., temperatures that can range from the solvent's freezing temperature to the solvent's boiling temperature. A given reaction can be carried out in one solvent or a mixture of more than one solvent. Depending on the particular reaction step, suitable solvents for a particular reaction step can be selected.

[0419] The compounds of these teachings can be prepared by methods known in the art of organic chemistry. The reagents used in the preparation of the compounds of these teachings can be either commercially obtained or can be prepared by standard procedures described in the literature. For example, compounds of the present invention can be prepared according to the method illustrated in the General Synthetic Schemes:

General Synthetic Schemes for Preparation of Compounds

[0420] The reagents used in the preparation of the compounds of this invention can be either commercially obtained or can be prepared by standard procedures described in the literature. In accordance with this invention, compounds in the genus may be produced by one of the following reaction schemes.

[0421] Compounds of the disclosure may be prepared according to the process outlined in Scheme 1-x.

##STR00074##

[0422] A suitably substituted compound of formula (1), a known compound or compound prepared by known methods wherein X^{sup.1} is an C_{sub.1-6} alkyl, is reacted with benzyl bromide in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like in the presence of a solvent such as ethanol, methanol, isopropanol, tetrahydrofuran, 1,4-dioxane, methylene chloride, N,N-dimethyl formamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (2). A compound of the formula (2) is reacted with a compound of the formula (3), a known compound or a compound prepared by known methods in which LG is a leaving group such as chlorine, bromine, iodine, mesylate, tosylate, and the like, in the presence of a base such as lithium diisopropylamide, sodium diisopropylamide, potassium diisopropylamide, lithium bis(trimethylsilyl)amide, sodium bis(trimethylsilyl)amide, potassium bis(trimethylsilyl)amide, sodium hydride, and the like in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, to provide a compound of the formula (4). A compound of the formula (4) is then treated with paraformaldehyde in the presence of an acid such as sulfuric acid, hydrochloric acid, and the like, in the presence of acetic acid, and optionally in an organic solvent such as methanol, ethanol, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (5). A compound of the formula (5) is then treated with a base such as sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, optionally with heating, and then treated with an acid such as sulfuric acid, hydrochloric acid, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, to provide a compound of the formula (6). A compound of the formula (6) is reacted with tert-butyldimethylchlorosilane in the presence of imidazole, in the presence of a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (7). Alternatively, a compound of the formula (6) is reacted with tert-butyldimethylsilyl trifluoromethanesulfonate in the presence of a base such as pyridine,

2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (7).

##STR00075##

[0423] A compound of the formula (7) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (8). A compound of the formula (8) is reacted with Di-tert-butyl dicarbonate in the presence of a base such as pyridine, 2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (9). A compound of the formula (9) is reacted with Tetra-n-butylammonium fluoride in the presence of solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (10).

##STR00076##

[0424] A compound of the formula (10) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (11). A compound of the formula (11) is reacted with a compound of the formula (12), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (13). A compound of the formula (13) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (14).

##STR00077##

[0425] A compound of the formula (14) is reacted with a compound of the formula (15), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (16).

##STR00078##

[0426] A compound of the formula (14) is reacted with a compound of the formula (17), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene

chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (18).

##STR00079##

[0427] A compound of the formula (14) is reacted with a compound of the formula (19), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (20).

##STR00080##

[0428] A compound of the formula (14) is reacted with a compound of the formula (21), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (22).

##STR00081##

[0429] A compound of the formula (14) is reacted with a compound of the formula (23), a known compound or a compound prepared by known methods, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (24).

##STR00082##

[0430] A compound of the formula (14) is reacted with a compound of the formula (25), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (26).

##STR00083##

[0431] A suitably substituted compound of formula (27), a known compound or compound prepared by known methods wherein X^{sup.1} is an C_{sub.1-6} alkyl, is reacted with a compound of the formula (28), a known compound or a compound prepared by known methods in which LG is a leaving group such as chlorine, bromine, iodine, mesylate, tosylate, and the like, in the presence of a base such as lithium diisopropylamide, sodium diisopropylamide, potassium diisopropylamide, lithium bis(trimethylsilyl)amide, sodium bis(trimethylsilyl)amide, potassium bis(trimethylsilyl)amide, sodium hydride, and the like in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, to provide a compound of the formula (29). A compound of the formula (29) is then treated with paraformaldehyde in the presence of an acid such as sulfuric acid, hydrochloric acid, and the like, in the presence of acetic acid, and optionally in an organic solvent such as methanol, ethanol, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (30). A compound of the formula (30) is then treated with a base such as sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, optionally with heating, and then treated with an acid such as sulfuric acid, hydrochloric acid, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, to provide a compound of the formula (31). A compound of the formula (31) is treated with carbon tetrabromide in the presence of

triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (32). A compound of the formula (32) is reacted with a compound of the formula (33), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (34).

##STR00084##

[0432] A compound of the formula (29) is then reacted with iodine in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbonate, potassium carbonate, lithium carbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, and the like to provide a compound of the formula (35). A compound of the formula (35) is reacted with a compound of the formula (36), a known compound or compound prepared by known methods, in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6 lutidine, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (37).

##STR00085##

[0433] A compound of the formula (4) is then reacted with iodine in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbonate, potassium carbonate, lithium carbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, and the like to provide a compound of the formula (38). A compound of the formula (38) is reacted with a compound of the formula (39), a known compound or compound prepared by known methods, in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6 lutidine, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (40). A compound of the formula (40) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (41).

##STR00086##

[0434] A compound of the formula (41) is reacted with a compound of the formula (42), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (43).

##STR00087##

[0435] A compound of the formula (41) is reacted with a compound of the formula (44), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene

chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (45).

##STR00088##

[0436] A compound of the formula (41) is reacted with a compound of the formula (46), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (47).

##STR00089##

[0437] A compound of the formula (41) is reacted with a compound of the formula (48), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (49).

##STR00090##

[0438] A compound of the formula (41) is reacted with a compound of the formula (50), a known compound or a compound prepared by known methods, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (51).

##STR00091##

[0439] A compound of the formula (41) is reacted with a compound of the formula (52), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (53).

##STR00092##

[0440] A compound of the formula (4) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (54). A compound of the formula (54) is reacted with a compound of the formula (55), a known compound or compound prepared by known methods, wherein X^{sup.2} is a halogen, in the presence of a solvent such as ethyl ether, tetrahydrofuran, 1,4-dioxane and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (56). A compound of the formula (56) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (57). A compound of the formula (57) is reacted with a reducing agent such as lithium borohydride, sodium borohydride, sodium cyanoborohydride and the like, in a solvent such as methanol, ethanol, isopropanol, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (58). A compound of the formula (58) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (59). A compound of the formula (59) is reacted with a compound of the formula (60), a known compound or a compound prepared by known methods, in

the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (61).

##STR00093##

[0441] A compound of the formula (62) is then converted to a compound of the formula (63), wherein LG is a mesylate, tosylate, nosylate, and the like, using methods that are known to one skilled in the art. Thus, a compound of the formula (62) is treated with a sulfonyl chloride such as methanesulfonyl chloride, toluenesulfonyl chloride p-nitrophenyl sulfonyl chloride, and the like, in the presence of a base such as triethylamine, diisopropyl amine, pyridine, 2,6-lutidine, and the like, in an organic solvent such as methylene chloride, dichloroethane, tetrahydrofuran, 1,4-dioxane, N,N-dimethylformamide, tetrahydrofuran, 1,4-dioxane and the like to provide a compound of the formula (63). A compound of the formula (63) is reacted with a compound of the formula (64), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (65).

##STR00094##

[0442] A compound of the formula (4) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (66). A compound of the formula (66) is reacted with a compound of the formula (67), a known compound or compound prepared by known methods, wherein X_{sup.2} is a halogen, in the presence of a solvent such as ethyl ether, tetrahydrofuran, 1,4-dioxane and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (68). A compound of the formula (68) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (69). A compound of the formula (69) is reacted with a reducing agent such as lithium borohydride, sodium borohydride, sodium cyanoborohydride and the like, in a solvent such as methanol, ethanol, isopropanol, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (70). A compound of the formula (70) is reacted with tert-butyldimethylchlorosilane in the presence of imidazole, in the presence of a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (71). Alternatively, a compound of the formula (70) is reacted with tert-butyldimethylsilyl trifluoromethanesulfonate in the presence of a base such as pyridine, 2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (71).

##STR00095##

[0443] A compound of the formula (71) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II),

palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (72). A compound of the formula (72) is reacted with Di-tert-butyl dicarbonate in the presence of a base such as pyridine, 2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (73). A compound of the formula (73) is reacted with Tetra-n-butylammonium fluoride in the presence of solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (74).

##STR00096##

[0444] A compound of the formula (74) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (75). A compound of the formula (75) is reacted with a compound of the formula (76), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (77). A compound of the formula (77) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (78).

##STR00097##

[0445] A compound of the formula (78) is reacted with a compound of the formula (79), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (80).

##STR00098##

[0446] A compound of the formula (78) is reacted with a compound of the formula (81), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (82).

##STR00099##

[0447] A compound of the formula (78) is reacted with a compound of the formula (83), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-

dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (84).

##STR00100##

[0448] A compound of the formula (78) is reacted with a compound of the formula (85), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (86).

##STR00101##

[0449] A compound of the formula (78) is reacted with a compound of the formula (87), a known compound or a compound prepared by known methods, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (88).

##STR00102##

[0450] A compound of the formula (78) is reacted with a compound of the formula (89), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (90).

##STR00103##

[0451] Diethanolamine (91) is reacted with 4-nitrobenzenesulfonyl chloride (NosCl) in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like in a solvent such as tetrahydrofuran, 1,4-dioxane, methylene chloride and the like to provide a compound of the formula (92). A compound of the formula (92) is then reacted with a compound of the formula (93), a known compound or one prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like, in a solvent such as acetonitrile, methanol, ethanol, dimethyl formamide, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (94). A compound of the formula (94) is reacted with a thiophenol in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbonate, potassium carbonate, lithium bicarbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, acetonitrile and the like, optionally in the presence of dimethylsulfoxide, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (95).

##STR00104##

[0452] A compound of the formula (96), a known compound or a compound prepared by known methods, is reacted with a compound of the formula (97), a known compound or a compound prepared by known methods in which X^{sup.3} is selected from the group consisting of chlorine, bromine, iodine, and methanetrifluorosulfonate, in the presence of a base such as sodium tert-butoxide, lithium tert-butoxide, potassium tert-butoxide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethyl amine, pyridine, 2,6-lutidine, and the like, in the presence of a palladium catalyst such as palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), tris(dibenzylideneacetone)dipalladium(0), and the like, in the presence of a solvent such as toluene, benzene, methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, N,N-dimethylformamide, and the like, optionally with heating, optionally with microwave irradiation, to

provide a compound of the formula (98). A compound of the formula (98) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (99).

##STR00105##

[0453] A compound of the formula (100), a known compound or a compound prepared by known methods, is reacted with a compound of the formula (101), a known compound or a compound prepared by known methods in which X^{sup.3} is selected from the group consisting of chlorine, bromine, iodine, and methanetrifluorosulfonate, in the presence of a base such as sodium tert-butoxide, lithium tert-butoxide, potassium tert-butoxide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethyl amine, pyridine, 2,6-lutidine, and the like, in the presence of a palladium catalyst such as palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), tris(dibenzylideneacetone)dipalladium(0), and the like, in the presence of a solvent such as toluene, benzene, methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, N,N-dimethylformamide, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (102). A compound of the formula (102) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (103).

##STR00106##

[0454] A compound of the formula (104), a known compound or a compound prepared by known methods in which X^{sup.4} is an C_{sub.1-6} alkyl, is reacted with di-tert-butyl dicarbonate in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like, in a solvent such as methanol, ethanol, methylene chloride, tetrahydrofuran, 1,4-dioxane, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (105). A compound of the formula (105) is reacted with a compound of the formula (106), a known compound or a compound prepared by known methods where in LG^{sup.1} is a leaving group such as chlorine, bromine, iodine, mesylate, tosylate, and the like, in the presence of a base such as lithium diisopropylamide, sodium diisopropylamide, potassium diisopropylamide, lithium bis(trimethylsilyl)amide, sodium bis(trimethylsilyl)amide, potassium bis(trimethylsilyl)amide, sodium hydride, and the like in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, to provide a compound of the formula (107).

##STR00107##

[0455] A compound of the formula (107) is reacted with a base such as sodium hydroxide, lithium hydroxide, potassium hydroxide, sodium carbonate, lithium carbonate, potassium carbonate, and the like in a solvent such as methanol, ethanol, isopropanol, tetrahydrofuran, 1,4-dioxane, N,N-dimethylformamide, and the like, optionally in the presence of water, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (108). A compound of the formula (108) is then reacted with iodine in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbonate, potassium carbonate, lithium bicarbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, and the like to provide a compound of the formula (109).

##STR00108##

[0456] A compound of the formula (109) is reacted with a compound of the formula (110), a known compound or compound prepared by known methods, in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6 lutidine, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (111). A compound of the formula (111) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (112).

##STR00109##

[0457] A compound of the formula (113) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (114). A compound of the formula (114) is reacted with a compound of the formula (115), a known compound or compound prepared by known methods, wherein X^{sup.5} is a halogen, in the presence of a solvent such as ethyl ether, tetrahydrofuran, 1,4-dioxane and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (116). A compound of the formula (116) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (117). A compound of the formula (117) is reacted with a reducing agent such as lithium borohydride, sodium borohydride, sodium cyanoborohydride and the like, in a solvent such as methanol, ethanol, isopropanol, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (118).

##STR00110##

[0458] A compound of the formula (7) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (119). A compound of the formula (119) is reacted with a compound of the formula (15), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (120). A compound of the formula (120) is reacted with tetra-n-butylammonium fluoride in the presence of solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (121). A compound of the formula (121) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (122). A compound of the formula (122) is reacted with a compound of the formula (12), a known compound or a compound prepared by known methods, in

the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (123).

[0459] The Examples provided below provide representative methods for preparing exemplary compounds of the present invention. The skilled practitioner will know how to substitute the appropriate reagents, starting materials and purification methods known to those skilled in the art, in order to prepare the compounds of the present invention.

EXAMPLES

[0460] The practice of the invention is illustrated by the following non-limiting examples. The Examples provided below provide representative methods for preparing exemplary compounds of the present invention. The skilled practitioner will know how to substitute the appropriate reagents, starting materials and purification methods known to those skilled in the art, in order to prepare the compounds of the present invention.

[0461] In the examples that follow, ¹H-NMR spectra were obtained on a Varian Mercury 300-MHz NMR. Purity (%) and mass spectral data were determined with a Waters Alliance 2695 HPLC/MS (Waters Symmetry C18, 4.6×75 mm, 3.5 μm) with a 2996 diode array detector from 210-400 nm.

##STR00111##

[0462] Example 1: Preparation of ethyl 1-benzylpiperidine-4-carboxylate: To a solution of ethyl piperidine-4-carboxylate (5.0 g, 31.8 mmol, 1.0 eq) and ethanol (15.0 mL), benzyl bromide (7.07 g, 31.8 mmol, 1.3 eq) was added dropwise at 0° C. Following, triethylamine (1.06 g, 10.5 mmol, 1.5 eq) was added in one portion while at 0° C. The resulting mixture was allowed to warm to RT and stir overnight. The reaction was concentrated in vacuo to remove the presence of ethanol. The resulting residue was suspended in a mixture of ethyl acetate: D.I. water (20 mL:20 mL). The organic layer was separated and the aqueous layer was extracted with ethyl acetate (2×10 mL). The combined extract was dried over Na₂SO₄, then filtered through a plug of silica gel and washed with ethyl acetate. The filtrate was concentrated in vacuo to give product that was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 7.41-7.20 (m, 5H), 4.14 (q, J=7.2 Hz, 2H), 3.51 (s, 2H), 2.87 (dt, J=3.5, 11.8 Hz, 2H), 2.29 (m, 1H), 2.04 (td, J=2.5, 11.4 Hz, 2H), 1.95-1.85 (m, 2H), 1.85-1.70 (m, 2H), 1.26 (t, J=7.1 Hz, 3H).

##STR00112##

[0463] Example 2: Preparation of methyl 4-allyltetrahydro-2H-pyran-4-carboxylate: This reaction was performed in oven-dried glassware under a nitrogen atmosphere. To a solution of lithium diisopropylamide (1M, 1.20 equiv) in dry tetrahydrofuran (4.16 mL), methyl tetrahydro-2H-pyran-4-carboxylate (0.5 g, 3.47 mmol, 1.0 equiv), in 5 mL dry THF, was added dropwise during 0.5 hours at -78° C. The mixture was allowed to stir at this temperature for 1 hr followed by the addition of allyl bromide (0.457 g, 3.78 mmol, 1.1 eq) dropwise. The reaction mixture was allowed to warm to RT over a 1 hr period. The reaction was quenched with 10% HCl (while cooling in ice bath) until acidic (pH=2). The organic layer was separated and the aqueous layer was extracted with ethyl acetate (3×10 mL). The extract was dried over Na₂SO₄ and then concentrated in vacuo to give product that was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 5.68-5.52 (m, 1H), 5.03-4.91 (m, 2H), 3.75 (dt, J=3.7, 11.8 Hz, 2H), 3.63 (s, 3H), 3.37 (td, J=2.1, 11.6 Hz, 2H), 2.21 (d, J=7.4 Hz, 2H), 2.03-1.95 (m, 2H), 1.53-1.40 (m, 2H).

##STR00113##

[0464] Example 3: Preparation of ethyl 4-allyl-1-benzylpiperidine-4-carboxylate: This reaction was performed in oven-dried glassware under a nitrogen atmosphere. To a solution of ethyl 1-

benzylpiperidine-4-carboxylate (6.24 g, 26.7 mmol, 1.0 eq) and dry THF (50 mL), lithium diisopropylamide (1M, 1.10 equiv) in dry tetrahydrofuran (29.3 mL) was added dropwise during 0.5 hours at -78°C . The mixture was allowed to stir at this temperature for 1 hr followed by the addition of allyl iodine (6.73 g, 3.78 mmol, 1.5 eq) dropwise. The reaction mixture was allowed to warm to RT and stir for 2 hr. The reaction was quenched with sat. aq. NH_4Cl until neutral pH (while cooling in ice bath). The organic layer was separated and the aqueous layer was extracted with ethyl acetate ($2 \times 50\text{ mL}$). The combined extract was dried over Na_2SO_4 , then filtered through a plug of silica gel and washed with ethyl acetate. The filtrate was concentrated in vacuo to give product that was used in the next step without further purification. $^1\text{H NMR}$ (400 MHz, CDCl_3) δ 7.37-7.20 (m, 5H), 5.78-5.62 (m, 1H), 5.10-4.97 (m, 2H), 4.17 (q, $J=7.1\text{ Hz}$, 2H), 3.47 (s, 2H), 2.78-2.64 (m, 2H), 2.28 (d, $J=7.4\text{ Hz}$, 2H), 2.18-2.03 (m, 4H), 1.61-1.46 (m, 2H), 1.26 (t, $J=7.1\text{ Hz}$, 3H).

##STR00114##

[0465] Example 4: Preparation of 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one: A mixture of glacial acetic acid (10.9 g, 180 mmol, 53.6 eq), paraformaldehyde (0.309 g, 10.3 mmol, 3.0 eq) and H_2SO_4 (0.191 g, 1.95 mmol, 0.57 eq) was stirred for 30 min at 70°C . before methyl 4-allyltetrahydro-2H-pyran-4-carboxylate (0.632 g, 3.43 mmol, 1.0 equiv) was added dropwise during 10 min. The reaction mixture was then maintained at $70-80^{\circ}\text{C}$. and allowed to stir overnight. Acetic acid was removed under reduced pressure and the reaction was quenched with 10% NaHCO_3 solution. The mixture was then extracted with ethyl acetate ($3 \times 10\text{ mL}$) and the combined organic phase was concentrated in vacuo to give a crude oil. The crude oil was used for next step without further purification.

[0466] A mixture of the crude oil (715 mg) and 30% NaOH (2.86 g NaOH, 4x crude oil) aqueous solution was refluxed for 2 hours. The mixture was cooled in an ice bath and excess 30% H_2SO_4 was added until acidic ($\text{pH} < 2$). The resulting mixture was extracted with ethyl acetate ($3 \times 25\text{ mL}$), the combined organic phase was washed with 10% NaHCO_3 , (50 mL), brine (50 mL), dried over Na_2SO_4 and concentrated in vacuo to give a crude product which was used in the next step without further purification. $^1\text{H NMR}$ (400 MHz, CDCl_3) δ 4.57 (m, 1H), 3.91 (dt, $J=4.5, 11.8\text{ Hz}$, 1H), 3.79 (dt, $J=4.5, 12.0\text{ Hz}$, 1H), 3.66 (t, $J=6.0\text{ Hz}$, 2H), 3.54-3.44 (m, 1H), 3.43-3.34 (m, 1H), 3.13 (b, 1H), 2.41 (dd, $J=6.1, 13.2\text{ Hz}$, 1H), 2.01-1.91 (m, 1H), 1.89-1.64 (m, 4H), 1.54-1.44 (m, 1H), 1.42-1.33 (m, 1H).

##STR00115##

[0467] Example 5: Preparation of 8-benzyl-3-(2-hydroxyethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A mixture of glacial acetic acid (78.1 g, 1.3 mol, 53.6 eq), paraformaldehyde (2.21 g, 73.5 mmol, 3.0 eq) and H_2SO_4 (3.63 g, 37 mmol, 1.5 eq) was stirred for 30 min at 70°C . before ethyl 4-allyl-1-benzylpiperidine-4-carboxylate (7.03 g, 24.5 mmol, 1.0 equiv) was added dropwise during 10 min. The reaction mixture was then maintained at $70-80^{\circ}\text{C}$. and allowed to stir overnight. Acetic acid was removed under reduced pressure and the reaction was quenched with 10% NaHCO_3 solution. The mixture was then extracted with ethyl acetate ($3 \times 40\text{ mL}$) and the combined organic phase was concentrated in vacuo to give a crude oil. The crude oil was used for next step without further purification.

[0468] A mixture of the crude oil (7.07 mg) and 30% NaOH (28 g NaOH, 4x crude oil) aqueous solution was refluxed for 2 hours. The mixture was cooled in an ice bath and excess 30% H_2SO_4 was added until acidic ($\text{pH} < 2$). The resulting mixture was the neutralized ($\text{pH}=8-9$) with sat. aq. NaHCO_3 solution and then extracted with ethyl acetate ($3 \times 100\text{ mL}$), the combined organic phase was dried over Na_2SO_4 and concentrated in vacuo to give a crude product which was used in the next step without further purification. $^1\text{H NMR}$ (400 MHz, CDCl_3) δ 7.39-7.22 (m, 5H), 4.65 (m, 1H), 3.83 (t, $J=5.6\text{ Hz}$, 2H), 3.54 (s, 2H), 2.95-2.84 (m, 1H), 2.83-2.73 (m, 1H), 2.42 (dd, $J=6.1, 13.0\text{ Hz}$, 1H), 2.30-2.07 (m, 4H), 2.00-1.84 (m, 3H), 1.75-1.59 (m, 2H), 1.58-1.48 (m, 1H).

##STR00116##

[0469] Example 6: Preparation of 8-benzyl-3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: To a solution of 8-benzyl-3-(2-hydroxyethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (10.0 g, 34.6 mmol, 1.0 eq.), imidazole (2.47 g, 36.3 mmol, 1.05 eq.) and dichloromethane (70 mL), was added a solution of tert-Butyldimethylsilyl chloride (1M, 5.47 g, 36.3 mmol, 1.05 eq.) in dichloromethane (36.3 mL). The reaction was allowed to stir at RT for 2 hr, before being quenched with D.I. water (50 mL). The organic layer was separated and the aqueous layer was extracted with dichloromethane (2×50 mL). The combined organic phase was dried over Na.sub.2SO.sub.4 and concentrated in vacuo to give a crude product which was further purified by column chromatography (Ethyl acetate/Hexanes, 0%~20%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.32-7.11 (m, 5H), 4.52 (m, 1H), 3.73-3.65 (m, 2H), 3.46 (s, 2H), 2.87-2.76 (m, 1H), 2.72 (dt, J=4.5, 11.8 Hz, 1H), 2.31 (dd, J=6.2, 12.9 Hz, 1H), 2.22-2.08 (m, 1H), 2.08-1.97 (m, 2H), 1.91-1.70 (m, 3H), 1.62 (dd, J=9.8, 12.8 Hz, 1H), 1.59-1.50 (m, 1H), 1.49-1.38 (m, 1H), 0.83 (s, 9H), 0.00 (s, 6H).

##STR00117##

[0470] Example 7: Preparation of tert-butyl 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A mixture of 8-benzyl-3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (4.77 g, 11.8 mmol, 1 eq.), Pd/C (954 mg, 20% wt) and MeOH (79 mL) was stirred at RT under 1 atm of H.sub.2 (filled balloon) overnight. The mixture was filtered through a plug of Celite, washed with MeOH (50 mL) and concentrated in vacuo to give a crude oil. The crude oil (3.78 g) was dissolved in dichloromethane (79 mL) and cooled to 0° C. before the addition of Di-tert-butyl dicarbonate (2.83 g, 13.0 mmol, 1.1 eq.) and trimethylamine (1.8 g, 17.7 mmol, 1.5 eq.). The reaction was allowed to warm to RT and stir for 45 min. At this point the reaction was diluted with sat. aq. NaHCO.sub.3 solution and then extracted with ethyl acetate (3×50 mL), the combined organic phase was dried over Na.sub.2SO.sub.4 and concentrated in vacuo to give a crude product which was used in the next step without further purification. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 4.57 (m, 1H), 3.91 (b, 1H), 3.77 (b, 1H), 3.73-3.66 (m, 2H), 3.17-3.05 (m, 1H), 3.04-2.93 (m, 1H), 2.31 (dd, J=6.2, 13.0 Hz, 1H), 1.96-1.81 (m, 2H), 1.81-1.64 (m, 3H), 1.59-1.48 (m, 1H), 1.48-1.32 (m, 10H), 0.83 (s, 9H), 0.00 (s, 6H).

##STR00118##

[0471] Example 8: Preparation of tert-butyl 3-(2-hydroxyethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: To a solution of tert-butyl 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (4.88 g, 11.8 mmol, 1 eq.) and THF (70 mL) was added tetra-n-butylammonium fluoride (3.24 g, 12.4 mmol, 1.05 eq.), using THF (10 mL) to complete transfer. The resulting solution was allowed to stir at RT for 30 min before being concentrated in vacuo to give a crude product which was further purified by column chromatography (MeOH/DCM, 0%~10%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 4.67 (m, 1H), 3.95 (dt, J=5.0, 13.6 Hz, 1H), 3.87-3.73 (m, 3H), 3.23-3.10 (m, 1H), 3.09-2.98 (m, 1H), 2.39 (dd, J=6.0, 13.0 Hz, 1H), 1.99-1.84 (m, 4H), 1.83-1.68 (m, 2H), 1.63-1.53 (m, 1H), 1.53-1.36 (m, 10H).

##STR00119##

[0472] Example 9: Preparation of 3-(2-bromoethyl)-2,8-dioxaspiro[4.5]decan-1-one: A solution of 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one (0.320 g, 1.60 mmol, 1 eq.) and THF (15 mL) was cooled to 0° C. before triphenylphosphine (0.630 g, 2.4 mmol, 1.5 eq.) and carbon tetrabromide (0.795 g, 2.4 mmol, 1.5 eq.) were sequentially added to the solution. The reaction solution was allowed to warm to RT and stir overnight. The resulting mixture was then filtered and concentrated in vacuo to give a crude mixture. This mixture was suspended in diethyl ether (50 mL) and filtered 2× using diethyl ether to wash the filter cakes. The final filtrate was loaded onto Celite in vacuo and further purified by column chromatography (ethyl acetate/hexanes, 0%~40%).

.sup.1H NMR (400 MHz, CDCl₃) δ 4.67 (m, 1H), 4.04 (dt, J=4.6, 11.8 Hz, 1H), 3.91 (dt, J=4.6, 12.1 Hz, 1H), 3.60 (m, 1H), 3.56-3.45 (m, 3H), 2.50 (dd, J=6.1, 12.9 Hz, 1H), 2.30-2.02 (m, 3H), 1.91 (m, 1H), 1.76 (dd, J=9.8, 13.0 Hz, 1H), 1.64-1.55 (m, 1H), 1.52-1.44 (m, 1H).

##STR00120##

[0473] Example 10: Preparation of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: The title compound was prepared according to the procedure for 3-(2-bromoethyl)-2,8-dioxaspiro[4.5]decan-1-one, except tert-butyl 3-(2-hydroxyethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate was substituted for 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one: .sup.1H NMR (400 MHz, CDCl₃) δ 4.68 (m, 1H), 3.97 (dt, J=5.0, 13.5 Hz, 1H), 3.83 (dt, J=5.0, 13.7 Hz, 1H), 3.54 (dd, J=5.3, 7.5 Hz, 2H), 3.27-3.14 (m, 1H), 3.13-3.01 (m, 1H), 2.42 (dd, J=6.0, 13.0 Hz, 1H), 2.31-2.20 (m, 1H), 2.20-2.09 (m, 1H), 2.01-1.90 (m, 1H), 1.89-1.78 (m, 1H), 1.74 (dd, J=9.8, 12.8 Hz, 1H), 1.66-1.56 (m, 1H), 1.54-1.36 (m, 10H).

##STR00121##

[0474] Example 11: Preparation of 8-benzyl-3-(2-bromoethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: A solution of 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one (2.07 g, 7.16 mmol, 1 eq.) and THF (70 mL) was cooled to 0° C. before triphenylphosphine (2.83 g, 10.8 mmol, 1.5 eq.) and carbon tetrabromide (3.58 g, 10.8 mmol, 1.5 eq.) were sequentially added to the solution. The reaction solution was allowed to warm to RT and stir overnight. The resulting mixture was then filtered and concentrated in vacuo to give a crude mixture. This mixture was suspended in diethyl ether (50 mL) and filtered 2× using diethyl ether to wash the filter cakes. The final filtrate was loaded onto Celite in vacuo and further purified by column chromatography on a C18 column. (ACN/H₂O, 0%~100%, w/0.1% formic acid). .sup.1H NMR (400 MHz, MeOD) δ 7.53 (b, 2H), 7.47 (b, 3H), 4.75 (m, 1H), 4.27 (s, 2H), 3.64-3.47 (m, 3H), 3.34 (m, 1H), 3.19 (b, 1H), 3.08 (b, 1H), 2.52 (m, 1H), 2.35-2.15 (m, 3H), 2.15-1.97 (m, 2H), 1.96-1.81 (m, 2H).

##STR00122##

[0475] Example 12: Preparation of 3-(2-(4-phenylpiperazin-1-yl)ethyl)-2,8-dioxaspiro[4.5]decan-1-one: A solution of 3-(2-bromoethyl)-2,8-dioxaspiro[4.5]decan-1-one (0.050 g, 0.190 mmol, 1 eq.), THF (4 mL) and 1-phenylpiperazine (0.065 g, 0.399 mmol, 2.1 eq.) was heated and stirred at 60° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was further purified by column chromatography (methanol/dichloromethane, 0%~10%). .sup.1H NMR (400 MHz, CDCl₃) δ 7.27 (m, 2H), 6.93 (d, J=8.3 Hz, 2H), 6.86 (t, J=7.3 Hz, 1H), 4.58 (m, 1H), 4.06 (dt, J=4.6, 11.9 Hz, 1H), 3.93 (dt, J=4.6, 12.0 Hz, 1H), 3.61 (m, 1H), 3.51 (m, 1H), 3.21 (t, J=5.0 Hz, 4H), 2.70-2.52 (m, 6H), 2.47 (dd, J=6.0, 12.8 Hz, 1H), 2.11 (m, 1H), 2.01-1.83 (m, 3H), 1.79 (dd, J=9.7, 13.1 Hz, 1H), 1.65-1.54 (m, 1H), 1.54-1.45 (m, 1H); MS (LC/MS, M+H⁺): 344.8.

##STR00123##

[0476] Example 13: Preparation of 3-(2-(4-(4-hydroxyphenyl)piperazin-1-yl)ethyl)-2,8-dioxaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 3-(2-(4-phenylpiperazin-1-yl)ethyl)-2,8-dioxaspiro[4.5]decan-1-one, except 4-(piperazin-1-yl)phenol was substituted for 1-phenylpiperazine: .sup.1H NMR (400 MHz, CDCl₃) δ 6.74 (d, J=8.9 Hz, 2H), 6.63 (d, J=8.9 Hz, 2H), 4.45 (m, 1H), 3.97 (dt, J=4.6, 11.8 Hz, 1H), 3.83 (dt, J=4.5, 12.3 Hz, 1H), 3.51 (m, 1H), 3.42 (m, 1H), 3.00 (t, J=4.7 Hz, 4H), 2.67-2.42 (m, 6H), 2.35 (dd, J=6.1, 12.1 Hz, 1H), 2.00 (m, 1H), 1.92-1.74 (m, 3H), 1.67 (dd, J=9.6, 12.9 Hz, 1H), 1.52-1.43 (m, 1H), 1.43-1.34 (m, 1H); MS (LC/MS, M+H⁺): 360.8.

##STR00124##

[0477] Example 14: Preparation of 8-benzyl-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of 8-benzyl-3-(2-bromoethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate (0.545 g, 1.37 mmol, 1 eq.), THF (13.7 mL), 1-(p-tolyl)piperazine (0.507 g, 2.88 mmol, 2.1 eq.) and triethylamine (0.107 g, 1.5 mmol, 1.1 eq.) was heated and stirred at 60° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that

was further purified by column chromatography (methanol/dichloromethane, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 7.39-7.23 (m, 5H), 7.10 (d, J=8.3 Hz, 2H), 6.87 (d, J=8.6 Hz, 2H), 4.53 (m, 1H), 3.54 (s, 2H), 3.17 (t, J=5.0 Hz, 4H), 2.95-2.85 (m, 1H), 2.84-2.75 (m, 1H), 2.70-2.49 (m, 6H), 2.40 (dd, J=6.2, 12.8 Hz, 1H), 2.30 (s, 3H), 2.27-2.05 (m, 3H), 2.01-1.79 (m, 3H), 1.76-1.58 (m, 2H), 1.58-1.46 (m, 1H); MS (LC/MS, M+H^{sup.}): 447.8.

##STR00125##

[0478] Example 15: Preparation of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A mixture of 8-benzyl-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (445 mg, 0.993 mmol, 1 eq.), Pd/C (90 mg, 20% wt) and EtOH (6.6 mL) was stirred at RT under 1 atm of H₂ (filled balloon) for 48 hrs. The mixture was filtered through a plug of Celite, washed with MeOH (50 mL) and concentrated in vacuo to give a crude oil. ¹H NMR (400 MHz, MeOD) δ 6.95 (d, J=8.2 Hz, 2H), 6.77 (d, J=8.5 Hz, 2H), 4.56 (m, 1H), 3.47 (m, 1H), 3.28-3.18 (m, 1H), 3.17-3.09 (m, 1H), 3.08-2.96 (m, 5H), 2.66-2.46 (m, 6H), 2.42 (dd, J=6.0, 13.0 Hz, 1H), 2.14 (s, 3H), 2.10-2.00 (m, 1H), 2.00-1.91 (m, 1H), 1.91-1.80 (m, 4H), 1.80-1.70 (m, 1H); MS (LC/MS, M+H^{sup.}): 357.8.

##STR00126##

[0479] Example 16: Preparation of methyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A solution of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (0.05 g, 0.14 mmol, 1 eq.), dichloromethane (2 mL) and triethylamine (0.44 g, 0.41 mmol, 3 eq.) was cooled to 0° C. before methyl chloroformate (0.027 g, 0.28 mmol, 2 eq.) was added to the solution. The reaction solution was allowed to warm to RT and stir for 3 hours. The reaction was diluted with dichloromethane and loaded onto Celite in vacuo and further purified by column chromatography on a C18 column. (ACN/H₂O, 0%~100%, w/0.1% formic acid). The resulting formate acid salt was dissolved in MeOH (2 mL) and Amberlite IRA-400(OH) resin was added. This mixture was allowed to stir at RT for 30 min and then filtered and concentrated in vacuo to afford pure free based product. ¹H NMR (400 MHz, CDCl₃) δ 7.08 (d, J=8.4 Hz, 2H), 6.85 (d, J=8.6 Hz, 2H), 4.59 (m, 1H), 4.02 (b, 1H), 3.85 (b, 1H), 3.72 (s, 3H), 3.35-3.24 (m, 1H), 3.23-3.10 (m, 5H), 2.71-2.50 (m, 6H), 2.38 (dd, J=6.0, 12.8 Hz, 1H), 2.28 (s, 3H), 2.07-1.72 (m, 5H), 1.69-1.47 (m, 2H); MS (LC/MS, M+H^{sup.}): 415.8.

##STR00127##

[0480] Example 17: Preparation of 8-acetyl-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for methyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except acetic anhydride was substituted for methyl chloroformate: ¹H NMR (400 MHz, CDCl₃) δ 7.08 (d, J=8.3 Hz, 2H), 6.85 (d, J=8.5 Hz, 2H), 4.60 (m, 1H), 4.19 (m, 0.5 H), 4.04-3.84 (m, 1H), 3.72 (m, 0.5H), 3.45-3.22 (m, 2H), 3.15 (t, J=4.8 Hz, 4H), 2.70-2.49 (m, 6H), 2.43-2.32 (m, 1H), 2.27 (s, 3H), 2.10 (s, 3H), 2.05-1.72 (m, 5H), 1.69-1.49 (m, 2H); MS (LC/MS, M+H^{sup.}): 399.8.

##STR00128##

[0481] Example 18: Preparation of N-methyl-1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-8-carboxamide: The title compound was prepared according to the procedure for methyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except N-methyl-1H-imidazole-1-carboxamide was substituted for methyl chloroformate: ¹H NMR (400 MHz, MeOD) δ 6.96 (d, J=8.3 Hz, 2H), 6.78 (d, J=8.2 Hz, 2H), 4.52 (m, 1H), 3.79 (dt, J=4.7, 13.8 Hz, 1H), 3.68 (dt, J=4.5, 13.8 Hz, 1H), 3.22 (m, 1H), 3.16-2.86 (m, 6H), 2.63 (s, 3H), 2.59-2.36 (m, 7H), 2.15 (s, 3H), 1.91-1.67 (m, 3H), 1.66-1.50 (m, 2H), 1.50-1.38 (m, 1H); MS (LC/MS, M+H^{sup.}): 414.8.

##STR00129##

[0482] Example 20: Preparation of tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A mixture of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.500 g, 1.38 mmol, 1 eq.), ACN (7 mL), 1-(2-

isopropylphenyl)piperazine (0.337 g, 1.65 mmol, 1.2 eq.) and K.sub.2CO.sub.3 (0.954 g, 6.9 mmol, 5 eq.) was heated and stirred at 80° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was further purified by column chromatography (methanol/dichloromethane, 0%~10%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.16 (dd, J=1.6, 7.4 Hz, 1H), 7.09-6.94 (m, 3H), 4.49 (m, 1H), 3.89 (b, 1H), 3.75 (b, 1H), 3.40 (sep, J=6.9 Hz, 1H), 3.09 (m, 1H), 2.98 (m, 1H), 2.81 (t, J=4.6 Hz, 4H), 2.65-2.39 (m, 5H), 2.30 (dd, J=6.1, 12.8 Hz, 1H), 1.99-1.60 (m, 5H), 1.51 (m, 1H), 1.46-1.29 (m, 11H), 1.12 (s, 3H), 1.10 (s, 6H); MS (LC/MS, M+H.sup.+): 485.8

##STR00130##

[0483] Example 21: Preparation of tert-butyl 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except 1,2,3,4-tetrahydroisoquinoline hydrochloride was substituted for 1-(2-isopropylphenyl)piperazine: .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.07-6.96 (m, 3H), 6.95-6.87 (m, 1H), 4.53 (m, 1H), 3.87 (b, 1H), 3.72 (b, 1H), 3.60-3.46 (m, 2H), 3.06 (m, 1H), 2.95 (m, 1H), 2.80 (t, J=5.8 Hz, 2H), 2.64 (t, J=6.0 Hz, 2H), 2.58 (t, J=7.3 Hz, 2H), 2.28 (dd, J=6.1, 12.9 Hz, 1H), 1.97-1.75 (m, 3H), 1.74-1.58 (m, 2H), 1.54-1.26 (m, 11H); MS (LC/MS, M+H.sup.+): 414.8

##STR00131##

[0484] Example 22: Preparation of tert-butyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A solution of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (1.5 g, 4.11 mmol, 1.1 eq.), THF (36 mL), 1-(p-tolyl)piperazine (0.660 g, 3.74 mmol, 1 eq.) and triethylamine (0.416 g, 4.11 mmol, 1.1 eq.) was heated and stirred at 70° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was further purified by column chromatography (methanol/dichloromethane, 0%~10%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.05 (d, J=8.3 Hz, 2H), 6.82 (d, J=8.5 Hz, 2H), 4.55 (m, 1H), 3.96 (m, 1H), 3.81 (m, 1H), 3.22-2.98 (m, 6H), 2.67-2.45 (m, 6H), 2.36 (dd, J=6.2, 12.9 Hz, 1H), 2.25 (s, 3H), 2.00-1.66 (m, 5H), 1.57 (m, 1H), 1.53-1.34 (m, 10H); MS (LC/MS, M+H.sup.+): 457.8

##STR00132##

[0485] Example 23: Preparation of tert-butyl 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A solution of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.5 g, 1.38 mmol, 1. eq.), THF (12 mL), 2-methyl-7-(piperazin-1-yl)-1H-benzo[d]imidazole (0.549 g, 1.65 mmol, 1.2 eq.) and triethylamine (0.500 g, 4.95 mmol, 3.5 eq.) was heated and stirred at 70° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was first purified by column chromatography (methanol/dichloromethane, 0%~10%). The resulting fractions were further purified by column chromatography on a C18 column. (ACN/H.sub.2O, 0%~100%, w/0.1% NH.sub.4OH) .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.16-6.97 (m, 2H), 6.65 (m, 1H), 4.54 (m, 1H), 3.98 (m, 1H), 3.83 (m, 1H), 3.40 (b, 4H), 3.17 (t, J=11.1 Hz, 1H), 3.06 (t, J=11.5 Hz, 1H), 2.68-2.42 (m, 9H), 2.36 (dd, J=6.2, 13.0 Hz, 1H), 1.99-1.66 (m, 5H), 1.58 (m, 1H), 1.54-1.33 (m, 10H); MS (LC/MS, M+H.sup.+): 497.8

##STR00133##

[0486] Example 24: Preparation of tert-butyl 3-(2-(4-(2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except 4-(2-(piperazin-1-yl)phenyl)morpholine was substituted for 2-methyl-7-(piperazin-1-yl)-1H-benzo[d]imidazole: .sup.1H NMR (400 MHz, CDCl.sub.3) δ 6.94-6.87 (m, 2H), 6.87-6.78 (m, 2H), 4.51 (m, 1H), 3.90 (m, 1H), 3.81-3.66 (m, 5H), 3.34-2.84 (m, 10H), 2.67-2.37 (m, 6H), 2.31 (dd, J=6.2, 12.9 Hz, 1H), 1.93-1.59 (m, 5H), 1.52

(m, 1H), 1.47-1.30 (m, 10H); MS (LC/MS, M+H.sup.+): 528.8

##STR00134##

[0487] Example 25: Preparation of 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: To a solution of tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.450 g, 0.930 mmol, 1 eq.) in dichloromethane (3 mL) at 0° C. was added trifluoroacetic acid (3 mL). The reaction was allowed to stir at RT for 30 min before being diluted with MeOH and concentrated in vacuo to afford the product as a TFA salt. The resulting TFA salt was dissolved in MeOH (2 mL) and Amberlite IRA-400(OH) resin was added. This mixture was allowed to stir at RT for 30 min and then filtered and concentrated in vacuo to afford pure free based product. .sup.1H NMR (400 MHz, MeOD) δ 7.16 (d, J=7.5 Hz, 1H), 7.07-6.93 (m, 3H), 4.50 (m, 1H), 3.44 (sep, J=6.9 Hz, 1H), 2.99 (dt, J=4.3, 12.6 Hz, 1H), 2.89 (dt, J=4.3, 13.2 Hz, 1H), 2.82 (t, J=4.7 Hz, 4H), 2.74-2.37 (m, 9H), 1.91-1.78 (m, 3H), 1.77-1.50 (m, 3H), 1.42 (m, 1H), 1.12 (s, 3H), 1.10 (s, 3H); MS (LC/MS, M+H.sup.+): 385.8

##STR00135##

[0488] Example 26: Preparation of 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one, except tert-butyl 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate was substituted for tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: .sup.1H NMR (400 MHz, MeOD) δ 7.18-7.09 (m, 3H), 7.09-7.03 (m, 1H), 4.64 (m, 1H), 3.68 (s, 2H), 3.25 (dt, J=5.0, 13.1 Hz, 1H), 3.10 (dt, J=5.0, 13.4 Hz, 1H), 2.99-2.88 (m, 3H), 2.85-2.61 (m, 5H), 2.55 (dd, J=6.1, 12.9 Hz, 1H), 2.09-1.95 (m, 3H), 1.90-1.70 (m, 3H), 1.63 (m, 1H); MS (LC/MS, M+H.sup.+): 314.8

##STR00136##

[0489] Example 27: Preparation of 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: A solution of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (0.05 g, 0.14 mmol, 1 eq.), dichloromethane (2 mL) and triethylamine (0.44 g, 0.41 mmol, 3 eq.) was cooled to 0° C. before methanesulfonyl chloride (0.032 g, 0.28 mmol, 2 eq.) was added to the solution. The reaction solution was allowed to warm to RT and stir for 3 hours. The reaction was diluted with dichloromethane and loaded onto Celite in vacuo and further purified by column chromatography on a C18 column. (ACN/H.sub.2O, 0%~100%, w/0.1% formic acid). .sup.1H NMR (400 MHz, DMSO) δ 7.01 (d, J=8.5 Hz, 2H), 6.82 (d, J=8.5 Hz, 2H), 4.59 (m, 1H), 3.58-3.45 (m, 1H), 3.44-3.33 (m, 1H), 3.13-2.95 (m, 5H), 2.88 (s, 3H), 2.86-2.78 (m, 1H), 2.62-2.31 (m, 7H), 2.19 (s, 3H), 1.97-1.57 (m, 7H); MS (LC/MS, M+H.sup.+): 435.8

##STR00137##

[0490] Example 28: Preparation of 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: .sup.1H NMR (400 MHz, MeOD) δ 7.34-7.27 (m, 1H), 7.23-7.08 (m, 3H), 4.67 (m, 1H), 3.70 (dt, J=4.8, 12.3 Hz, 1H), 3.61-3.47 (m, 2H), 3.40-3.09 (m, 11H), 3.04 (m, 1H), 2.88 (s, 3H), 2.56 (dd, J=6.4, 12.8 Hz, 1H), 2.29-2.10 (m, 2H), 2.05 (m, 1H), 1.96-1.80 (m, 3H), 1.75 (m, 1H), 1.24 (s, 3H), 1.22 (s, 3H); MS (LC/MS, M+H.sup.+): 463.7

##STR00138##

[0491] Example 29: Preparation of 8-(methylsulfonyl)-3-(2-(4-(2-morpholinophenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(4-(2-morpholinophenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-

azaspiro[4.5]decan-1-one: .sup.1H NMR (400 MHz, MeOD) δ 7.13-6.90 (m, 4H), 4.67 (m, 1H), 3.86 (t, J=4.6 Hz, 4H), 3.70 (dt, J=4.7, 12.3 Hz, 1H), 3.61-3.08 (m, 16H), 3.04 (m, 1H), 2.88 (s, 3H), 2.56 (dd, J=5.9, 13.0 Hz, 1H), 2.29-2.11 (m, 2H), 2.05 (m, 1H), 1.97-1.81 (m, 3H), 1.75 (m, 1H); MS (LC/MS, M+H.sup.+): 507.2

##STR00139##

[0492] Example 30: Preparation of 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: .sup.1H NMR (400 MHz, MeOD) δ 7.47-7.26 (m, 2H), 7.04 (d, J=6.7 Hz, 1H), 4.69 (m, 1H), 3.79-3.28 (m, 12H), 3.14 (m, 1H), 3.01 (m, 1H), 2.87 (s, 3H), 2.79 (s, 3H), 2.58 (dd, J=5.9, 12.9 Hz, 1H), 2.38-2.14 (m, 2H), 2.12-1.98 (m, 1H), 1.96-1.80 (m, 3H), 1.75 (m, 1H); MS (LC/MS, M+H.sup.+): 476.2

##STR00140##

[0493] Example 31: Preparation of 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: .sup.1H NMR (400 MHz, MeOD) δ 7.31-7.21 (m, 3H), 7.20-7.15 (m, 1H), 4.67 (m, 1H), 4.26 (s, 2H), 3.70 (dt, J=4.9, 12.3 Hz, 1H), 3.53 (dt, J=5.2, 12.5 Hz, 1H), 3.39 (t, J=6.2 Hz, 2H), 3.31-3.10 (m, 5H), 3.03 (m, 1H), 2.88 (s, 3H), 2.56 (dd, J=6.0, 12.9 Hz, 1H), 2.30-2.12 (m, 2H), 2.05 (m, 1H), 1.97-1.80 (m, 3H), 1.75 (m, 1H); MS (LC/MS, M+H.sup.+): 392.7.

##STR00141##

[0494] Example 32: Preparation of 8-(phenylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except benzenesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.69 (m, 2H), 7.53 (m, 1H), 7.46 (m, 2H), 7.00 (d, J=8.3 Hz, 2H), 6.75 (d, J=8.5 Hz, 2H), 4.44 (m, 1H), 3.46 (m, 1H), 3.22 (m, 1H), 3.10 (t, J=4.7 Hz, 4H), 2.97 (m, 1H), 2.87 (m, 1H), 2.75-2.44 (m, 6H), 2.19 (s, 3H), 2.12 (dd, J=6.1, 13.0 Hz, 1H), 2.02-1.77 (m, 4H), 1.72-1.53 (m, 3H); MS (LC/MS, M+H.sup.+): 498.2.

##STR00142##

[0495] Example 33: Preparation of 8-((4-methoxyphenyl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 4-methoxybenzenesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.62 (d, J=9.0 Hz, 2H), 6.99 (d, J=8.3 Hz, 2H), 6.91 (d, J=8.9 Hz, 2H), 6.75 (d, J=8.6 Hz, 2H), 4.45 (m, 1H), 3.80 (s, 3H), 3.42 (m, 1H), 3.18 (m, 1H), 3.04 (t, J=4.9 Hz, 4H), 2.95 (m, 1H), 2.86 (m, 1H), 2.57-2.39 (m, 6H), 2.19 (s, 3H), 2.11 (dd, J=6.1, 12.9 Hz, 1H), 2.01-1.86 (m, 2H), 1.86-1.50 (m, 5H); MS (LC/MS, M+H.sup.+): 528.2.

##STR00143##

[0496] Example 34: Preparation of 8-((4-chlorophenyl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 4-chlorobenzenesulfonyl chloride was substituted for

methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.56 (d, J=8.6 Hz, 2H), 7.37 (d, J=8.5 Hz, 2H), 6.93 (d, J=8.4 Hz, 2H), 6.69 (d, J=8.6 Hz, 2H), 4.40 (m, 1H), 3.37 (m, 1H), 3.10 (m, 1H), 3.02 (t, J=4.7 Hz, 4H), 2.96 (m, 1H), 2.87 (m, 1H), 2.59-2.34 (m, 6H), 2.13 (s, 3H), 2.06 (dd, J=5.9, 12.9 Hz, 1H), 1.95-1.81 (m, 2H), 1.81-1.67 (m, 2H), 1.66-1.49 (m, 3H); MS (LC/MS, M+H.sup.+): 532.2.

##STR00144##

[0497] Example 35: Preparation of 8-((tetrahydro-2H-pyran-4-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except tetrahydro-2H-pyran-4-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.08 (d, J=8.3 Hz, 2H), 6.84 (d, J=8.5 Hz, 2H), 4.60 (m, 1H), 4.08 (dd, J=3.6, 11.5 Hz, 2H), 3.80 (m, 1H), 3.57-3.42 (m, 4H), 3.41-3.30 (m, 3H), 3.22-3.06 (m, 5H), 2.69-2.47 (m, 6H), 2.30 (dd, J=6.1, 12.9 Hz, 1H), 2.27 (s, 3H), 2.03-1.65 (m, 9H); MS (LC/MS, M+H.sup.+): 506.2.

##STR00145##

[0498] Example 36: Preparation of 8-(thiophen-2-ylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except thiophene-2-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.53 (dd, J=1.2, 5.0 Hz, 1H), 7.46 (dd, 1.3, 3.8 Hz, 1H), 7.06 (dd, J=3.8, 5.0 Hz, 1H), 6.99 (d, J=8.4 Hz, 2H), 6.75 (d, J=8.5 Hz, 2H), 4.46 (m, 1H), 3.49 (m, 1H), 3.24 (m, 1H), 3.12-2.97 (m, 5H), 2.92 (m, 1H), 2.61-2.41 (m, 6H), 2.19 (s, 3H), 2.14 (dd, J=6.0, 13.0 Hz, 1H), 2.04-1.88 (m, 2H), 1.88-1.73 (m, 2H), 1.73-1.57 (m, 3H); MS (LC/MS, M+H.sup.+): 504.1.

##STR00146##

[0499] Example 37: Preparation of 4-((1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-8-yl)sulfonyl)benzonitrile: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 4-cyanobenzenesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.89 (d, J=8.1 Hz, 2H), 7.85 (d, 7.8 Hz, 2H), 7.08 (d, J=8.2 Hz, 2H), 6.84 (d, J=8.5 Hz, 2H), 4.57 (m, 1H), 3.53 (m, 1H), 3.24 (m, 2H), 3.20-3.07 (m, 5H), 2.68-2.48 (m, 6H), 2.28 (s, 3H), 2.21 (dd, J=6.0, 13.0 Hz, 1H), 2.08-1.97 (m, 2H), 1.97-1.65 (m, 5H); MS (LC/MS, M+H.sup.+): 523.2.

##STR00147##

[0500] Example 38: Preparation of 8-((6-chloroimidazo[2,1-b]thiazol-5-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 6-chloroimidazo[2,1-b]thiazole-5-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.90 (d, J=4.5 Hz, 1H), 7.11-7.02 (m, 3H), 6.84 (d, J=8.6 Hz, 2H), 4.57 (m, 1H), 3.74 (m, 1H), 3.52 (m, 1H), 3.26 (m, 1H), 3.20-3.06 (m, 5H), 2.67-2.47 (m, 6H), 2.32-2.21 (m, 4H), 2.13-1.97 (m, 2H), 1.97-1.82 (m, 2H), 1.82-1.65 (m, 3H); MS (LC/MS, M+H.sup.+): 578.1.

##STR00148##

[0501] Example 39: Preparation of 8-(((methylsulfonyl)methyl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared

according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except (methylsulfonyl)methanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ¹H NMR (400 MHz, CDCl₃) δ 7.09 (d, J=8.3 Hz, 2H), 6.86 (d, J=8.6 Hz, 2H), 4.61 (m, 1H), 4.45 (s, 2H), 3.89 (m, 1H), 3.66 (m, 1H), 3.45 (m, 1H), 3.34 (m, 1H), 3.23 (s, 3H), 3.17 (t, J=4.9 Hz, 4H), 2.73-2.52 (m, 6H), 2.37 (dd, J=6.1, 12.9 Hz, 1H), 2.28 (s, 3H), 2.15-1.85 (m, 4H), 1.85-1.65 (m, 3H); MS (LC/MS, M+H^{sup.}): 514.2.

##STR00149##

[0502] Example 40: Preparation of 2-((1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-8-yl)sulfonyl)acetonitrile: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyanomethanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ¹H NMR (400 MHz, CDCl₃) δ 7.09 (d, J=8.3 Hz, 2H), 6.86 (d, J=8.5 Hz, 2H), 4.64 (m, 1H), 4.02-3.91 (m, 3H), 3.69 (m, 1H), 3.58 (m, 1H), 3.48 (m, 1H), 3.18 (t, J=4.9 Hz, 4H), 2.73-2.54 (m, 6H), 2.34 (dd, J=6.1, 13.0 Hz, 1H), 2.29 (s, 3H), 2.13-2.02 (m, 2H), 2.02-1.89 (m, 2H), 1.89-1.71 (m, 3H); MS (LC/MS, M+H^{sup.}): 461.2.

##STR00150##

[0503] Example 41: Preparation of 8-(propylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 1-propanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ¹H NMR (400 MHz, CDCl₃) δ 7.09 (d, J=8.3 Hz, 2H), 6.85 (d, J=8.4 Hz, 2H), 4.61 (m, 1H), 3.73 (m, 1H), 3.48-3.35 (m, 2H), 3.34-3.25 (m, 1H), 3.16 (t, J=4.8 Hz, 4H), 2.92 (m, 2H), 2.69-2.50 (m, 6H), 2.30 (dd, J=6.0, 12.9 Hz, 1H), 2.28 (s, 3H), 2.07-1.65 (m, 9H), 1.07 (t, J=7.4 Hz, 3H); MS (LC/MS, M+H^{sup.}): 464

##STR00151##

[0504] Example 42: Preparation of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-8-((trifluoromethyl)sulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except trifluoromethanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ¹H NMR (400 MHz, CDCl₃) δ 6.96 (d, J=8.3 Hz, 2H), 6.72 (d, J=8.5 Hz, 2H), 4.50 (m, 1H), 3.87 (m, 1H), 3.70-3.23 (b, 3H), 3.03 (t, J=4.9 Hz, 4H), 2.56-2.38 (m, 6H), 2.19 (dd, J=6.2, 12.9 Hz, 1H), 2.15 (s, 3H), 1.98-1.86 (m, 2H), 1.86-1.52 (m, 5H); MS (LC/MS, M+H^{sup.}): 490

##STR00152##

[0505] Example 43: Preparation of 8-(isopropylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 2-propanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ¹H NMR (400 MHz, CDCl₃) δ 6.95 (d, J=8.2 Hz, 2H), 6.72 (d, J=8.6 Hz, 2H), 4.47 (m, 1H), 3.67 (m, 1H), 3.44 (m, 1H), 3.35 (m, 1H), 3.25 (m, 1H), 3.12-2.94 (m, 5H), 2.58-2.36 (m, 6H), 2.18 (dd, J=6.0, 12.9 Hz, 1H), 2.14 (s, 3H), 1.92-1.71 (m, 4H), 1.71-1.45 (m, 3H); MS (LC/MS, M+H^{sup.}): 464

##STR00153##

[0506] Example 44: Preparation of 8-(cyclopropylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyclopropanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.08 (d, J=8.3 Hz, 2H), 6.85 (d, J=8.5 Hz, 2H), 4.61 (m, 1H), 3.78 (m, 1H), 3.50 (m, 1H), 3.37 (m, 1H), 3.28 (m, 1H), 3.16 (t, J=4.9 Hz, 4H), 2.70-2.50 (m, 6H), 2.36-2.25 (m, 5H), 2.10-1.98 (m, 2H), 1.98-1.65 (m, 5H), 1.23-1.11 (m, 2H), 1.07-0.95 (m, 2H); MS (LC/MS, M+H.sup.+): 462

##STR00154##

[0507] Example 45: Preparation of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-8-((3,3,3-trifluoropropyl)sulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3,3,3-trifluoropropane-1-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 6.95 (d, J=8.3 Hz, 2H), 6.72 (d, J=8.5 Hz, 2H), 4.49 (m, 1H), 3.64 (m, 1H), 3.38-3.27 (m, 2H), 3.23 (m, 1H), 3.09-2.93 (m, 6H), 2.59-2.36 (m, 8H), 2.17 (dd, J=5.9, 13.0 Hz, 1H), 2.15 (s, 3H), 1.96-1.85 (m, 2H), 1.85-1.56 (m, 5H); MS (LC/MS, M+H.sup.+): 518

##STR00155##

[0508] Example 46: Preparation of 8-(isobutylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except isobutanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.08 (d, J=8.3 Hz, 2H), 6.85 (d, J=8.5 Hz, 2H), 4.61 (m, 1H), 3.70 (m, 1H), 3.45-3.33 (m, 2H), 3.29 (m, 1H) 3.16 (t, J=4.8 Hz, 4H), 2.79 (dd, J=2.2, 6.6 Hz, 2H), 2.70-2.49 (m, 6H), 2.37-2.20 (m, 5H), 2.07-1.97 (m, 2H), 1.97-1.67 (m, 5H), 1.12 (d, J=6.7 Hz, 6H); MS (LC/MS, M+H.sup.+): 478

##STR00156##

[0509] Example 47: Preparation of 8-(cyclopentylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyclopentanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 6.95 (d, J=8.3 Hz, 2H), 6.72 (d, J=8.6 Hz, 2H), 4.47 (m, 1H), 3.64 (m, 1H), 3.42-3.22 (m, 3H), 3.18 (m, 1H) 3.03 (t, J=4.9 Hz, 4H), 2.57-2.37 (m, 6H), 2.18 (dd, J=6.1, 12.9 Hz, 1H), 2.15 (s, 3H), 1.95-1.42 (m, 15H); MS (LC/MS, M+H.sup.+): 490

##STR00157##

[0510] Example 48: Preparation of 8-(cyclohexylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyclohexanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 6.95 (d, J=8.3 Hz, 2H), 6.72 (d, J=8.4 Hz, 2H), 4.47 (m, 1H), 3.66 (m, 1H), 3.38 (m, 1H), 3.27 (m, 1H), 3.18 (m, 1H), 3.03 (t, J=4.9 Hz, 4H), 2.77 (tt, J=3.4, 12.0 Hz, 1H), 2.59-2.36 (m, 6H), 2.19 (dd, J=6.0, 12.9 Hz, 1H), 2.15 (s, 3H), 2.06-1.94 (b, 2H), 1.91-1.72 (m, 6H), 1.72-1.49 (m, 4H), 1.37 (qd, J=3.3, 12.3 Hz, 2H), 1.23-0.99 (m, 3H); MS (LC/MS, M+H.sup.+): 504

##STR00158##

[0511] Example 49: Preparation of 8-(ethylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except ethanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.08 (d, J=8.2 Hz, 2H), 6.85 (d, J=8.6 Hz, 2H), 4.61 (m, 1H), 3.75 (m, 1H), 3.51-3.35 (m, 2H), 3.30 (m, 1H), 3.15 (t, J=4.9 Hz, 4H), 2.98 (q, J=7.4 Hz, 2H), 2.70-2.48 (m, 6H), 2.31 (dd, J=6.2, 13.0 Hz, 1H), 2.28 (s, 3H), 2.07-1.96 (m, 2H), 1.96-1.66 (m, 5), 1.38 (t, J=7.4 Hz, 3H); MS (LC/MS, M+H.sup.+): 450

##STR00159##

[0512] Example 50: Preparation of 8-(pyridin-3-ylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except pyridine-3-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0% ~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 9.00 (d, J=2.3 Hz, 1H), 8.84 (dd, J=1.5, 4.8 Hz, 1H), 8.06 (dt, J=1.9, 8.0 Hz, 1H), 7.5 (dd, J=4.9, 7.9 Hz, 1H), 7.08 (d, J=8.3 Hz, 2H), 6.84 (d, J=8.5 Hz, 2H), 4.55 (m, 1H), 3.56 (m, 1H), 3.29 (m, 1H), 3.24-3.05 (m, 6H), 2.66-2.47 (m, 6H), 2.27 (s, 3H) 2.09 (dd, J=6.0, 13.0 Hz, 1H), 2.09-1.96 (m, 2H), 1.96-1.64 (m, 5H); MS (LC/MS, M+H.sup.+): 499

##STR00160##

[0513] Example 51: Preparation of 8-((1-methyl-1H-pyrazol-4-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 1-methyl-1H-pyrazole-4-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.75 (s, 1H), 7.71 (s, 1H), 7.07 (d, J=8.4 Hz, 2H), 6.84 (d, J=8.6 Hz, 2H), 4.55 (m, 1H), 3.95 (s, 3H), 3.48 (m, 1H), 3.23 (m, 1H), 3.13 (t, J=4.9 Hz, 4H), 3.02 (m, 1H), 2.93 (m, 1H), 2.67-2.48 (m, 6H), 2.27 (s, 3H), 2.22 (dd, J=6.1, 12.9 Hz, 1H), 2.11-1.96 (m, 2H), 1.96-1.64 (m, 5H); MS (LC/MS, M+H.sup.+): 502

##STR00161##

[0514] Example 52: Preparation of 8-((1H-imidazol-4-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 1H-imidazole-4-sulfonyl chloride was substituted for methanesulfonyl chloride: .sup.1H NMR (400 MHz, MeOD) δ 7.88 (s, 1H), 7.74 (s, 1H), 7.09 (d, J=8.4 Hz, 2H), 6.90 (d, J=8.4 Hz, 2H), 4.58 (b, 1H), 3.67 (m, 1H), 3.53 (m, 1H), 3.40-3.05 (m, 10H), 2.99 (m, 1H), 2.85 (m, 1H), 2.38 (dd, J=5.7, 12.9 Hz, 1H), 2.26 (s, 3H), 2.19-1.91 (m, 3H), 1.90-1.72 (m, 3H), 1.72-1.58 (m, 1H); MS (LC/MS, M+H.sup.+): 488

##STR00162##

[0515] Example 53: Preparation of 8-(furan-2-ylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except furan-2-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0% ~10%): .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.59 (s, 1H), 7.07 (d, J=8.4 Hz, 2H), 7.02 (d, J=3.4 Hz, 1H), 6.84 (d, J=8.4 Hz, 2H), 6.52 (dd, J=1.7, 3.3 Hz, 1H), 4.56 (m, 1H), 3.70 (m, 1H), 3.47 (m, 1H), 3.24 (m, 1H), 3.19-3.07 (m, 5H), 2.68-2.46 (m, 6H), 2.33-2.18 (m, 4H), 2.07-1.79 (m, 4H),

1.79-1.60 (m, 3H); MS (LC/MS, M+H.sup.+): 488

##STR00163##

[0516] Example 54: Preparation of 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A mixture of 8-benzyl-3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (3.25 g, 8.04 mmol, 1 eq.), Pd/C (0.65 g, 20% wt) and MeOH (54 mL) was stirred at RT under 1 atm of H.sub.2 (filled balloon) overnight. The mixture was filtered through a plug of Celite, washed with MeOH (50 mL) and concentrated in vacuo to give a crude oil. The crude oil (2.43 g) was dissolved in dichloromethane (50 mL), followed by addition of triethylamine (5.55 mL, 38.8 mmol, 5 eq.). Cooled to 0° C. before the addition of methanesulfonyl chloride (2.83 g, 13.0 mmol, 1.1 eq.). The reaction was allowed to warm to RT and stir for 30 min. At this point the reaction was diluted with D.I. water and then extracted with dichloromethane (3×50 mL), the combined organic phase was dried over Na.sub.2SO.sub.4 and concentrated in vacuo to give a crude product which was further purified by column chromatography (EtOAc/DCM, 0%~10%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 4.61 (m, 1H), 3.75-3.65 (m, 2H), 3.60 (m, 1H), 3.37-3.18 (m, 3H), 2.75 (s, 3H), 2.21 (dd, J=6.0, 13.0 Hz, 1H), 2.02-1.91 (m, 2H), 1.91-1.62 (m, 5H), 0.83 (s, 9H), 0.00 (s, 6H).

##STR00164##

[0517] Example 55: Preparation of 3-(2-hydroxyethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for tert-butyl 3-(2-hydroxyethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for tert-butyl 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate and the product was initially purified by column chromatography on a C18 column. (ACN/H.sub.2O, 0%~100%, w/0.1% formic acid), followed by purification on a silica gel column (MeOH/DCM, 0%~10%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 4.73 (m, 1H), 3.84 (t, J=5.5 Hz, 2H), 3.68 (m, 1H), 3.41-3.32 (m, 2H), 3.29 (m, 1H), 2.83 (s, 3H), 2.33 (dd, J=6.0, 13.0 Hz, 1H), 2.10-2.00 (m, 2H), 2.10-1.71 (m, 6H).

##STR00165##

[0518] Example 56: Preparation of 3-(2-bromoethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of 3-(2-hydroxyethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one (0.890 g, 3.21 mmol, 1 eq.) and dichloromethane (12 mL) was cooled to 0° C. before triphenylphosphine (1.26 g, 4.81 mmol, 1.5 eq.) and carbon tetrabromide (1.6 g, 4.81 mmol, 1.5 eq.) were sequentially added to the solution. The reaction solution was allowed to warm to RT and for 4 hrs. The resulting mixture was then filtered and concentrated in vacuo to give a crude mixture. This mixture was suspended in diethyl ether (50 mL) and filtered 2× using diethyl ether to wash the filter cakes. The final filtrate was loaded onto Celite in vacuo and further purified by column chromatography (EtOAc/DCM, 0%~40%). .sup.1H NMR (400 MHz, MeOD) δ 4.72 (m, 1H), 3.67 (m, 1H), 3.54 (dd, J=5.3, 7.6 Hz, 2H), 3.44-3.25 (m, 3H), 2.82 (s, 3H), 2.34 (dd, J=6.0, 12.9 Hz, 1H), 2.31-2.21 (m, 1H), 2.21-2.10 (m, 1H), 2.10-1.98 (m, 2H), 1.88-1.71 (m, 3H).

##STR00166##

[0519] Example 57: Preparation of 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of 3-(2-bromoethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one (50 mg, 0.147 mmol, 1 eq.), 1-(4-methoxyphenyl)-piperazine (59.33 mg, 0.308 mmol, 2.1 eq.) and Acetonitrile (2 mL) was microwaved for 1 hour at 120° C. The solvent was then evaporated in vacuo and the product was suspended in 15 mL of saturated NaHCO.sub.3 and extracted in dichloromethane (3×15 mL). The combined organic phase was dried over Na.sub.2SO.sub.4 and concentrated in vacuo to give a crude mixture that was then dissolved in dichloromethane and purified by column chromatography (methanol/dichloromethane, 0%~10%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.4 (d, J=9.2 Hz, 2H), 6.5 (d, J=9.2 Hz, 2H), 4.61 (m, 1H), 3.77 (s, 3H), 3.67 (m, 1H), 3.36 (m, 2H), 3.29 (m, 1H),

3.1 (t, J=7.1 Hz, 4H), 2.8 (s, 3H), 2.62 (m, 4H), 2.56 (t, J=7.1 Hz, 2H), 2.29 (dd, J=7.2, 6 Hz, 1H), 2.05 (m, 2H), 1.99-1.71 (m, 6H); MS (LC/MS, M+H.sup.+): 452.

##STR00167##

[0520] Example 58: Preparation of 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that 1-(4-trifluoromethylphenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 7.5 (d, J=8.76 Hz, 2H), 6.94 (d, J=8.7 Hz, 2H), 4.64 (m, 1H), 3.68 (m, 1H), 3.46-3.28 (m, 7H), 2.83 (s, 3H), 2.67-2.55 (m, 6H), 2.3 (dd, J=7.1, 6.1 Hz, 1H), 2.11-2.01 (m, 2H), 2.0-1.74 (m, 5H); MS (LC/MS, M+H.sup.+): 490

##STR00168##

[0521] Example 59: Preparation of 4-(4-(2-(8-(methylsulfonyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-3-yl)ethyl)piperazin-1-yl)benzonitrile: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that triethylamine (0.06 mL, 0.44 mmol, 3 eq.) was added in the microwave mixture and 1-(4-cyanophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 7.49 (d, J=8.4 Hz, 2H), 6.85 (d, J=8.4 Hz, 2H), 4.61 (m, 1H), 3.65 (m, 1H), 3.43-3.26 (m, 7H), 2.81 (s, 3H), 2.64-2.52 (m, 6H), 2.28 (dd, J=6.8, 5.9 Hz, 1H), 2.02 (m, 2H), 1.96-1.74 (m, 5H); MS (LC/MS, M+H.sup.+): 447

##STR00169##

[0522] Example 60: Preparation of 8-(methylsulfonyl)-3-(2-(4-(4-nitrophenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that it was microwaved for 1.5 hours at 120° C. and 1-(4-nitrophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 8.13 (d, J=9.3 Hz, 2H), 6.82 (d, J=9.3 Hz, 2H), 4.62 (m, 1H), 3.66 (m, 1H), 3.46-3.36 (m, 4H), 3.36-3.28 (m, 3H), 2.8 (s, 3H), 2.67-2.5 (b, 6H), 2.28 (dd, J=6.8, 5.9 Hz, 1H), 2.08-1.98 (m, 2H), 1.95-1.72 (m, 5H); MS (LC/MS, M+H.sup.+): 467

##STR00170##

[0523] Example 61: Preparation of 3-(2-(4-(4-chlorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that it was microwaved for 2 hours at 120° C. and 1-(4-chlorophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 7.19 (d, J=8.9 Hz, 2H), 6.82 (d, J=8.9 Hz, 2H), 4.6 (m, 1H), 3.65 (m, 1H), 3.42-3.26 (m, 3H), 3.15 (t, J=4.9 Hz, 4H), 2.8 (s, 3H), 2.66-2.52 (m, 6H), 2.27 (dd, J=6.8, 5.9 Hz, 1H), 2.07-1.98 (m, 2H), 1.97-1.7 (m, 5H); MS (LC/MS, M+H.sup.+): 456

##STR00171##

[0524] Example 62: Preparation of 3-(2-(4-(4-iodophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-chlorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that and 1-(4-iodophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 7.49 (d, J=8.8 Hz, 2H), 6.65 (d, J=8.8 Hz, 2H), 4.58 (m, 1H), 3.64 (m, 1H), 3.4-3.24 (m, 3H), 3.15 (t, J=4.8 Hz, 4H), 2.79 (s, 3H), 2.62-2.5 (m, 6H), 2.26 (dd, J=6.8, 6 Hz, 1H), 2.0-1.96 (m, 2H), 1.95-1.69 (m, 5H); MS (LC/MS, M+H.sup.+): 548

##STR00172##

[0525] Example 63: Preparation of 3-(2-(4-(4-fluorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and

purified according to the procedure for 3-(2-(4-(4-chlorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that and 1-(4-fluorophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 6.94 (m, 2H), 6.85 (m, 2H), 4.59 (m, 1H), 3.65 (m, 1H), 3.4-3.24 (m, 3H), 3.1 (t, J=4.8 Hz, 4H), 2.8 (s, 3H), 2.65-2.5 (m, 6H), 2.27 (dd, J=6.7, 5.9 Hz, 1H), 2.0 (m, 2H), 1.97-1.69 (m, 5H); MS (LC/MS, M+H⁺): 440.

##STR00173##

[0526] Preparation of 3-morpholino-4-nitrobenzonitrile: A solution of 3-fluoro-4-nitrobenzonitrile (0.6 g, 3.61 mmol, 1 eq.) and morpholine (0.629 g, 7.22 mmol, 2 eq.) in dimethyl sulfoxide (6.57 mL) was heated at 60° C. for 4 hours. The reaction solution was diluted with water 20 mL and extracted with ethyl acetate (3×20 mL). The combined organic phase was dried over Na₂SO₄ and concentrated in vacuo to give a crude product which was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 7.82 (d, J=8.3 Hz, 1H), 7.40 (d, J=1.5 Hz, 1H), 7.33 (dd, J=1.6, 8.3 Hz, 1H), 3.86 (m, 4H), 3.11 (m, 4H); MS (LC/MS, M+H⁺): 234

##STR00174##

[0527] Preparation of 4-(5-methyl-2-nitrophenyl)morpholine: The title compound was prepared according to the procedure for 3-morpholino-4-nitrobenzonitrile, except 2-fluoro-4-methyl-1-nitrobenzene was substituted for 3-fluoro-4-nitrobenzonitrile. ¹H NMR (400 MHz, CDCl₃) δ 7.76 (d, J=8.2 Hz, 1H), 6.93 (b, 1H), 6.88 (d, J=8.3 Hz, 1H), 3.86 (m, 4H), 3.06 (m, 4H), 2.40 (s, 3H); MS (LC/MS, M+H⁺): 223

##STR00175##

[0528] Preparation of 3-morpholino-4-nitrophenol: The title compound was prepared according to the procedure for 3-morpholino-4-nitrobenzonitrile, except 3-fluoro-4-nitrophenol was substituted for 3-fluoro-4-nitrobenzonitrile. ¹H NMR (400 MHz, MeOD) δ 7.90 (d, J=9.0 Hz, 1H), 6.56 (d, J=2.4 Hz, 1H), 6.49 (dd, J=2.5, 9.0 Hz, 1H), 3.83 (m, 4H), 3.02 (m, 4H); MS (LC/MS, M+H⁺): 225

##STR00176##

[0529] Preparation of 4-(2-nitro-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine: A solution of 3-morpholino-4-nitrophenol (1.34 g, 5.98 mmol, 1 eq.), 2-(trimethylsilyl)ethoxymethyl chloride (1.05 g, 6.28 mmol, 1.05 eq.) and N,N-diisopropylethylamine (2.31 g, 17.9 mmol, 3 eq.) in dichloromethane (30.0 mL) was stirred at 25° C. for 16 hours. The reaction solution was diluted with 40 mL of water and extracted with dichloromethane (3×40 mL). The combined organic phase was dried over Na₂SO₄ and concentrated in vacuo to give a crude product which was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 7.93 (d, J=9.7 Hz, 1H), 6.68 (m, 2H), 5.24 (s, 2H), 3.85 (m, 4H), 3.74 (m, 2H), 3.04 (m, 4H), 0.94 (m, 2H), 0.00 (s, 9H); MS (LC/MS, M+H⁺): 355

##STR00177##

[0530] Preparation of 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine: A mixture of 4-(5-methyl-2-nitrophenyl)morpholine (1.58 g, 7.11 mmol, 1 eq.), Pd on carbon (316 mg, 20% wt) and methanol (72 mL) was stirred at 25° C. under 1 atm of H₂ (filled balloon) for 48 hours. The mixture was filtered through a plug of Celite, washed with methanol (50 mL) and concentrated in vacuo to give the crude intermediate, 4-methyl-2-morpholinoaniline.

[0531] (((4-nitrophenyl)sulfonyl)azanediyl)bis(ethane-2,1-diyl) bis(4-nitrobenzenesulfonate) (1.0 g, 1.5 mmol, 1 eq.), 4-methyl-2-morpholinoaniline (0.346 g, 1.8 mmol, 1.2 eq.), N,N-diisopropylethylamine (1.55 g, 12.0 mmol, 4 eq.) and acetonitrile (4.7 mL) were mixed in a microwave reaction vial (10 mL) fitted with a no-invasive vial cap. The reaction vials containing the mixture were reacted in the microwave for 1 h at 175° C. After 1 h, the solvent was removed under reduced pressure. The residue was dissolved in dichloromethane and washed with HCl (10%,

3×30 mL) and saturated NaHCO₃ (40 mL). The organic phase was dried over Na₂SO₄ and concentrated in vacuo to afford the crude product. The title compound was purified by column chromatography on a silica gel column. (ethyl acetate/dichloromethane, 0%~10%): ¹H NMR (400 MHz, CDCl₃) δ 8.44 (d, J=8.9 Hz, 2H), 8.02 (d, J=9.0 Hz, 2H), 6.82 (m, 2H), 6.74 (b, 1H), 3.71 (t, J=4.2 Hz, 4H), 3.24 (b, 8H), 3.05 (m, 4H), 2.29 (s, 3H); MS (LC/MS, M+H^{sup.}.): 447

##STR00178##

[0532] Preparation of 3-morpholino-4-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)benzonitrile: The title compound was prepared according to the procedure for 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine, except 3-morpholino-4-nitrobenzonitrile was substituted for 4-(5-methyl-2-nitrophenyl)morpholine and 4-amino-3-morpholinobenzonitrile for 4-methyl-2-morpholinoaniline. ¹H NMR (400 MHz, CDCl₃) δ 8.43 (d, J=8.8 Hz, 2H), 8.01 (d, J=8.8 Hz, 2H), 7.30 (dd, J=1.8, 8.2 Hz, 1H), 7.14 (d, J=1.8 Hz, 1H), 6.90 (d, J=8.3 Hz, 1H), 3.73 (t, J=4.5 Hz, 4H), 3.36 (m, 4H), 3.26 (m, 4H), 3.02 (t, J=4.3 Hz, 4H); MS (LC/MS, M+H^{sup.}.): 458

##STR00179##

[0533] Preparation of 4-(2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine: The title compound was prepared according to the procedure for 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine, except 4-(2-nitro-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine was substituted for 4-(5-methyl-2-nitrophenyl)morpholine and 2-morpholino-4-((2-(trimethylsilyl)ethoxy)methoxy)aniline for 4-methyl-2-morpholinoaniline. ¹H NMR (400 MHz, CDCl₃) δ 8.41 (d, J=8.8 Hz, 2H), 8.01 (d, J=8.7 Hz, 2H), 6.80 (d, J=8.7 Hz, 1H), 6.67 (dd, J=2.7, 8.7 Hz, 1H), 6.60 (d, J=2.6 Hz, 1H), 5.14 (s, 2H), 3.74 (m, 2H), 3.68 (t, J=4.4 Hz, 4H), 3.23 (b, 4H), 3.16 (b, 4H), 3.03 (b, 4H), 0.95 (m, 2H), 0.00 (s, 9H); MS (LC/MS, M+H^{sup.}.): 579

##STR00180##

[0534] Preparation of 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine: Potassium carbonate (1.5 g, 10.8 mmol 12 eq.) was added to a mixture of acetonitrile and dimethylsulfoxide (CH₃CN/DMSO 49:1, 2.4 mL) and heated to 50° C. Thiophenol (0.988 g, 8.96 mmol, 10 eq.) was added dropwise via syringe to the mixture with stirring. After 30 minutes a solution of 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine (0.410 g, 0.896 mmol, 1 eq.) in acetonitrile and dimethyl sulfoxide (acetonitrile/dimethyl sulfoxide 49:1, 4.5 mL) was added dropwise. The reaction mixture was stirred for 3 hours, quenched with excess NaOH solution (40%) and concentrated under reduced pressure. The residue was extracted with dichloromethane (5×30 mL) and the organic phase was dried over MgSO₄, and concentrated in vacuo to give a crude oil. The oil was purified by reverse phase chromatography (acetonitrile in H₂O, gradient from 1%~100% with 0.1% formic acid) to afford the formic acid salt of the desired piperazine. The salt was dissolved in dichloromethane, washed with saturated NaHCO₃ solution, and the organic phase concentrated in vacuo to provide the product. ¹H NMR (400 MHz, CDCl₃) δ 6.87-6.76 (m, 2H), 6.71 (s, 1H), 3.84 (t, J=4.5 Hz, 4H), 3.18 (b, 4H), 3.07 (b, 4H), 2.98 (b, 4H), 2.29 (s, 3H); MS (LC/MS, M+H^{sup.}.): 262

##STR00181##

[0535] Preparation of 3-morpholino-4-(piperazin-1-yl)benzonitrile: The title compound was prepared according to the procedure for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine, except 3-morpholino-4-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)benzonitrile was substituted for 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine. ¹H NMR (400 MHz, CDCl₃) δ 7.24 (dd, J=1.8, 8.2 Hz, 1H), 7.08 (d, J=1.8 Hz, 1H), 6.88 (d, J=8.4 Hz, 1H), 3.81 (t, J=4.6 Hz, 4H), 3.21 (b, 4H), 3.11 (b, 4H), 3.00 (b, 4H); MS (LC/MS, M+H^{sup.}.): 273

##STR00182##

[0536] Preparation of 4-(2-(piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)

phenyl)morpholine: The title compound was prepared according to the procedure for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine, except 4-(2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine was substituted for 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine. ¹H NMR (400 MHz, CDCl₃) δ 6.83 (d, J=8.6 Hz, 1H), 6.67 (dd, J=2.7, 8.5 Hz, 1H), 6.59 (d, J=2.8 Hz, 1H), 3.82 (t, J=4.7 Hz, 4H), 3.74 (m, 2H), 3.18 (b, 4H), 3.10-2.92 (b, 8H), 0.95 (m, 2H), 0.00 (s, 9H); MS (LC/MS, M+H^{sup.}.): 394

##STR00183##

[0537] Preparation of tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A solution of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.05 g, 0.138 mmol, 1 eq.), 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine (0.044 g, 0.166 mmol, 1.2 eq.), and triethylamine (0.070 g, 0.69 mmol, 5 eq.) in acetonitrile (2 mL) was microwaved at 120° C. for 1 hour. The reaction mixture was filtered and the filtrate was concentrated in vacuo to give a crude product. The title compound was purified by column chromatography on a silica gel column. (methanol/dichloromethane, 0%~10%) ¹H NMR (400 MHz, CDCl₃) δ 6.87-6.76 (m, 2H), 6.71 (s, 1H), 4.59 (m, 1H), 3.98 (m, 1H), 3.90-3.73 (m, 5H), 3.31-2.94 (m, 10H), 2.70-2.47 (m, 6H), 2.39 (dd, J=6.2, 12.8 Hz, 1H), 2.28 (s, 3H), 2.03-1.70 (m, 5H), 1.60 (m, 1H), 1.55-1.38 (m, 10H); MS (LC/MS, M+H^{sup.}.): 543

##STR00184##

[0538] Preparation of tert-butyl 3-(2-(4-(4-cyano-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except 3-morpholino-4-(piperazin-1-yl)benzonitrile was substituted for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine. ¹H NMR (400 MHz, CDCl₃) δ 7.26 (dd, J=1.8, 8.3 Hz, 1H), 7.09 (d, J=1.6 Hz, 1H), 6.89 (d, J=8.3 Hz, 1H), 4.57 (m, 1H), 3.96 (m, 1H), 3.89-3.73 (m, 5H), 3.40-2.98 (m, 10H), 2.72-2.45 (m, 6H), 2.37 (dd, J=6.1, 12.8 Hz, 1H), 2.03-1.67 (m, 5H), 1.58 (m, 1H), 1.54-1.38 (m, 10H); MS (LC/MS, M+H^{sup.}.): 554

##STR00185##

[0539] Preparation of 8-(methylsulfonyl)-3-(2-(4-(2-morpholino-4-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except 4-(2-(piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine was substituted for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine and 3-(2-bromoethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one for tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate. ¹H NMR (400 MHz, CDCl₃) δ 6.83 (d, J=8.6 Hz, 1H), 6.66 (dd, J=2.7, 8.6 Hz, 1H), 6.59 (d, J=2.7 Hz, 1H), 5.14 (s, 2H), 4.61 (m, 1H), 3.81 (t, J=4.4 Hz, 4H), 3.74 (m, 2H), 3.66 (m, 1H), 3.35 (m, 2H), 3.26 (m, 1H), 3.21-2.94 (b, 8H), 2.80 (s, 3H), 2.70-2.41 (m, 6H), 2.30 (dd, J=5.6, 12.9 Hz, 1H), 2.08-1.67 (m, 7H), 0.95 (m, 2H), 0.00 (s, 9H); MS (LC/MS, M+H^{sup.}.): 653

##STR00186##

[0540] Preparation of 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.077 g, 0.142 mmol, 1 eq.) in trifluoroacetic acid:dichloromethane (1:3, 2 mL) was allowed to stir at 25° C. for 30 minutes. The reaction solution was diluted with methanol (2 mL) and concentrated in vacuo to give a crude intermediate as a trifluoroacetic acid salt. The resulting material was dissolved in dichloromethane (10 mL) and washed with sat. NaHCO₃(aq.) solution (10 mL). The aqueous layer was backwashed with dichloromethane (2×10 mL) and the combined organic layers were dried over Na₂SO₄ and concentrated in vacuo to give a crude intermediate as

a free base.

[0541] The resulting free base (0.052 g, 0.114 mmol, 1 eq.) was dissolved in dichloromethane (2 mL) and cooled to 0° C., and trimethylamine (0.058 g, 0.57 mmol, 5 eq.) and methanesulfonyl chloride (0.026, 0.228 mmol, 2 eq.) were added. The reaction solution was allowed to stir at 25° C. for 30 minutes and then concentrated in vacuo to give a crude solid. The title compound was purified by column chromatography on a silica gel column. (methanol/dichloromethane, 0%~10%) .sup.1H NMR (400 MHz, CDCl.sub.3) δ 6.79-6.68 (m, 2H), 6.64 (s, 1H), 4.54 (m, 1H), 3.75 (t, J=4.4 Hz, 4H), 3.59 (m, 1H), 3.37-2.88 (b, 11H), 2.74 (s, 3H), 2.69-2.33 (b, 6H), 2.27-2.16 (m, 4H), 2.02-1.92 (m, 2H), 1.92-1.62 (m, 5H); MS (LC/MS, M+H.sup.+): 521

##STR00187##

[0542] Preparation of 4-(4-(2-(8-(methylsulfonyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-3-yl)ethyl)piperazin-1-yl)-3-morpholinobenzonitrile: The title compound was prepared according to the procedure for 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one, except tert-butyl 3-(2-(4-(4-cyano-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate was substituted for 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 7.29 (dd, J=1.8, 8.3 Hz, 1H), 7.12 (d, J=1.7 Hz, 1H), 6.91 (d, J=8.3 Hz, 1H), 4.63 (m, 1H), 3.85 (t, J=4.3 Hz, 4H), 3.67 (m, 1H), 3.55-2.96 (b, 11H), 2.83 (s, 3H), 2.75-2.40 (b, 6H), 2.30 (dd, J=6.0, 12.9 Hz, 1H), 2.12-1.99 (m, 2H), 1.99-1.71 (m, 5H); MS (LC/MS, M+H.sup.+): 532

##STR00188##

[0543] Preparation of 3-(2-(4-(4-hydroxy-2-morpholinophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: To a small vial, 8-(methylsulfonyl)-3-(2-(4-(2-morpholino-4-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (0.190 g, 0.291 mmol, 1 eq.) was added and dissolved in hexamethylphosphoramide (1.5 mL). Tetra-n-butylammonium fluoride trihydrate (0.230 g, 0.873 mmol, 3 eq.) was added, followed by 300 mg of 4 Å molecular sieves. The reaction mixture was stirred at 60° C. for 48 hours before being filtered and concentrated in vacuo to give a crude oil. The title compound was purified by column chromatography on a silica gel column. (methanol/dichloromethane, 0%~10%) .sup.1H NMR (400 MHz, CDCl.sub.3) δ 6.70 (d, J=9.2 Hz, 1H), 6.40-6.32 (m, 2H), 4.53 (m, 1H), 3.75 (t, J=4.3 Hz, 4H), 3.58 (m, 1H), 3.38-2.84 (b, 11H), 2.74 (s, 3H), 2.70-2.33 (b, 6H), 2.21 (dd, J=5.9, 12.8 Hz, 1H), 2.01-1.80 (m, 4H), 1.79-1.60 (m, 3H); MS (LC/MS, M+H.sup.+): 523.

Formulations

[0544] The present invention also relates to compositions or formulations which comprise the 5-hydroxytryptamine receptor 7 activity modulators according to the present invention. In general, the compositions of the present invention comprise an effective amount of one or more compounds of the disclosure and salts thereof according to the present invention which are effective for providing modulation of 5-hydroxytryptamine receptor 7 activity; and one or more excipients.

[0545] For the purposes of the present invention the term “excipient” and “carrier” are used interchangeably throughout the description of the present invention and said terms are defined herein as, “ingredients which are used in the practice of formulating a safe and effective pharmaceutical composition.”

[0546] The formulator will understand that excipients are used primarily to serve in delivering a safe, stable, and functional pharmaceutical, serving not only as part of the overall vehicle for delivery but also as a means for achieving effective absorption by the recipient of the active ingredient. An excipient may fill a role as simple and direct as being an inert filler, or an excipient as used herein may be part of a pH stabilizing system or coating to insure delivery of the ingredients safely to the stomach. The formulator can also take advantage of the fact the compounds of the present invention have improved cellular potency, pharmacokinetic properties, as

well as improved oral bioavailability.

[0547] The present teachings also provide pharmaceutical compositions that include at least one compound described herein and one or more pharmaceutically acceptable carriers, excipients, or diluents. Examples of such carriers are well known to those skilled in the art and can be prepared in accordance with acceptable pharmaceutical procedures, such as, for example, those described in *Remington's Pharmaceutical Sciences*, 17th edition, ed. Alfonso R. Gennaro, Mack Publishing Company, Easton, PA (1985), the entire disclosure of which is incorporated by reference herein for all purposes. As used herein, "pharmaceutically acceptable" refers to a substance that is acceptable for use in pharmaceutical applications from a toxicological perspective and does not adversely interact with the active ingredient. Accordingly, pharmaceutically acceptable carriers are those that are compatible with the other ingredients in the formulation and are biologically acceptable.

Supplementary active ingredients can also be incorporated into the pharmaceutical compositions.

[0548] Compounds of the present teachings can be administered orally or parenterally, neat or in combination with conventional pharmaceutical carriers. Applicable solid carriers can include one or more substances which can also act as flavoring agents, lubricants, solubilizers, suspending agents, fillers, glidants, compression aids, binders or tablet-disintegrating agents, or encapsulating materials. The compounds can be formulated in conventional manner, for example, in a manner similar to that used for known 5-hydroxytryptamine receptor 7 activity modulators. Oral formulations containing a compound disclosed herein can comprise any conventionally used oral form, including tablets, capsules, buccal forms, troches, lozenges and oral liquids, suspensions or solutions. In powders, the carrier can be a finely divided solid, which is an admixture with a finely divided compound. In tablets, a compound disclosed herein can be mixed with a carrier having the necessary compression properties in suitable proportions and compacted in the shape and size desired. The powders and tablets can contain up to 99% of the compound.

[0549] Capsules can contain mixtures of one or more compound(s) disclosed herein with inert filler(s) and/or diluent(s) such as pharmaceutically acceptable starches (e.g., corn, potato or tapioca starch), sugars, artificial sweetening agents, powdered celluloses (e.g., crystalline and microcrystalline celluloses), flours, gelatins, gums, and the like.

[0550] Useful tablet formulations can be made by conventional compression, wet granulation or dry granulation methods and utilize pharmaceutically acceptable diluents, binding agents, lubricants, disintegrants, surface modifying agents (including surfactants), suspending or stabilizing agents, including, but not limited to, magnesium stearate, stearic acid, sodium lauryl sulfate, talc, sugars, lactose, dextrin, starch, gelatin, cellulose, methyl cellulose, microcrystalline cellulose, sodium carboxymethyl cellulose, carboxymethylcellulose calcium, polyvinylpyrrolidone, alginic acid, acacia gum, xanthan gum, sodium citrate, complex silicates, calcium carbonate, glycine, sucrose, sorbitol, dicalcium phosphate, calcium sulfate, lactose, kaolin, mannitol, sodium chloride, low melting waxes, and ion exchange resins. Surface modifying agents include nonionic and anionic surface modifying agents. Representative examples of surface modifying agents include, but are not limited to, poloxamer 188, benzalkonium chloride, calcium stearate, cetostearyl alcohol, cetomacrogol emulsifying wax, sorbitan esters, colloidal silicon dioxide, phosphates, sodium dodecylsulfate, magnesium aluminum silicate, and triethanolamine. Oral formulations herein can utilize standard delay or time-release formulations to alter the absorption of the compound(s). The oral formulation can also consist of administering a compound disclosed herein in water or fruit juice, containing appropriate solubilizers or emulsifiers as needed.

[0551] Liquid carriers can be used in preparing solutions, suspensions, emulsions, syrups, elixirs, and for inhaled delivery. A compound of the present teachings can be dissolved or suspended in a pharmaceutically acceptable liquid carrier such as water, an organic solvent, or a mixture of both, or a pharmaceutically acceptable oils or fats. The liquid carrier can contain other suitable pharmaceutical additives such as solubilizers, emulsifiers, buffers, preservatives, sweeteners, flavoring agents, suspending agents, thickening agents, colors, viscosity regulators, stabilizers, and

osmo-regulators. Examples of liquid carriers for oral and parenteral administration include, but are not limited to, water (particularly containing additives as described herein, e.g., cellulose derivatives such as a sodium carboxymethyl cellulose solution), alcohols (including monohydric alcohols and polyhydric alcohols, e.g., glycols) and their derivatives, and oils (e.g., fractionated coconut oil and *arachis* oil). For parenteral administration, the carrier can be an oily ester such as ethyl oleate and isopropyl myristate. Sterile liquid carriers are used in sterile liquid form compositions for parenteral administration. The liquid carrier for pressurized compositions can be halogenated hydrocarbon or other pharmaceutically acceptable propellants.

[0552] Liquid pharmaceutical compositions, which are sterile solutions or suspensions, can be utilized by, for example, intramuscular, intraperitoneal or subcutaneous injection. Sterile solutions can also be administered intravenously. Compositions for oral administration can be in either liquid or solid form.

[0553] Preferably the pharmaceutical composition is in unit dosage form, for example, as tablets, capsules, powders, solutions, suspensions, emulsions, granules, or suppositories. In such form, the pharmaceutical composition can be sub-divided in unit dose(s) containing appropriate quantities of the compound. The unit dosage forms can be packaged compositions, for example, packeted powders, vials, ampoules, prefilled syringes or sachets containing liquids. Alternatively, the unit dosage form can be a capsule or tablet itself, or it can be the appropriate number of any such compositions in package form. Such unit dosage form can contain from about 1 mg/kg of compound to about 500 mg/kg of compound, and can be given in a single dose or in two or more doses. Such doses can be administered in any manner useful in directing the compound(s) to the recipient's bloodstream, including orally, via implants, parenterally (including intravenous, intraperitoneal and subcutaneous injections), rectally, vaginally, and transdermally.

[0554] When administered for the treatment or inhibition of a particular disease state or disorder, it is understood that an effective dosage can vary depending upon the particular compound utilized, the mode of administration, and severity of the condition being treated, as well as the various physical factors related to the individual being treated. In therapeutic applications, a compound of the present teachings can be provided to a patient already suffering from a disease in an amount sufficient to cure or at least partially ameliorate the symptoms of the disease and its complications. The dosage to be used in the treatment of a specific individual typically must be subjectively determined by the attending physician. The variables involved include the specific condition and its state as well as the size, age and response pattern of the patient.

[0555] In some cases it may be desirable to administer a compound directly to the airways of the patient, using devices such as, but not limited to, metered dose inhalers, breath-operated inhalers, multidose dry-powder inhalers, pumps, squeeze-actuated nebulized spray dispensers, aerosol dispensers, and aerosol nebulizers. For administration by intranasal or intrabronchial inhalation, the compounds of the present teachings can be formulated into a liquid composition, a solid composition, or an aerosol composition. The liquid composition can include, by way of illustration, one or more compounds of the present teachings dissolved, partially dissolved, or suspended in one or more pharmaceutically acceptable solvents and can be administered by, for example, a pump or a squeeze-actuated nebulized spray dispenser. The solvents can be, for example, isotonic saline or bacteriostatic water. The solid composition can be, by way of illustration, a powder preparation including one or more compounds of the present teachings intermixed with lactose or other inert powders that are acceptable for intrabronchial use, and can be administered by, for example, an aerosol dispenser or a device that breaks or punctures a capsule encasing the solid composition and delivers the solid composition for inhalation. The aerosol composition can include, by way of illustration, one or more compounds of the present teachings, propellants, surfactants, and co-solvents, and can be administered by, for example, a metered device. The propellants can be a chlorofluorocarbon (CFC), a hydrofluoroalkane (HFA), or other propellants that are physiologically and environmentally acceptable.]

[0556] Compounds described herein can be administered parenterally or intraperitoneally. Solutions or suspensions of these compounds or a pharmaceutically acceptable salts, hydrates, or esters thereof can be prepared in water suitably mixed with a surfactant such as hydroxyl-propylcellulose. Dispersions can also be prepared in glycerol, liquid polyethylene glycols, and mixtures thereof in oils. Under ordinary conditions of storage and use, these preparations typically contain a preservative to inhibit the growth of microorganisms.

[0557] The pharmaceutical forms suitable for injection can include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. In some embodiments, the form can be sterile and its viscosity permits it to flow through a syringe. The form preferably is stable under the conditions of manufacture and storage and can be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (e.g., glycerol, propylene glycol and liquid polyethylene glycol), suitable mixtures thereof, and vegetable oils.

[0558] Compounds described herein can be administered transdermally, i.e., administered across the surface of the body and the inner linings of bodily passages including epithelial and mucosal tissues. Such administration can be carried out using the compounds of the present teachings including pharmaceutically acceptable salts, hydrates, or esters thereof, in lotions, creams, foams, patches, suspensions, solutions, and suppositories (rectal and vaginal).

[0559] Transdermal administration can be accomplished through the use of a transdermal patch containing a compound, such as a compound disclosed herein, and a carrier that can be inert to the compound, can be non-toxic to the skin, and can allow delivery of the compound for systemic absorption into the blood stream via the skin. The carrier can take any number of forms such as creams and ointments, pastes, gels, and occlusive devices. The creams and ointments can be viscous liquid or semisolid emulsions of either the oil-in-water or water-in-oil type. Pastes comprised of absorptive powders dispersed in petroleum or hydrophilic petroleum containing the compound can also be suitable. A variety of occlusive devices can be used to release the compound into the blood stream, such as a semi-permeable membrane covering a reservoir containing the compound with or without a carrier, or a matrix containing the compound. Other occlusive devices are known in the literature.

[0560] Compounds described herein can be administered rectally or vaginally in the form of a conventional suppository. Suppository formulations can be made from traditional materials, including cocoa butter, with or without the addition of waxes to alter the suppository's melting point, and glycerin. Water-soluble suppository bases, such as polyethylene glycols of various molecular weights, can also be used.

[0561] Lipid formulations or nanocapsules can be used to introduce compounds of the present teachings into host cells either in vitro or in vivo. Lipid formulations and nanocapsules can be prepared by methods known in the art.

[0562] To increase the effectiveness of compounds of the present teachings, it can be desirable to combine a compound with other agents effective in the treatment of the target disease. For example, other active compounds (i.e., other active ingredients or agents) effective in treating the target disease can be administered with compounds of the present teachings. The other agents can be administered at the same time or at different times than the compounds disclosed herein.

[0563] Compounds of the present teachings can be useful for the treatment or inhibition of a pathological condition or disorder in a mammal, for example, a human subject. The present teachings accordingly provide methods of treating or inhibiting a pathological condition or disorder by providing to a mammal a compound of the present teachings including its pharmaceutically acceptable salt) or a pharmaceutical composition that includes one or more compounds of the present teachings in combination or association with pharmaceutically acceptable carriers. Compounds of the present teachings can be administered alone or in combination with other

therapeutically effective compounds or therapies for the treatment or inhibition of the pathological condition or disorder.

[0564] Non-limiting examples of compositions according to the present invention include from about 0.001 mg to about 1000 mg of one or more compounds of the disclosure according to the present invention and one or more excipients; from about 0.01 mg to about 100 mg of one or more compounds of the disclosure according to the present invention and one or more excipients; and from about 0.1 mg to about 10 mg of one or more compounds of the disclosure according to the present invention; and one or more excipients.

Procedures

[0565] The following procedures can be utilized in evaluating and selecting compounds as 5-hydroxytryptamine receptor 7 activity modulators. [0566] Radiolabel Binding Studies for Serotonin 5HT.sub.7 receptors, method 1:

[0567] A solution of the compound of the disclosure to be tested is prepared as a 1-mg/ml stock in Assay Buffer or DMSO according to its solubility. A similar stock of the reference compound chlorpromazine is also prepared as a positive control. Eleven dilutions (5×assay concentration) of the compound of the disclosure and chlorpromazine are prepared in the Assay Buffer by serial dilution to yield final corresponding assay concentrations ranging from 10 pM to 10 μM.

[0568] A stock concentration of 5 nM [³H]LSD (lysergic acid diethyl amide) is prepared in 50 mM Tris-HCl, 10 mM MgCl.sub.2, 1 mM EDTA, pH 7.4 (Assay Buffer). Aliquots (50 μl) of radioligand are dispensed into the wells of a 96-well plate containing 100 μl of Assay Buffer. Duplicate 50-μl aliquots of the compound of the disclosure test and chlorpromazine positive control reference compound serial dilutions are added.

[0569] Membrane fractions of cells expressing recombinant 5HT.sub.7 receptors (50 L) are dispensed into each well. The membranes are prepared from stably transfected cell lines expressing 5HT.sub.7 receptors cultured on 10-cm plates by harvesting PBS-rinsed monolayers, resuspending and lysing in chilled, hypotonic 50 mM Tris-HCl, pH 7.4, centrifuging at 20,000×g, decanting the supernatant and storing at −80° C.; the membrane preparations are resuspended in 3 ml of chilled Assay Buffer and homogenized by several passages through a 26 gauge needle before using in the assay.

[0570] The 250-μl reactions are incubated at room temperature for 1.5 hours, then harvested by rapid filtration onto 0.3% polyethyleneimine-treated, 96-well filter mats using a 96-well Filtermate harvester. Four rapid 500-μl washes are performed with chilled Assay Buffer to reduce non-specific binding. The filter mats are dried, then scintillant is added to the filters and the radioactivity retained on the filters is counted in a Microbeta scintillation counter.

[0571] Raw data (dpm) representing total radioligand binding (i.e., specific+non-specific binding) are plotted as a function of the logarithm of the molar concentration of the competitor (i.e., test or reference compound). Non-linear regression of the normalized (i.e., percent radioligand binding compared to that observed in the absence of test or reference compound) raw data is performed in Prism 4.0 (GraphPad Software) using the built-in three parameter logistic model describing ligand competition binding to radioligand-labeled sites:

$$y = \text{bottom} + \frac{(\text{top} - \text{bottom})}{(1 + 10^{\times -\log IC_{\text{sub.50}}})}$$

where bottom equals the residual radioligand binding measured in the presence of 10 M reference compound (i.e., non-specific binding) and top equals the total radioligand binding observed in the absence of competitor. The log IC.sub.50 (i.e., the log of the ligand concentration that reduces radioligand binding by 50%) is thus estimated from the data and used to obtain the K_i by applying the Cheng-Prusoff approximation:

$$[00001] K_i = IC_{50} / (1 + [\text{ligand}] / K_D)$$

where [ligand] equals the assay radioligand concentration and K_D equals the affinity constant of the radioligand for the target receptor.

[0572] Compounds of the disclosure are also screened at a single concentration of 10 μ M using the same method described for the Radiolabel Binding Studies for Serotonin 5HT.sub.7 receptors to determine the percent inhibition of [.sup.3H]LSD binding.

Radiolabel Binding Studies for Serotonin 5-HT₇ Receptors, Method 2:

[0573] A solution of the compound of the disclosure to be tested is prepared as a 1-mg/ml stock in Assay Buffer or DMSO according to its solubility. A similar stock of the reference compound chlorpromazine is also prepared as a positive control. Eleven dilutions (5 $\times$ assay concentration) of the compound of the disclosure and chlorpromazine are prepared in the Assay Buffer by serial dilution to yield final corresponding assay concentrations ranging from 10 pM to 10 μ M.

[0574] A stock concentration of 5 nM [.sup.3H]-5-Hydroxytryptamine ([.sup.3H]-5HT) is prepared in 50 mM Tris-HCl, 10 mM MgCl.sub.2, 1 mM EDTA, pH 7.4 (Assay Buffer). Aliquots (50 μ l) of radioligand are dispensed into the wells of a 96-well plate containing 100 μ l of Assay Buffer. Duplicate 50- μ l aliquots of the compound of the disclosure test and chlorpromazine positive control reference compound serial dilutions are added.

[0575] Membrane fractions of cells expressing recombinant 5HT.sub.7 receptors (50 L) are dispensed into each well. The membranes are prepared from stably transfected cell lines expressing 5HT.sub.7 receptors cultured on 10-cm plates by harvesting PBS-rinsed monolayers, resuspending and lysing in chilled, hypotonic 50 mM Tris-HCl, pH 7.4, centrifuging at 20,000 $\times$ g, decanting the supernatant and storing at -80 $^{\circ}$ C.; the membrane preparations are resuspended in 3 ml of chilled Assay Buffer and homogenized by several passages through a 26 gauge needle before using in the assay.

[0576] The 250- μ l reactions are incubated at room temperature for 1.5 hours, then harvested by rapid filtration onto 0.3% polyethyleneimine-treated, 96-well filter mats using a 96-well Filtermate harvester. Four rapid 500- μ l washes are performed with chilled Assay Buffer to reduce non-specific binding. The filter mats are dried, then scintillant is added to the filters and the radioactivity retained on the filters is counted in a Microbeta scintillation counter.

[0577] Raw data (dpm) representing total radioligand binding (i.e., specific+non-specific binding) are plotted as a function of the logarithm of the molar concentration of the competitor (i.e., test or reference compound). Non-linear regression of the normalized (i.e., percent radioligand binding compared to that observed in the absence of test or reference compound) raw data is performed in Prism 4.0 (GraphPad Software) using the built-in three parameter logistic model describing ligand competition binding to radioligand-labeled sites:

$$[00002] y = \text{bottom} + [(\text{top} - \text{bottom}) / (1 + 10^{x - \log IC_{50}})]$$

where bottom equals the residual radioligand binding measured in the presence of 10 μ M reference compound (i.e., non-specific binding) and top equals the total radioligand binding observed in the absence of competitor. The log IC.sub.50 (i.e., the log of the ligand concentration that reduces radioligand binding by 50%) is thus estimated from the data and used to obtain the K_i by applying the Cheng-Prusoff approximation:

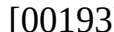
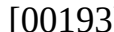
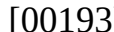
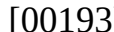
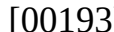
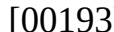
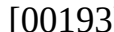
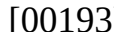
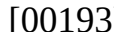
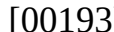
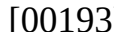
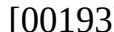
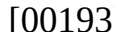
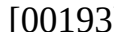
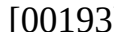
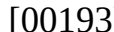
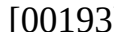
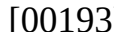
$$[00003] K_i = IC_{50} / (1 + [\text{ligand}] / K_D)$$

where [ligand] equals the assay radioligand concentration and K_D equals the affinity constant of the radioligand for the target receptor.

[0578] Compounds of the disclosure are also screened at a single concentration of 10 μ M using the same method described for the Radiolabel Binding Studies for Serotonin 5HT.sub.7 receptors to determine the percent inhibition of [.sup.3H]-5HT binding.

[0579] Results for representative compounds according to the present invention are listed in Table 11.

TABLE-US-00011 TABLE 11 Radiolabel Binding Studies for Serotonin 5HT.sub.7 receptors results for exemplary compounds of the disclosure 5-HT.sub.7 5-HT₇ % inhib IC.sub.50 Entry Structure @10 uM (nm) 1 [00189] 96.5 18 2 [00190] 88.8 149 3 [00191] 90 81 4 [00192] 88.5 122 5

[00193]  91.6 102 6 [00194]  90.5 93 7 [00195]  N.D. 34 8 [00196]  100 46 9 [00197]  N.D. 303 10 [00198]  N.D. 64 11 [00199]  N.D. 77 12 [00200]  N.D. 108 13 [00201]  N.D. 96 14 [00202]  N.D. 47 15 [00203]  N.D. 131 16 [00204]  N.D. 56 17 [00205]  N.D. 116 18 [00206]  N.D. 86 19 [00207]  N.D. 461 20 [00208]  N.D. 67 21 [00209]  N.D. 75 22 [00210]  N.D. 202 N.D. = not determined

Functional Serotonin 5HT.sub.7 assay, method 1:

[0580] Cell lines stably expressing human 5HT.sub.7 receptors are seeded in 96-well, poly-L-lysine-coated plates 48 hours prior to the assay (40,000 cells per well) in Dulbecco's Modified Eagle Medium (DMEM) containing 5% dialyzed serum. Twenty hours prior to the assay, the medium is changed to serum-free DMEM. On the day of the assay, the DMEM is washed and replaced with 30 μ l of assay buffer (1 $\times$ Krebs-Ringer bicarbonate glucose buffer, 0.75 mM IBMX, pH 7.4). A 10-min pre-incubation is performed in a 37-degree centigrade, humidified incubator. Then, the cells are stimulated by addition of 30 μ l of 2 $\times$ dilutions of compounds of the disclosure or chlorpromazine (final concentrations ranging from 0.1 nM to 10 M, each concentration assayed in triplicate). A positive control (100 μ M forskolin) is also included. Accumulation of cAMP is allowed to continue for 15 min, after which the buffer is removed and the cells are lysed with Cell Lysis Buffer (CatchPoint cAMP Assay Kit, Molecular Devices). Next, the lysates are transferred to 96-well, glass-bottom plates coated with goat anti-rabbit IgG and adsorbed with rabbit anti-cAMP (Molecular Devices). Following a 5 minute incubation, horseradish peroxidase-cAMP conjugate is added (Molecular Devices) and a 2-hour incubation is performed at room temperature. Then, after three washes with Wash Buffer (Molecular Devices), Stoplight Red substrate (Molecular Devices), reconstituted in Substrate Buffer (Molecular Devices) containing freshly-added 1 mM H.sub.2O.sub.2, is added and, after a 15-min incubation at room temperature, fluorescence is measured (excitation 510-545 nm, emission 565-625 nm). For each assay, a cAMP calibration curve is generated and controls without lysate and without antibody are included.

[0581] For agonist tests, raw data (maximum fluorescence, fluorescence units) for each concentration of the compounds of the disclosure or chlorpromazine are normalized to the basal (vehicle-stimulated) fluorescence (reported as fold increase over basal) and plotted as a function of the logarithm of the molar concentration of the drug (i.e., test or reference compound). Non-linear regression of the normalized data is performed in Prism 4.0 (GraphPad Software) using the built-in three parameter logistic model (i.e., sigmoidal concentration-response) describing agonist-stimulated activation of one receptor population:

$$y = \text{bottom} + \frac{(\text{top} - \text{bottom})}{(1 + 10^{\log EC_{50} - x})}$$

where bottom equals the best-fit basal fluorescence and top equals the best-fit maximal fluorescence stimulated by the compound of the disclosure or chlorpromazine. The log EC₅₀ (i.e., the log of the drug concentration that increases fluorescence by 50% of the maximum fluorescence observed for the compound of the disclosure or chlorpromazine) is thus estimated from the data, and the EC₅₀ (agonist potency) is obtained. To obtain an estimate of the relative efficacy of the test compound (Rel. Emax), its best-fit top is compared to and expressed as a ratio of that for the chlorpromazine (Rel. Emax of the reference agonist is 1.00).

[0582] To ascertain whether compounds of the disclosure are antagonists, a double-addition paradigm is employed. First, 30 μ l of a compound of the disclosure (20 M) is added (10 M final concentration) and a 15 minute incubation is performed. Then, 30 μ l of chlorpromazine (3 $\times$; EC₉₀) is added (final concentration of agonist is EC₃₀) and cAMP accumulation is allowed to proceed for 15 minutes. The samples are then processed for cAMP measurements as detailed

above. Measurements of chlorpromazine-induced cAMP accumulation are compared to the signals elicited by the chlorpromazine following addition of vehicle instead of test compound and expressed as a ratio. 'Hits' (compounds that antagonize chlorpromazine-stimulated increases in baseline-normalized fluorescence by at least 50%) are then characterized by a modified Schild analysis.

[0583] For modified Schild analysis, a family of chlorpromazine concentration-response isotherms is generated in the absence and presence of graded concentrations of test compound (added 15 min prior to reference agonist). Theoretically, compounds that are competitive antagonists cause a dextral shift of agonist concentration-response isotherms without reducing the maximum response to agonist (i.e., surmountable antagonism). However, on occasion, factors such as non-competitive antagonism, hemiequilibria, and/or receptor reserve cause apparent insurmountable antagonism. To account for such deviations, we apply the modified Lew-Angus method to ascertain antagonist potency (Christopoulos et al., 1999). Briefly, equieffective concentrations of agonist (concentrations of agonist that elicit a response equal to the EC_{sub}.25% of the agonist control curve) are plotted as a function of the compound of the disclosure concentration present in the wells in which they were measured. Non-linear regression of the baseline-normalized data is performed in Prism 4.0 using the following equation:

$$[00004] pEC_{25\%} = -\log([B] + 10^{-pK}) - \log c$$

where EC_{25%} equals the concentration of agonist that elicits a response equal to 25% of the maximum agonist control curve response and [B] equals the antagonist concentration; K, c, and s are fit parameters. The parameter s is equal to the Schild slope factor. If s is not significantly different from unity, pK equals pK_B; otherwise, pA₂ is calculated (pA₂=pK/s). The parameter c equals the ratio EC_{sub}.25%/ [B].

Functional Efficacy Assay for 5-HT_{SUB.7} Receptors Method 2:

[0584] Functional efficacy of the compounds of the disclosure on 5-HT_{sub.7} serotonin receptors were measured in a cell based cAMP enzyme fragment complementation assay using the HitHunter cAMP assay (DiscoverX). Cells stably expressing human 5HT_{sub.7} receptors were plated in 96-well plates at 4000 cells/well, 16-20 hours prior to assay in growth media (Ultraculture medium, 2 mM GlutaMax and G418 1 mg/mL. Serial dilutions of the agonist, 5-Carboxamidotryptamine (5-CT), were prepared in a final concentration range of 10 M to 10 nM. Compounds of the disclosure were prepared in 3-fold serial dilutions to obtain a final concentration range of 10 M to 0.1 nM. Compounds of the disclosure are tested for agonist activity in the absence of 5-CT and antagonist activity in the presence of 5-CT. For the cAMP assay, the protocol was followed according to the instructions provided by the supplier. Briefly, cells were incubated with a compound of the disclosure for 30 minutes at 37° C. prior to addition of EC_{sub.70} concentration of 5-CT. After an additional 30 minutes, cAMP antibody/cell lysis solution was added (20 L/well) and incubated for 60 minutes at room temperature. cAMP XS+EA reagent is added (20 L/well) and incubated for 2 hours at room temperature. Luminescence was read on the Envision Multilabel plate reader.

[0585] The disclosures of each and every patent, patent application, and publication cited herein are hereby incorporated herein by reference in their entirety.

[0586] While this invention has been disclosed with reference to specific embodiments, it is apparent that other embodiments and variations of this invention may be devised by others skilled in the art without departing from the true spirit and scope of the invention. The appended claims are intended to be construed to include all such embodiments and equivalent variations.

Claims

1. A compound having formula (I): ##STR00211## or a pharmaceutically acceptable salt thereof, wherein: R is selected from the group consisting of COR_{sup.2}, CO_{sub.2}R_{sup.2a}, and SO_{sub.2}R_{sup.2d}; R_{sup.2} is selected from the group consisting of C_{sub.1-6} linear alkyl, C_{sub.3-}

- 7 branched alkyl, and C.sub.3-7 cycloalkyl; R.sup.2a is selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, and C.sub.3-7 cycloalkyl; R.sup.2d is selected from the group consisting of C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl; R.sup.3 is phenyl or substituted phenyl, wherein the substituents are selected from the group consisting of hydroxyl, halo, cyano, C.sub.1-6 alkoxy, C.sub.1-6 linear alkyl, C.sub.3-7 branched alkyl, C.sub.3-7 cycloalkyl, and heterocyclyl; and n is 1, 2, 3, or 4.
2. The compound of claim 1, wherein R.sup.2, R.sup.2a, or R.sup.2d is C.sub.1-6 linear alkyl.
 3. The compound of claim 1, wherein R is COR.sup.2.
 4. The compound of claim 3, wherein R.sup.2 is C.sub.1-6 linear alkyl.
 5. The compound of claim 3, wherein R.sup.2 is C.sub.3-7 branched alkyl.
 6. The compound of claim 3, wherein R.sup.2 is C.sub.3-7 cycloalkyl.
 7. The compound of claim 1, wherein R is CO.sub.2R.sup.2a.
 8. The compound of claim 7, wherein R.sup.2a is C.sub.1-6 linear alkyl.
 9. The compound of claim 7, wherein R.sup.2, is C.sub.3-7 branched alkyl.
 10. The compound of claim 7, wherein R.sup.2, is C.sub.3-7 cycloalkyl.
 11. The compound of claim 1, wherein R is SO.sub.2R.sup.2d.
 12. The compound of claim 1, wherein R.sup.2d is C.sub.1-6 linear alkyl.
 13. The compound of claim 1, wherein R.sup.2d is C.sub.3-7 branched alkyl.
 14. The compound of claim 1, wherein R.sup.2d is C.sub.3-7 cycloalkyl.
 15. The compound of claim 1, wherein R.sup.3 is selected from the group consisting of 4-hydroxyphenyl, 3-fluorophenyl, 4-fluorophenyl, 3-chlorophenyl, 4-chlorophenyl, 3,4-dichlorophenyl, 4-fluoro-3-chlorophenyl, 4-cyanophenyl, 2-methoxyphenyl, 2-methylphenyl, 3-methylphenyl, 4-methylphenyl, 2-isopropylphenyl, 4-trifluoromethylphenyl, 2-morpholinophenyl, and 4-methyl-2-morpholinophenyl.
 16. The compound of claim 15, wherein R.sup.3 is 4-fluorophenyl.
 17. The compound of claim 1, wherein the compound is selected from a compound of Table 1, or a pharmaceutically acceptable salt thereof.
 18. The compound of claim 1, wherein the compound is selected from a compound of Table 9, or a pharmaceutically acceptable salt thereof.
 19. The compound of claim 1, wherein the compound is selected from a compound of Table 11, or a pharmaceutically acceptable salt thereof.
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